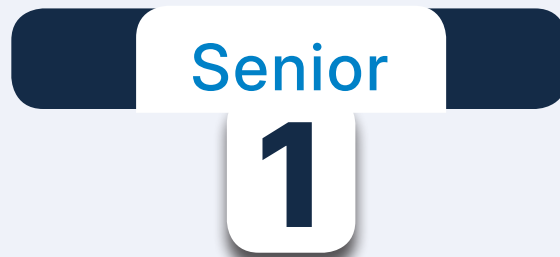




MATHEMATICS

Student's book



Version Edited in 2026

Experimental version

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FOREWORD

Dear student,

The Rwanda Basic Education Board (REB) is honoured to present the Mathematics book for Senior One (S1). This Textbook is designed to support the effective implementation of the Competence-Based Curriculum (CBC); ensuring learner-centered teaching practices that align with Rwanda's vision of empowering learners to reach their full potential, actively participate in society, and pursue meaningful careers.

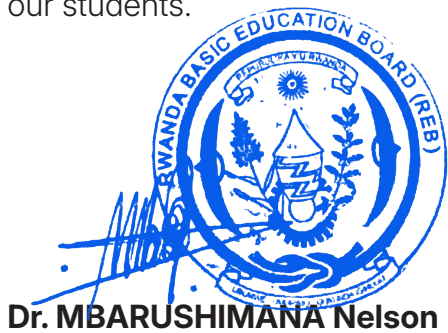
As Rwanda advances towards building a knowledge-based economy, education plays a pivotal role. Teachers are central to this transformation, guiding students to connect classroom knowledge with real-life experiences, thus fostering individual growth and national development.

This student book serves as a comprehensive resource for teaching Mathematics at Senior one in lower secondary level. The subject is an integral part of the CBC, aimed at equipping students with essential knowledge, skills, values, and attitudes necessary for becoming responsible and competent citizens. Through this subject, students are encouraged to develop problem-solving abilities and contribute positively to their communities.

In pursuit of quality education, the Government of Rwanda emphasizes the importance of using teaching and learning materials that are aligned with the curriculum.

Although the teacher's guide provides answers to all pupil's activities, teachers are encouraged to independently work through the exercises to better understand and evaluate students' responses.

I sincerely thank everyone who contributed to the refinement of this textbook. Special recognition goes to the Curriculum Officers who coordinated the entire process. We thank all partners, lecturers, teachers, and individuals who were involved in the process from start to the end. Every feedback was highly valued and will help improve learning outcomes of our students.



Dr. MBARUSHIMANA Nelson

Director General

ACKNOWLEDGEMENT

I would like to extend my sincere appreciation to everyone who contributed to the refinement of the Mathematics student's book for Senior One (S1). This achievement would not have been possible without the active involvement of various stakeholders.

Special thanks go to Rwanda Basic Education Board (REB) staff, subject experts, university lecturers and teachers for their valuable work in the adaptation of this textbook.

I am also grateful to all our education development partners whose support was instrumental throughout the process.



MURUNGI Joan

Head of CTLR department

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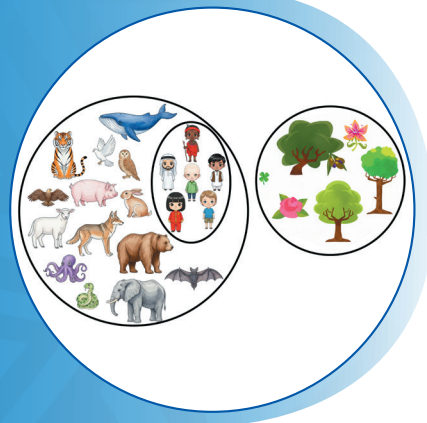
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UNIT 1

SETS



Key Unit Competence:

Use set notations, Venn diagrams and relations in solving related problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Define sets, cardinality, subsets (with formulas), and represent relationship and operations using Venn and Papygram diagrams.
- Explain ordered pairs, Cartesian products, domains, ranges, graphs, relations and functions.

Skills

- Represent sets using roster and set-builder notation, construct Venn diagrams, and perform set operations with proper notation.
- Compute Cartesian products, analyze relations for reflexivity, symmetry, and transitivity, and classify equivalence relations.
- Classify mappings as injective, surjective, bijective, or neither

Attitudes and values

- Value diagrams (Venn, Papygram) for simplifying complex ideas and develop logical approaches to problem-solving.
- Demonstrate confidence in representing mathematical relationships in multiple forms and collaborate effectively in group activities.

INTRODUCTION

This unit introduces the fundamental ideas of groups or **sets** and **relations** among members or **elements** of the set, which form the foundation of many topics in Mathematics. The purpose of this unit is to help you understand how objects can be grouped, compared, and related to one another. These skills are important for developing logical thinking, problem-solving, and mathematical reasoning, which will be used in higher levels of mathematics and in everyday life. In this unit, you will study: Set concept and relations.

The key words to be explained in this unit are the following: Set, element, membership ($\in$), empty set ($\emptyset$), singleton set, finite set, infinite set, cardinal number, subset, number of subsets, universal set, complement, union ($\cup$), intersection ($\cap$), difference, symmetric difference, Venn diagram, equal sets, equivalent sets, relation, ordered pair, Cartesian product, and particular relations: mapping and functions.



Introductory activity

Observe the picture and answer to the question that follows.



How can you group objects such as books, clothes or food in a store?

What is the importance of grouping things?

1.1 Meaning of the set and its representation



Activity 1.1

- 1) Identify and list any ten items in your home that can be grouped together.
- 2) Name the groups you have identified in step 1 above.

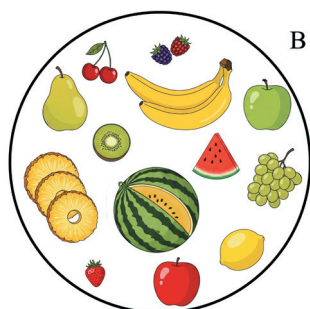


Content Summary

A Set: A set is group of items with a common feature. Most things in life have common features. Such features are used to group things together. A set is a well-defined collection of distinct objects considered as a whole. We write a set using a capital letter.

Examples of sets: A set of animals, a set of houses, set of cars and so on. An object or item in a set is called a member or **an element** of that set.

Example 1:



B is a set of fruits; Then, watermelon, banana, etc are the elements of the set B.

Example 3: The Set of courses a student is taking: Mathematics, English, Biology are members of that set.

Example 4: The Set of Unique Letters in the Word "MISSISSIPPI: M, I, S, P are members of that set.

Representation of a set:

There are three forms commonly used to describe or represent a set: **Statement form**, **Roster or tabular form** and **the Set builder form**.

1. Statement form

The set is described in words. It is also called **the descriptive form**.

Examples:

- a) A is the set of all even natural numbers less than 10. Its elements are 2, 4, 6, and 8.
- b) B is the set of vowels in the English alphabet.
- c) C is the set of planets in our solar system.
- d) The set of alphabets a, b, c, d and e. These letters can be described using a statement as follows: The set of first five letters of the alphabets.

- e) The set of the numbers 2, 4, 6, 8, 10. This can be described as: The set of first five even numbers or the first five natural numbers divisible by 2.

2. Roster or tabular form:

All elements are listed within curly bracket {}, separated by commas. Order doesn't matter, repetition is not accepted.

Example:

Consider the set of all the letters in the word Mathematics. This set can be described by listing all the elements in it. If we let this set be N , then $N = \{m, a, t, h, e, i, c, s\}$.

Note that this form of describing sets is known as **Roster or tabular form**.

In the roster form, **each element is written only once**. For example, in the word mathematics, the letters m, a and t, each appears twice but we list each of them only once.

For large regular patterns, ellipsis (...) may be used:

$D = \{1, 3, 5, \dots, 99\}$: The set of odd numbers from 1 to 99.

3. Set builder form

In this form, we describe a set by using a rule or an algebraic formula or property which defines only the set we want: $\{x / \text{condition on } x\}$ and read as: "The set of all x such that the condition holds.»

Example 1:

$A = \{\text{The integer } x: 1 < x < 10\}$ is read as the set of integers x such that x lies between 1 and 10.

Thus $A = \{2, 3, 4, 5, 6, 7, 8, 9\}$. When a set is described using an algebraic expression, we say we are using set builder notation.

The symbol ":" in the statement $\{\text{The natural numbers } x: 1 < x < 10\}$ is used to denote the statement "such that".

Example 2:

Describe set T in words, given that $T = \{x : x = 2n - 1, \text{ where } n = 1, 2, 3, \dots\}$

Solution

n	1	2	3	4	5	6	7	...
x	1	3	5	7	9	11	13	...

We see that **T is the set of odd numbers** and $T = \{1, 3, 5, 7, 9, 11, 13, \dots\}$.

Example 3: Sets A and B are subsets of the set of natural numbers such that $A = \{x: 1 \leq x \leq 7\}$ and

$B = \{x: 2 \leq x \leq 20 \text{ and } x \text{ is an even number}\}$.

List the members of: (i) set A (ii) set B.

Solution

$A = \{x: 1 \leq x \leq 7\}$ means that x is a positive natural number between 1 and 7 inclusive.

$A = \{x: 1 \leq x \leq 7\} = \{1, 2, 3, 4, 5, 6, 7\}$

$B = \{x: 2 \leq x \leq 20, x \text{ is an even number}\}$ means all even numbers between 2 and 20 inclusive.

$B = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$.

$B = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$ and $A = \{1, 2, 3, 4, 5, 6, 7\}$.

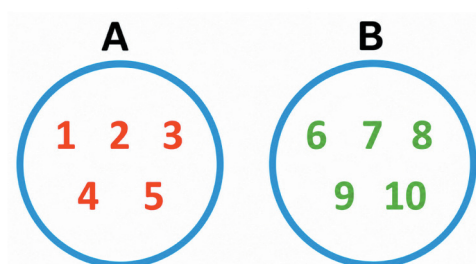
Sets A and B have some common members; these are 2, 4 and 6.

4. Venn Diagram

A set can be represented by a Venn diagram or a pictorial representation using closed curve (usually a circle).

Example:

Suppose we have two sets A and B where $A = \{1, 2, 3, 4, 5\}$ and $B = \{6, 7, 8, 9, 10\}$. We can see that the two sets A and B have no element in common. Therefore, they will be represented as disjoint circles. The Venn diagram for this data will be represented as:



APPLICATION ACTIVITY 1.1

- 1) Identify the five sets and describe them in a similar form.
- 2) Carry out the following activities:
 - a) Write down the first ten members of the set of even counting numbers.
 - b) List and write down any five members of the set of carnivorous animals.
- 3) Given sets $J = \{1, 2, 3, 4\}$, $K = \{2, 3, 5, 7\}$, $L = \{1, 2, 5, 8\}$, $M = \{3, 4, 5, 8\}$ and $N = \{9, 10\}$, represent them in different Venn diagrams.

1.2 The special sets: Empty set, singleton, disjoint, finite and infinite sets



Activity 1.2

Read and answer to the questions:

In the school store:

- A cupboard has **no chalk**
- A drawer has **one ruler**
- A box has **20 exercise books**
- The road outside has **endless stones**.

Question:

1. How do we name a group or set with **no items**?
2. How do we name a group or set with **only one item**?
3. How do we name a group or set with **countable items**?
4. How do we name a group or a set with items we **cannot finish to count**?



Content Summary

We have:

Type of Set	Definition	Number of Elements
Empty set	A set with no elements	0
Singleton	A set with exactly one element	1
Finite set	A set with a limited number of elements	Limited, countable
Infinite set	A set with unlimited number of elements	Unlimited, uncountable

Sets which have no common members are called **disjoint sets**. For example, the set of alphabets and the set of natural numbers are disjoint sets.

Example 1

Match each description in Column A with the correct type of set in Column B.

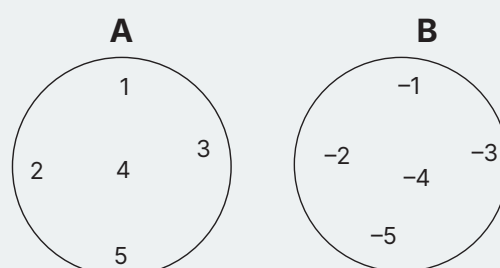
Column A	Column B
1. Set of classroom with no pupils during holidays	A. Infinite set
2. Set of Rwandan road vehicles in Kigali	B. Singleton set
3. Set of the President of USA	C. Empty set
4. Set of even numbers	D. Finite set

Solution $1 \rightarrow C$ (Empty set) $3 \rightarrow B$ (Singleton set) $2 \rightarrow D$ (Finite set) $4 \rightarrow A$ (Infinite set)**Example 2:**

Discuss if the set A and B are disjoint sets. Suppose $A = \{1, 2, 3, 4, 5\}$ and $B = \{-1, -2, -3, -4, -5\}$.

Solution

The members of set A and those of set B are very different. No members of A belongs to set B and vice versa. Set A and Set B are said to be disjoint and can be illustrated in the following diagram.

**APPLICATION ACTIVITY 1.2**

- 1) State whether each of the following is an **empty set, singleton set, finite set, or infinite set**.
 - a) The set of students in a classroom at 10:00 pm.
 - b) The set of head teachers of your school this year.
 - c) The set of provinces of Rwanda.
 - d) The set of natural numbers.
 - e) The set of cows in a farmer's kraal in your District.
 - f) The set of meter markings on a measuring tape going on forever.
- 2) Write True (T) or False (F).
 - a) A singleton set has two elements.
 - b) The set of students in a school is a finite set.
 - c) The set of stars in the sky is a finite set.
 - d) The empty set has one element.
 - e) The set of integers is an infinite set.

1.3 Membership of a set and cardinal number of a set



Activity 1.3

- 1) Write down the set of four legged domestic animals you know.
- 2) Is a cow a member of that set? If so, how can you represent this in set notation?
- 3) Is a lion a member of the above set? If so, how can you represent this in set notation?



Content Summary

We use symbol $\in$ to mean "is a member of" and $\notin$ to mean "is not a member of".

Example 1:

Given the set $A = \{1, 3, 5, 7, 9, 11\}$, $5 \in A$ is read as 5 is a member of set A. $8 \notin A$ read as 8 is not a member of the set A.

Consider set $T = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$.

The number of elements (n) in the set T is 9. This is written as $n(T) = 9$. The number of elements in the set is called the **cardinality** or **cardinal number** of the set.

Suppose $n(A)$ means the number of elements in set A.

If D = set of days of the week, then $n(D) = 7$.

Suppose F = set of two legged cows. Then: $n(F) = 0$. This means the set F has no members. Therefore, set F has no cardinal number i.e. 0 is the cardinal number of the set F .

Example 2:

- a) List down the elements of the set G of vowels in the word "algebra".
- b) Describe set B using words, given that $B = \{1, 3, 5, 7 \dots\}$.
- c) List all the elements of set A given that $A = \{x: -3 < x < 8, \text{ and } x \text{ is an integer}\}$ and give the cardinal number of the set A .
- d) Complete by $\in$ or $\notin$:
 - i) $4 \underline{\hspace{1cm}} A$
 - ii) $8 \underline{\hspace{1cm}} A$
 - iii) $10 \underline{\hspace{1cm}} A$
 - iv) $3 \underline{\hspace{1cm}} A$
 - v) $7 \underline{\hspace{1cm}} A$

Solution:

- a) $G = \{a, e\}$
- b) B is the set of all odd numbers.
- c) Set $A = \{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7\}$. $n(A) = 10$.

d) Given that Set $A = \{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7\}$,

- | | |
|--------------------|---------------|
| i) $4 \in A$ | iv) $3 \in A$ |
| ii) $8 \notin A$ | v) $7 \in A$ |
| iii) $10 \notin A$ | |



APPLICATION ACTIVITY 1.3

- Find the cardinal number of each set. $A = \{\text{pen, book, ruler, eraser}\}$ $B = \{2, 4, 6, 8, 10\}$, $C = \{a, e, i, o, u\}$, $D = \{7\}$ $E = \{\text{Monday, Tuesday, Wednesday}\}$.
- If $n(A) = 6$ and 2 elements are removed, what is the new cardinal number?
- If $n(B) = 4$ and 3 elements are added, what is $n(B)$ now?
- Can the cardinal number of a set be negative? Explain.
- $A = \{2, 4, 8, 10, \dots, 20\}$, $B = \{5, 10, 15, 20, 25, 30, \dots, 50\}$. Complete by $\in$ or $\notin$:
i) $4 \underline{\hspace{1cm}} A$ ii) $8 \underline{\hspace{1cm}} B$ iii) $10 \underline{\hspace{1cm}} A$ iv) $3 \underline{\hspace{1cm}} B$ v) $45 \underline{\hspace{1cm}} B$

1.4 Subset of a set and number of subsets



Activity 1.4

Consider the set $A = \{\text{lion, cow, buffalo, goat, pig, dove, tilapia, weaver bird, crocodile, pigeons, snake}\}$.

- Discuss and write down other sets that can be formed from set A above.
- Name the sets you formed in step 1 above.



Content Summary

Subset of a set:

A set B is a subset of a set A if every element of B is also an element of A . In other words, B is a subset of the set A if whenever $a \in B$, then $a \in A$.

The empty set denoted by $\{\}$ or $\emptyset$ is also a subset of any set. To show a subset, we use the symbol $\subset$. This means, B is a subset of A is written as $B \subset A$.

Example 1: Let us consider A as the set of domestic animals.

The set $B = \{\text{cow, goat, pig}\}$ is a set containing members which are all the domestic animals. B is a subset of the set A . we write it by: $B \subset A$.

Similarly, we can get the set $M = \{\text{lion, weaver bird, pigeon}\}$. It has the members which are not domestic animals. Therefore, the set M is not a subset of the set A . we write it by: $M \not\subset A$.

A set formed by some elements or all the elements from a given set is called a subset.

Natural numbers are counting numbers from 1 to infinity. The set of natural numbers is represented as set $\mathbb{N} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, \dots\}$.

Even numbers are numbers divisible by 2. Set E of even numbers is represented as set $E = \{2, 4, 6, 8, \dots\}$.

Therefore, the set of even numbers is a subset of the set of natural numbers. This can be represented as $E \subset \mathbb{N}$.

Odd numbers are numbers which are not divisible by 2. The set of odd numbers is represented by $D = \{1, 3, 5, 7, 9, 11, 13, 15, \dots\}$.

The set of odd numbers is a subset of the set of natural numbers which is written as $D \subset \mathbb{N}$.

A prime number is a number which is divisible by one and itself only. In other words, prime numbers have only two factors. The set P of prime numbers is represented as set $P = \{2, 3, 5, 7, 11, 13, 17, \dots\}$. The set of prime numbers is a subset of natural numbers represented as $P \subset \mathbb{N}$.

Number of subsets for a set:

If the number of elements of the set A is n, then the number of subsets of A is given by $N = 2^n$.

Example 2: To be done as a practical activity:

Mutesi has 4 friends: Peter, Amanda, Keza and Innocent. Mutesi needs to organize a Birthday Ceremony. She needs to know the number of different ways she can invite her friends. Help her to know that number taking note that deciding to invite no one of those friends is one way.

Solution:

Let the set of friends be: $F = \{\text{Peter, Amanda, Keza, Innocent}\}$. Number of friends $n(F) = 4$.

Let: P = Peter, A = Amanda, K = Keza, I = Innocent. We can list ways of inviting them: Choosing whom to invite is the same as choosing a subset of F:

- She can decide to invite any one of them: 1 way.
- She can invite one friend: 4 ways: $\{P\}, \{A\}, \{K\}, \{I\}$
- She can invite 2 of them: 6 ways: $\{P, A\}, \{P, K\}, \{P, I\}, \{A, K\}, \{A, I\}, \{K, I\}$
- She can invite 3 of them: 4 ways: $\{P, A, K\}, \{P, A, I\}, \{P, K, I\}, \{A, K, I\}$
- She can invite all of them: 1 way: $\{P, A, K, I\}$.

We see that the total is 16 and $N = 2^4 = 2 \times 2 \times 2 \times 2 = 16$.

Example 3: Given set $B = \{a, b, c\}$, list all the subsets of set B in your exercise book.

Solution:

The subsets of set B are $\{\}$, $\{a\}$, $\{b\}$, $\{c\}$, $\{a, b\}$, $\{a, c\}$, $\{b, c\}$ and $\{a, b, c\}$.

The number of subsets $= 2^3 = 8$.

Example 4:

Use the symbol $\in$, $\notin$, $\subset$ or $\not\subset$.

a) $3 \underline{\hspace{1cm}} \{1, 3, 6, 9\}$

d) $75 \underline{\hspace{1cm}} \{5, 10, 15, 20, 25, \dots\}$

b) $\{o, a, b, c, d, i\} \underline{\hspace{1cm}} \{a, b, c, d, e, f, g\}$

e) $\{9, 15\} \underline{\hspace{1cm}} \{3, 9, 12, \dots, 30\}$

c) $\{\} \underline{\hspace{1cm}} \{3, 5\}$

Solution:

a) $3 \in \{1, 3, 6, 9\}$

b) $\{o, a, b, c, d, i\} \not\subset \{a, b, c, d, e, f, g\}$

c) $\{\} \subset \{3, 5\}$ Empty set is a subset for every set.

d) $75 \in \{5, 10, 15, 20, 25, \dots\}$ set of multiples of 5.

e) $\{9, 15\} \subset \{3, 9, 12, \dots, 30\}$ set of multiples of 3.

Example 5

$A = \{2, 4, 8, 12\}$

$C = \{9, 19\}$ and

$B = \{5, 10, 15, 20, \dots, 50\}$,

$D = \{1, 3, 5, 7, 9, \dots, 25\}$.

Comment using True or False.

a) $A \subset D$

c) $C \subset D$

b) $A \subset B$

d) $D \not\subset C$

Solution:

a) $A \subset D$ False

c) $C \subset D$ True

b) $A \subset B$ False

d) $D \not\subset C$ True.



APPLICATION ACTIVITY 1.4

1) Copy and complete this table:

Set	Subsets	Number of Subsets
$\{\}$	$\{\}$	1
$\{1\}$		
$\{1, 2\}$		
$\{1, 2, 3\}$		

2) Consider another set B such that

$B = \{\text{HIV, Gonorrhoea, Syphilis, Trichomonas, Genital Herpes}\}$

Discuss and write down all the subsets of B. How many subsets have you obtained?

3) A certain set B has 32 subsets. How many elements are there in the set B?

1.5 Representing sets using Venn diagrams



Activity 1.5

1) Name animals you know. Write them on the board or in the exercises' book.

2) Classify them into mammals, birds, and fish.

Discuss:

- Some animals may belong to more than one group (e.g., animals that can live in water and on land).
- Draw overlapping circles on the board for the groups. Place animals in the correct part of the circles:
 - Mammals → first circle M.
 - Birds → second circle B.
 - Fish → third circle F.



Content Summary

- Venn Diagram: A diagram drawn using circles to show how items in different groups relate.
- Set: A collection of objects with something in common.
- Circle: Represents a set.
- Overlap area: Shows items that belong to more than one set.
- Items outside the circles: Belong to none of the sets shown.

NB: Venn diagrams help us organize information visually. They can be used to compare groups, show relationships, and identify similarities and differences.

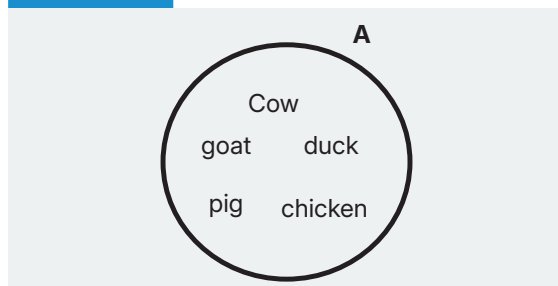
Example 1:

Set of animals in a farm:

$A = \{\text{cow, goat, pig, chicken, duck}\}$.

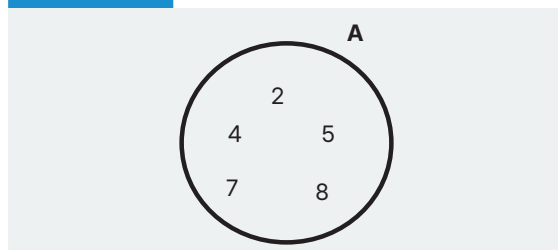
Represent set A on a Venn diagram.

Solution:



Example 2: Given set $A = \{2, 4, 5, 7, 8\}$, represent set A on a Venn diagram.

Solution:



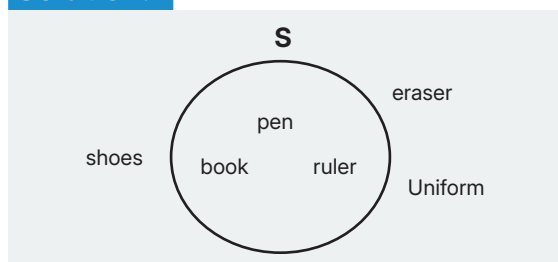
Example 3: Set: $S = \{\text{pen, book, ruler}\}$. **Other items:** eraser, uniform, shoes

Step 1: Draw a circle labelled S.

Step 2: Place pen, book, ruler inside the circle.

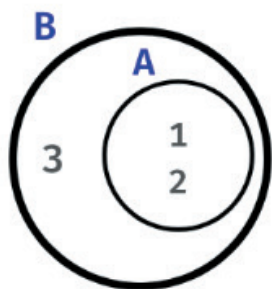
Step 3: Place eraser, uniform, shoes outside the circle.

Solution:



Example 4

Given the set A and B



- Write A and B in the roster form (list of elements).
- Complete using $\subset$ or $\not\subset$
 - $A \underline{\hspace{1cm}} B$
 - $B \underline{\hspace{1cm}} A$
- Complete:
 - $n(A) = \underline{\hspace{1cm}}$
 - $n(B) = \underline{\hspace{1cm}}$

Solution

- $A = \{1, 2\}$. $B = \{1, 2, 3\}$.
- Complete using $\subset$ or $\not\subset$
 - $A \subset B$
 - $B \not\subset A$.
- Number of elements:
 - $n(A) = 2$
 - $n(B) = 3$



APPLICATION ACTIVITY 1.5

Draw a Venn diagram that represents the following:

- 1) Set of fruits in a basket: {apple, banana, mango}. Other items: orange, tomato, carrot are not in the basket.
- 2) Set of students wearing uniform: {John, Mary, Grace}. Other students: Paul, Peter do not.

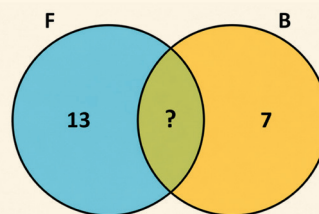
1.6 Intersection of sets



Activity 1.6

- 1) In a class of 30 students where each student plays either basketball or football: 18 students play football and 12 play Basketball. Let F represent the set of students who play Football, B the set of students who play Basketball.

How many students play both Football and Basketball if 7 students play **only basketball**?



- 2) Consider the sets $A = \{6, 7, 8, 9, 10\}$, $B = \{1, 2, 3, 4, 5, 6, 7\}$, $C = \{2, 4, 6, 8, 10, 12\}$ and $D = \{1, 3, 5, 7, 9\}$.

List the common elements between the following sets:

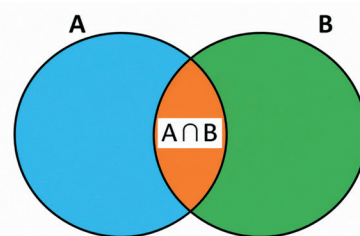
- | | |
|-------------|-------------------|
| (a) A and B | (c) B and D |
| (b) B and C | (d) A and B and C |



Content Summary

The intersection of two or more sets is the set of elements that are common to those given sets. In other words, it is the set of elements that belong to every set at the same time.

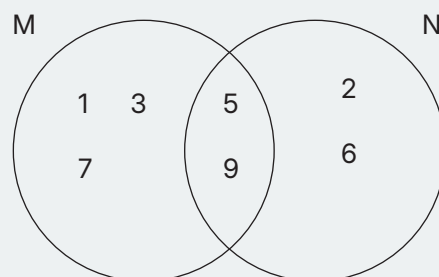
The intersection of the set A and B is written as $A \cap B$ and read " **A intersection B**". On our diagram, $A \cap B$ is the common region between A and B and it is shared with orange colour.



Example 1: Given sets $M = \{1, 3, 5, 7, 9\}$ and $N = \{2, 5, 6, 9\}$; find $M \cap N$. Represent set M and N on a Venn diagram.

Solution:

$$M \cap N = \{1, 3, 5, 7, 9\} \cap \{2, 5, 6, 9\} = \{5, 9\}.$$



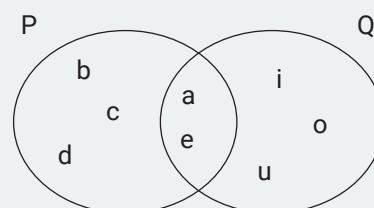
on the Venn diagram, of intersection of sets overlaps each other as shown in the example.

Example 2: Given that set P = The set of the first 5 letters of the alphabet and set Q = the set of all the vowels;

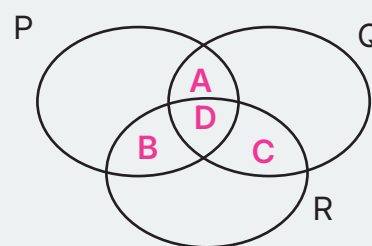
- List the elements of set P and Q.
- Find $P \cap Q$.
- Draw a Venn diagram to illustrate set $P \cap Q$.

Solution:

- $P = \{a, b, c, d, e\}$ and $Q = \{a, e, i, o, u\}$
- $P \cap Q = \{a, b, c, d, e\} \cap \{a, e, i, o, u\} = \{a, e\}$
- The Venn diagram is as shown:



If there are three sets with intersection, they can be represented on a Venn diagram as shown:



- Region **A** represents intersection of sets P and Q only.
- Region **B** represents intersection of sets P and R only.
- Region **C** represents intersection of sets Q and R only.
- Region **D** represents the intersection of all the three sets P, Q and R which is written as $(P \cap Q \cap R)$.

The unlabelled regions represent the elements of each set which is not common with the other sets.

Example 3: You are given the following subsets of the set of integers.

- $A = \{x: x \text{ is a prime number less than } 20\}$,
- $B = \{x: x \text{ is an odd number less than } 15\}$
- $C = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Draw a Venn diagram to represent the intersection of sets A, B and C.

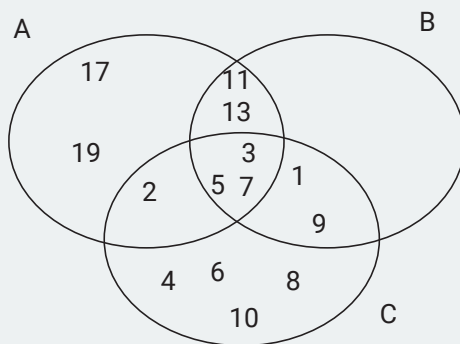
Solution

First, list the elements of set A, B and C as follows; $A = \{2, 3, 5, 7, 11, 13, 17, 19\}$,
 $B = \{1, 3, 5, 7, 9, 11, 13\}$, $C = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ Then list all the common elements between 2 sets $A \cap B = \{3, 5, 7, 11, 13\}$,

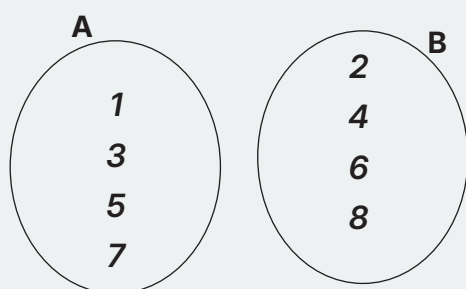
$A \cap C = \{2, 3, 5, 7\}$, $B \cap C = \{1, 3, 5, 7, 9\}$

Then list all the common elements between 2 sets, $A \cap B \cap C = \{3, 5, 7\}$.

The Venn diagram is as follows:



Note: When intersection of two sets A and B is an empty set, i.e. $A \cap B = \emptyset$, the set A and B are called **disjoint sets**.

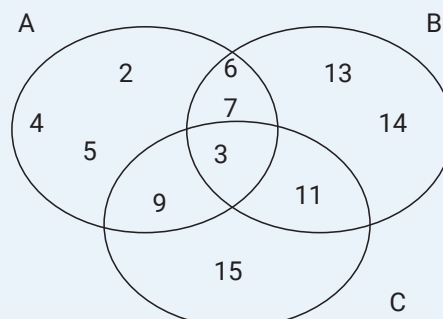
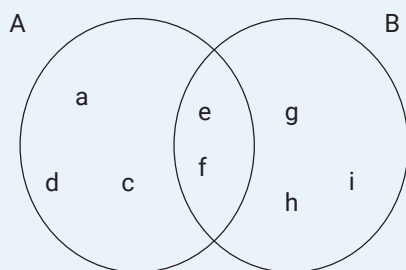


$A = \{1, 3, 5, 7\}$ $B = \{2, 4, 6, 8\}$. We see that $A \cap B = \emptyset$. A and B are disjoint sets.



APPLICATION ACTIVITY 1.6

- 1) List elements of the sets in the Venn diagrams below and for each case, provide the set $A \cap B$.



- 2) Given sets $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{3, 6, 9, 12, 15\}$ and $C = \{9, 10, 11, 12, 13, 14, 15, 16, 17\}$, draw Venn diagrams for the following sets and identify elements for:

- a) $A \cap B$ (b) $A \cap C$
c) $B \cap C$ (d) $A \cap B \cap C$

1.7 Union of sets



Activity 1.7

- 1) In a classroom, there are Students who like **meat**: $M = \{\text{Karabo, Rugira, Kanani, Bwiza, Keza}\}$, Students who like **the bread**: $B = \{\text{Giraneza, Kanyarwanda, Kamariza, Cyusa}\}$

Make a set of all students who:

- a) Like **both** pizza **and** bread.
b) Like **pizza or bread or both**.

- 2) You are given the universal set $U =$ set of natural numbers below 12 and sets $A = \{4, 5, 6, 7, 8\}$, $B = \{0, 1, 3, 4, 5\}$, $C = \{2, 4, 6, 8, 10\}$ and $D = \{1, 3, 5, 7, 11\}$.

Discuss and write down the following sets:

- a) The elements of set A combined with the set B.
b) The elements of set C which are not in the universal set.
c) The elements of set D which are not in the universal set.
d) The elements of sets A and C which are not in the universal set.



Content Summary

The **union** of two sets A and B is a set that contains **all elements** that belong to at least one of the sets A or B (or both). **A union B** is written as $A \cup B$.

Definition in set-builder notation: $A \cup B = \{x | x \in A \text{ or } x \in B\}$.

In a Venn diagram, the union is represented by shading the entire area covered by both sets. You realize that, elements of two or more sets can be put together to form a set. The set formed is known as the union of sets. If there are common elements, they should be written once.

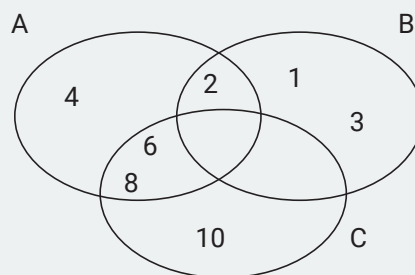
Example:

- a) Given the following sets $A = \{a, b, c, d, e, f\}$ and $B = \{a, b, c, h, i\}$, find: (i) $n(A)$, (ii) $n(B)$, (iii) $(A \cup B)$ (iv) $n(A \cup B)$
- b) If $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3\}$, $C = \{6, 8, 10\}$, $D = \{2, 3, 6\}$, find:
- (i) $n(A)$ (iv) $n(A) + n(B)$, (vii) $n(A \cup B \cup C)$,
 (ii) $n(B)$ (v) $n(A) + n(C) - n(B)$, (viii) $A \cup C \cup D$
 (iii) $n(C)$, (vi) $A \cup B \cup C$ (ix) $n(A \cup C \cup D)$,
 (x) Draw Venn diagrams to represent $A \cup B \cup C$ and $A \cup C \cup D$

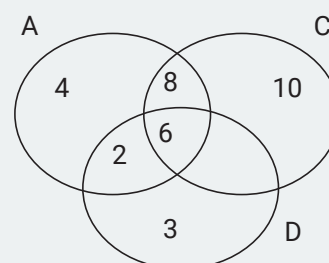
Solution:

- a) (i) $n(A) = 6$, (ii) $n(B) = 5$,
 (iii) $A \cup B = \{a, b, c, d, e, f\} \cup \{a, b, c, h, i\} = \{a, b, c, d, e, f, h, i\}$,
 (iv) $n(A \cup B) = 8$.

- b) (i) $n(A) = 4$ (ii) $n(B) = 3$, (iii) $n(C) = 3$,
 (iv) $n(A) + n(B) = 4 + 3 = 7$
 (v) $n(A) + n(C) - n(B) = 4 + 3 - 3 = 4$
 (vi) $A \cup B \cup C = \{2, 4, 6, 8\} \cup \{1, 2, 3\} \cup \{6, 8, 10\} = \{1, 2, 3, 4, 6, 8, 10\}$
 (vii) $n(A \cup B \cup C) = 7$
 (viii) $A \cup C \cup D = \{2, 4, 6, 8\} \cup \{6, 8, 10\} \cup \{2, 3, 6\} = \{2, 3, 4, 6, 8, 10\}$
 (ix) $n(A \cup C \cup D) = 6$
 (x) $A \cup B \cup C$



This represents $A \cup B \cup C$



This represents $A \cup C \cup D$



APPLICATION ACTIVITY 1.7

- Given the sets $A = \{a, b, c, d\}$, $B = \{a, e, i, o\}$, $C = \{b, e, d, f\}$ and $D = \{a, d, e, g\}$: Find:
 - $A \cup B$
 - $B \cup C$
 - $A \cup B \cup C$
 - $A \cup C \cup D$
 - $C \cup D \cup B$
 - $C \cup D$
- Find the union of the following sets:
 - $A =$ Set of even numbers from 0 to 20, $B =$ Set of integers greater than -2 but less than 9, $C =$ Set of prime numbers between 1 and 7.
 - $X = \{\text{stool, chair, table, bed}\}$, $Y = \{\text{chair, beans, lemon}\}$
- Draw a Venn diagram for the pair of sets P and Q such that $P = \{1, 3, 5, 7, 11, 13\}$, $Q = \{3, 6, 9, 12, 15\}$. Work out:
 - $(P \cap Q)$ and $(P \cup Q)$
 - i) $n(P)$, ii) $n(Q)$, iii) $n(P) + n(Q)$ iv) $n(P \cap Q)$ and $n(P \cup Q)$.
 - Verify that: $n(P \cup Q) = n(P) + n(Q) - n(P \cap Q)$.
- For each case from (a) to (d), draw a Venn diagram with 3 overlapping sets A , B and C .
Then shade: a) $A \cap B$. b) $B \cup C$ c) $A \cup C$ d) $B \cap C$ e) $A \cap B \cap C$.

1.8 Simple and symmetric differences of sets



Activity 1.8

Given set $A = \{\text{mangoes, pawpaw, oranges}\}$, set $B = \{\text{apples, pawpaw, strawberry}\}$, Find:

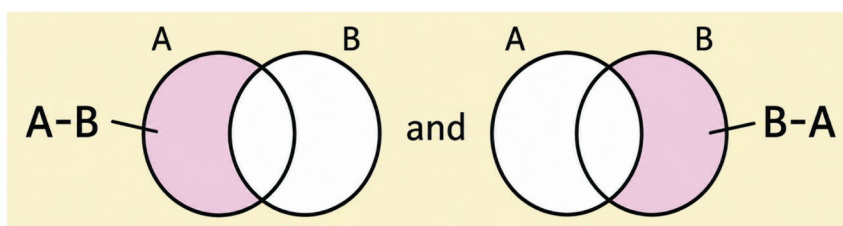
- Set of elements of A which do not belong to set B .
- Set of elements of B which do not belong to set A .
- Union of the set found in (a) and the set found in (b).



Content Summary

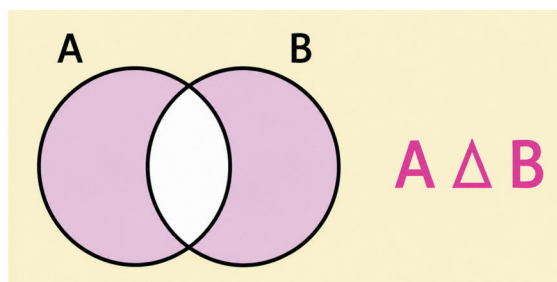
Simple difference between sets A and B written as $A-B$ or A/B is the set of the elements of set A which are not in set B . Similarly, $B-A$ or B/A is difference between sets B and A . This is the set of elements that are in set B and not in set A .

$A-B$ and $B-A$ can be shown using a Venn diagram as follows:



The union of sets $A-B$ and $B-A$ is known as the *symmetric difference* between sets A and B .

It is written in symbols as $A \Delta B$ to mean $A \Delta B = (A-B) \cup (B-A)$. We read $A \Delta B$ as "**A symmetric difference B**" or "**A delta B**". It can be shown using a Venn diagram as illustrated here:



Example 1:

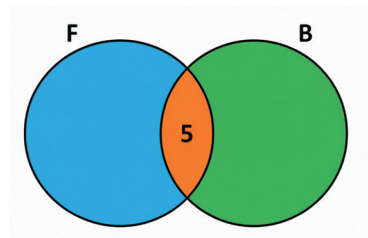
Given that $A = \{3, 4, 5, 6, 7, 8\}$ and $B = \{2, 4, 8, 12\}$, find: a) $A - B$ b) $B - A$ c) $A \Delta B$

Solution:

- a) $A - B = \{3, 5, 6, 7\}$
- b) $B - A = \{2, 12\}$
- c) $A \Delta B = (A-B) \cup (B-A) = \{2, 3, 5, 6, 7, 12\}$

Example 2:

In a class of 25 students where each student plays either basketball or football: 18 students play football and 12 play Basketball. Let F represent the set of students who play Football, B the set of students who play Basketball. There are 5 students who play both games.



- a) How many students play one game only?
- b) Complete $n(F \Delta B) = \underline{\hspace{2cm}}$

Solution

$$n(F) = 18 \text{ and } n(B) = 12. \quad n(F \Delta B) = n(F-B) + n(B-F)$$

$$\text{However, } n(B \cap F) = 5.$$

$$\text{Therefore, } n(B-F) = n(B) - n(B \cap F) = 12 - 5 = 7$$

$$\text{And } n(F-B) = n(F) - n(B \cap F) = 18 - 5 = 13.$$

$$\text{Therefore } n(F \Delta B) = n(F-B) + n(B-F) = 7 + 13 = 20.$$



APPLICATION ACTIVITY 1.8

- 1) Given the sets $P = \{1, 2, 3, 4, 5, 6\}$ and $Q = \{2, 4, 7, 8\}$, find:
 - a) $P - Q$, b) $Q - P$, c) $Q \Delta P$
- 2) If set $A = \{\text{tomatoes, onions, green, pepper}\}$ and $B = \{\text{onions, potatoes, rice}\}$: find:
 - a) $A - B$ (b) $B - A$ c) $A \Delta B$
- 3) Sets $C = \{3, 4, 5, 6, 7\}$ and $D = \{5, 6, 7\}$. Find:
 - a) $C - D$, b) $D - C$ c) $C \Delta D$ d) $n(C \Delta D)$.

1.9 Universal set and complement of sets



Activity 1.9

You are given the set ε = set of natural numbers below 12 and sets $A = \{4, 5, 6, 7, 8\}$, $B = \{0, 1, 3, 4, 5\}$, $C = \{2, 4, 6, 8, 10\}$ and $D = \{1, 3, 5, 7, 11\}$.

In pairs, write down the following sets:

- The elements of set ε which are not in the set A.
- The elements of set ε which are not in the set C.
- The elements of set ε which are not in the set D.
- The elements of sets A and C which are not in the set ε .



Content Summary

A universal set is a collection of all elements or members of all the related sets, known as its subsets. In a particular context:

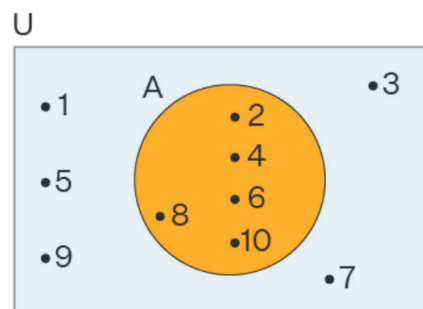
- Universal set is the superset of all sets.
- All sets are subsets of universal set.

A set that contains all the subsets under consideration is known as a universal set. A universal set is denoted by the symbol ε or U .

For example, the set of all real numbers is the universal set in the context of sets of rational numbers, irrational numbers, integers, whole numbers, natural numbers, etc.

In the universal set U below, we can see that the universal set U has these elements

$\{1, 2, 3, \dots, 10\}$, and the subsets of this universal set are the set $A = \{2, 4, 6, 8, 10\}$ and $\{1, 3, 5, 7, 9\}$.



The complement of a set A noted as A' is the set of all elements in the universal set that are not members of a given set. This is $U - A$. In the previous diagram the region of A' is shaded with the blue colour.

For a subset A of the universal set (U), its complement is represented as A' which includes the elements of the universal set which are not the elements of set A. The universal set consists of a set of all elements of all its related subsets, whereas the empty set contains no elements of the subsets.

The complement of the universal set is an empty set (null set), denoted by ' $\{\}$ ' or the symbol " $\emptyset$ ".

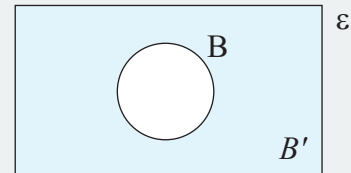
Examples:

1. $\varepsilon = \{1, 2, 3, 4, 5\}$ and set $A = \{3, 5\}$. Then the complement of $A = \{1, 2, 4\}$ written as $A' = \{1, 2, 4\}$.
2. Given that: $\varepsilon = \{a, b, c, d, \dots, z\}$, B = set of letters in the word congratulations. Describe the set B' and show set B and B' in a Venn diagram.

Solution:

Draw a rectangle and label it with ε to show the universal set. Within the rectangle draw set B as a subset of ε .

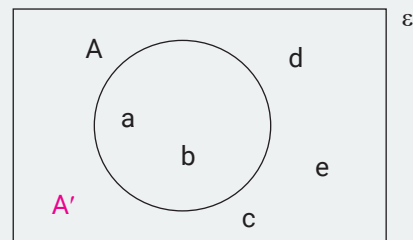
Shade the region within the universal set but outside set B and label it B' . B' is the set of all the alphabets except those in set B .



3. Given that $\varepsilon = \{a, b, c, d, e\}$ and $A = \{a, b\}$, find A'

Solution:

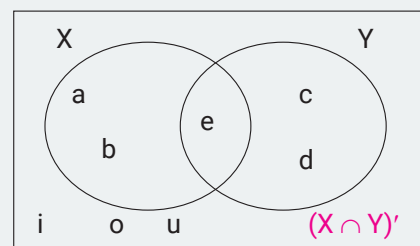
$\varepsilon = \{a, b, c, d, e\}$, $A = \{a, b\}$ thus, $A' = \{c, d, e\}$. This can be represented on a Venn diagram as shown.



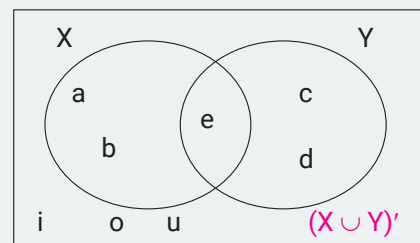
4. If $\varepsilon = \{a, e, i, o, u, c, d\}$, sets $X = \{a, b, e\}$ and $Y = \{c, d, e\}$, find: (a) $(X \cap Y)'$ (b) $(X \cup Y)'$

Solution

- a) $\varepsilon = \{a, e, i, o, u, c, d\}$, $X = \{a, b, e\}$ and $Y = \{c, d, e\}$ thus, $X \cap Y = \{e\}$ and therefore, $(X \cap Y)' = \{a, i, o, u, c, b, d\}$.



- b) $\varepsilon = \{a, e, i, o, u, c, d\}$, sets $X = \{a, b, e\}$ and $Y = \{c, d, e\}$. So, $(X \cup Y) = \{a, b, c, d, e\}$. Then, $(X \cup Y)' = \{i, o, u\}$



5. Find the members of the set B if the universal set $\varepsilon = \{11, 12, 13, 14, \dots, 19\}$ and $B' = \{12, 14, 15, 18\}$.

Solution

$\varepsilon = \{11, 12, 13, 14, 15, 16, 17, 18, 19\}$. Elements of B' are in red colour.

Given that B' and B give ε , we find that $B = \{11, 13, 16, 17, 19\}$.

**APPLICATION ACTIVITY 1.9**

- 1) If $A = \{m, n, k\}$ and $C = \{f, g, h, i, j, k, l, m\}$, find: (a) $(A \cap C)'$ (b) $(A \cup C)'$
- 2) If $\varepsilon = \{2, 3, 4, 5, 6\}$ and $A = \{2, 4, 6\}$. Find A' .
- 3) If $A = \{\text{counting numbers from 1 to 15}\}$ and $B = \{\text{even numbers from 2 to 14}\}$, find (i) $(A \cup B)'$ (ii) $(A \cap B)'$
- 4) Given sets $A = \{3, 5\}$, $B = \{7, 9, 11, 13\}$, $C = \{3, 5, 7\}$ and $\varepsilon = \{3, 5, 7, 9, 11, 13\}$. Use Venn diagrams to show the following sets by shading the appropriate regions:
 (i) A' (ii) $(A \cap B)'$, (iii) B' (iv) $(A \cup B)'$, (v) $(A \cap C)'$ (vi) $(B \cap C)'$

1.10 Comparison of sets: equal sets, equivalent sets**1.10.1 Equivalent sets****Activity 1.10.1**

Consider set $A = \{1, 2, 3\}$ $B = \{\text{Monday, Tuesday, Wednesday, Thursday}\}$, $C = \{a, b, c\}$. Tell your class partner, which sets have the same number of elements. Name them as equivalent sets.

**Content Summary**

If two sets P and Q are such that $n(P) = n(Q)$ but the elements are different, the two sets are called equivalent sets.

Example 1: Given the set A and B :



$A = \{a, b, c\}$, $B = \{1, 2, 4\}$, the elements in the set A are different from the elements in the set B . But $n(A) = n(B) = 3$

Therefore, A and B are equivalent sets. Two or more sets are said to be equivalent if they have the same number of elements.



APPLICATION ACTIVITY 1.10.1

- 1) Show that the set $G = \{7, 3, 2, 1\}$ is equivalent to the set $M = \{a, z, w, n\}$.
- 2) In the set of natural numbers, let P be the set of prime numbers less than 10 and O the set of odd numbers less than 10. Are P and O equivalent sets?

1.10.2 Equal sets



Activity 1.10.2

Observe the following sets A , B and C :

$A = \{\text{chair, table, desk}\}$, $C = \{\text{Pencil, book, pen}\}$, $D = \{\text{Table, chair, desk}\}$. Discuss with the colleague and select the sets which are equal and the sets which are equivalent. Discuss the difference between equal and equivalent sets.



Content Summary

Two or more sets that contain exactly the same elements are called equal sets.

A and B are equal sets if every element of A is in B (so $A \subseteq B$) and every element of B is in A (so $B \subseteq A$). Therefore, $A = B$.

Note: Equal sets are also equivalent sets because they have same **cardinality** (same number of elements for finite sets).

Example 1: The set $N = \{1, 2, 3, 4, 6\}$, the set $P = \{6, 2, 4, 3, 1\}$. N and P contain the same elements and $N = P \Rightarrow n(P) = n(N) = 5$.

Note: For infinite sets, "equivalent" means they have the same cardinality (e.g., $\mathbb{N}$ and $\mathbb{Z}$ are equivalent but not equal).

Example 2: State whether the following sets are equal or equivalent:

- a) $A =$ set of prime numbers less than 20. $B =$ set of factors of 36.
- b) $R =$ set of letters of the word *Mississippi*. $Q =$ the set of letters in the word *minister*.

Solution:

- a) $A = \{1, 3, 5, 7, 11, 13, 17, 19\}$ $B = \{1, 2, 3, 4, 6, 9, 12, 36\}$. The set A and B are equivalent because they have the same number of elements, but those elements are not totally the same.
- b) $R = \{M, i, s, p\}$. $Q = \{m, i, n, s, t, e, r\}$. The sets R and Q are different, they are not equal nor equivalent.



APPLICATION ACTIVITY 1.10.2

State whether the pairs of sets in questions 1-6 below are equivalent, equal or neither.

- 1) $\{3, 2, 1\}$ and $\{1, 2, 6\}$
- 2) $\{a, b, z\}$ and $\{z, b, a\}$
- 3) $\{r, s, t\}$ and $\{r, s, m, t\}$
- 4) $\{\text{Leah, Carol}\}$ and $\{\text{Carol, Leah}\}$.
- 5) $\{\text{cup}\}$ and $\{\text{plate, cup}\}$.
- 6) $\{\text{Set of positive integers}\}$ and $\{\text{Set of natural numbers}\}$

1.11 General problems involving Venn diagrams of sets



Activity 1.11

In a class, 15 students play cricket, 11 play hockey, 6 play both games and everyone plays at least one of the mentioned games. Find the total number of students in the class.



Content Summary

Main steps for solving set problems:

1. Understand the problem context
2. Represent the sets clearly using the diagrams for visualization
3. Identify relationships between sets.
4. Apply set operations and solve.

Examples 1: All the 120 students at Kabaya Secondary School learn French or English or both. 75 Learn French and 60 Learn English. How many students learn both languages?

Solution:

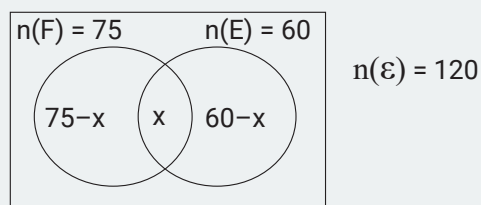
Let F be the set of students who study French,
 E : the set of students who study English. $n(\epsilon) = 120$, $n(F) = 75$, $n(E) = 60$, $n(F \cap E) = x$

$$75 - x + x + 60 - x = 120$$

$$135 - x = 120$$

$$-x = -15$$

$x = 15$. The number of those who study both languages is 15.





APPLICATION ACTIVITY 1.11

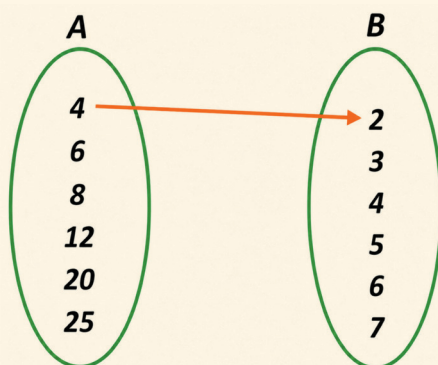
- 1) Out of 17 teachers in a school, 10 teach Economics and 9 teach Mathematics. The number of teachers who teach both subjects is twice that of those who teach none of the subjects. With the aid of a Venn diagram, Find the number of teachers who teach:
 - a) Both subjects
 - b) None of the subject
 - c) Only one subject
- 2) In a class of 40 students, 26 play football and 20 play volleyball. 17 students play both games. How many students play none of the games at all?
- 3) In a group of 20 girls, 16 play football, 12 play hockey and 2 do not play either games. How many play both games?

1.12 Introduction to relations



Activity 1.12

- 1) Write down as many as possible the biological relations you have with your family members.
- 2) Write down any four relations you have among your classmates.
- 3) Are there any students in your class that you are biologically related to? If any, write down their names and the relations you have with them.
- 4) The set A and the set B are represented in the diagram below.



The relation "**.... is the multiple of....**" is defined from the set A to the set B. Use arrows to match numbers that respect that relation. The example is given "*4 is the multiple of 2*" and we can write the ordered pair (4,2). Write all ordered pairs you found according to that relation.



Content Summary

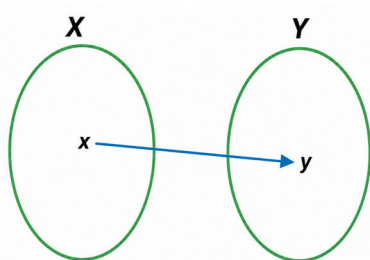
Meaning of relation: From this activity, you have realized that you have many relations. You have used statements such as 'Uwimana is my aunt and so on. In a statement such as 'Mucyo is the uncle of Mutesi'; the phrase "...**is the uncle of**..." indicates that there is a biological connection between Mucyo and Mutesi. This tells us that relation is a connection between two or more things.

A **relation** is a **connection, association, or correspondence** between elements of the same set or between elements of two different sets.

Examples of relations: is a father of..., ...is a cousin of..., ...is a multiple of..., ...is a factor of..., ...is a square of..., ...is taller than..., ...is seated next to and so on.

As we have seen, a binary relation R links the elements of a set X to the elements of another set Y . This means that through a relation R , a member x contained in the set X ($x \in X$) is related to a member y contained in the set Y ($y \in Y$).

This statement is generally abbreviated as xRy .



Note that order in this symbolic representation is important hence $xRy \neq yRx$.

- A **relation** is a **connection between elements of two sets**.
- It is written using **ordered pairs** like (x, y) .
- The **first element** comes from the first set.
- The **second element** comes from the second set.
- If the relation is defined in one set A (R defined from A to A), the ordered pair (x, y) is such that $x \in A$ and $y \in A$.

Example 1:

Let Set X = Set of boys and Set Y = set of fathers. This means that a boy (x) in Set X is a son of a father y in set Y . We can represent this relation R i.e. "is a son of" as xRy . But, the father y is not the son of x .

Example 2:

The relation R : "is the half of..." is defined from the set $A = \{1, 2, 3, 4, 6\}$ to the set $B = \{4, 6, 8, 10\}$.

- Write all ordered pairs of the relation R .
- Write R in form of a set.

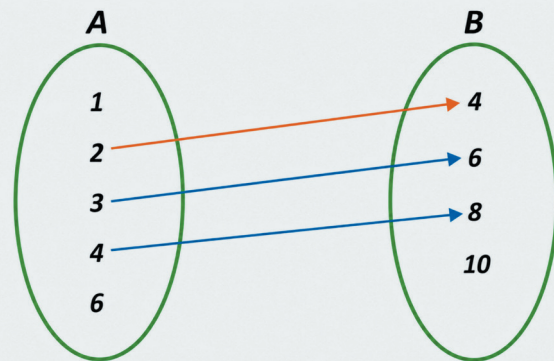
Solution:

The relation R is "is the half of" from A to B .
That means: for $a \in A$ and $b \in B$, $(a, b) \in R$ if $b = 2a$.

Respecting this relation, we have the ordered pairs: $(2,4)$, $(3,6)$ and $(4,8)$.

In the **set form of the relation R** , we have:
 $R = \{(2,4), (3,6), (4,8)\}$.

Example: Set $X = \{\text{daughters}\}$, Set $Y = \{\text{mothers}\}$; The relation "**is a daughter of**" between sets X and Y is represented as $R = \{(x, y) \in X \times Y \mid x \text{ is a daughter of } y\}$.

**APPLICATION ACTIVITY 1.12**

- 1) In the set $A = \{1, 2, 3\}$, the relation $R: \{(a,b): b = a + 1\}$. Represent the relation R in the following table:

Element a of A	Element $a+1$ of A	The ordered pair (a,b)
1	$1 + 1 = 2$	$(1,2)$
2		
3		

- 2) Write the relation of students and their favourite subjects as shown in the following table:

Student	Subject	Ordered pair
Anne	Maths	(Anne, Math)
John	English	...
Paul	Maths	...

- 3) In a given class, we can form the subsets A and B : $A = \{\text{Jean, Amina}\}$, $B = \{\text{Mathematics, English}\}$. Jean studies Mathematics. Amina studies English.
- Write the relation.
 - How many elements are in the relation?

1.13 Ordered pairs



Activity 1.13

Draw two separate circles (Venn diagrams): Circle A for Boys: {John, Eric, Paul}, Circle B for Bag colours: {Red, Blue, white}. Each boy can choose one bag.

Discuss:

- How can we show the bag that can be chosen by each boy?
- How can we write the matching?



Content Summary

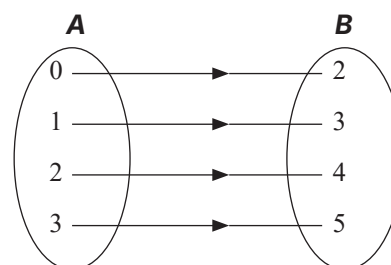
Ordered pair:

As the name suggests, an ordered pair is an entity comprising of two objects or elements put in a specified order. Suppose x and y are two objects. The ordered pair of x and y is denoted by (x, y) . The first element x is known as the first coordinate (or **first component**), while the second element y is known as the second coordinate (or **second component**).

The order of the elements is significant because $(x, y) \neq (y, x)$.

Examples of ordered pairs are $(1, 2)$; (father, son); (child, age); (a, b) and so on.

Example: Consider the relation R : "is two less than", shown in the arrow diagram.



We can write down a set of order pairs as $R = \{(0, 2), (1, 3), (2, 4), (3, 5)\}$.

"is 2 less than"



APPLICATION ACTIVITY 1.13

- Let the set $A = \{1, 2, 3\}$ and $B = \{2, 4, 6\}$. A relation R is defined from the set A to the set B by: "is the half of".
 - List all ordered pairs in R .
 - Write R in set notation.
- Given the sets $A = \{2, 4\}$ and $B = \{3, 5\}$. List **all possible ordered pairs** from A to B .
- Let the sets $S = \{\text{Cow, Goat, Chicken}\}$ and $T = \{\text{Milk, Meat, Eggs}\}$. Form a relation showing what each animal gives.

- 4) Let us define the relation R : "is less or equal to" ($\leq$) defined in the set $A = \{1, 3, 5, 4\}$. Do you think that $2 \leq 2$? If yes, $2R2$ and we have the ordered pair $(2, 2)$. Then, write all ordered pairs that respects this relation.

1.14 Cartesian product



Activity 1.14

Draw two separate circles (Venn diagrams): Set A (Shoes colour): {Black, Brown} and Set B (Sizes): {38, 39}. Each shoe colour can be paired with each size.

Discuss in group the following questions:

- How many different pairs can we make?
- How do we write these pairs?



Content Summary

Let X and Y be two sets. The Cartesian product of sets X and Y , denoted by $X \times Y$ is the set of the ordered pairs (x, y) , where $x \in X$ and $y \in Y$. This is expressed in symbols as:

$X \times Y = \{(x, y) / x \in X \text{ and } y \in Y\}$. (The symbol $/$ means such that).

Example 1: If Set $A = \{a, b\}$, and Set $B = \{2, 3, 4\}$, find the Cartesian Product $A \times B$.

Solution:

$A \times B$ is the set of all possible ordered pairs whose first component is a member of set A and the second component is a member of B

$A \times B = \{(a, 2), (a, 3), (a, 4), (b, 2), (b, 3), (b, 4)\}$.

Order matters: $A \times B \neq B \times A$

Number of elements: $n(A \times B) = n(A) \times n(B)$

Example 2: Number of Ordered Pairs Given: $A = \{p, q, r\}$, $B = \{0, 1\}$

Solution:

$A \times B = \{(p, 0), (p, 1), (q, 0), (q, 1), (r, 0), (r, 1)\}$.

$n(A \times B) = 3 \times 2 = 6$.

Example 3: Given that $A = \{1, 3, 5\}$.

- Find the number of elements in $A \times A$.
- List all ordered pairs for $A \times A$.

Solution

- a) $n(A \times A) = 3 \times 3 = 9$
- b) $A = \{1, 3, 5\}$. Thus, all ordered pairs are $A \times A = \{(1,1), (1,3), (1,5), (3,3), (3,1), (3,5), (5,1), (5,3), (5,5)\}$.

**APPLICATION ACTIVITY 1.14**

- 1) Sets: Students = {Anna, Paul}, Subjects = {Math, Science}. Find all ordered pairs.
- 2) Sets: Shops = {Shop A, Shop B}; Products = {Rice, Sugar}. Find the Cartesian product.
- 3) Given that the Cartesian product $A \times B$ is the set of all ordered pairs (x, y) where $x \in A$ and $y \in B$, Find the Cartesian products of the following two sets:
 - a) Set $A = \{a, b\}$, Set $B = \{2, 3\}$
 - b) Set $C = \{m, n\}$, Set $F = \{5, 6, 7\}$
 - c) Set $D = \{p, q, r\}$, Set $E = \{1, 2, 3\}$.
- 4) Given that $M = \{1, 2\}$, $N = \{4, 6\}$ and $P = \{6, 8\}$. Find $M \times (N \cup P)$.

1.15 Properties of relations**Activity 1.15**

- 1) Using the relation R: "**is a descendant of**": Musinga is a descendant of Rwabugiri and Rwabugiri is a descendant of Rwogera.
 - Can a person be his/her own descendant?
 - If Musinga is a descendant of Rwabugiri, is Rwabugiri a descendant of Musinga?
 - Do you think that Musinga is also a descendent of Rwogera?

**Content Summary**

A relation R on a set X can have one or more of the following properties: Reflexive, symmetric and transitive.

a) Reflexive: The relation R is said to be reflexive if for all values x , xRx . This means that every element is related to itself. For all $x \in X$, xRx .

Example of reflexive relations:

- i) "equals to" e.g. $x = x$; $2 = 2$
- ii) "less than or equal to" e.g. $x \leq x$; $2 \leq 2$

In a reflexive relation, we have arrows for all values in the domain **pointing back to themselves** e.g. $3 \leq 3$ implies $3R3$.

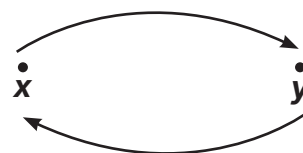


b) Symmetric: The relation R is said to be symmetric if x is related to y , then y is related to x . In other words, for all values x and y , xRy implies yRx . This means that the relation and its inverse are the same.

Example of a symmetric relations:

i) "is equal to" since we can say $x = y$ and $y = x$.

In an arrow diagram for a Symmetric relation, we have an **opposite arrow for every arrow**.



ii) "is married to" : if Alice is married to Bob, then Bob is married to Alice.

Example of antisymmetric relation:

i) "greater than" For example, when $x > y$, y is not greater than x .

ii) "is taller than" (if Alice is taller than Bob, and Bob is taller than Carol, then Alice is taller than Carol).

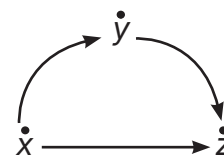
c) Transitive:

The relation R is said to be transitive If x is related to y and y is related to z , then x is related to z . In other words, if for all values x , y and z , xRy and yRz implies xRz .

Examples of a transitive relation:

i) "is greater than" e.g. if $x > y$ and $y > z$, it implies $x > z$.

ii) "is less than" e.g. if $x < y$ and $y < z$, it implies $x < z$. In an arrow diagram, every arrow between two values x and y , and y and z , has an arrow going straight from x to z .



For example, i) $2 < 3$, ii) $3 < 5$ then, $2 < 5$

d) Antisymmetric: The relation R is said to be antisymmetric if for all values x and y , xRy and yRx is possible only in case $x = y$. Otherwise, when xRy , y is not related to x .



APPLICATION ACTIVITY 1.15

1) Consider the relation "is a sibling of" among students in a school.

- Is it reflexive? (Does every student have themselves as a sibling?)
- Is it symmetric? (If Alice is a sibling of Bob, is Bob a sibling of Alice?)
- Is it transitive? (If Alice is a sibling of Bob, and Bob is a sibling of Carol, is Alice a sibling of Carol?)

- 2) Let $X = \{1,2,3\}$ and the relation $R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$. Determine whether R is: (a) Reflexive, (b) Symmetric, (c) Transitive.
- 3) The relation "is a factor of" on the set $\{2,4,8\}$: 2 is a factor of 4, 4 is a factor of 8. Is 2 a factor of 8? Is the relation transitive?
- 4) Consider the set S = set of all students in your class. The relation "has the same birth month as."
- Explain why this relation is reflexive.
 - Explain why it is symmetric.
 - Explain why it is transitive.
- 5) Let $A = \{1,2,3\}$ and $R = \{(1,1), (2,2), (3,3), (1,2), (2,3), (1,3)\}$. Determine whether R is:
- Reflexive
 - Symmetric
 - Transitive
 - Antisymmetric
- 6) State whether the following statements are **True (T)** or **False (F)**.
- Every reflexive relation is symmetric.
 - If $(a,b) \in R$ and $(b,a) \in R$, then R is symmetric.
 - If $(a,b) \in R$ and $(b,c) \in R$, then for R to be transitive, (a,c) must be in R .
 - In an antisymmetric relation, it is possible to have (a,b) and (b,a) with $a \neq b$.
- 7) Let $A = \{a,b,c\}$ and $R = \{(a,a), (b,b), (c,c), (a,b), (b,c)\}$. Answer by true or false >
- R is Reflexive
 - R is Symmetric
 - R is Transitive
 - R is Antisymmetric

1.16 Equivalence relation



Activity 1.16

The school was organizing a school sports day. All students had to wear **jerseys of the same color** in the **same House** (e.g., *Blue House, Red House, Green House, Yellow House*).

For example, the following 6 students are assigned to houses:

- Alice → Blue House
- Bob → Blue House
- Carol → Red House
- David → Red House
- Emma → Green House
- Frank → Green House

Now the relation is defined: "**is in the same house as.**"

Verify if this relation is:

- **Reflexive:** Is Alice in the same House as herself?
- **Symmetric:** If Alice is in the same House as Bob, is Bob in the same House as Alice?
- **Transitive:** If Alice is in the same House as Bob, and Bob is in the same House as Carol, are Bob and Carol in the same House?

**Content Summary**

An equivalence relation is a relation that is reflexive, symmetric and transitive. Generally, an equivalence relation is denoted by $\sim$

Example 1: Prove that " $=$ " is an equivalence relation.

Solution:

Reflexive: it is true that $x = x$ for all values. Therefore, $=$ is reflexive.

Symmetric: If $x = y$, it is also true that $y = x$. Therefore, $=$ is symmetric

Transitive: If $x = y$ and $y = z$, this means that $x = z$. Thus ' $=$ ' is transitive.

Therefore ' $=$ ' is an equivalence relation.

Other equivalence relations include: is the same age as, lives in the same district as, Has the same birth month as, has the same mother as, is in the same school class as, etc.

Example 2

(Homework to be presented):

Given the relation P: "is parallel to" in the set of lines in the plane. Determine whether P is:

- a) Reflexive b) Symmetric c) Transitive d) An equivalence relation.

**APPLICATION ACTIVITY 1.16**

- 1) Let $A = \{1, 2, 3\}$ and $R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$. Determine whether R is: a) Reflexive, b) Symmetric, c) Transitive, d) An equivalence relation
- 2) Let $A = \{a, b, c\}$ and $R = \{(a,a), (b,b), (c,c), (a,b), (b,c)\}$. Check if R is an equivalence relation.
- 3) Let $A = \{1, 2, 3\}$ and $R = \{(1,1), (2,2), (3,3), (1,2)\}$. Add the **minimum ordered pairs** to make R an equivalence relation.
- 4) Let $A = \{a, b\}$ and $R = \{(a,a), (a,b)\}$. Make R an equivalence relation.
- 5) Let $A = \{1, 2, 3, 4\}$ and $R = \{(x,y) \mid x - y \text{ is even}\}$. Show that R is an equivalence relation.

1.17 Papygram diagram



Activity 1.17

Given the set F for family members such that $F = \{\text{Father, Mother, First-born, Second-born, third born}\}$ where their ages are: Father: 45 years, Mother: 42 years, First-born: 18 years, Second-born: 15 years, third born: 10 years. The **relation**: «is older than» is defined in the set A (from A to A).

- Complete all ordered pairs for that relation in the set A .
- Draw a circle or a Venn diagram with elements of the set A . In this set, use arrows to join a first coordinate and second coordinate of each ordered pair.

What type of diagram do you get?



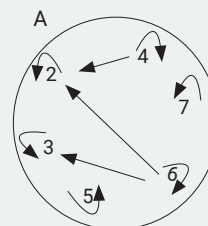
Content Summary

A **papygram** is a **circular diagram** used to visually represent **relations** between elements within a **single set**. It uses **arrows** to show how each element relates to others according to a specific rule or condition.

Example

- Draw a papygram to show the relation "is a multiple of" in the set of numbers $A = \{2, 3, 4, 5, 6, 7\}$

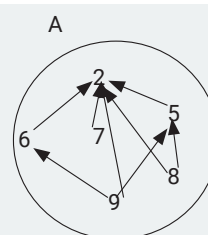
Solution:



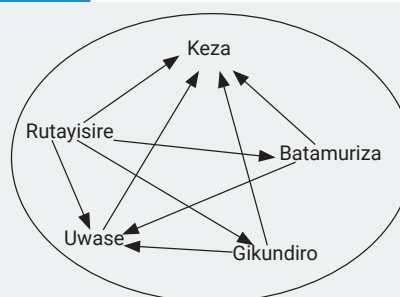
The arrow which goes back to the same number means a number is a multiple of itself.

- Draw a papygram to illustrate the relation "exceeds by more than 2" for $A = \{2, 5, 6, 7, 8, 9\}$
- Draw a papygram to show the relation "is older than" for the following relation; Keza is 12 years old, Batamuriza is 14 years old, Rutayisire is 15 years old, Uwase is 13 years, and Gikundiro is 14 years old.

Solution:



Solution:

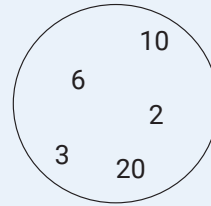




APPLICATION ACTIVITY 1.17

1) Draw a papygram to show the relation 'is a factor of' in the set {2, 3, 4, 6, 7, 8}.

2) Complete the arrow diagram to show the relation "is a multiple of".



3) The following are weights of some students in a senior one class at given school.

Student	Weight
Teta	50 kg
Uwimana	36 kg
Ishimwe	47 kg
Akaliza	40 kg
Bwiza	60 kg

Draw a papygram to show the relation "is heavier than" for the students mentioned above.

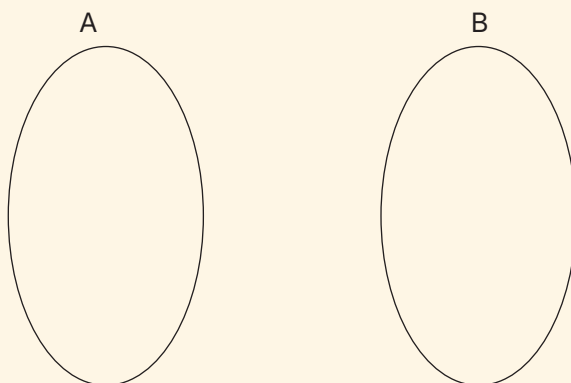
1.18 Mapping



Activity 1.18

In pairs,

- Write down set A of the first six natural numbers.
- Form another set B by squaring each number in set A above.
- Write the numbers in set A in a circle and their squares in set B in a different circle on the same level side by side.



- Use arrows to match the numbers in set A to their squares in set B.

Do you find a first coordinate with 2 second coordinates? How would you call such matching?



Content Summary

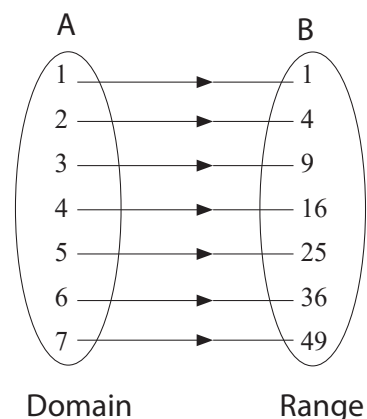
A **mapping** is a **rule** or **process** that associates **each element** of one set (domain) with **one element** of another set (range).

Consider the example below;

$$A = \{1, 2, 3, 4, 5, 6, 7\}$$

B = set of the first seven square numbers.

By using the operation of “squaring” each member of set A produces the members of set B. We say that, the operation of “squaring” maps set A onto set B as shown below.



From A you have squared the elements in set A to give rise to the elements in set B. We say that squaring elements in set A maps onto set B. Your diagram showed the mapping. Set A is called the **domain** and set B is called the **range**.

Domain is sometimes referred to as the **input** and the range as the **output**.



APPLICATION ACTIVITY 1.18

- 1) Set $M = \{1, 2, 3, 4, 5\}$ is mapped onto set N by the relation “multiply by 4”. List the elements of set N and map M onto N.
- 2) A set A maps onto set B by the operation “multiply by 3 and add 1”. The elements of set A are $\{5, 6, 7, 8, 9\}$
 - a) List the element of set B.
 - b) Map set A onto B.

1.19 Domain, range and graph of a mapping



Activity 1.19

Carry out the following activities.

1. Given set $A = \{1, 2, 3, 4, 5, 6\}$ and the relation “multiply by 2 and add 1”, Find set B which is the range.
2. Using sets A and B, order them in pairs such that an element in set A is paired to an element it is mapped onto in set B.
3. What name would you call such a pairing in step 2 above?
4. Draw a Cartesian plane in your graph books and plot the pair on points in step 3 above. Let the domain be in the x – axis and the range be in the y – axis.
5. Use a straight line to join the points you plotted in step 4 above.
6. What name would you call such a graph?



Content Summary

A **relation** from set A to set B is any subset of the Cartesian Product $A \times B$. The **domain of a relation** is the set of all first elements of the ordered pairs in the relation.

A relation R from A to B is a subset of the Cartesian Product $A \times B$.

The difference between relation and mapping is that for a mapping, every element of the domain is associated with one and only one element of the codomain.

$$\text{Domain } (D) = \{x \mid (x, y) \in R\}$$

The **range** of a mapping is the set of all second coordinates of the ordered pairs in the mapping. $\text{Range } (R) = \{y \mid (x, y) \in R\}$.

The graph of a mapping is the set of all points in the Cartesian plane that represent the ordered pairs of the mapping.

Example: Consider the domain $A = \{0, 1, 3, 4\}$ Set A is mapped onto set B by the relation $x \rightarrow 2x$. Find the range, the ordered pair and hence draw the graph of the relation.

Solution:

The range of values will be: $x \rightarrow 2x$

$$\text{If } x = 0 \rightarrow 2 \times 0 = 0$$

$$x = 1 \rightarrow 2 \times 1 = 2$$

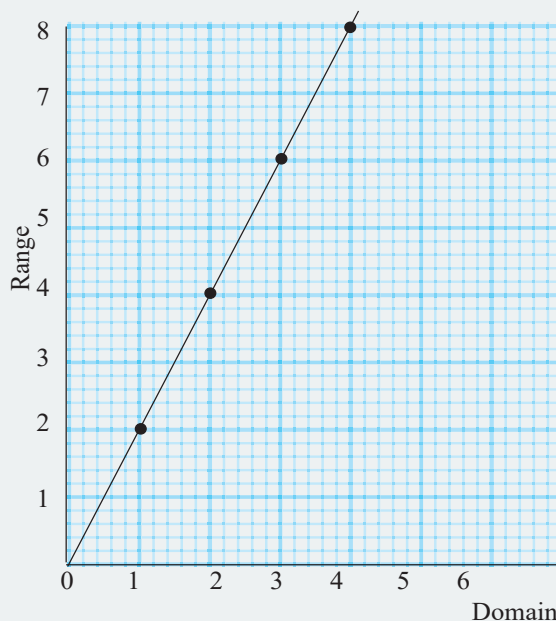
$$x = 3 \rightarrow 2 \times 3 = 6$$

$$x = 4 \rightarrow 2 \times 4 = 8$$

$$\text{Range} = \{0, 2, 6, 8\}$$

Ordered pairs: $(0, 0)(1, 2)(3, 6)(4, 8)$

The graph takes the following shape.



APPLICATION ACTIVITY 1.19

1) Let $R = \{(1,3), (2,5), (3,7), (4,9)\}$

a) Find the domain b) Find the range c) Plot the graph of the relation

2) Given $A = \{1,2,3,4\}$, $B = \{2,4,6,8,10\}$ $R = \{(1,2), (2,4), (3,6), (4,8)\}$

a) Write the domain b) Write the range c) Does 10 belong to the range? Why?

3) A relation shows the number of notebooks a student has:

$R = \{(\text{Eric}, 2), (\text{Mary}, 3), (\text{Paul}, 2), (\text{Anna}, 4)\}$ a) Write the domain b) Write the range

1.20 Particular relations

1.20.1 One to one, many to one mapping and one to many relation



Activity 1.20.1

Work out.

- 1) By using internet and AI find the definition of one to one, one to many, many to one, and many to many relations.
- 2) A school assigns each student a unique ID card with a specific code of his/her name. What name would you call such a mapping? Is it one to one or many to one mapping or one to many relation?



Content Summary

A **one-to-one mapping** from the set A to the set B is a mapping in which **different elements of A are mapped to different elements of B**.

Example: The diagram below shows the relation “is half of” between the following sets: $A = \{2, 4, 6, 8\}$ and $B = \{4, 8, 12, 16\}$.

- i) Present this relation
- ii) What name would you call such a mapping?

Many-to-one mapping: Two or more different inputs can share the same output.

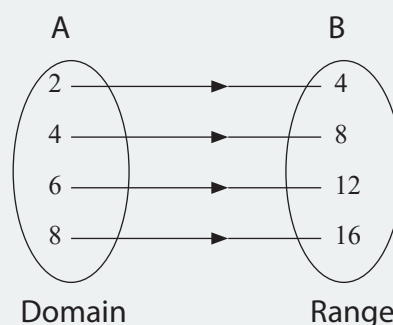
Real-life examples: Students $\rightarrow$ Favorite fruit, People $\rightarrow$ Birth month, Numbers $\rightarrow$ Even/Odd classification.

Example: Some S1 students obtained the following grades:

Name	Grades
Phocus	B
John	B
Sandra	C
Dorice	C
Peter	A

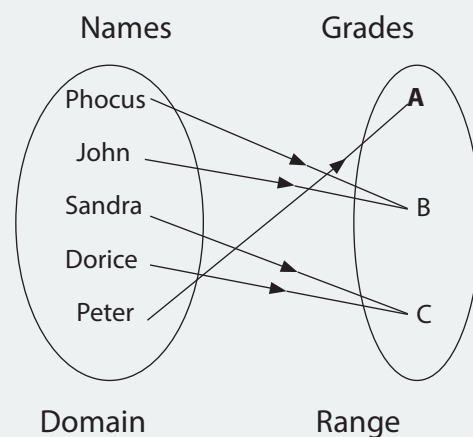
- a) Use the names as the domain and the grades as the range to show the mapping.
- b) What name would you call such a mapping?

Solution:



This mapping is one-to-one.

Solution:



This relation is many to one mapping.

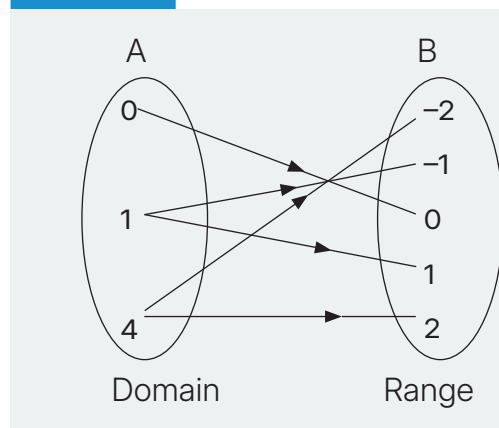
One-to-many relation is a relation where a **single element in the domain is associated with two or more elements in the codomain**.

Example: Use a diagram to illustrate the relation “is square of ” from A to B where $A = \{0, 1, 4\}$ and $B = \{-2 \leq x \leq 2\}$.

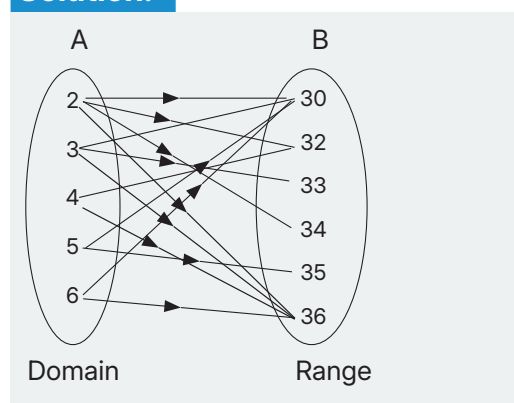
Many to many relation: one element in the domain can be associated with several elements in the codomain and one element in the codomain can also be associated with several elements in the domain.

Example: Show the mapping of the relation “is a factor of” for the sets $A = \{2, 3, 4, 5, 6\}$ and $B = \{30, 32, 33, 34, 35, 36\}$.

Solution:



Solution:



APPLICATION ACTIVITY 1.20.1

- 1) Set $M = \{1, 2, 3, 4, 5\}$ is mapped onto set N by the relation “multiply by 4”. List the elements of set N and map M onto N .
- 2) A set A maps onto set B by the operation “multiply by 3 and add 1”. The elements of set A are $\{5, 6, 7, 8, 9\}$ (a) List the element of set B . (b) Map set A onto B . (c) What type of mapping is this?

1.20.2 Functions



Activity 1.20.2

In the table below, a relation is given,

a) Complete the second coordinate for each given first coordinate:

Relation	First coordinate	Second coordinate	First coordinate	Second coordinate	First coordinate	Second coordinate
(a) $x \rightarrow 2x^2$,	-1		1		2	
(b) $y \rightarrow y^2 + 1$	-1		1		2	
(c) $m \rightarrow \frac{m}{2} + 5$	-1		1		2	

b) Is there a real number taken as first coordinate that has no second coordinate?

c) Is there a real number taken as first coordinate that has more than one second coordinate?



Content Summary

You have realized that each relation is a mapping because every element in the domain has one image in the range. In addition, every object on the domain has **one and only one image** in the range. This type of mapping is called a **function**.

A function is a **special type of mapping** where **each element** in the domain is associated with exactly **one element** in the range.

For the function f , every input x has **only one output** $f(x)$. Denoted as $f:A \rightarrow B$ / $x \rightarrow f(x)$ is a real number.

Mappings, which are functions, are one – to – one mappings and many – to-one mappings.

Example 1: Consider a function $f : x \rightarrow f(x) = 4x$. One way of writing this is $x \rightarrow 4x$. This is read as : f is the function which maps x onto $4x$. The most common and important notation is $f(x) = 4x$ read as f of x equals $4x$.

Example 2: Given the function $f(x) = 4x + 2$, find $f(2)$.

Solution:

$$f(x) = 4x + 2, \quad f(2) = 4 \times 2 + 2 = 8 + 2 = 10. \quad f(2) = 10.$$

Example 3: The function $g(y) = 3y^2 + 2y$. Find $g(2)$.

Solution:

$$g(y) = 3y^2 + 2y.$$

$$g(2) = 3(2 \times 2) + 2(2) = 3 \times 4 + 4 = 16, \text{ Hence, } g(2) = 16.$$

Example 4: The function $h(x) = ax^2 - 2$ and $h(2) = 18$. Find:

- (a) the value of a . (b) $h(-1)$ (c) $h(3)$

Solution:

a) $h(x) = ax^2 - 2$, $h(2) = a(2 \times 2) - 2 = 4a - 2$. Since $h(2) = 18$, we have $4a - 2 = 18$.

Then, $4a - 2 = 18$ implying $4a = 20$ Therefore, $a = 5$.

b) Substitute the value of a $h(x) = ax^2 - 2$. So, $h(-1) = 5(-1 \times -1) - 2 = 3 \Rightarrow h(-1) = 3$

c) $h(x) = ax^2 - 2$. So, $h(3) = 5(3 \times 3) - 2 = 5 \times 9 - 2 = 43 \Rightarrow h(3) = 43$.

Note: Since each value x is allowed only one value $y=f(x)$ is a function. We can think of a function as a **machine** that “eats” x -values and spits back y -values so that the machine only spits out one output for any input.

Example: $y=f(x)=x-6$

**Types of functions:**

There are three **types of functions** (mappings where each input has exactly one output), but they differ in how inputs relate to outputs.

Type	Also Called	Key property
Injection	One-to-one	No two inputs map to same output
Surjection	Onto	Every output is used (has an input)
Bijection	One-to-one correspondence	Both injection and surjection

a) Injection (One-to-One): A function $f:A \rightarrow B$ is **injective** if **different inputs always give different outputs**. No two elements in domain map to same element in range.

Each output gets at most one arrow to it.

For injection:

- Each element of the **domain** has **one unique image**.
- **No image is shared** by two different elements.
- The **codomain may have extra elements** that are not used.

Example: $f(x)=2x$ (from integers to integers). Different $x \rightarrow$ different $2x$

Range: only even numbers (not all integers)

b) Surjection (Onto): A function $f: A \rightarrow B$ is **surjective** if **every element in the codomain B has at least one pre-image in A**. The range = the codomain. All outputs are “covered”. Every output has at least one arrow pointing to it.

Example: $f(x) = \sqrt[3]{x}$ (from the set of real numbers to the set of real numbers). Every real number has a cube root.

c) Bijection (One-to-one correspondence): A function $f: A \rightarrow B$ is bijective if it is both injective and surjective. It has a perfect “pairing” between domain A and codomain B. Same number of elements in A and B (if finite).

Example: $f(x)=x+1$ (from integers to integers)

Different $x \rightarrow$ different $x+1$ (injective)

Every integer $y=f(x)$ has pre-image that is equal to $y-1$ (it is surjective).



APPLICATION ACTIVITY 1.20.2

1) For each mapping below, state whether it is **one-to-one (injective)** or **not**, and **give a reason**.

a) $A=\{1,2,3\}$, $B=\{a,b,c\}$: $1 \rightarrow a$, $2 \rightarrow b$, $3 \rightarrow c$

b) $A=\{x,y,z\}$, $B=\{1,2\}$: $x \rightarrow 1$, $y \rightarrow 1$, $z \rightarrow 2$

c) $A=\{p,q,r\}$, $B=\{4,5,6,7\}$, $6p \rightarrow 4$, $q \rightarrow 5$, $r \rightarrow 6$

2) The function $g(y) = 3y^2 + 2y$.

a) Find $g(5)$.

b) is it injective or surjective in the set of real numbers?



END OF UNIT ASSESSMENT

Assessing knowledge and understanding:

1. Explain a set and give one example.
2. Explain the following special sets: empty set and singleton set.
3. What is meant by the cardinal number of a set?
4. State the meaning of an ordered pair.
5. Explain the following relations: Reflexive, Symmetric, Transitive

Assessing skills

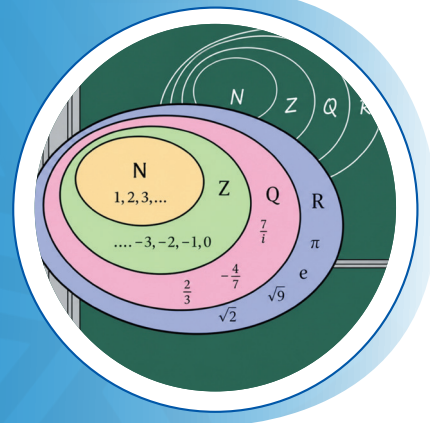
6. If $A = \{1, 2, 3, 4\}$, find all its subsets.
7. Let $A = \{1, 3, 5\}$, $B = \{3, 4, 5\}$. Find: (a) $A \cap B$, (b) $A \cup B$, (c) $A - B$
8. Given $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{2, 4, 6\}$. Find A' (complement).
9. If $A = \{1, 2, 3\}$ and $B = \{a, b\}$, find $A \times B$.
10. Set $A = \{2, 4, 6\}$ maps to set B by "multiply by 2". Find: (a) Set B (b) Ordered pairs.
11. In a class of 40 students, 18 like Mathematics, 20 like Science, and 8 like both. Find how many like neither subject.
12. In a family of 50 members, 28 like rice, 22 like beans and 10 like both. How many like only rice?
13. Students are assigned lockers: $A \rightarrow L1, B \rightarrow L1, C \rightarrow L2$. Is this a one-to-one mapping?
14. In a school, every student has exactly one ID number, and no two students share the same ID. What type of mapping is this?
15. In a village, children are grouped into football teams by age. Several children belong to the same team. Which type of mapping is formed?

Assessing attitudes and values

16. Which topic in this unit (**Sets, Venn diagrams, Relations, or Functions**) felt most useful to you? Why?
17. Did learning about sets helped you solve problems outside of Mathematics?
☐ Yes ☐ A little ☐ Not yet. Explain your answer.

UNIT 2

SETS OF NUMBERS



Key Unit Competence:

Use the properties of sets of numbers in solving real life problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain and identify the symbols used for different sets of numbers (natural, integers, rational, irrational, and real).
- Describe the relationships among the different sets of numbers.
- Describe the operations of addition, subtraction, multiplication, and division in each number set.
- Explain the properties of the four operations (commutative, associative, distributive, identity, and inverse) in real numbers.

Skills

- Explain that rational and irrational numbers together form the set of real numbers.
- Represent sets of numbers using set notation, Venn diagrams, and number lines, and accurately perform operations within and across different number sets.
- Convert between fractions and decimals (including recurring decimals) and apply the properties of operations to simplify numerical expressions.
- Compare and order rational and irrational numbers and solve real-life and mathematical problems involving the four operations and number sets.

Attitudes and values

- Show accuracy, orderliness, logical thinking, and confidence when performing number operations and solving problems.
- Work cooperatively in groups and respect others' ideas during discussions.
- Demonstrate persistence and curiosity while exploring numerical patterns.

INTRODUCTION

This unit covers natural numbers, integers and operations, rational and decimal numbers (including recurring decimals), irrational numbers, and real numbers. It also includes the relationships between these sets and the properties of the four basic operations (addition, subtraction, multiplication, and division) in the set of real numbers.

Key Words or concepts of the Unit are mainly the natural numbers, subsets, integers, rational numbers, fractions, decimal numbers, recurring decimals, irrational numbers, real numbers, number sets, operations, properties of operations, addition, subtraction, multiplication, division and number relationships.



Introductory activity

Consider the following numbers $3, 0, -5, \frac{1}{2}, 4.5, \sqrt{2}, -1, 7, \frac{2}{3}, 0.25$.

- 1) Where do you usually see or use such numbers in daily life? (Think about money, temperature, counting objects, measuring, etc.).
- 2) Group the numbers in any way to makes sense. Then, try to answer to these questions:
 - Which numbers are used for counting?
 - Which numbers have fractions or decimals?
 - Which numbers are negative?
 - Which numbers cannot be written as fractions?

2.1 Sets of natural numbers



Activity 2.1

Carry out the following activities:

- Count the following objects in your class:
 - (a) desks.
 - (b) Mathematics textbooks in your class.
 - (c) pencils in your class.
 - (d) students in your class.
- Write the following numbers in words:
 - (a) 12 679
 - (b) 789 000
 - (c) 12 345 798
 - (d) 363 879 987
- What name do you call such numbers?



Content Summary

Natural numbers are the most basic numbers in mathematics, used for counting objects (like 1 apple, 2 apples, 3 apples) and for ordering (like first, second, third).

The set of natural numbers is denoted by $\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$

On a number line, natural numbers are represented as follows:



The dotted lines and the arrowhead indicate that the numbers can go on forever in the direction shown. We therefore say, they are infinite (have no end). The set of natural numbers has infinite number of elements: $n(\mathbb{N}) = \infty$.

Example 1: Write the place value of each digit of the year 1994.

Solution:

The year 1994 has four digits. The place value of 1 is thousands, the first 9 is hundreds, the second 9 is tens and 4 is ones.



APPLICATION ACTIVITY 2.1

- 1) Write the following numbers in words: (a) 917 405 (b) 80 165 (c) 439 357 890.
- 2) True or False:
 - a) All integers greater than zero are natural numbers.
 - b) The number 0 is always considered as natural number.

2.2 Subset of natural numbers



Activity 2.2

Carry out the following activities:

- a) You are given the set of natural numbers between 0 and 20.
- b) Use a mathematical dictionary or internet to define the terms: even, odd and prime numbers.
- c) From the set described in part (a) above:
 - i) Draw the subset of odd numbers.
 - ii) Draw the subset of even numbers.
 - iii) Draw the subset of prime numbers.
 - iv) How many even numbers are prime numbers?
 - v) How many odd numbers are prime numbers?



Content Summary

- a) **Even numbers:** Numbers which are divisible by 2 or numbers which are multiples of 2.
- b) **Odd numbers:** Numbers which leave a remainder of 1 when divided by 2.
- c) **Prime numbers:** Numbers which are divisible by 1 and themselves only.

Example: From the set of natural numbers between 0 and 20, give the even numbers, odd numbers and prime numbers

Solution:

The even numbers from 0 to 20 are 2, 4, 6, 8, 10, 12, 14, 16, 18 and 20. The odd numbers between 0 and 20 are 1, 3, 5, 7, 9, 11, 13, 15, 17 and 19. The prime numbers between 0 and 20 are 2, 3, 5, 7, 11, 13, 17 and 19



APPLICATION ACTIVITY 2.2

- 1) Set $\varepsilon = \{1, 2, 3, 4, 5, 6, 7, 8, \dots, 50\}$. Write down the subsets of set A which are;
 - (a) Even numbers
 - (b) Odd numbers
 - (c) Prime numbers
 - (d) Represent the above in a Venn diagram.
- 2) Given $\varepsilon = \{1, 4, 8, 11, 16, 25, 49, 53, 75\}$, list the elements of the following subsets
 - (a) Even numbers
 - (b) Odd numbers
 - (c) Prime numbers
 - (d) Represent the above on a Venn diagram

2.3 Operations on natural numbers and their properties



Activity 2.3

Add, subtract, multiply and divide the following pairs of natural numbers

- | | |
|--------------|------------------|
| a) 20 and 30 | d) 8 and 7 |
| b) 14 and 17 | e) 1000 and 1999 |
| c) 12 and 48 | |



Content Summary

- a) **Addition:** The symbol for addition is (+). Given natural numbers a , b , and c . $a + b = c$, c is the sum of numbers a and b .

Properties of addition: Given numbers a and b , the sum of a and b is equal to the sum b and a .

1. $a + b = b + a = c$. For example, $24 + 46 = 46 + 24 = 70$. This is called the commutative property of addition. When you interchange the terms, the sum remains the same.
2. $(a + b) + c = a + (b + c)$. For example, $(32 + 52) + 47 = 32 + (52 + 47)$. This is called the associative property of addition. The associative property allows regrouping of the terms in any order and the sum remains the same at the end of the operation.
3. Given the natural number a , $a + 0 = 0 + a$. From example $0 + 67 = 67 + 0$. 0 is called the identity element of addition.

- b) **Subtraction:** the symbol for subtraction is (-). Given that $x - y = t$, t represents the **difference** of the two terms. For example, $54 - 24$ is not equal to $24 - 54$. We say that **subtraction is not commutative**. Similarly, $(45 - 23) - 21 \neq 45 - (23 - 21)$.

This means that subtraction is not associative.

- c) **Multiplication:** The symbol for multiplication is $\times$. If the natural numbers $a \times b = c$ then a is the multiplier, b the multiplicand and c is the product of the two **factors** (multiplier and multiplicand).

The properties of multiplication:

1. If you are given two natural numbers a and b , then $a \times b = b \times a$. For example, $28 \times 22 = 22 \times 28$. This is called the **commutative property** of multiplication.
2. For natural numbers a , b and c , $a \times (b \times c) = (a \times b) \times c$. For example, $11 \times (12 \times 13) = (11 \times 12) \times 13$. This is called the **associative property** of multiplication.

3. Any natural number multiplied by one gives that natural number. For example, $64 \times 1 = 64$.

1 is called the **identity** element of multiplication.

4. Note that $a(b + c) = ab + ac$. This is called the **distributive property** of multiplication over addition.

d) **Division:** The division of natural numbers has specific properties, but unlike addition or multiplication, division is more limited: it is **not always closed, not always commutative, and not always associative**.

Examples: Work out:

a) $1\,234 + 110 + 72 + 4$

c) $863 \times 56 \times 12$

b) $502\,341 - 51\,021$

d) $612 \div 23 =$

Solution:

- a) Arrange the numbers in vertical form aligning the numbers according to their place value:

$$\begin{array}{r} 1\,234 \\ 110 \\ 72 \\ + 4 \\ \hline 1\,420 \end{array}$$

- b) Arrange the numbers vertically and work them out.

$$\begin{array}{r} 502\,341 \\ - 51\,021 \\ \hline 451\,320 \end{array}$$

- c) First multiply 863 by 56, then multiply the resulting product by 12.

$$\begin{array}{r} 863 \\ \times 56 \\ \hline 5178 \\ + 43150 \\ \hline 48328 \end{array} \quad \text{then } \rightarrow \quad \begin{array}{r} 48328 \\ \times 12 \\ \hline 96656 \\ + 483280 \\ \hline 579936 \end{array}$$

- d)

$$\begin{array}{r} 26 \\ 23 \overline{) 612} \\ \underline{- 46} \quad \text{..... } 23 \times 2 = 46 \\ 152 \\ \underline{- 138} \quad \text{..... } 23 \times 6 = 138 \\ 14 \quad \text{..... remainder} \end{array}$$

Thus, $612 \div 23 = 26 \text{ rem } 14$



APPLICATION ACTIVITY 2.3

- 1) Work out the following:

a) $45 + (50 + 30)$, b) $18100 + 940 + 2693$, c) $(8400 + 2200) + 3340$

- 2) Work out the following:

(a) $456 - 225$, (b) $100\,000 - 39\,400$,
(c) $4\,824 - 1\,793$, (d) $6\,453 - 2\,141 - 1\,340$

- 3) Copy and complete the subtractions and replace them by the missing digit to make the calculations correct.

$$\begin{array}{r} \text{a) } 586 \\ - 2\Box4 \\ \hline 352 \end{array}$$

$$\begin{array}{r} \text{b) } 1\Box5\Box \\ - 289 \\ \hline 1265 \end{array}$$

$$\begin{array}{r} \text{c) } 4\Box\Box6 \\ - 358\Box \\ \hline 869 \end{array}$$

$$\begin{array}{r} \text{d) } 8\Box4\Box7 \\ - 638\Box \\ \hline 79075 \end{array}$$

- 4) Multiply the following:

(a) $20 \times 30 \times 40$

(b) $15 \times (35 \times 20)$

(c) $(32 \times 40) \times 20$

(d) $943 \times 16 \times 11$

- 5) Expand the following where $y = 3$ and $x = 2$:

(a) $7(20 + y)$

(b) $13(xy + y)$,

(c) $xy(3x + 2y + 3)$,

(d) $2y2x(3 \times 2 + 4y)$

- 6) Evaluate: (a) $412 \div 6 =$, (b) $991 \div 31 =$,

(c) $56\,560 \div 40 =$, (d) $3\,219\,312 \div 47 =$

- 7) In Ruhengeri, a farmer harvested 34 500 kg of potatoes in the first season and 24 750 kg of potatoes in the second season. Find the total harvest.

- 8) In a city, there were 45 600 girls in secondary schools and 39 540 boys. Find the total number of students in the city.

- 9) A farmer requires 37 posts placed 5 m apart to fence one side of his farm. How many posts would he require if the posts were 4 m apart?

2.4 The set of integers and its subsets



Activity 2.4

- 1) Using a Mathematics dictionary and internet, define what an integer is.
- 2) What integer would you give to each of the following situation?
 - a) A fish which is 50 m below the water level if the water level is considered as 0.
 - b) Temperature of the room which is hot at 42°C .
 - c) A boy who is 2 m below the ground level in a hole if the level of the ground is considered as 0.
 - d) A bird which is 3 m high on a tree if the level of the ground is considered as 0.



Content Summary

The set of integers is represented by $\mathbb{Z}$. Integers are whole numbers which are either **negative** or a **positive** and include zero. $W = \{0, 1, 2, 3, \dots\}$

The set of positive integers is $\mathbb{Z}^+ \{+1, +2, +3, +4, \dots\}$ and the set of negative integers $\mathbb{Z}^-$ is $\{\dots, -5, -4, -3, -2, -1\}$. Integers can be represented on a number line as shown below.



Note: When you move towards the left of the number line, numbers become smaller and smaller. This means -5 is smaller than -2 and -100 is less than -1.

We can represent this as $-5 < -2$ and $-100 < -1$.

Example: When measuring temperature, the value of the temperatures of the body or surrounding can be negative or positive. The normal body temperature is about $+37^\circ\text{C}$ and the temperature of the freezing mercury is about -39°C .



The thermometer shows the temperature using integers.

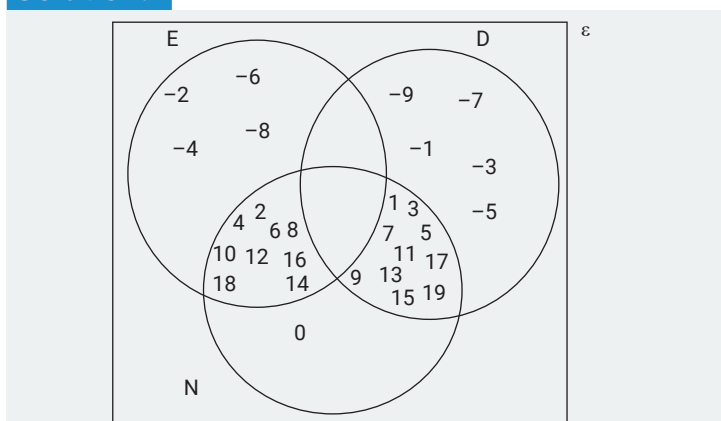
Integers have several subsets such as, set of even numbers, set of odd numbers, set of prime numbers, set of negative numbers and so on.

Subset Type	Symbol	Example
Integers	$\mathbb{Z}$	$\{..-2, -1, 0, 1, 2..\}$
Positive integers	$\mathbb{Z}^+$	$\{..1, 2, 3, \dots\}$
Negative integers	$\mathbb{Z}^-$	$\{-1, -2, -3, \dots\}$
Even integers	E	$\{...-4, -2, 0, 2, 4...\}$
Odd integers	O	$\{...-5, -3, -1, 1, 3...\}$
Whole numbers	W^+	$\{0, 1, 2, 3, \dots\}$

Example:

Given $\varepsilon = \{\text{integers between } -10 \text{ and } 20\}$ draw a Venn diagram to show the following subsets of ε :

Set E = set of positive and negative even numbers, set D = set of positive and negative odd integers, set $\mathbb{N}$ = Set of natural numbers.

Solution:**APPLICATION ACTIVITY 2.4**

- 1) Given $\epsilon = \{\text{integers between } -10 \text{ and } 20\}$ draw a Venn diagram to show the following subsets of ϵ :
Set D = set of positive and negative odd numbers, set E = positive and negative even integers and set P = set of prime numbers.
- 2) Given that $\epsilon = \{-4 \text{ to } +22\}$, show on a Venn diagram the subsets P, E and D.
Given that P = {prime numbers}, E = {even numbers}, D = {odd numbers}
- 3) Given $\epsilon = \{-6 \text{ to } +28\}$ and Sets E = {even numbers}, D = {odd numbers} and N = {natural numbers} P = {prime numbers}
Show in a Venn diagram with: a) E, D, N and ϵ b) P, D, N and ϵ

2.5 Operations in the set of integers and their properties**Activity 2.5**

Given a set of integer cards:

Pick two cards, e.g., -4 and +7, and find the sum.

Repeat with other examples: $-5 + (-3)$, $+6 + (-2)$, $0 + 5$.

Discuss **patterns and rules** you notice (positive + negative, negative + negative, etc.).



Content Summary

a) Addition of integers is the operation of **finding the sum** of two or more integers. For any two integers a and b : $a+b \in \mathbb{Z}$.

NOTE: $+3$ is the additive inverse of -3 . Therefore, when we add a number and its additive inverse, we get zero (0) as the result.

Rules: When adding integers:

1. Same signs $\rightarrow$ **add and keep the sign.**
2. Different signs $\rightarrow$ **subtract and keep the sign of the bigger number.**
3. Adding zero does not change a number.

Examples: a) $5+3=8$ b) $(-4)+(-6)=-10$... c) $7 + (+2) = 9$, d) $-9 + 0 = -9$

b) Subtraction is the operation of finding the difference between two integers. It is the addition of the opposite.

Rules: To subtract integers:

1. Change the sign of the number being subtracted
2. Then apply the rules of addition

Examples:

- | | |
|--------------------|----------------------|
| 1) $8-5=8+(-5)=3$ | 3) $-3-5=-3+(-5)=-8$ |
| 2) $6-(-4)=6+4=10$ | 4) $-7-(-2)=-7+2=-5$ |

c) Multiplication is the operation of **repeated addition** of the same integer.
 $A \times b \in \mathbb{Z}$

Sign Rules: To multiply integers

- | | |
|--|--|
| 1) Negative $\times$ Negative = Positive | 5) Same signs $\rightarrow$ positive |
| 2) Positive $\times$ Negative = Negative | 6) Different signs $\rightarrow$ negative |
| 3) Negative $\times$ Positive = Negative | 7) Any number $\times 0 = 0$ |
| 4) Positive $\times$ Positive = Positive | |

Examples

- 1) $4 \times 3 = 12$ 2) $(-5) \times (-2) = 10$ 3) $6 \times (-3) = -18$ 4) $0 \times (-8) = 0$

d) Division is the operation of **sharing or grouping objects equally**.

When we divide a by the integer b (where $b \neq 0$), the result of $\frac{a}{b}$ is **not always an integer**.

Sign rules:

- | | |
|--|---|
| a) Negative $\div$ Negative = Positive | d) Positive $\div$ Negative = Negative |
| b) Positive $\div$ Positive = Positive | e) Same signs $\rightarrow$ positive |
| c) Negative $\div$ Positive = Negative | f) Different signs $\rightarrow$ negative |
| | g) Division by 0 is not defined in $\mathbb{R}$. |

Examples:

- 1) $12 \div 3 = 4$ 2) $(-20) \div (-5) = 4$ 3) $18 \div (-6) = -3$ 4) $-15 \div 5 = -3$

**APPLICATION ACTIVITY 2.5**

- 1) Work out

a) $(+3) + (-2)$	d) $(+3) + (+4) + (-2)$
b) $(+7) + (+3)$	e) $(+2) + (-8) + (-3)$
c) $(-6) + (-2) + (-1)$	
- 2) What is the difference between the following temperatures?

a) 15°C and -6°C	d) $+5^{\circ}\text{C}$ and $+9^{\circ}\text{C}$
b) -8°C and -3°C	e) $+20^{\circ}\text{C}$ and -1°C
c) -10°C and $+10^{\circ}\text{C}$	
- 3) Work out each of the following integers:

a) $(+5) \times (-2)$	g) $(-4)3(-4)^2$
b) $(-10) \times (-10)$	h) $(+5) \times (-6) \times (-1)$
c) $(-12) \times (+5)$	i) $(-3^2) \times (-2)^2$
d) $(+6) \times (+5)$	j) $(-5) \times (+20)$
e) $(-6) \times (+6)$	
f) $(-2) \times (+5) \times (-3)$	
- 4) Work out the following:

a) $(-28) \div (-7)$	d) $(-48) \div (-6)$
b) $(+55) \div (-11)$	e) $(-36) \div (+9)$
c) $(+19) \div (-3)$	f) $(-100) \div (-10)$

2.6 Set of decimal numbers**Activity 2.6**

Write the following numbers in words:

- a) 1 b) 2.5 c) 3 d) 4.75 e) 5

Discuss the following questions:

- Which numbers look different from the others?
- What is special about numbers like **2.5** and **4.75**?
- Have you ever seen numbers written with a **dot** in real life?



Content Summary

Decimal fractions are those fractions whose denominators are powers of ten. A number in which there are decimal fractions is usually written with a dot between the whole number part and the decimal fraction part. This dot is called a **decimal point**.

Thus $543 \frac{7}{10}$ can be written as 543.7 and $543 \frac{1}{4} = 543 \frac{25}{100}$ can be written as 543.25.

These are examples of mixed decimals, i.e as they contain a whole number and a decimal fraction.

Types of decimal numbers:

a) Terminating decimals: End after a finite number of digits.

Examples: 0.5; 1.25; 2.75

b) Recurring (Repeating) decimals: Have digits that repeat.

Examples: 0.333...; 0.666...

Place Value of Decimals: tenths, hundredths, thousandths using a place-value chart.

Example 1: In 2.375: 3 → tenths, 7 → hundredths. 5 → thousandths

Example 2: Give the place values of digits in the decimal number 4093.2518

Solution:

Thousands	Hundreds	Tens	Ones	Decimal point	Tenths	hundredths	Thousandths	Ten thousandths
4	0	9	3	.	2	5	1	8

Note: The place values of the digits in a decimal number decrease by 1 from the decimal point to the right. Each place value to the right of the decimal point is one-tenth of the preceding place value. A decimal number can either be written as a fraction or in decimal form.

The decimal fraction $\frac{3}{1000}$ can be written as 0.003 read as **three thousandths** or zero point zero, zero three. The digit 3 is in the **thousandths place value**.



APPLICATION ACTIVITY 2.6

- 1) Give the place value of the digit underlined.
 - i) 21.062 ii) 0.983 15
- 2) Write these decimal numbers by using fractions
 - i) 18.42 iii) 0.96
 - ii) 8.31 iv) 8.03

2.7 Addition, subtraction, multiplication and division of decimals



Activity 2.7

- 1) Add, subtract, multiply and divide the following pairs of numbers
 - a) 0.245 and 0.8 c) 0.894 and 0.04
 - b) 0.44 and 0.12 d) 0.94692 and 0.214
- 2) What rules do you need to observe when carrying out the operations on decimal numbers?
- 3) Compare your results with those of other members in your class.



Content Summary

Addition and subtraction of decimals: When adding or subtracting decimal numbers arrange them vertically, aligning the decimal points. In so doing, digits with the same place value get aligned vertically and are then added or subtracted.

Example: Work out: (a) $10.5 + 9.7 + 0.62$ (b) $15.81 - 9.36$

Solution:

(a) $\begin{array}{r} 10.5 \\ 9.7 \\ + 0.62 \\ \hline 20.82 \end{array}$	(b) $\begin{array}{r} 15.81 \\ - 9.36 \\ \hline 6.45 \end{array}$
---	--

Multiplication of decimals: The decimals are multiplied in the same way as whole numbers. Arrange the numbers vertically with the digits right below each and multiply. The number of decimal places in the product is the sum of the decimal places in the numbers being multiplied (factors).

Example: Multiply: 0.48×0.006

Solution:

$$\begin{array}{r} 0.48 \quad \rightarrow \text{has 2 decimal places} \\ \times 0.006 \quad \rightarrow \text{has 3 decimal places} \\ \hline 0.00288 \quad \rightarrow \text{has 5 decimal places} \end{array}$$

Note: When the number of non-zero digits in the product is less than the required number of decimals, zero(s) are inserted in the product as in Example above, so as to give the number of decimal places required.

Division of decimals: in dividing a decimal by a whole number, we do it the same way as with two whole numbers but we ensure that the decimal point is inserted in the answer

Example: Simplify:
 $0.303\ 52 \div 8$ vertically aligned with the divided.

$$\begin{array}{r}
 0.037\ 94 \\
 8 \overline{) 0.303\ 52} \\
 \underline{-24} \\
 63 \\
 \underline{-56} \\
 75 \\
 \underline{-72} \\
 32 \\
 \underline{32} \\
 0
 \end{array}$$

In dividing a decimal number with another decimal, multiply the divisor and dividend by a power of 10 to make the divisor a whole number then divide as usual. Similarly, When dividing a decimal by a power of 10, count the zeros in the divisor, and move the decimal point an equal number of places to the left.

Example: Evaluate: (a) $0.376\ 2 \div 0.09$, (b) $82.13 \div 100$

Solution

$$\text{a) } \frac{0.3762}{0.09} = \frac{0.3762 \times 100}{0.09 \times 100} = \frac{37.62}{9} = 4.18$$

$$\text{b) } 82.13 \div 100 = 0.821\ 3$$

Mixed operations on decimals: an expression of decimal numbers may include more than one operation. Just like integers and fractions, the order of operation would be as follows: "BODMAS".

Perform the operations within the brackets.

Perform by division followed by multiplication.

Perform the addition followed by subtraction.

Example: Find the value of each of the following:

$$\text{a) } 5.4 + 3.2 - 8.2$$

$$\text{b) } \frac{0.3 \times 0.08}{0.05}$$

$$\text{c) } 23.97 + 2(11.4 - 6.03) - 7.8$$

Solution:

$$\text{a) } 5.4 + 3.2 - 8.2 = 8.6 - 8.2 = 0.4 \quad (\text{Addition followed by subtraction})$$

$$\text{b) } \frac{0.3 \times 0.08}{0.05} = \frac{0.024 \times 100}{0.05 \times 100} = \frac{2.4}{5} = 0.48$$

$$\begin{aligned}
 \text{c) } 23.97 + 2(11.4 - 6.03) - 7.8 &= 23.97 + 2(5.37) - 7.8 \text{ (brackets first)} \\
 &= 23.97 + 10.74 - 7.8 \text{ (Multiplication follows)} \\
 &= 34.71 - 7.8 \text{ (finally subtraction)} \\
 &= 26.91
 \end{aligned}$$



APPLICATION ACTIVITY 2.7

- 1) Find the value of:

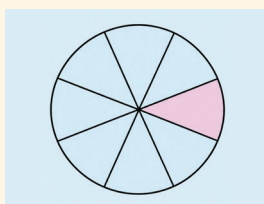
a) $+ 0.6 - 4.07$	d) $69.2 + 331.9 - 39.8$
b) $97.32 - 41.1 + 30.6 - 8.7$	e) $- 7.48 + 3.903$
c) $0.67 + 7.6 - 0.76$	f) $18.42 - 8.31 - 0.96 + 8.03$
- 2) What is the area of a rectangular steel plate 10.5 cm long and 7 cm wide?
- 3) Karera walks at the rate of 44 paces in 0.5 minutes. If each pace is 8.75 cm long; find the time he takes to walk 2.31 km.
- 4) Twenty five sheets of aluminium of a gauge have a total thickness of 18.75 cm. What is the thickness of each sheet?
- 5) Keza spends $\frac{1}{3}$ of her salary on accommodation and $\frac{2}{5}$ of the remainder on food. What fraction is left for other purposes? (express your answer in decimal form).

2.8 Set of rational numbers: Fractional numbers

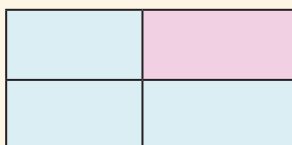


Activity 2.8

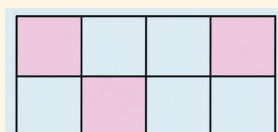
Study the following figures and answer the questions that follow.



What fraction does the shaded part represent for the whole circle?



What is the fraction of the unshaded part?



What fraction is represented by the shaded part?

What other name would you give to fractions? How many categories of fractions do you know? Name and describe each of them.



Content Summary

The **fractions** are **rational numbers**. Fractions are expressed in the form $\frac{a}{b}$ where a and b are integers and the value b should not be equal to zero i.e. $b \neq 0$.

As $\frac{2}{3}$ is a fraction, then 2 is called the **numerator** and 3 is called the **denominator**.

There are **2 types** of fractions: **proper fractions and improper fractions**. Proper fraction is where the numerator is less than the denominator.

Example is $\frac{11}{20}$. Improper fraction is where the numerator is greater than the denominator.

Example $\frac{17}{5}$. This fraction can be written as a mixed fraction as $3\frac{2}{5}$.

Equivalent fractions: Fractions that **represent the same value** even if they have different numerators and denominators.

Example: $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$

Simplifying Fractions: Reducing a fraction to its **lowest terms** by dividing numerator and denominator by their **greatest common factor (GCF)**.

Example: $\frac{6}{8} \rightarrow$ divide both by 2 $\rightarrow \frac{3}{4}$

Comparing Fractions: Use **common denominators** or **convert to decimals** to compare fractions.

Example: $\frac{3}{4} > \frac{2}{3}$ (because $\frac{9}{12} > \frac{8}{12}$).

Example: Tick True or false

- | | |
|---|--|
| a) The number 5 is a rational number.
True ✓ / False | f) $-\frac{2}{3}$ is a rational number.
True ✓ / False |
| b) The number $\sqrt{4}$ is a rational number.
True / False ✓ | g) $\sqrt{9}$ is a rational number.
True / False ✓ |
| c) The decimal $0.\bar{3}$ is a rational number.
True ✓ / False | h) 0.101001000100001... is a rational number.
True / False ✓ |
| d) 0.75 is a rational number.
True ✓ / False | i) $\frac{\sqrt{16}}{2}$ is a rational number.
True ✓ / False |
| e) π (approximately 3.14159...) is a rational number.
True ✓ / False | |



APPLICATION ACTIVITY 2.8

- 1) Differentiate the Improper fraction and proper fraction.
- 2) By using figure differentiate the numerator to denominator.
- 3) A chocolate bar is divided into 8 equal pieces. John eats 3 pieces. What fraction of the chocolate has John eaten? What fraction is left?

2.9 Addition, subtraction, multiplication and division of rational numbers



Activity 2.9

Work out the following fractions:

1. $\frac{1}{2} + \frac{1}{3}$

2. $1\frac{2}{3} + \frac{1}{2}$

3. $3\frac{5}{6} - 2\frac{3}{7}$

4. $\frac{6}{10} - \frac{4}{9}$

What method did you use when working out?



Content Summary

The addition or subtraction of fraction with Same Denominator: Keep the denominator and **add or subtract the numerators**.

$$\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}$$

$$\frac{a}{b} - \frac{c}{b} = \frac{a-c}{b}$$

With the different denominators, proceed as follows:

- i) Find the **Least Common Denominator (LCD)**,
- ii) Convert fractions,
- iii) Add or subtract the numerators,
- iv) Keep the denominator,
- v) Simplify.

Example:

1) $\frac{2}{3} + \frac{1}{4} = \frac{2 \times 4}{12} + \frac{1 \times 3}{12} = \frac{8+3}{12} = \frac{11}{12}$ LCD of 3 and 4 is 12:

2) $3 + \frac{2}{5} = \frac{15}{5} + \frac{2}{5} = \frac{17}{5} = 3\frac{2}{5}$

3) To add: $4\frac{3}{4} + 3\frac{1}{2}$

We will first convert the mixed fractions to single fractions.

$$\frac{19}{4} - \frac{7}{2} \quad (\text{LCM of 4 and 2 is 4})$$

$$\frac{19}{4} - \frac{14}{4} = \frac{19-14}{4} = \frac{5}{4} = 1\frac{1}{4}$$

Multiplication: when multiplying fraction, the numerator will multiply with the numerator and the denominator will multiply with the denominator.

The rule working is as follows $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$

When there is a common factor existing between the numerator and the denominator, the common factor cancels out by division.

The rule $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$ applies.

Example: work out

1. $\frac{2}{5} \times \frac{3}{4} =$

2. $4\frac{1}{2} \times 3\frac{2}{9} =$

3. $\frac{2}{5} \div \frac{1}{2} =$

4. $8\frac{3}{4} \div 4\frac{3}{3} =$

Solution

1. $\frac{2}{5} \times \frac{3}{4} = \frac{2 \times 3}{5 \times 4} = \frac{6}{20} = \frac{3}{10},$

2. $4\frac{1}{2} \times 3\frac{2}{9} = \frac{9}{2} \times \frac{29}{9} = \frac{\cancel{9}^1}{2} \times \frac{29}{\cancel{9}_1} = \frac{29}{2} = 14\frac{1}{2}$

3. $\frac{2}{5} \div \frac{1}{2} = \frac{2}{5} \times \frac{2}{1} = \frac{2 \times 2}{5 \times 1} = \frac{4}{5},$

4. $8\frac{3}{4} \div 4\frac{3}{3} = \frac{35}{4} \div \frac{15}{3} = \frac{35}{4} \times \frac{3}{15} = \frac{35 \times 1}{4 \times 5} = \frac{7}{4} = 1\frac{3}{4}$

NB: to work out fractions which involve more than one operation, we use a rule. This rule is called **BODMAS**. i.e. B = Brackets O = Of D = Division M = Multiplication A = Addition S = Subtraction The operations are worked out as they appear in the BODMAS.

Example: Work out $\frac{3}{5} + \left(2\frac{1}{2} - \frac{2}{3}\right) \div \frac{5}{6}$

Solution:

Step 1: Work out the brackets $\left(2\frac{1}{2} - \frac{2}{3}\right) = \frac{5}{2} - \frac{2}{3} = \frac{15-4}{6} = \frac{11}{6}$

Step 2: Work out on the operation of division $\frac{11}{6} \div \frac{5}{6} = \frac{11}{6} \times \frac{6}{5} = \frac{11}{5}$

Step 3: Lastly, work out on addition operator. $\frac{3}{5} + \frac{11}{5} = \frac{3+11}{5} = \frac{14}{5} = 2\frac{4}{5}$



APPLICATION ACTIVITY 2.9

Work out the following fractions

1. $1\frac{1}{3} + 3\frac{1}{2}$

4. $2\frac{3}{4} - 1\frac{4}{5}$

7. $5\frac{2}{3} \div 4\frac{1}{4}$

2. $2\frac{4}{5} + 1\frac{6}{7}$

5. $\frac{15}{6} \times \frac{4}{5}$

8. $2\frac{1}{2} \div \frac{1}{2} \div \frac{9}{70} \div \frac{4}{15}$

3. $7\frac{5}{6} - 4\frac{7}{8}$

6. $6\frac{7}{8} \times 2\frac{6}{11}$

9. $\frac{3\frac{1}{2} - 1\frac{5}{6} \times \frac{3}{11}}{1\frac{3}{4} + 7\frac{2}{3} \div 3\frac{5}{6}}$

2.10 Conversion between decimals and fractions



Activity 2.10

Divide the following fractions (numerator by denominator) to obtain a decimal number.

Use the long division method to divide. (i) $\frac{1}{2}$, (ii) $\frac{4}{5}$, (iii) $\frac{12}{1}$, (iv) $\frac{1}{3}$

- What do you notice?
- Is there a fraction that is not ending when you divide? What do you call such a fraction?



Content Summary

a) Changing fractions to decimals:

When you divide a fraction such as $\frac{1}{3}$ the same number(s) repeat itself without ending. This number is called a **recurring decimal**.

Example Change the following fraction to decimals $\frac{1}{7}$

Solution:

$$\begin{array}{r}
 0.142857 \\
 7 \overline{) 1.0} \\
 \underline{- 7} \\
 30 \\
 \underline{- 28} \\
 20 \\
 \underline{- 14} \\
 60 \\
 \underline{- 56} \\
 40 \\
 \underline{- 35} \\
 50 \\
 \underline{- 49} \\
 + 1
 \end{array}
 \quad \frac{1}{7} = 0.142857...$$

b) Changing decimals to fractions: This can be done by looking at the number of decimal places.

Example: $0.4 = \frac{4}{10} = \frac{2}{5}$

a) Changing recurring decimals to fractions: is the process of rewriting a repeating decimal number as a ratio of two integers by using algebraic methods so that the repeating part is eliminated. In other words, it means expressing a recurring decimal in the form: $\frac{a}{b}, b \neq 0$

b) where a and b are integers.

Example: Convert to fraction: (i) $0.\overline{16}$ (ii) $0.\overline{213}$

Solution:

(i) $0.\overline{16}$ Let $y = 0.1666\dots$ (1)

Multiplying 100 both sides, we get,

$$100y = 16.666\dots \quad (2)$$

Subtracting (1) from (2) we get,

$$100y = 16.666$$

$$-y = 0.1666$$

$$99y = 16.5$$

Multiply by 10 to remove the decimal place.

$$990y = 165. \text{ Hence } y = \frac{165}{990} = \frac{1}{6}$$

(ii) $0.\overline{213}$

$$\text{Let } y = 0.213\ 213\ 213\dots\dots\dots (1)$$

Multiply (1) by 1000

$$1000y = 213.213\ 213\dots\dots\dots (2)$$

Subtract (1) from (2)

$$1000y = 213.213213$$

$$-y = 0.213213$$

$$999y = 213 \quad y = \frac{213}{999} = \frac{71}{333}$$

$$\text{Therefore, } 0.\overline{213} = \frac{71}{333}$$

NB: A recurring decimal e.g. $0.222\dots$, $0.12323\dots$, $0.213213\dots$ etc are normally written in short form as $0.\dot{2}$, $0.1\dot{2}3$, $0.2\dot{1}3$ etc. The dot is put above the recurring digit(s).

Note: As conclusion, we see that all integers and decimal numbers are elements of the set $\mathbb{Q}$ of rational numbers. This means that $\mathbb{Z} \subset \mathbb{D} \subset \mathbb{Q}$.



APPLICATION ACTIVITY 2.10

1) Express the following fractions into decimals.

a) $\frac{3}{8}$ b) $\frac{8}{9}$, c) $\frac{1}{7}$, d) $\frac{4}{9}$

2) Convert the following decimals to fractions.

a) 0.2 (b) 0.62 (c) 0.012 (d) 0.001 (e) 1.4 (f) 1.2

3) Convert the following recurring decimals to fractions. a) $0.\dot{5}$, b) $0.7\dot{2}$, c) $0.1\dot{3}$, d) $0.\dot{7}1\dot{7}$

2.11 Irrational numbers



Activity 2.11

Discuss and carry out the following:

1. Find: (a) $\sqrt{8}$ (b) $\sqrt{\frac{2}{5}}$

2. Can you express $0.24156735\dots$ as a fraction?

What do you notice? Is it possible to get the ratio in step 1 and 2 above?

How do you call such numbers?



Content Summary

Irrational numbers: are numbers which do not have exact roots neither can they be expressed as fractions.

- An **irrational number cannot** be expressed as a fraction.
- Its decimal **never ends** and **does not repeat**.

Examples:

- $\pi = 3.141592\dots$ (infinite, non-repeating)
- $\sqrt{2} = 1.414213\dots$ (infinite, non-repeating)
- Every number is either **rational** or **irrational**.
- Rational numbers include **integers, fractions, and repeating or terminating decimals**.
- Irrational numbers include **non-repeating, non-terminating decimals** and certain roots and constants like π

Solution:

Examples: Find without calculator $\sqrt{12}$.

12 is estimated between 9 and 16. Therefore $\sqrt{12}$ is between 3 and 4.

Number	9	12	16
Square root	3	$\sqrt{12}$	4

We can see that 3 is less than $\sqrt{12}$ and $\sqrt{12}$ is less than 4. We can represent it as $3 < \sqrt{12} < 4$. To be more exact, consider the decimals between 3 and 4. The number 12 is less than half way between 9 and 16, the decimals should be less than half-way between 3 and 4.

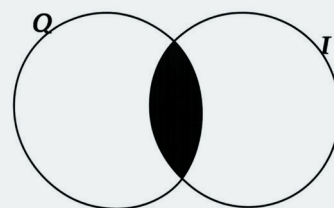
Let us try 3.4 and 3.5. $(3.4)^2 = 11.56$ and $(3.5)^2 = 12.25$. More accurate is $(3.45)^2 = 11.9025$ and $(3.47)^2 = 12.409$ so $3.45 < \sqrt{12} < 3.47$

Two significant figures, $\sqrt{12} = 3.4$.

Irrational numbers can be represented on a number line as follows:



Note: The set $\mathbb{Q}$ of rational numbers and the set I of Irrational numbers have no common elements. They are disjoint sets. $\mathbb{Q} \cap I = \emptyset$, $\mathbb{Q} \not\subset I$ and $I \not\subset \mathbb{Q}$.





APPLICATION ACTIVITY 2.11

From the list below, **write only the irrational numbers**:

a) $\sqrt{2}$

c) 0.25

e) $-\sqrt{7}$

g) $0.\dot{6}$

b) $\sqrt{16}$

d) π

f) 3

h) $\sqrt{11}$

2.12 The set of real numbers



Activity 2.12

Look at numbers in everyday life:

- Temperature: 36.5°C 36.5
- Money: 250. 75 FRW
- Count of students: 30
- Square root: $\sqrt{2}$
- π in circle calculations: 3.1416

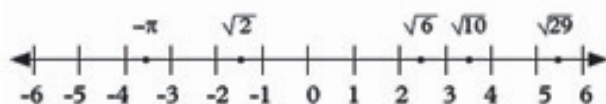
Discuss in group

- Which numbers are whole numbers?
- Which numbers have decimals?
- Are there numbers that **cannot be written as fractions**?
- What is the set that combines all numbers used in **real life**?



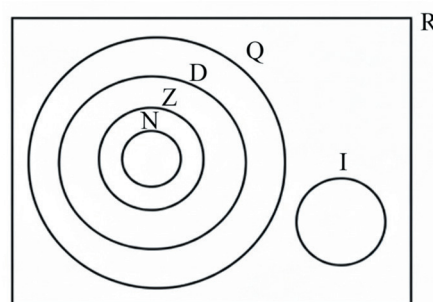
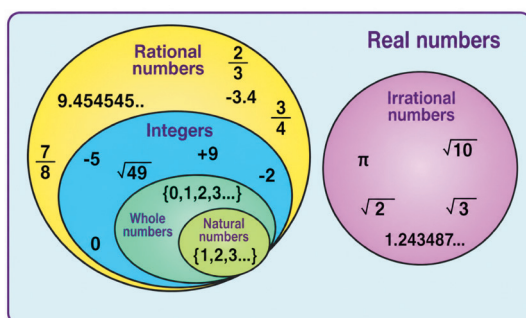
Content Summary

The set of real numbers is denoted by $\mathbb{R}$. Real numbers are represented on a number line as infinite points or they are set of decimal numbers found on a number line. This is illustrated on the number line below.



Note: All these sets are subsets of real numbers that is natural numbers, integers, rational numbers, irrational numbers.

- Every number you can think of in real life is a real number.
- They form the foundation of algebra, geometry, and calculus.
- This can be shown in the Venn diagram below.



Example: Place each number in the correct group: Natural (N), Integer (Z), Rational (Q), Irrational (I), Real (R)

Numbers: 3, -5, $\frac{1}{4}$, $\sqrt{2}$, 0, 4.5, π

Number	Set
3	N, Z, Q, R
-5	Z, Q, R
$\frac{1}{4}$	Q, R
$\sqrt{2}$	I, R
0	Z, Q, R
4.5	Q, R
π	I, R



APPLICATION ACTIVITY 2.12

- Show the following real numbers on the number line
a) -4, 1.7, π , $\sqrt{2}$, 3.8, -1, b) -2, 1.1, $-\pi$, $\sqrt{6}$, -2.8, 13.
- Given that $\sqrt{2} = 1.414213$, $\sqrt{3} = 1.732050$ and $\sqrt{6} = 2.443489$, find:
a) $\sqrt{6} - \sqrt{2}$ (b) $\sqrt{3} + \sqrt{2}$ (c) $\sqrt{3} - \sqrt{2}$ (d) $\sqrt{6} + \sqrt{2}$ (e) $\sqrt{6} - \sqrt{3}$
- Given $\varepsilon = \{-2, 3, 8, 16, 25, 27, 36, 37, \sqrt{6}, \sqrt{11}, 2.3, 81\}$, list and show on a Venn diagram the subsets Z, Q and N where: Z = {integers}, Q = {rational numbers} and N = {natural numbers}

2.13 The relationship between sets of numbers



Activity 2.13

Draw six boxes of sets of numbers and complete these numbers appropriately:

$3, -5, 0, \frac{1}{4}, 2.75, \sqrt{2}, \pi$

Then answer the following questions

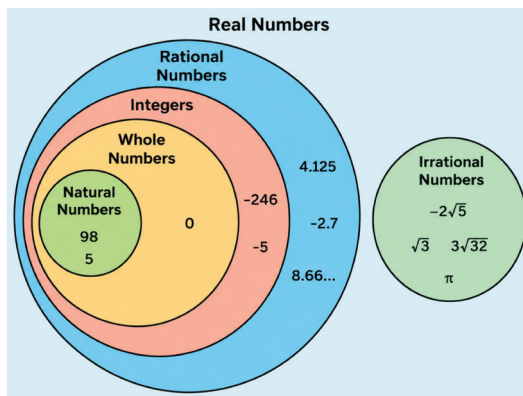
- Is every natural number an integer?
- Are all integers decimals?
- Are all decimals rational?
- Do rational and irrational numbers belong to one larger set?



Content Summary

We have learnt the following sets of numbers:

- 1) **Natural Numbers** or the set of **counting numbers**: $N = \{1, 2, 3, 4, 5, \dots\}$
- 2) All **whole numbers**: Positive numbers including zero.
- 3) **Integers (Z)**, $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- 4) **Decimal Numbers (D)**: Numbers written in **decimal form with a finite number of decimal places**. **Examples**: 0.5, 2.75, -4.2
- 5) **Rational Numbers (Q)**: Numbers that can be written as a **fraction**:
 $Q = \{\frac{a}{b} : a, b \in Z, b \neq 0\}$
Examples: $\frac{3}{4}, -2, 0.25, 1.5$
- 6) **Irrational Numbers (I)**: Numbers that **cannot be written as fractions** and have **non-terminating, non-repeating decimals**.
Examples: $\sqrt{2}, \sqrt{3}, \pi$
- 7) **Real Numbers (R)**: All **rational and irrational numbers together**: $R = Q \cup I$.



Meaning:

- Natural numbers are part of integers.
- Integers are part of rational numbers.
- Decimal numbers are also rational numbers.
- Rational and irrational numbers together form real numbers.

Real numbers include **all numbers on the number line**. $N \subset Z \subset D \subset Q \subset R$ and $I \subset R$



APPLICATION ACTIVITY 2.13

1) Classify each number into N, Z, D, Q, I, R:

- | | | |
|---------|------------------|---------|
| a) 6 | d) $\frac{1}{2}$ | f) 0 |
| b) -4 | e) $\sqrt{5}$ | g) -3.2 |
| c) 0.75 | | |

2) True or False

- | | | |
|------------------|------------------|----------------------------------|
| a) $N \subset Z$ | c) $D \subset Q$ | e) $I \subset R$ |
| b) $Z \subset Q$ | d) $Q \subset I$ | f) Every real number is rational |

3) Fill in the Blanks

- The set of integers is denoted by _____
- All decimal numbers are _____ numbers.
- $\sqrt{3}$ belongs to the set of _____ numbers.
- The largest set of numbers is called _____.

4) Multiple Choice

- Which set contains all others?
A. N B. Z C. Q D. R
- Which number is irrational?
A. 0.5 B. -2 C. $\sqrt{7}$ D. $\frac{3}{4}$
- Which of the following is a decimal number?
A. $\sqrt{2}$ B. $\frac{1}{3}$ C. 2.25 D. π
- Which statement is correct?
A. $Q \subset Z$ B. $N \subset Z$ C. $R \subset Q$ D. $I \subset Q$

2.14 Properties of four operations in the set of real numbers



Activity 2.14

Answer to these questions:

- What do you notice about $7 + 0$?
- What do you notice about 5×1 ?
- Compare $(2 + \sqrt{3}) + 4$ and $2 + (\sqrt{3} + 4)$. Are they equal?



Content Summary

A. Closure Property of 4 operations in the set of real numbers:

The result of addition, subtraction, multiplication, or division of two real numbers (except division by zero) is also a real number.

Examples: $2 + 3 = 5 \in \mathbb{R}$ and $-4 \times \frac{1}{2} = -2 \in \mathbb{R}$

B. Commutative property of Addition and multiplication

Addition: $a + b = b + a$

Multiplication: $a \times b = b \times a$

Subtraction and Division are not commutative.

Examples: $3 + 5 = 5 + 3$ and $2 \times (-4) = (-4) \times 2$

Subtraction and division of real numbers are NOT commutative.

C. Associative Property Addition and multiplication

– Addition: $(a + b) + c = a + (b + c)$

– Multiplication: $(a \times b) \times c = a \times (b \times c)$

Subtraction and Division are not associative.

Example: $(2 + 3) + 4 = 2 + (3 + 4)$

D. Distributive property:

Multiplication has a distributive property over addition and subtraction of real numbers.

$$a(b + c) = ab + ac$$

$$a(b - c) = ab - ac.$$

For example, if $a = 2$, $b = 3$ and $c = 5$, then

$$\begin{aligned} 2(3 + 5) &= (2 \times 3) + 2 \times 5 \\ 16 &= 16 \end{aligned}$$

$$\begin{aligned} 2(3 - 5) &= (2 \times 3) - (2 \times 5) \\ -4 &= -4 \end{aligned}$$

E. Identity operators on real numbers: An identity operator on a real number leaves the value of the number unchanged.

Consider the real number a

(i) **Zero (0)** is an identity in addition and subtraction operator on real numbers.

$$a + 0 = a, \quad a - 0 = a$$

For example, if $a = 4$, then $4 + 0 = 4$ and $4 - 0 = 4$

(ii) **One (1)** is an identity in multiplication and division on real numbers.

$$a \times 1 = a, \quad a \div 1 = a, \quad a^1 = a$$

For example, if $a = 2$, then $2 \times 1 = 2$,

$$2 \div 1 = 2 \text{ and } 2^1 = 2$$

j) Inverse Property: Additive inverse: $a + (-a) = 0$

Multiplicative inverse: $a \times \left(\frac{1}{a}\right) = 1$ ($a \neq 0$)

Example: $7 + (-7) = 0$ and $2 \times \frac{1}{2} = 1$



APPLICATION ACTIVITY 2.14

- 1) Write **TRUE** or **FALSE**.
 - a) $a+b=b+a$ for all real numbers a and b .
 - b) $(a-b)-c=a-(b-c)$ for all real numbers.
 - c) $a \times 1 = a$ for all real numbers.
 - d) Every real number has a multiplicative inverse.
 - e) $a(b+c)=ab+ac$
- 2) Choose the correct answer.
 - a) Which property is shown by: $8+5=5+8$

A. Associative B. Commutative C. Distributive D. Inverse
 - b) Which is the additive identity in $\mathbb{R}$?

A. 1 B. -1 C. 0 D. -2
 - c) The multiplicative inverse of -5 is:

A. 5 B. -5 C. $\frac{1}{5}$ D. $-\frac{1}{5}$
 - d) Which statement is correct?

A) $a-b=b-a$ B) $a \div b = b \div a$ C) $a+b=b+a$ D) $a \div 1=1$
- 3) A shopkeeper multiplies the price of one book (**300 FRW**) by **1** to get the total for one book. What property is used?
- 4) A learner calculates: $5(200+300) = 5 \times 200 + 5 \times 300$ Which property is applied?



END OF UNIT ASSESSMENT

Assessing knowledge and understanding:

1. Explain the set of natural numbers
2. Write three examples of integers.
3. What is a rational number?
4. Explain a decimal number.
5. What is an irrational number?
6. Explain the set of real numbers ($\mathbb{R}$).
7. State the relationship between the set $\mathbb{N}$ and $\mathbb{Z}$.
8. State the relationship between the set $\mathbb{Q}$ and $\mathbb{R}$.
9. Write TRUE or FALSE:
 - a) All natural numbers are integers.
 - b) All integers are natural numbers.
 - c) All rational numbers are real numbers.
 - d) All real numbers are irrational numbers.
 - e) 0 is an integer.
10. By using example give the properties of operations in real numbers.

Assessing skills

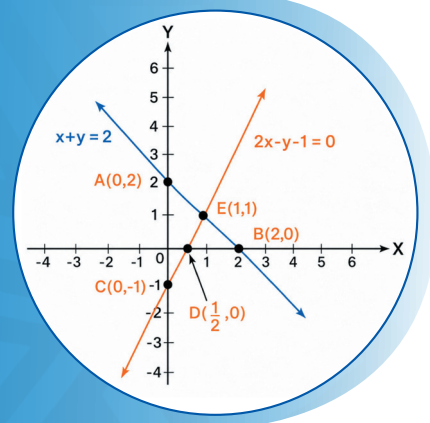
11. Work out the following: (a) $\frac{(-a) \times (-b)}{(-c)}$, b) $(-2) \times (+3y)$, c) $(+18x) \div (-3x)$.
12. Find the values of
 - (a) $\{9 + (-2) \times (-15)\} \times (-2 + 7) \div 3$
 - b) $\frac{-6 + (-5) + 8 \times -2}{-4 + (-2)}$
 - c) $-3 \times 23 + (-5) \times (-1) - 8 \times (-4)$
13. Evaluate (a) $0.45 + (2.45 - 1.009) \div 0.2 \div 0.008$
 (b) $0.547 \times 0.12 + 2.1 - 0.46 \div 0.007$
 (c) $16.57 - (12.432 \div 0.004) \times (16 \times 0.4) + 1.16$

Assessing Attitudes and values

14. A trader bought 15 sacks of maize and later sold 7 sacks. He also had a debt of 3 sacks.
 - a) Write the three numbers used in this story.
 - b) Show the set to which each number belongs.
 - c) Explain why understanding positive and negative numbers is important in business
15. Mention one way rational or decimal numbers are used in your daily life.

UNIT 3

LINEAR FUNCTIONS, EQUATIONS AND INEQUALITIES



Key Unit Competence:

Apply linear equations and inequalities in solving real-life problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain the number line and locate integers, fractions, and points using ordered pairs.
- Identify the Cartesian plane, its axes, and the coordinates of a point.
- Explain linear functions, their graphs, and the meaning of gradient (slope).
- Describe, write, and explain how to solve linear equations in one unknown.
- Explain inequality symbols, represent and solve linear inequalities, and relate them to real-life situations

Skills

- Locate and plot real numbers and points accurately.
- Draw linear graphs and find gradient and intercepts.
- Solve and represent linear equations correctly.
- Apply linear equations and graphs to real-life problems.
- Solve and represent linear inequalities accurately

Attitudes and values

- Show accuracy, carefulness, and logical thinking in all work.
- Develop patience, persistence, and self-confidence in problem solving.
- Appreciate the use of algebra and graphs in daily life.
- Work cooperatively and share ideas respectfully.
- Maintain neatness, organization, and correct use of symbols

INTRODUCTION

This unit aims to help you understand and use number lines, coordinates, linear graphs, equations, and inequalities to describe, solve, and interpret mathematical and real-life problems. The concepts in this unit are used in business (profit, loss, cost and sales), transport (distance–time graphs), construction (design and measurements), science (relationships between quantities), budgeting, mobile data and electricity consumption, and decision making using conditions and limits (inequalities).



Introductory activity

A student uses a motorcycle taxi to go to school. The cost is: 200 FRW for the first 1 km, then 150 FRW for each additional km. The student has 2,000 FRW in his pocket. Discuss the following:

1. How much does the trip cost for 2 km? 5 km?
2. Is there a relationship between distance and cost?
3. What is the distance that can be covered by the student using 2,000 FRW?
4. How can we represent this situation using numbers or a graph?

3.1 The number line



Activity 3.1

- Draw a long number line on the ground.
- Mix up number cards and each student randomly selects a number card and goes to stand on the number of the line that matches his/her card.

Then, answer to the following questions:

- 1) Who is to the left of 3?
- 2) Who is between 0 and 5?
- 3) Move three steps to the right of 2. Where are you now?

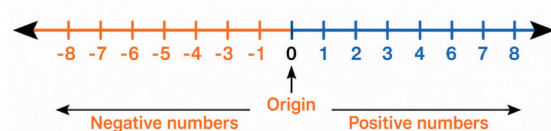


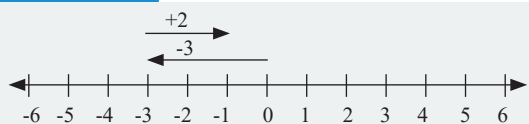
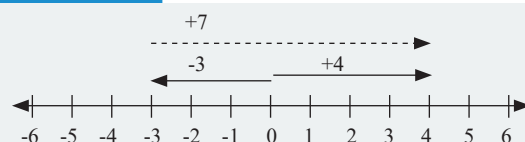
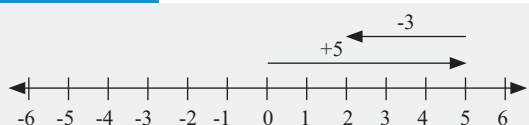
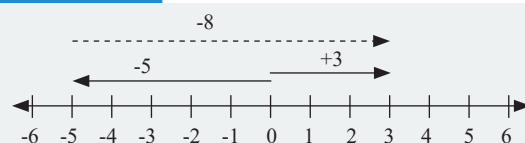
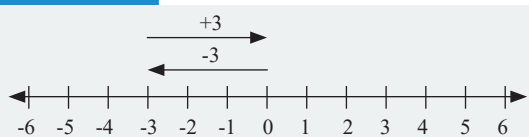
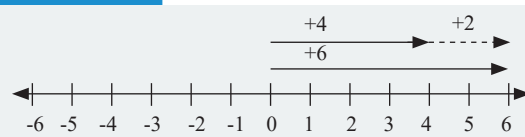
Content Summary

A number line is a straight line on which numbers are represented as points at equal intervals. It can include whole numbers, integers, fractions, decimals, and even negative numbers.

Key points:

- 1) Zero is usually placed at the center.
- 2) Numbers to the right are greater.
- 3) Numbers to the left are smaller.



Example:1. Work out $(-3) + (+2)$ **Solution:**4. Work out $(+4) - (-3)$ **Solution:**2. Work out $(+5) + (-3)$ **Solution:**5. Work out $(-5) - (+3)$ **Solution:**3. Work out $(-3) + (+3)$ **Solution:**6. Work out $(+6) - (+4)$ **Solution:****APPLICATION ACTIVITY 3.1**

1) Work out the following integers on a number line.

(a) $(-5) - (-4)$

(b) $(-5) - (+5)$

(c) $(-7) - (-9)$

(d) $(+3) - (+2)$

(e) $(+8) - (-3)$

(f) $(-1) - (-3)$

2) Try to locate the following numbers on one number line: $\sqrt{5}$; $\frac{3}{8}$; 4 ; $\frac{\sqrt{2}}{2}$.

3.2. Position of a point on a plane surface



Activity 3.2

- 1) Draw a number line and on it illustrate:
-7, -3, 1, 5, 9, 13, -2.5, 6.5
- 2) In the following gridded paper, the drawing of a dog is in 2E Cell. Where is the hen?

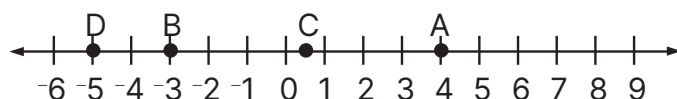
G					
F					
E					
D					
C					
B					
A					
	1	2	3	4	5



Content Summary

If we mark points on the number line, we can use real numbers to locate where they are on that number line.

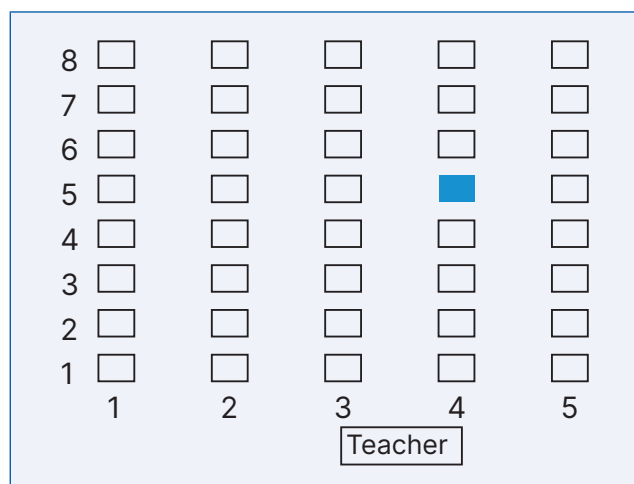
Example:



A is 4 units to the right of zero and B is 3 units to the left of zero. This can be shortened to A (4) and B (-3). Similarly, C is the point C(1/2) and D (-5). A (4), B (-3), C(1/2) and D(-5) give the positions of A, B, C and D on the number line with reference to zero (0).

The **position of a point on a plane surface** is a way of showing exactly **where** the point is located on that flat surface.

Here is a scenario:



Each rectangle represents a desk. Let us describe the position of the shaded desk.

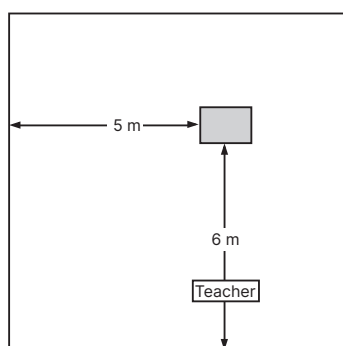
You might describe the desk as being in the fourth column and fifth row. If we agreed to write this as (4, 5), it would mean 4 columns across from the left-hand side and then 5 rows up (i.e. from the front towards the back).

Do (5, 4) and (4,5) mean the same position?

The order of the numbers inside brackets is important, it is the same as the ordered pairs.

If all the desks in the classroom were removed and then you were asked to put your desk back in the room, exactly where it was before, how would you do it?

If you knew it was 5 metres from one wall and 6 metres from another, could you do it then? How many places on your classroom floor are 5 m from one wall and 6 m from another? We have to decide which walls we are going to use.



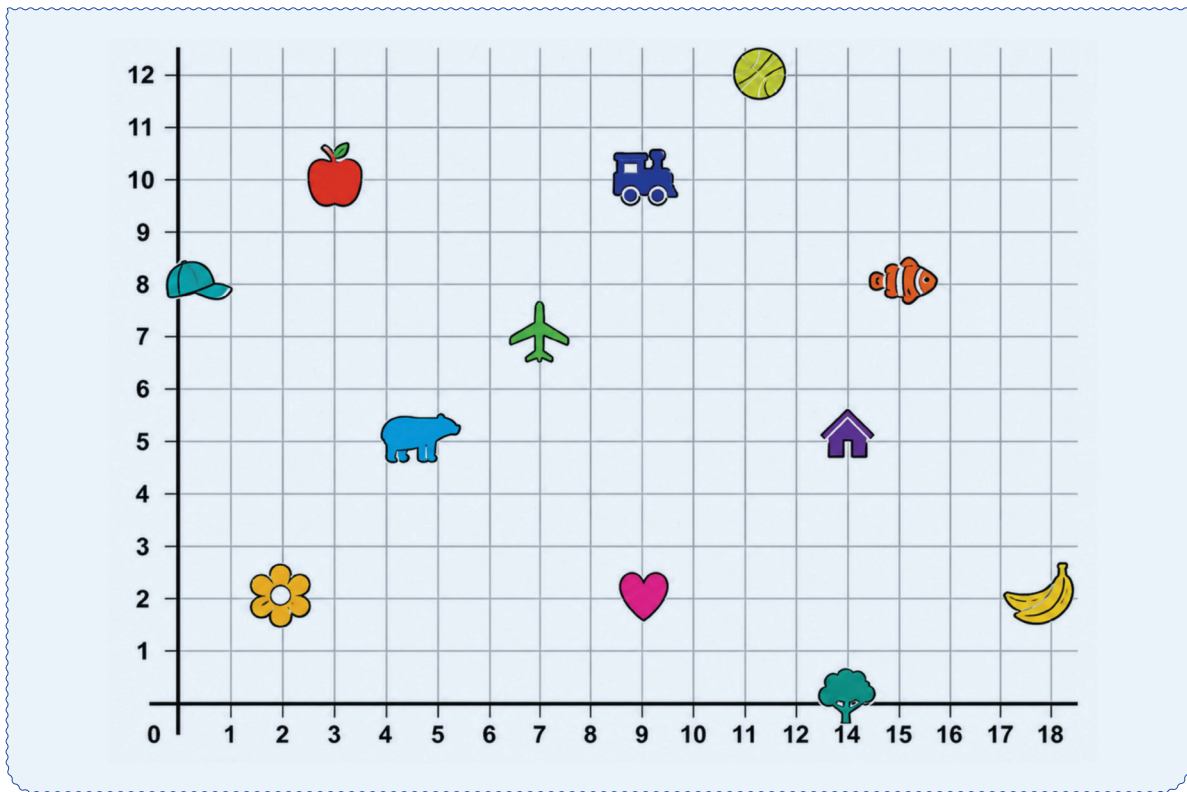
This figure shows one way in which you might fix the position of your desk. We could say that the desk has an ordered pair (5, 6), to mean five metres from the left-hand wall and six metres from the front wall. From the above discussion, you should have concluded the following: to locate a specific point.

- i) You need a reference point.
- ii) You need the direction.



APPLICATION ACTIVITY 3.2

- 1) Draw a number line from -5 to 5 . On it, mark the points A (3), B (-1), C ($-4\frac{1}{2}$) and D ($3\frac{1}{2}$).
- 2) In the following figure, the apple is at the point (3,10). Use ordered pairs to locate the position of : a) the airplane b) The hat c) The heart.



3.3 Drawing and labelling axes



Activity 3.3

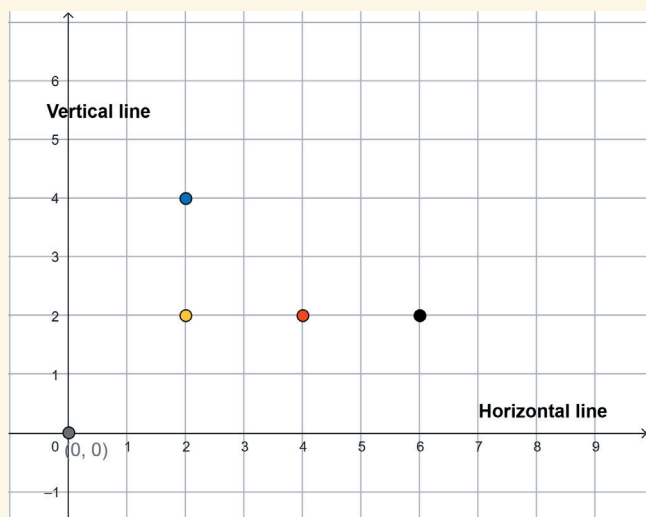
Marbles of the different colours are deposited on a gridded surface. Discuss how you can explain and represent the **position** of each marble: Red marble, Black marble, Blue marble and yellow marble.

The head teacher asks:

Draw two perpendicular lines on that gridded surface (horizontal and vertical lines),

Mark their intersection point as 0 and use the grids to show numbers on those lines as if you have number lines

How can you locate the position of each marble on that gridded surface?





Content Summary

a) Axes:

Axes are two perpendicular number lines used to locate points on a plane. They intersect at a point O with coordinate (0,0) and it is called the **origin**.

b) Types of Axes: one is the **Horizontal axis** called the **x-axis**. The other is the **Vertical axis** called the **y-axis**

c) Origin: The point where the x-axis and y-axis meet. It is written as **O (0, 0)**.

d) Direction of Axes:

- On the **x-axis**: Right direction indicates positive (+). The left direction indicates negative (-).
- On the **y-axis**: Upward direction indicates positive (+). Downward direction indicates negative (-)

e) Labelling axes

- Use equal distance representing 1 unit on both axes.
- Label numbers correctly on each axis. The distance of one unit (from 0 to 1) is maintained when labelling other points such as from 1 to 2, from 2 to 3 and so on. You can use a ruler and a pair of compasses. Always write **x-axis** and **y-axis** clearly.

Example 1: Which axis is horizontal and which is vertical?

Solution:

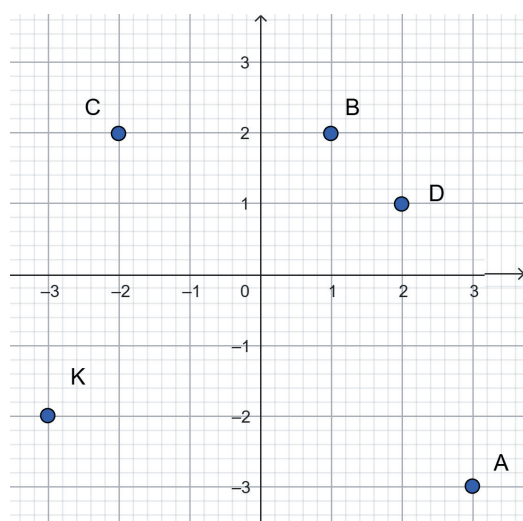
Horizontal line → **x-axis** and Vertical line → **y-axis**

Example 2: Which direction is positive on the y-axis?

Solution:

Upward direction is **positive (+)** on the y-axis.

Example 2 Name coordinate of the position of A, B, C, D and K



A(3,-3), B(1,2), C(-2,2), D(2,1) and K(-3,-2).



APPLICATION ACTIVITY 3.3

- 1) A student draws two crossing and perpendicular lines (the vertical and the horizontal one). How should she label each line?
- 2) Why are axes important in real life?
- 3) A learner labels the vertical axis as x-axis. Is this correct?
- 4) Draw axes and locate point M(4,1), P(0,5) and W(3, -1).

3.4 Cartesian plane and coordinates of a point



Activity 3.4

Draw a plane of desks in your classroom. Identify a pair of suitable walls to act as the vertical and horizontal axes. Using the columns and rows of desks, describe the position of your desk, and those of other members of your group. Compare your answers with the rest of the class. Use actual measurements, and a suitable scale to locate position of your desk.



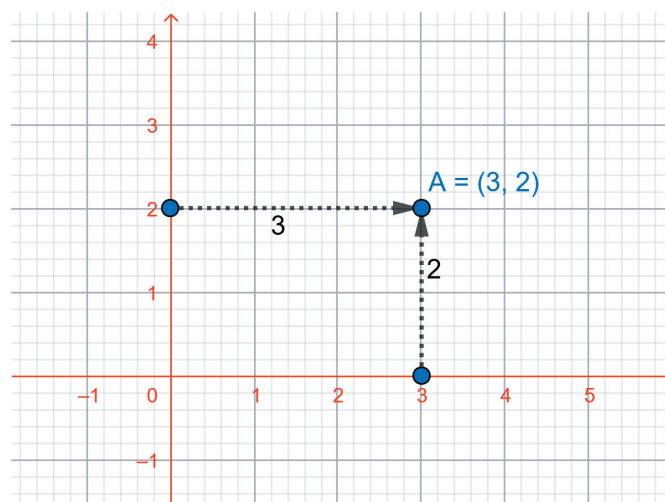
Content Summary

The positions of points on a line are found by using a number line, and are written down using a single number e.g. P(-7).

The positions of a point on a plane surface is found using two fixed number lines (i.e. two directions), usually perpendicular, and each point is described using two numbers that form an ordered pair. The perpendicular lines are called axes and the plane they make is called Cartesian plane.

The **ordered pairs** that describe the positions of points on a Cartesian plane are called Cartesian coordinates.

The first number is called the x-coordinate, and it gives the distance of the point from the origin in the direction of the x-axis.



The second number is called the y-coordinate and gives the distance of the point from the origin in the direction of the y-axis. For example, in $A(3, 2)$: 3 is the x-coordinate and 2 is the y-coordinate.

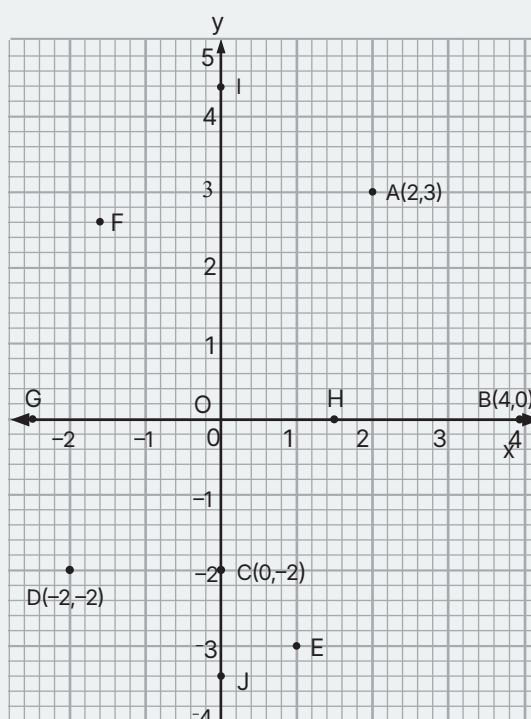
Plotting points: To plot a point means to mark its position on a Cartesian plane.

Procedure:

- Start at the origin.
- Move along the x-axis the number of steps, and in the direction given by the x-coordinate of the point.
- Move up or down parallel to the y-axis the number of steps, and in the direction given by the y-coordinate of the point.

Example: Starting from the zero point, O. represent these points in Cartesian plane: A is 2 units to the right and 3 units up, written as $A(2, 3)$. B is 4 units to the right and 0 units up, written as $B(4, 0)$. C is 0 units to the right and 2 units down, written as $C(0, -2)$. D is 2 units to the left and 2 units down, written as $D(-2, -2)$

Solution:

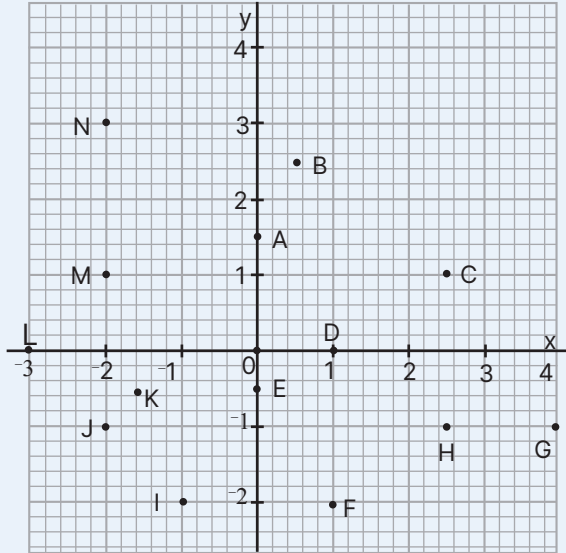




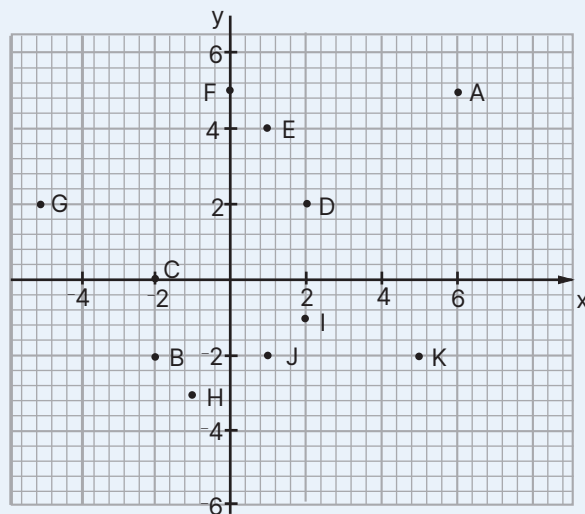
APPLICATION ACTIVITY 3.4

In this exercise work in pairs:

- 1) Name the points in figure below which have the following coordinates. B (0, -0.5); A(0, 1.5); C(-1, -2); E(0.5, 2.5); G(-2, -1); and I(2.5, 1);



- 2) What are the coordinates of the points A, B, C, D, E, F, G, H, I, J, K in figure below?



- 3) **Homework:** Using a scale of '1 cm represents 2 units' on both axes, plot the following points. A (8, 10), B (-8, -10), C (3, -5), D (-6, 9), E (-4, -7), F (1, 8), G (2, 0), H (0, -6), I (-2.4, 5.2), J (-4, 3.8), K (0, 6.6), L (0.8, -7.8).

3.5 Graph of a linear function



Activity 3.5

A water vendor charges a **fixed starting fee of 200 Frw**, plus **50 Frw for every jerrycan** delivered.

Two students, Alice and John, want to compare the total cost depending on the number of jerrycans.

- 1) Discuss "If Alice buys 1 jerrycan and John buys 3, who pays more and why?"
- 2) Fill this short table:

Number of jerrycans (x)	Starting fees	Cost of jerrycans	Total cost (y)
0	200	$0 \times 50 = 0$	$200 + 0 = -$
1	200	$1 \times 50 = 50$	$200 + 50 = -$
2	-	-	-
3	-	-	-
x	-	-	-

- 3) For each value x, find the corresponding value of y and write the ordered pair (x,y).
- 4) Draw a Cartesian plan and locate the 4 points M(0,y), P(1,y), Q(2,y) and S(3,y).
- 5) What type of graph do you find when you join those point? A line, or a curve?



Content Summary

A **linear function** is a rule that shows a relationship between variables in the form: **$y = mx + c$** Where:

- **m** = slope or gradient (the steepness of the line)
- **x** is a variable
- **c** = y-intercept (the value of y when $x = 0$)

When you join points (x,y) obtained you find the graph in form of a **straight line**. This is the reason why the equation is called a **linear function**.

In general, a variable means:

- i) A quantity which is not constant i.e. a quantity able to assume different numerical values under different conditions.
- ii) In a function such as **$y = mx + c$** where **m** and **c** are constants, x and y are variables.
- iii) The value of **y** depends on the value of **x** i.e. the value of **y** is determined by the value of **x**.
- iv) Thus: y is the dependent variable and x the independent variable.
- v) It is possible to draw a line called graph of the equation $y = mx + c$ using points (x,y) obtained using a table of values:

x	0	1	2	$\frac{-c}{m}$
$y = mx + c$	c	$(m \times 1) + c$	$(m \times 2) + c$	0
Point	$(0, c)$	$(1, (m \times 1) + c)$	$(2, (m \times 2) + c)$	$\frac{-c}{m}, 0$

Examples

1. Consider the equation $y = x + 2$ (a linear equation in x and y). Represent it in a cartesian plane.

Solution:

For every value of x , there is a corresponding value of y . x and y are called variables. The equation gives the connection (or relation) between the variables.

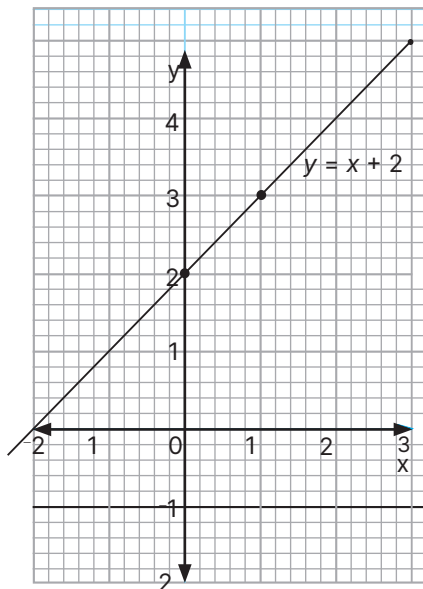
For example, if $x = 0$, $y = 2$; if $x = 1$, $y = 3$; if $x = 3$, $y = 5$, etc. These values can be written as ordered pairs of corresponding x and y values as: $(0, 2)$, $(1, 3)$, $(3, 5)$, etc.

x	0	1	2	3
$y = x + 2$	2	3	4	5

If these pairs are plotted as points on the Cartesian plane, they give the graph of the equation

$y = x + 2$, as shown in the table and the figure below:

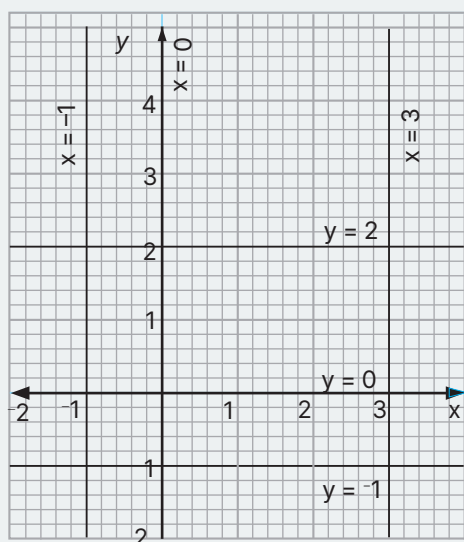
Since the points lie on a straight line, they are joined using a straight edge or ruler.



We find that for $x=0$, $y=2$. The point $(0,2)$ is where the line crosses the y -axis.

For $y=0$, we see that $x = -2$. The point $(-2,0)$ is the point where the line crosses the x -axis. This point is called x -intercept.

2. On a different pair of axes, draw the lines whose equations are $x = 4$, $x = -3$, $y = 3$, $y = -4$, $y = 5$, $y = 2\frac{1}{2}$, $y = 0$

Solution:**APPLICATION ACTIVITY 3.5**

- On a Cartesian plane, draw and label the lines whose equations are:
(a) $x = 5$ (b) $y = -2$ (c) $y = 3.5$ (d) $x = -4$
- For each of the following lines, find the coordinates of any two points on the line. Use the points to draw the lines whose equations are:
(a) $y = x + 1$ (b) $y = 5x - 1$ (c) $y + \frac{1}{4}x = 2$ (d) $2x - y = 3$
- Copy and complete this table for the equation $3y + 2x = 21$

x	-6	-3	0	3	6
y	11	-	-	-	3

- Using a scale of 2 cm to represent 1 unit on both axes, draw the graph of $3y + 2x = 21$
- Use your graph to find the value of: (i) y when $x = -1.5$ (ii) x when $y = 4$.

3.6 Intercepts: The y and x-intercept**Activity 3.6**

A simple linear equation has the equation: $y = 2x - 4$.

Plot that line in the Cartesian plane and discuss:

- "What happens to the line when $y = 0$?"
- "What happens when $x = 0$?"



Content Summary

- **The x-intercept** is the point where a graph **crosses the x-axis**.
 - At this point, **$y = 0$** .
 - It shows the value of **x** when the function touches or cuts the x-axis.
 - For the line whose equation is $y = mx + c$, the point of coordinates $(-c/m, 0)$ is called **x-intercept**.

Example:

For the function: $y = 2x - 4$

Set **$y = 0$** :

$$0 = 2x - 4 \Rightarrow x = 2$$

So, the **x-intercept** = **$(2, 0)$** .

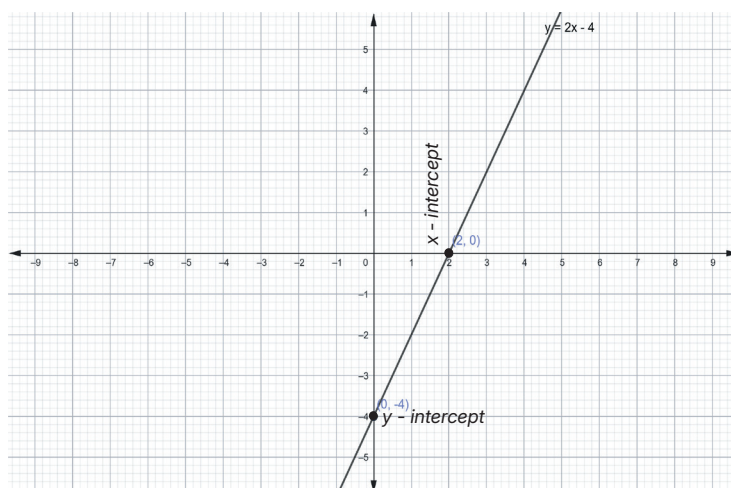
- **The y-intercept** is the point where a graph **crosses the y-axis**.

Note; For $y = mx + c$, the **y-intercept** is the point of coordinates $(0, c)$.

- At this point, **$x = 0$** .
- It shows the value of **y** when the function touches or cuts the y-axis.

Example:

- For the function: $y = 2x - 4$.
- Set **$x = 0$** :
- $y = 2(0) - 4 = -4$
- So, the **y-intercept** = **$(0, -4)$** .



Note:

Intercept for $y = mx + c$	Meaning	Condition	Coordinates look like
x-intercept	Where the graph crosses x-axis	$y = 0$	$(\frac{-c}{m}, 0)$
y-intercept	Where the graph crosses y-axis	$x = 0$	$(0, c)$



APPLICATION ACTIVITY 3.6

Find the y or x-intercept of the following without drawing the graphs:

(a) $y = 3x + 7$

(e) $32x + 16y - 15 = 23$

(h) $2y + x = 7$

(b) $7 - 2x = 4y$

(f) $-10(x + 3) = 0.5y$

(i) $5x + 6y - 4 = 0$

(c) $4y + x - 8 = 0$

(g) $y = 5x - 4$

(j) $2y - 8 = 7x$

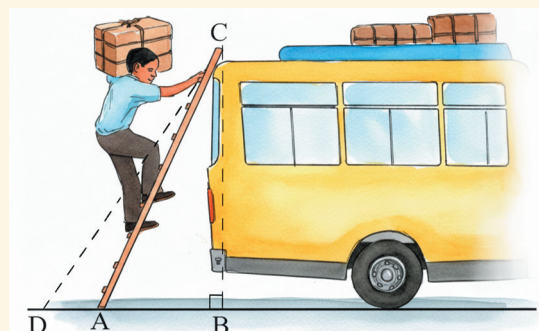
(d) $2y + 13$

3.7 The slope or gradient of a linear function



Activity 3.7

- 1) A bus conductor wishes to move some luggage to the roof rack of his bus. To get onto the roof, he uses an extendable ladder. Discuss the good position the bus conductor can use the extendable ladder and give the reason: Will he use the position BC? AC? or AC? Explain.



- 2) In the same cartesian plane, draw the lines with equations:

i) $y = 0.05x + 2$ ii) $y = x + 2$ iii) $y = 3x + 2$ iv) $y = -3x + 2$

How do the directions of lines differ depending on the coefficient m of x in the equation $y = mx + c$?



Content Summary

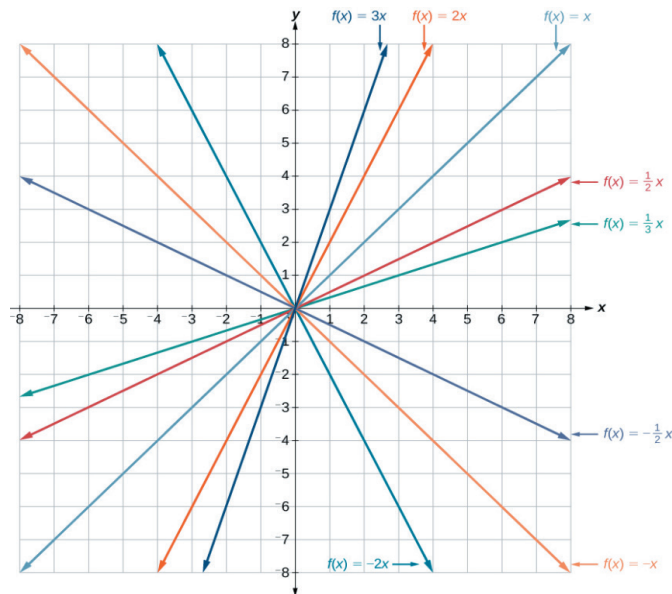
The **slope** of a straight line measures its steepness and indicates the direction in which the line rises or falls. It is defined as the ratio of the **vertical change** to the **horizontal change** between any two points on the line.

When riding on a bicycle, a slope transforms cycling by introducing gravity as the dominant opposing force. On flat ground, effort mainly overcomes rolling and air resistance, but an inclination requires the cyclist to exert significantly greater power to work against gravity's pull, necessitating more energy, and higher torque.



A **positive slope** means the line rises from left to right, while a **negative slope** means it falls from left to right. A **zero slope** corresponds to a horizontal line, and an **undefined slope** corresponds to a vertical line. The slope is an important concept in coordinate geometry as it describes the rate of change of one variable with respect to another and is widely used in graphing linear equations and solving real-life problems involving rates and trends.

Every straight line has a slope with respect to the horizontal axis. The measure of the slope is called the **gradient**.

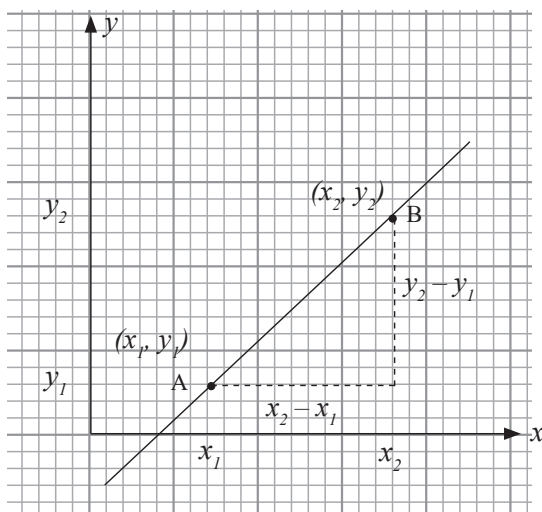


If equation of the line is $y = mx + c$, the gradient is m

In the Cartesian plane, the gradient of a line is the measure of its slope or inclination to the x -axis. It is defined as the **ratio of the change** in y -coordinate (vertical) to the change in the x -coordinate (horizontal).

$$\text{gradient} = \frac{\text{change in } y\text{-coordinate}}{\text{change in } x\text{-coordinate}}$$

Consider a line passing through the points A (x_1, y_1) and B (x_2, y_2).



By definition, gradient/slope is given by

or

$$\text{gradient} = \frac{\text{change in } y\text{-coordinate}}{\text{change in } x\text{-coordinate}} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}$$

Example 1

Find the gradient of a line passing through the points (2, 5) and (7, 9).

Solution:

(2, 5) and (7, 9)

(x_1, y_1) and (x_2, y_2)

$$\begin{aligned}\text{Gradient} &= \frac{\text{Change in } y\text{-coordinate}}{\text{Change in } x\text{-coordinate}} \\ &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{9 - 5}{7 - 2} \\ &= \frac{4}{5}\end{aligned}$$

$$\begin{aligned}\text{Or Gradient} &= \frac{\text{Change in } y\text{-coordinate}}{\text{Change in } x\text{-coordinate}} \\ &= \frac{y_1 - y_2}{x_1 - x_2} \\ &= \frac{5 - 9}{2 - 7} \\ &= \frac{-4}{-5} = \frac{4}{5}\end{aligned}$$

Example 2

Given that line AB passes through the points A(-3, -4) and B(-1, -1); while line CD passes through the points C(-3, 5) and D(8, 1). Find the gradients of line segments:

(a) AB (b) CD

Solution:

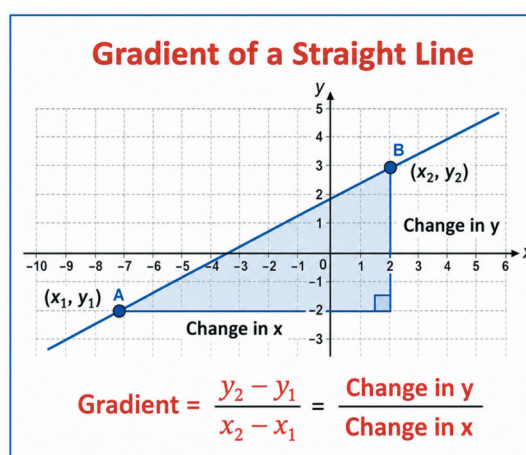
A(-3, -4) and B(-1, -1)

$$\begin{aligned}\text{Gradient } AB &= \frac{\text{Change in } y\text{-coordinate}}{\text{Change in } x\text{-coordinate}} \\ &= \frac{-4 - (-1)}{-3 - (-1)} = \frac{-3}{-2} \\ &= \frac{3}{2}\end{aligned}$$

C(-3, 5) and D(8, 1).

$$\begin{aligned}\text{Gradient } CD &= \frac{\text{change in } y\text{-coordinate}}{\text{change in } x\text{-coordinate}} \\ &= \frac{1 - 5}{8 - (-3)} = \frac{-4}{11}\end{aligned}$$

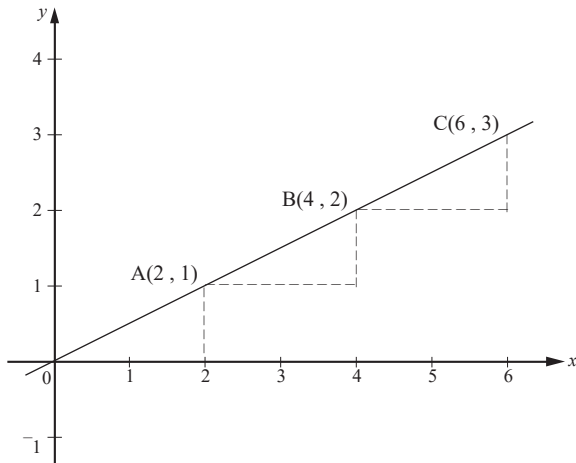
The gradient can be observed in the following way:



When you observe a line, the ratio, $\frac{\text{Vertical distance}}{\text{Horizontal distance}}$ is a measure of **steepness** or slope. It is known as **gradient**. Gradient measures how steep an incline is.

Note that gradient is higher for a steeper incline.

Example 1: Find the gradient of the following line:



Solution:

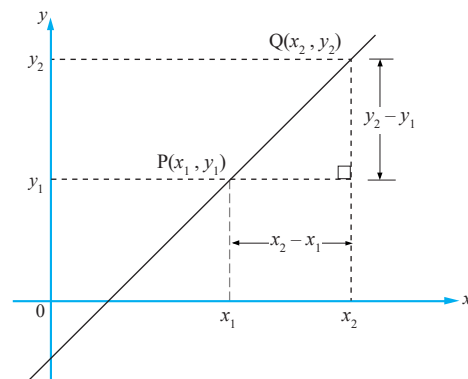
The gradient =

$$\frac{\text{Vertical distance}}{\text{Horizontal distance}} = \frac{3}{6} = \frac{1}{2}$$

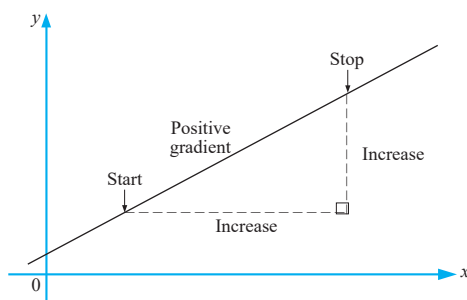
Example 2:

The gradient for a line passing by two points

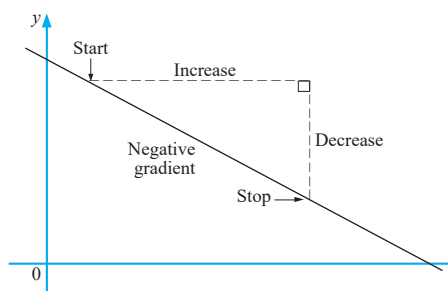
$$P(x_1, y_1) \text{ and } Q(x_2, y_2) \text{ is } \frac{y_2 - y_1}{x_2 - x_1}$$



In the same way, the following graphs show how we can find the gradient of lines:



The gradient is positive



The gradient is negative

Examples: Calculate the gradients of the line segments joining

- a) (2, 5) and (4, 7) b) (0, 4) and (2, 0).

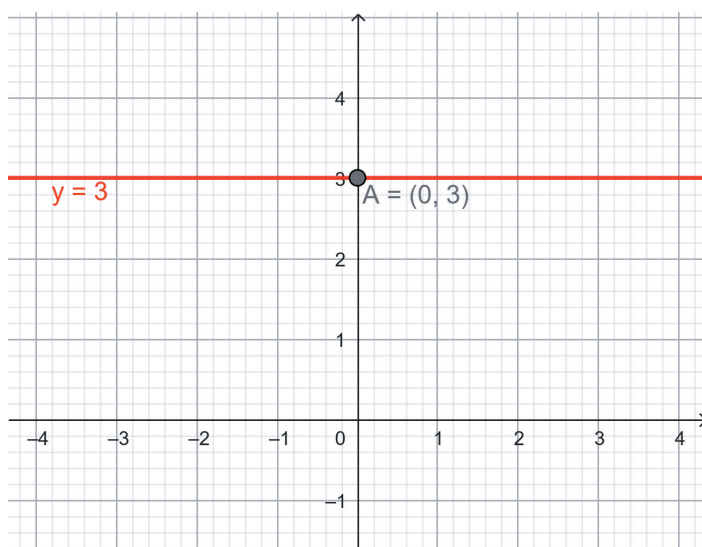
Solution:

a) The gradient = $\frac{y_2 - y_1}{x_2 - x_1} = \frac{7-5}{4-2} = \frac{2}{2} = 1$; The gradient is positive then the line is increasing.

b) The gradient = $\frac{y_2 - y_1}{x_2 - x_1} = \frac{0-4}{2-0} = \frac{-4}{2} = -2$

The gradient is negative then the line is decreasing.

Example 3: i) The gradients of the line $y = 3$ is 0.

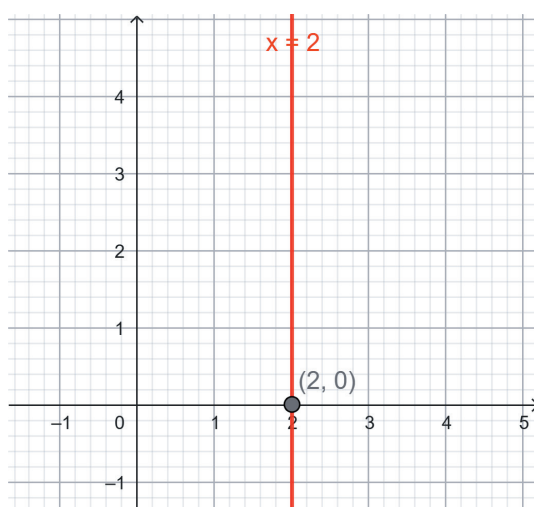


i.e. vertical distance = 0 horizontal distance = c where $c \neq 0$

$$\frac{\text{Vertical distance}}{\text{Horizontal distance}} = \frac{0}{c} = 0$$

Therefore, we conclude that **the gradient of horizontal line is 0.**

ii) For the line $x = 2$, this is a vertical line



If we consider two points of the line for example $(2,0)$ and $(2,2)$,

$$\text{the gradient} = m = \frac{2-0}{2-2} = \frac{2}{0} \notin \mathbb{R}$$

$$\text{Gradient} = \frac{\text{Vertical distance}}{\text{Horizontal distance}} = \frac{c}{0} \notin \mathbb{R}$$

This quantity is not defined in the set of real number. The gradient of the line $x = 2$ is not defined i.e. does not exist since division by zero is not possible.

We conclude that **the gradient for vertical lines** (" $x = k$, where k is a constant") **is not defined**.

Example 1

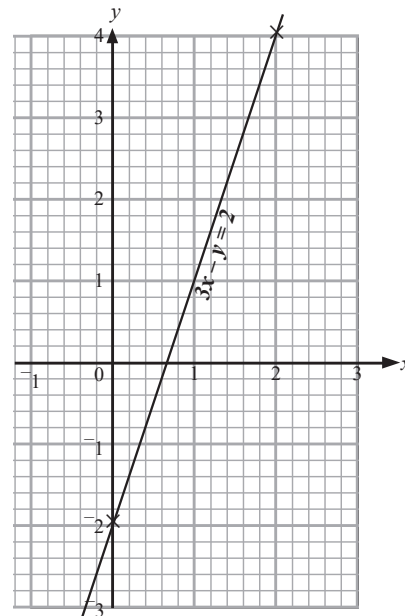
Find the gradient of the line $3x - y = 2$ and draw the line on squared paper. At what value does the line $3x - y = 2$ cut the y -axis?

Solution

$3x - y = 2$ is equivalent to $y = 3x - 2$. Comparison with $y = mx + k$ gives gradient $m = 3$.

The following figure shows the line $3x - y = 2$.

The value at which the line cuts the y -axis is referred to as y -intercept. In this example is the point $(0, -2)$.



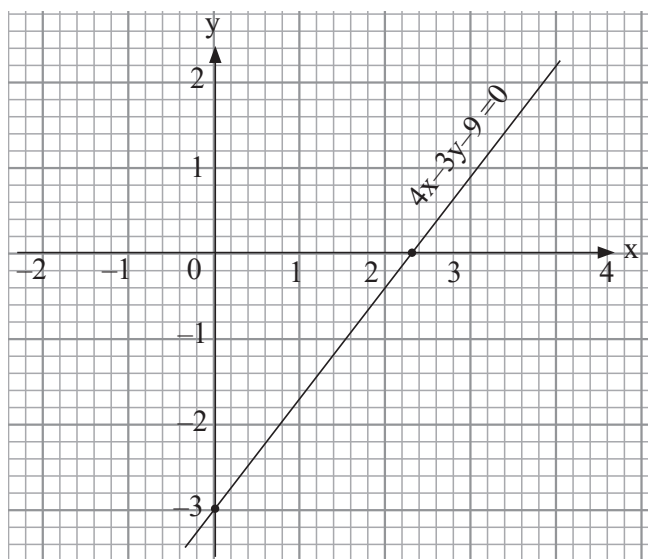
Example 2

Find the gradient and y -intercept of the line whose equation is $4x - 3y - 9 = 0$. Sketch the line.

Solution

$4x - 3y - 9 = 0$ is equivalent to $y = \frac{4}{3}x - 3$.

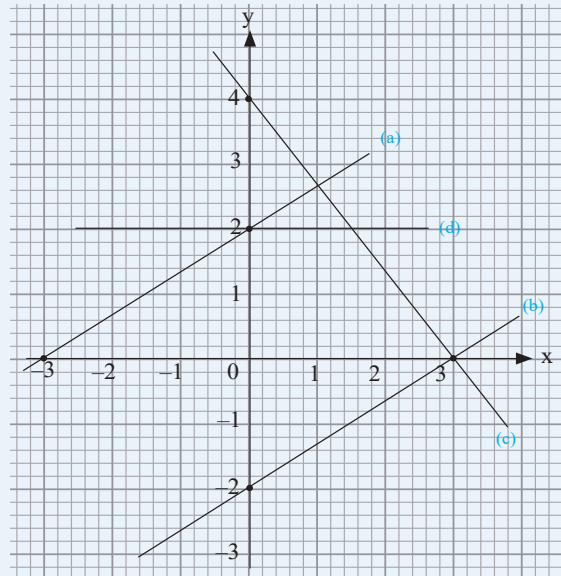
Comparison with $y = mx + k$ gives gradient, $m = \frac{4}{3}$, y -intercept, $k = -3$



APPLICATION ACTIVITY 3.7

- 1) In each of the following cases, determine the gradient and y-intercept by writing the equation in the form $y = mx + k$. Sketch the line.
 - (a) $5x = 2y$ (b) $y = 2x + 1$
 - (c) $2x + y = 3$ (d) $4x - 2y + 3 = 0$
 - (e) $2x + 3y = 3$ (f) $5 = 5x - 2y$
- 2) For each of the following pairs of points, find the change in the **x**-coordinate and the corresponding change in the **y**-coordinate. Hence find the gradients of the lines passing through them.
 - (a) (0, 2) and (3, 4) (b) (0, 2) and (5, 0)
 - (c) (-2, -2) and (2, 0) (d) (-1, -2) and (1, 8)
 - (e) (-1, 2) and (3, -2) (f) (0, -2) and (0, 3)
 - (g) (2, -8) and (-2, 8) (h) (-2, 0) and (3, 0)
- 3) Find the y-intercepts of the lines with the given gradients and passing through the given points.
 - (a) 3, (2, 6) (b) $\frac{5}{3}$, (-2, 3) (c) -2, (7, 4) (d) -1, (2, 4)
 - (e) 0, (-3, -2) (f) Undefined, (1, 3)

- 4) Write down, in the gradient-intercept form, the equations of lines (a), (b), (c) and (d) in Graph 6.8.



3.8 Like terms and unlike terms of algebraic expressions



Activity 3.8

There are 40 cattle (c), 16 sheep (s) and 32 goats (g) on a farm.

Write down an algebraic expression for the animals above if 5 goats are added.

Is there any way of grouping same animals together?

Discuss, then make a presentation to class.



Content Summary

Like terms have the same variable and same power.

Unlike terms have different variables or powers.

Only **like terms** can be added or subtracted.

Simplifying algebraic expressions means **combining like terms**.

Example 1: Simplify: $6abp - 7ab - 6abp + 8ab$

Solution:

$$\begin{aligned} \text{First collect like terms and simplify} &= (6abp - 6abp) + (-7ab + 8ab) \\ &= (0abp) + (1ab) = ab \end{aligned}$$

Example 2: Simplify: $8mn + 6pq - 5mn + 3pq$

Solution:

$$\begin{aligned} \text{First collect like terms and simplify} &= 8mn + 6pq - 5mn + 3pq = (8mn - 5mn) + (6pq + 3pq) \\ &= 3mn + 9pq \end{aligned}$$

Example 3: Simplify: $5x + 4y + 10b - r + r$

Solution:

First collect like terms and simplify $= 5x + 4y + 10b - r + r = 5x + 4y + 10b$.

**APPLICATION ACTIVITY 3.8**

Simplify these expressions:

- a) $4x + 5x =$
- b) $2y + 4y - 2y =$
- c) $2p + 4q + 3p - 2p =$
- d) $9mno - 6mno + 3xy - 2ab + 3xy + 6ab =$
- e) $5xyz - 3xyz + 2xzy =$
- f) A rectangle has length $2a - d$ and width $4a - 3d$. Find its perimeter.
- g) A square has sides of $3x + 6m$. Find its perimeter.

3.9 Substituting algebraic expressions with addition and subtraction

**Activity 3.9**

- 1) If $x = 5$ and $y = 4$, evaluate: $x + 6y$
- 2) If $w=5$, $x=3$ and $y=6$ find the values of wxy

Explain your answers to the colleague.

**Content Summary**

Substitution means replacing a variable with a number.

Always use brackets when substituting.

Then add or subtract to get the final value.

Substitution helps us find the numerical value of algebraic expressions.

Example 1 If $x = 2$, $y = 1$ and $z = 7$, simplify: $y + z - x$.

Solution:

Substitute then simplify: $y + z - x = 1 + 7 - 2 = 8 - 2 = 6$

Example 2 If $x = 2$, $y = 1$ and $z = 7$, simplify $y + x + z$.

Solution:

Substitute then simplify: $y + x + z = 1 + 2 + 7 = 3 + 7 = 10$



APPLICATION ACTIVITY 3.9

Work out the following: If $x = 10$, $y = 2$ and $t = 5$, $p = 6$, $q = 8$, $r = 4$; evaluate the following:

- | | | |
|--------------------|----------------|--------------------|
| a) $x - 3$ | f) $p + q + y$ | k) $t + x + p - r$ |
| b) $t + y$ | g) $x + y + t$ | l) $y + p + t$ |
| c) $y + q - t$ | h) $q + r - y$ | m) $x + r + p$ |
| d) $q + r - p$ | i) $x + r + p$ | n) $y + r - p$ |
| e) $p + x + r - t$ | j) $q + p - r$ | o) $x + r - p$ |

3.10 Substituting algebraic expressions involving multiplication



Activity 3.10

The side of a square is x .

- Find the perimeter.
- What is its area?
- What is the perimeter, and area if $x = 5$ m?



Content Summary

To work out algebraic expressions with multiplication,

Use brackets when substituting.

Multiply after substituting.

expand then substitute the values.

Simply to obtain the answer.

This method helps calculate unknown values easily.

Example 1: If $a = 3$, $b = 4$, $c = 5$ and $d = 6$ Simplify $ab + cd$.

Solution:

Expand, substitute then simplify:

$$ab + cd = (a \times b) + (c \times d) = (3 \times 4) + (5 \times 6) = 12 + 30 = 42$$

Example 2: 2 If $a = 3$, $b = 4$, $c = 5$ and $d = 6$ Simplify abd

Solution:

Expand, substitute then simplify: $abd = (a \times b \times d) = (3 \times 4 \times 6) = 12 \times 6 = 72$



APPLICATION ACTIVITY 3.10

Work out the following: If $p = 6$, $q = 8$, $r = -4$, $s = 10$ and $t = -3$, evaluate:

- | | | | |
|----------------|-----------------|---|---------------|
| (a) pqs | (e) $rs - 4t$ | (i) $9(2p + 3r)$ | (m) $3st$ |
| (b) $qxq + 2p$ | (f) $pr + qt$ | (j) $2st - 4(2r)$ | (n) $6r + st$ |
| (c) $st - 2r$ | (g) $5r - 4s$ | (k) $310qs + 56pr$ | |
| (d) prt | (h) $7qr - 3st$ | (l) $r \times r + 2t \times t \times t$ | |

3.11 Substituting algebraic expressions involving division



Activity 3.11

If $\frac{x}{2}$ a fraction, discuss in group:

- What does $\frac{x}{2}$ mean?
- If $x=10$, what is $\frac{x}{2}$?
- How do we find the answer?



Content Summary

To work out substituting algebraic expressions involving division, substitute into the equation, simply then divide.

Example 1: If $k = 4$, $l = 6$, $m = -3$, $n = 10$ and $p = -5$ Simplify $\frac{kl}{m}$

Solution:

$$\text{Expand, substitute then divide.} = \frac{k \times l}{m} = \frac{4 \times 6}{-3} = 24 \div -3 = -8$$

Example 2: If $k = 4$, $l = 6$, $m = -3$, $n = 10$ and $p = -5$ Simplify $\frac{2mn}{p}$

Solution:

$$\text{Expand, substitute then divide.} = \frac{2 \times m \times n}{p} = \frac{2 \times (-3 \times 10)}{-5} = (2 \times -30) \div -5 = -60 \div -5 = +12$$



APPLICATION ACTIVITY 3.11

Work out the following: If $v = 2$, $w = 14$, $x = -4$, $y = 18$ and $z = -9$.

(a) $\frac{y}{x}$ (b) $\frac{y}{v}$ (c) $\frac{y}{z}$ (d) $\frac{wx}{z}$ (e) $\frac{2vy}{x}$ (f) $\frac{xy}{2z}$

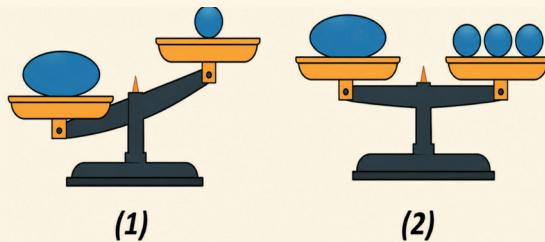
3.12 Linear equations



Activity 3.12

An equation consists two parts which are equal to one another as shown below.

- Get a beam balance.
- Place small stones on one side and some fruits like oranges or mangoes on the other.
- Do they balance?
- If not, remove some items on the heavier side or add some items on the lighter side to balance the two sides.



- Now, given $x + 2 = 18$, what number must be put in place of x so that the two sides are equal?
- Explain how you get the number.



Content Summary

A **linear equation** is a mathematical statement that shows a relationship between variables and can be written in the form of an equation where the **highest power of the variable is 1**.

Meaning of a Linear Equation: A linear equation is an equation that can be written in the form: $ax+b=0$ or $ax+b=c$ where: x is the variable, a , b , and c are real numbers, $a \neq 0$

Example: $6x + 12 = 0$

Linear Equations in one Variable: These equations contain **one unknown** and have **one solution**.

Example: $2x+3=11$

Solving linear equation: Solving linear equations means to find the value of the unknown that verifies the equality.

To solve $2x+3=11$ means to find the value of x that makes $2x+3=11$ true when it is substituted by x .

Example 1: For $2x+3=11$, we see that when we substitute x by 4, we find that $(2 \times 4) + 3 = 11$. Therefore, solving $2x+3=11$ is to find that $x = 4$.

Example 2: verify if $x=3$ is the solution of the equation $1+5x=22-2x$

Solution:

We substitute x by 3 in: $1 + 5x = 22 - 2x$.

We find: LHS: $1+5x=1+5 \times 3=1+15=16$

RHS: $22-2x=22-2 \times 3=22-6=16$.

We see that LHS = RHS. Therefore, $x=3$ is the solution of equation $1 + 5x = 22 - 2x$. In the next lesson, we will see how to solve linear equations.

Example 3: Basing on the definition of equation, chose the correct equation in the following: (1) $1 + 3 = 4$ (2) $x + 3 = 4$.

Solution:

Based on the definition of an **equation**: A statement that asserts the equality of two mathematical expressions, containing at least one variable or unknown, the correct choice is: (2) $x + 3 = 4$. The expression $1 + 3 = 4$ is simply a *true arithmetic statement* (an identity with no variable), not an equation in the algebraic sense.

Applications of Linear Equations:

Linear equations are used to represent real-life situations such as:

Fair sharing and budgeting: **Example:** You have 8000Frw to buy books and pens. Each book costs 1200 and each pen costs 400. If you buy 5 books, the number of pens you can buy is x . Therefore, the cost of books is $1200 \text{Frw} \times 5 = 6000 \text{Frw}$. The cost of pens is: $400 \times x = 400x$.

The equation that shows the total cost is: **$6000 + 400x = 8000$** .

Distance, time, and speed problems: **Example:** A car travels at a constant speed of 60 km/h. The distance to be covered after the time x is 50 km. The equation that can be solved is: Distance = Speed $\times$ Time or $50 = 60 \times x$ or **$60x = 50$** .

Cost and profit calculations: **Example:** A vendor buys apples at 90Frw each and sells them at 135Frw each. If the number of apples to be sold is m she can make the profit. How many apples must be sold to make a profit of 900Frw.

The profit after selling one apple is $135 \text{Frw} - 90 \text{Frw} = 45 \text{Frw}$. To realize a profit of 900Frw, the vendor will use the equation $45 \times m = 900$ which is: **$45m = 900$** .



APPLICATION ACTIVITY 3.12

1) Which of the following are equations?

- (a) $3 - 5 + 1x$ (b) $6 - 5 = 1$ (c) $5 + 2 - 6x$
 (d) $4 \times 6x$ (e) $3 - 9x$ (f) $16 - 9x = 3 + 4x$
 (g) $4 + 17 - 3x$ (h) $6 \times 4x = 24$ (i) $17 = 13 + 4x$

2) State whether the following are true or false.

- (a) $300 + 600 = 700 + 200$ (b) $70 - 30 = 60 - 20$ (c) $60x + 20 = 60$
 (d) $2 = 2x - 3$ (e) $13 - 17 = +7 - 3$ (f) $(-24) \div (+3) = (-8)$

3) A student used 4000Frw to buy notebooks. If the cost of one notebook is 500Frw and given that the student buys x notebooks, choose the correct equation that shows the cost used.

- a) $4000x = 500$ b) $500x = 4000$ c) $500/x = 4000$ d) $500 + x = 4000$

3.13 Solve linear equations with one unknown



Activity 3.13

Work out

I am thinking of a number x . When I **add 4** to this number, the answer becomes **10**."

- a) Write the equation that expresses the situation.
 b) What do you think the hidden number x could be?
 c) How can we solve the equation to find the hidden number x ?



Content Summary

The value of the unknown that makes an equation true is called the **solution** of the equation. **To solve an equation** means to find the value of the unknown which makes the equation true.

Steps used to Solve linear equation in one unknown:

Step 1: Simplify both sides

Clear parentheses using the distributive property: $a(b+c) = ab + ac$

Combine like terms on each side.

Step 2: Move variable terms to one side

Use addition or subtraction to get all terms with the variable (x) on one side of the equation and constant terms on the other.

Example: $3x + 5 = 2x + 9$

Subtract $2x$ from both sides: $3x - 2x + 5 = 2x - 2x + 9 \Rightarrow x + 5 = 9$

Step 3: Isolate the variable term

Use addition or subtraction to move constants away from the variable term.

Example: $x + 5 = 9$

Subtract 5: $x + 5 - 5 = 9 - 5 \Rightarrow x = 4$

Step 4: Solve for the variable

If the variable is multiplied by a coefficient a (for example $a \cdot x = b$), divide both sides by that coefficient a .

Example: $5x = 20$

Divide by 5: $\frac{5x}{5} = \frac{20}{5} \Rightarrow x = 4$

Step 5: Check your solution

Substitute the solution back into the original equation to verify both sides are equal.

Example: As $x = 4$, replace x by 4 and check if the LHS = RHS in $3x + 5 = 2x + 9$. We have:

LHS: $3 \times 4 + 5 = 12 + 5 = 17$. RHS: $2 \times 4 + 9 = 8 + 9 = 17$.

As the LHS = RHS, the solution $x = 4$ is correct. The **solution set** $S = \{4\}$.

Special cases:

- **Infinite solutions:** Simplifying leads to a true statement without the variable (e.g., $3 = 3$). In this case, The **solution set** $S =$ the set of all real numbers $= \mathbb{R}$
- **No solution:** Simplifying leads to a false statement (e.g., $2 = 5$). In this case, the solution set S is empty. $S = \emptyset = \{\}$.

Notes:

- 1) Adding to both sides a number such as -4 , means the same as subtracting 4 from both sides.
- 2) Multiplying both sides by a number such as $\frac{1}{3}$ means the same as dividing both sides by 3.

Example 1: Solve the equations: (a) $x + 6 = 10$ (b) $x - 4 = 14$ (c) $2x = 14$

Solution:

a) $x + 6 = 10$. We are required to find a number to which we add 6 to get 10.

$$x + 6 = 10 \Rightarrow x + 6 - 6 = 10 - 6 \Rightarrow x = 4$$

The solution is $x = 4$. $S = \{4\}$.

b) $x - 4 = 14$. We are required to find a number from which we subtract 4 to get 14.

$$x - 4 = 14 \Rightarrow x - 4 + 4 = 14 + 4 \Rightarrow x = 18$$

The solution is $x = 18$. $S = \{18\}$.

c) $2x = 14$. We are required to find a number which when multiplied by 2 gives 14.

$$2x = 14 \Rightarrow \frac{2x}{2} = \frac{14}{2} \Rightarrow x = 7$$

The solution is $x = 7$. $S = \{7\}$.

Example 2: Solve the equation $8x - 6 = 5x + 9$.

Solution:

$$8x - 6 = 5x + 9$$

Add 6 to both sides: $8x - 6 + 6 = 5x + 9 + 6$ then $8x = 5x + 15$

Subtract $5x$ from both sides: $8x - 5x = 5x - 5x + 15$ then $3x = 15$

Divide both sides by 3: $\frac{3x}{3} = \frac{15}{3}$ i.e. $x = 5$. $S = \{5\}$.



APPLICATION ACTIVITY 3.13

Solve the following equations using the balance method

1. (a) $x + 113 = 153$

(b) $24 = x + 13$

(c) $x + 8 = -12$

(d) $x + -7 = -19$

2. (a) $x - 17 = 37$

(b) $11 = x - 7$

(c) $x - 2 = -4$

(d) $x - -2 = -7$

3. (a) $x - 72 = -24$

(b) $-x + 72 = -30$

(c) $4 = x - 5$

(d) $15 + 2x = -3x$

3.14 Solve linear equations with one unknown involving fractions



Activity 3.14

- 1) Imagine you have a cake cut into **4 equal parts**. You eat **one part**, which is $\frac{1}{4}$ of **the cake**, and you are left with $\frac{3}{4}$ of the cake." How do we write $\frac{1}{4}$ of something **x** using a fraction?
- 2) If you eat $\frac{1}{2}$ of a given number **x** of bananas, you are left with 10 bananas.
 - a) write the equation that represent that situation.
 - b) Solve the equation to find the total number of bananas you had before.



Content Summary

Expressions such as $\frac{a+b}{c}$ and $\frac{x-4}{x+1}$ are called algebraic fractions. In order to solve equations involving fractions, we must remember how to use equivalent fractions using the LCM of denominators in the entire expression.

Example: Solve $\frac{2}{5}y + \frac{3}{4} = 10 + \frac{y}{2}$

Solution:

$\frac{2}{5}y + \frac{3}{4} = 10 + \frac{y}{2}$ We must write the expression using the common denominator.

That is the LCM (5, 4 and 2) = 20.

Therefore, let us multiply both sides by 20:

$$20\left(\frac{2}{5}y + \frac{3}{4}\right) = 20\left(10 + \frac{y}{2}\right)$$

$$8y + 15 = 200 + 10y$$

Subtracting $10y$ from both sides: $8y + 15 - 10y = 200 + 10y - 10y$

$$\text{We get: } -2y + 15 = 200$$

Subtracting 15 from both sides: $-2y + 15 - 15 = 200 - 15$

$$\text{We get: } -2y = 185$$

Dividing both sides by -2 : $\frac{-2y}{-2} = \frac{185}{-2}$ then, $y = -\frac{185}{2} = -92\frac{1}{2}$

$$S = \left\{-\frac{185}{2}\right\}.$$



APPLICATION ACTIVITY 3.14

Solve the following equations using the balance method

1) a) $\frac{x}{6} - 20 = 10$, b) $9 - \frac{x}{2} = 5$, c) $\frac{k}{5} = 0$, d) $p - 2\frac{1}{2} = 6\frac{1}{2}$

2) a) $-3\frac{3}{4} = x + 1\frac{2}{5}$, b) $4\frac{1}{2} = 5q - \frac{1}{4}$, c) $2p - 8 = p - 3$, d) $t + 7 = 17 - 4t$

3.15 Solve linear equations with one unknown involving brackets or a factor



Activity 3.15

A basket contains some identical apples. The number of apples is **three times a certain number x** . After adding **4 more apples**, the total becomes **19 apples**.

- The number of apples was three times the unknown number x , how can we write that using brackets or multiplication?
- Write the equation that represents this situation after adding 4 more apples.
- Solve the equation to *find the hidden number x* .



Content Summary

To solve linear equations with one unknown involving brackets or a factor, proceed as follows:

- **Expand the brackets:** Use the distributive property $a(b+c)=ab+ac$ to remove any parentheses.

Identities below can sometimes appear: **Remember that:**

- $a + (b + c) = a + b + c$
- $a + (b - c) = a + b - c$
- $a - (b + c) = a - b - c$
- $a - (b - c) = a - b + c$

- A positive number $\times$ a positive number = a positive number
- A positive number $\times$ a negative number = a negative number.
- A positive number $\div$ a negative number = a negative number.
- A positive number $\div$ a positive number = a positive number.

- **Combine like terms** (if necessary): Simplify both sides by adding/subtracting similar terms.
- **Use addition or subtraction:** Move variable terms to one side and constant terms to the other by adding/subtracting the same amount on both sides.
- **Use multiplication or division:** Isolate the variable by multiplying or dividing both sides by the same nonzero number.
- **Check the solution:** Substitute the value back into the original equation to verify it satisfies the equality

Example 1:

Solve the equations: $9x - (4x - 3) = 11 + 2(2x - 1)$.

Solution:

$$9x - (4x - 3) = 11 + 2(2x - 1)$$

Removing brackets: $9x - 4x + 3 = 11 + 4x - 2$

Simplifying both sides: $5x + 3 = 4x + 9$

Subtracting $4x$ and 3 from both sides:

$$5x - 4x = 9 - 3$$

Then, $x = 6$

Example 2: $\frac{5x + 2}{4} - \frac{3}{2} = \frac{7x + 1}{3}$

Solution:

Multiply both sides by 12 (LCM of 4, 2 and 3)

$$12x \left\{ \left(\frac{5x + 2}{4} \right) - \frac{3}{2} \right\} = 12x \left(\frac{7x + 1}{3} \right)$$

i.e. $3(5x + 2) - 6 \times 3 = 4(7x + 1)$

Removing brackets: $15x + 6 - 18 = 28x + 4$

Simplifying LHS: $15x - 12 = 28x + 4$

Subtracting $28x$ from both sides and adding 12 to both sides:

$$15x - 28x = 4 + 12$$

i.e. $-13x = 16$

Dividing both sides by -13 :

$$\frac{-13}{-13}x = \frac{16}{-13} \quad \text{i.e. } x = -\frac{16}{13}$$

**APPLICATION ACTIVITY 3.15**

Solve the following equations:

1) $4d + (5 - d) = 17$

2) $12m + (1 - 7m) = 23$

3) $23 = 7 - (3 - 4t)$

4) $(8w - 7) + (5w + 13) = 0$

5) $24 - (5 + 3x) = 8x + (4 - 5x)$

6) $3(5x - 1) = 4(3x + 2)$

7) $5(3c + 4) - 3(4c + 7) = 0$

8) $4(3k - 1) = 11k - 3(k - 4)$

9) $4(3x - 5) - 7(2x + 3) + 2(5x + 11) = 5$

10) $7(5x - 3) + 10 = 2(3x - 5) + 3(5 - 7x)$

11) $\frac{x + 1}{2} - \frac{x - 2}{3} = \frac{1}{8}$

12) $\frac{1}{8} - \frac{14y - 3}{2} = \frac{y - 4}{4}$

13) $\frac{x}{5} + 1\frac{1}{2}x - \frac{11}{20} = \frac{5x - 1}{3}$

3.16 Solving problems involving linear equations



Activity 3.16

Discuss each problem scenario here below before solving:

1. A number increased by 7 gives 25. What is the number?
2. Three times a number minus 4 is 14. Find the number.
3. A farmer bought 5 identical hens for 20,000Frw. How much is one hen?
4. A rectangle has a perimeter of 30 cm. Its length is 2 cm more than its width. Find the width.

Then answer these questions:

- i) How can we represent the relationships in algebra?
- ii) What equation can we form?



Content Summary

Solving Problems Using Linear Equations:

- | | |
|--------------------------------------|--|
| a) Read the problem carefully. | e) Turn the relationships into an equation. |
| b) Identify the unknown. | f) Solve the linear equation. |
| c) Let the unknown be x | g) Write the final answer using correct units. |
| d) Translate the words into algebra. | h) Check your answer in the original problem. |

Examples:

- 1) The sum of two numbers is 120 and their difference is 18. Find the two numbers.

Solution:

Let the smaller number be x . Then, the larger number is $x + 18$.

Sum of the two numbers = $x + (x + 18)$.

$$\text{i.e. } x + (x + 18) = 120$$

$$x + x + 18 = 120$$

$$2x + 18 = 120$$

$$2x = 102$$

$$x = 51$$

Thus, the smaller number is 51 and the larger number is $51 + 18 = 69$.

- 2) Tom and Mary share 450 FRW so that Mary gets 54 FRW less than Tom. Find their shares.

Solution:

Let Tom's share be x FRW. Then, Mary's share is $(x - 54)$ FRW.

Together, they have 50 FRW.

$$\therefore x + x - 54 = 450$$

$$2x = 450 + 54$$

$$2x = 504 \Rightarrow x = 252$$

Thus, Tom's share is 252 FRW and Mary's share is 54 FRW less

i.e. $(252 - 54)$ FRW = 198 FRW.

**APPLICATION ACTIVITY 3.16**

Read the problem, form related equation and solve it:

1. When I double a number and add 17, the result is 59. What is the number?
2. When a number is added to 4 times of itself, the result is 30. Find the number.
3. The difference of two numbers is 5 and their sum is 19. Find the two numbers.
4. Mr Ali has 7 marbles less than Mohammed and they have 29 between them. How many does each boy have?
5. When a number is doubled and 4 added, the result is the same as when it is tripled and 9 subtracted. Find the number.
6. The length of a rectangle is 3 times as long as it is wide. The total length round its boundary is 56 cm. Find its length and width.
7. Erick is twice as old as Peter, John is 3 years younger than Erick. The sum of their ages is 32. Find their individual ages.

3.17 Linear inequality and the number line**Activity 3.17**

Read and solve: Amina and John are comparing their pocket money. Amina has 1,500 FRW and John has 2,000 FRW. Then Amina says: "I have less than John." And John says: "I have more than Amina."

1. Use symbols to write: 1,500 is less than 2,000.
2. Use symbols to write: 2,000 is greater than 1,500.
3. Can we show their pocket money on a number line? Where should 1,500 be placed compared to 2,000 on the number line?



Content Summary

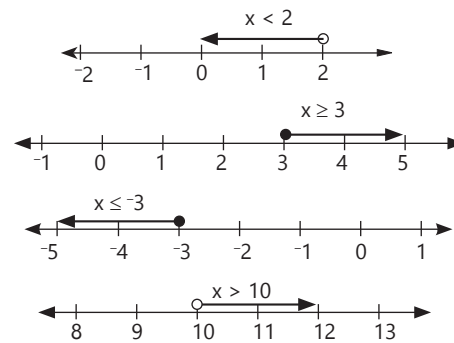
The symbols of inequalities

- **Greater than**, e.g. $6 > 2$ means 6 is greater than 2.
- **Less than**, e.g. $3 < 7$ means 3 is less than 7.
- **Greater than or equal to** (or 'not less than').
- **meaning 'less than or equal to'**, ($\leq$ or 'not greater than').

Statements containing these symbols are called inequalities or in equations.

For example, $x < 2$, $x \geq 3$, $y \leq -3$, $y > 10$ are inequalities and can be illustrated on a number line as shown below.

These are also called simple inequality statements.

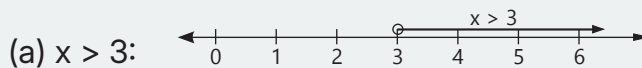


Example

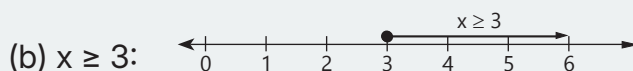
Illustrate each of the following on a number line:

- (a) $x > 3$ (b) $x \geq 3$ (c) $x < -2$ (d) $x \leq -2$

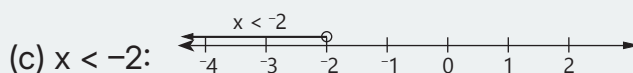
Solution:



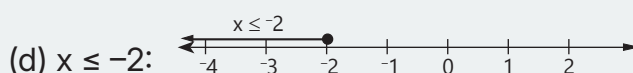
The number 3 is not included in the list of numbers to the right of 3. The heavy arrow shows that the values of x go on without end. The open dot (o) is used to indicate that 3 is not included.



number 3 is included in the list of the required numbers. The closed dot (•) is used to show that 3 is part of the list.



the number -2 is not included.



the number -2 is included..



APPLICATION ACTIVITY 3.17

1) Rewrite each of the following:

statements using either $<$, $\leq$, $>$ or $\geq$ instead of the words.

- | | |
|------------------------|--------------------------------------|
| a) 5 is less than 7 | d) -2 is greater than -3 |
| b) 5 is greater than 2 | e) x is greater than or equal to 4 |
| c) -1 is less than 0 | f) y is less than or equal to -5 |

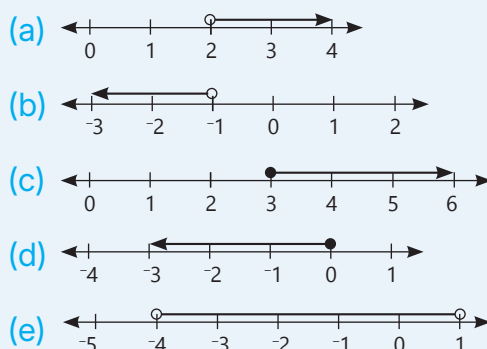
2) Copy and complete each of the following statements by inserting the correct symbol, $=$, $<$ or $>$.

- (a) $7 + 3$ ___ 11 (b) $6 - 3$ ___ 5 (c) $13 + 2$ ___ 16 (d) $12 - 4$ ___ 3
 (e) 3×4 ___ 8 (f) -4×3 ___ 12 (g) -6×-2 ___ -6×2

3) Illustrate each of the following inequalities on a number line.

- (a) $x > -5$ (b) $x < 0$ (c) $x \geq -3$ (d) $x \leq 4$ (e) $x > -0.5$ (f) $x \leq 2.5$

4) Write each the following using inequality



3.18 Solve linear inequalities in one unknown and represent the solution as interval



Activity 3.18

Work out: **Two students Eric and Marc, run a race.** Eric can run **less than** 100 meters in **20 seconds**. Marc can run **more than** 100 meters in **20 seconds**.

- If "less than 100 m" is written using symbols, what does it look like?
- If "more than 100 m" is written using symbols, what does it look like?
- How can we represent distances on a number line to compare them?
- How can we represent distances on intervals?



Content Summary

A **linear inequality** is like a linear equation, but it uses inequality symbols instead of the equal sign. **Example:** $2x+5 < 11$.

The representation of solution as an Interval

- Linear inequalities use symbols: $<$, $>$, $\leq$, $\geq$
- Solving inequalities follows the same steps as solving equations.
- If you multiply or divide by a negative number, reverse the sign.
- Solutions can be represented on a number line.
- Solutions can also be written as an interval:
- $(a, b) \rightarrow a$ and b not included
- $[a, b] \rightarrow a$ and b included
- Use ∞ or $-\infty$ for unbounded intervals

Example:

1) Solve: $2x+5 < 11$

Subtract 5: we have $2x < 6$

Divide by 2: $x < 3$

Interval: $(-\infty, 3)$

$S = (-\infty, 3)$

2) Solve: $5-x \leq 9$

Subtract 5: $-x \leq 4$

Multiply by -1 (reverse sign): $x \geq -4$

Interval: $[-4, \infty)$

$S = [-4, \infty)$



APPLICATION ACTIVITY 3.18

Solve the following inequality and express answers as interval:

1. $x+7 > 12$

3. $7-2x < 1$

5. $-5x > 25$

2. $3x-2 \leq 10$

4. $4(3-x) \geq 8$

3.19 Solve linear inequalities in one unknown and represent the solution on a number line



Activity 3.19

Work out.

A school library only allows students who are aged 15 years or older to enter.

- Who students are allowed to enter the library?
- Which ages are NOT allowed to enter?
- Which ages ARE allowed to enter?
- How can we write "ages 12 or more" using a mathematical symbol?
- How can we show these ages on a number line?



Content Summary

We defined the inequality “ x is greater than y written as $x > y$ ” to mean $x - y$ is positive. While “ x is less than y written as $x < y$ ” to mean $x - y$ is negative.

Example: Solve the inequality: $3x - 2 < 10 + x < 2 + 5x$

Solution:

$$3x - 2 < 10 + x < 2 + 5x$$

Split the inequality into two simultaneous inequalities as:

$$3x - 2 < 10 + x \dots\dots\dots(i)$$

$$\text{and } 10 + x < 2 + 5x \dots\dots\dots(ii)$$

Solve each separately.

$$3x - 2 < 10 + x$$

$$\Rightarrow 3x - x < 10 + 2$$

$$\Rightarrow 2x < 12$$

$$\Rightarrow x < 6 \dots\dots\dots(iii)$$

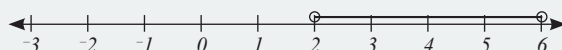
$$10 + x < 2 + 5x$$

$$\Rightarrow 10 - 2 < 5x - x$$

$$\Rightarrow 8 < 4x$$

$$\Rightarrow 2 < x \dots\dots\dots(iv)$$

Combine (iii) and (iv) to get $2 < x < 6$ This is represented on a number line



$$S = (2, 6)$$



APPLICATION ACTIVITY 3.19

Solve the following inequalities and represent the solutions on number lines.

- | | |
|------------------------|-----------------------|
| 1) (a) $x + 4 > 11$ | (b) $x - 6 \leq 5$ |
| 2) (a) $2x - 8 \leq 4$ | (b) $3x + 4 > 19$ |
| 3) (a) $3 > 4x - 2$ | (b) $7 \leq 5x + 12$ |
| 4) (a) $3 - 2x < 5$ | (b) $4 - 5x \geq -11$ |

3.20 Solve linear inequalities with one unknown involving fractions and represent the solution on a number line



Activity 3.20

Work out: Amina and Peter are sharing a 1-litre bottle of juice. Amina says she wants more than half of the juice. Peter says he wants at least one-third of the juice.

- Who is asking for a larger share?
- Which fraction is bigger:
- How can we compare these two fractions using inequality symbols?
- If Amina wants more than $\frac{1}{2}$ of the juice, how can we write this using inequality notation?
- How can we show all amounts greater than $\frac{1}{2}$ on a number line?



Content Summary

To solve the fractions in Inequalities:

- Clear fractions by multiplying by the LCM of the denominators.
- Always keep track of signs.
- Only flip the inequality sign when multiplying or dividing by a negative number.

Example: Solve the inequality $\frac{x+6}{2} > \frac{x-2}{3} + 5$

Solution:

In order to solve the inequality, we need to remove the denominator and then simplify the terms in the inequality.

Thus $\frac{x+6}{2} > \frac{x-2}{3} + 5$ multiply all the terms by the LCM

$$6\left(\frac{x+6}{2}\right) > 6\left(\frac{x-2}{3} + 5\right)$$

$$3(x+6) > 2(x-2) + 30$$

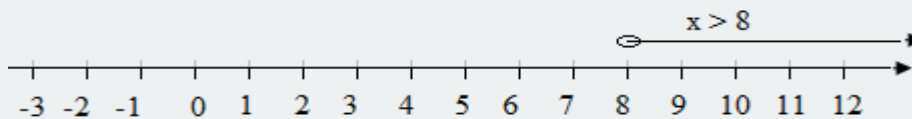
$$3x + 18 > 2x - 4 + 30 \text{ expand the terms}$$

$$3x + 18 > 2x + 26$$

$$3x - 2x > 26 - 18$$

$$x > 8$$

$$S = (8, +\infty)$$



APPLICATION ACTIVITY 3.20

1) State whether each statement is **True** or **False**. Explain briefly.

i) If $x > 4$, then $\frac{x}{2} > 2$

ii) If $\frac{x}{3} < 2$, then $x > 6$.

iii) Multiplying an inequality by a negative number keeps the sign the same.

iv) Subtracting 1 from both sides of $x \geq 5$ gives $x \geq 4$

2) Solve and represent on number line

1. $\frac{3x-1}{4} \leq 5$

3. $7 < \frac{2x+1}{3}$

2. $\frac{x}{5} - 2 > -1$

4. $\frac{x-10}{3} \geq -4$

3.21 Solve linear inequalities with one unknown involving brackets and represent the solution on a number line



Activity 3.21

Work out.

If you buy 3 packets of juice, each costing the same price, and add a fixed service fee of 200 FRW, your total bill will be more than 2,000 FRW.

Let the cost of one packet be x FRW.

- Write an inequality that represents this situation and find out the possible price of one packet.
- How can the total cost be expressed using brackets?
- What does “more than” mean in inequalities?
- How can you solve the inequality to find possible values of x ?



Content Summary

A linear inequality with one unknown involving brackets is an inequality where the unknown variable appears inside parentheses/brackets.

How to solve them:

- Expand the brackets using distributive property. $a(b + c) = ab + ac$
- Simplify the expression by combining like terms.
- Isolate the variable as in solving linear equations:
- Add/subtract terms
- Multiply/divide by a positive number
- If you multiply or divide by a negative number, the inequality sign reverses.
- Represent the solution on a number line using:
 - Open circle for $>$ or $<$
 - Closed circle for $\geq$ or $\leq$
 - Arrow in the direction of allowed values.

Example: Solve and represent on a number line:

1. $3(x + 2) > 15$

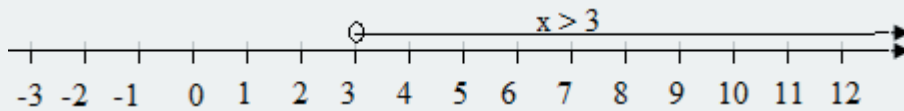
Solution:

Expand: $3x + 6 > 15$

Subtract 6: $3x > 9$

Divide by 3: $x > 3$

Number line: open circle at 3, arrow to the right.



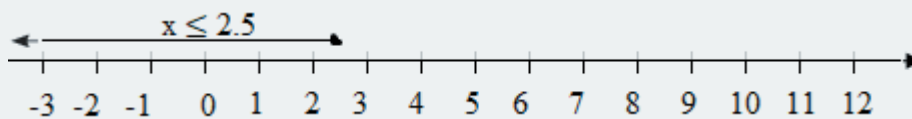
$$2.5(2x - 1) \leq 20$$

Expand: $10x - 5 \leq 20$

Add 5: $10x \leq 25$

Divide by 10: $x \leq 2.5$

Number line: closed circle at 2.5, arrow to the left.



APPLICATION ACTIVITY 3.21

1) Solve the following inequalities

- i) $4(x - 3) > 8$
- ii) $3(5x + 5) \leq 30$
- iii) $-3(x + 1) < 9$
- iv) $5(x - 1) < 15$

2) Real life problems

- a) A taxi driver charges a fixed starting fee of 500 RWF plus 300 RWF per km. If the total cost is *more than 2,000 RWF*, write and solve the inequality to find the minimum distance a passenger must travel.
- b) A teacher buys x notebooks. The price for each notebook is 700 RWF, and she pays an additional packaging fee of 400 RWF. If she spends **at most 5,000 RWF**, find the possible number of notebooks she can buy.

3.22 Model and solve problems using linear inequalities



Activity 3.22

Think of a situation in your life where you have had to use your knowledge and skills of inequalities to solve a problem.

- Share your experience with your partner.
- Discuss with your partner whether there was any other way to handle your problem.
- Would you like to share the same with the rest of the class?



Content Summary

Linear inequalities are mathematical statements that compare two expressions using inequality symbols such as $<$, $>$, $\leq$, $\geq$ instead of an equal sign. They are used to model real-life situations where quantities are not exactly equal but are limited by conditions like *more than*, *at least*, *at most*, or *less than*.

To model problems using linear inequalities, we:

- Identify the unknown quantity and represent it with a variable.
- Translate key words into inequality symbols:
 - *More than* $\rightarrow >$
 - *Less than* $\rightarrow <$
 - *At least* $\rightarrow \geq$
 - *At most* $\rightarrow \leq$
- Write an inequality that represents the situation.

Example

Peter wants to order for some special tools for his business. The cost of each tool is 2 398 RWF. The cost of delivering the whole order is 1 499 RWF. How many of these tools can he order, if he has only 15 000 RWF to spend?

Solution:

Cost of each tool = 2 398 RWF

Let the number of tools ordered be n

Cost of n tools is $2398 \text{ RWF} \times n =$
 $2\,398 \text{ RWF } n$

Cost of delivering this tool is 1 499 RWF

Total cost will be $2\,398 \text{ FRW } n + 1\,499 \text{ RWF}$

Since total cost cannot exceed 15 000 RWF

we have $2\,398 n + 1\,499 \leq 15\,000$

$\Rightarrow 2398 n \leq 15\,000 - 1\,499$

$2398 n \leq 13501$

$n \leq \frac{13501}{2398} = 5.63$

Since one cannot buy part of a tool, then the number of tools used must be a whole number $n = 5$



APPLICATION ACTIVITY 3.22

1. Mohammed scored 60% in a mathematics quiz. When a physics score is doubled, the score is still less than the score in mathematics. What is the maximum marks she could have scored in physics?
2. Joan's age is 25 less than her father's age. If her father is 50 years old, what is the maximum age (to the nearest whole number) that she could have been seven years ago.



END OF UNIT ASSESSMENT

A. Assessing knowledge and understanding (KU)

- | | |
|--------------------------------------|---|
| a) Define a number line. | d) What is the gradient of a straight line? |
| b) What is the Cartesian plane? | e) Define the x-intercept and y-intercept. |
| c) State the coordinates of a point. | |

B. assessing skills (S)

- Solve: $3x - 7 = 8$.
- Solve: $\frac{x}{4} - 2 = 3$.
- Solve: $2(x + 3) = 18$.
- Draw and label the axes, then plot the point $(-2, 3)$.
- Given $A(1,2)$ and $B(5,10)$, find the gradient of AB.
- Solve the inequality and express the answer as an interval: $4x - 6 > 10$.
- Solve and represent on a number line: $\frac{x}{3} + 2 \leq 5$.
- Solve the inequality: $3(x - 2) \geq 9$.
- Graph the linear function $y = 2x + 1$.

C. Assessing attitudes and values: Chose the correct answer.

a) Why is accuracy important when drawing graphs?

- It makes the graph look more colorful.
- It ensures the data is represented correctly and prevents misinterpretation.
- It helps the graph get drawn faster.
- It is only important for graded assignments.

b) How do inequalities help in real life?

- They limit people's choices.
- They help represent constraints in budgeting, scheduling, and resource allocation.
- They are only used in advanced mathematics.
- They make equations more complicated than necessary.

c) Why is it important to label axes clearly when representing solution of equation or inequality?

- To fill empty space on the graph.
- To make sure anyone reading the graph understands what is being shown.
- Because the teacher requires it.

D) Labels are optional if the graph is simple.

d) Give one value shown when solving equations step-by-step.

A) Speed

B) Logical reasoning and systematic thinking

C) Memorization

D) Guesswork

e) Explain how number lines promote positive learning habits.

A) They allow students to avoid writing numbers.

B) They visually organize information, helping in understanding order and relationships between numbers.

C) They are only useful for primary school students.

D) They make math more abstract and challenging.

f) How can working with coordinates encourage teamwork or communication skills?

A) By letting one person do all the work.

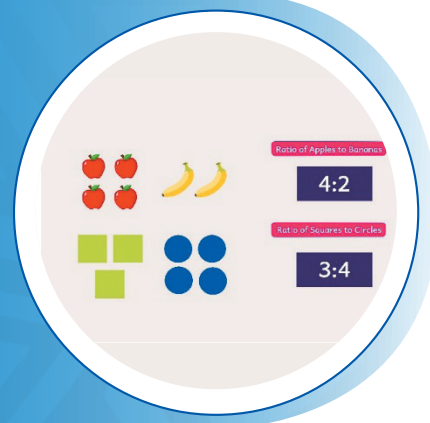
B) By requiring clear communication to describe positions and collaborate on plotting points accurately.

C) By encouraging silent independent work.

D) By using computers instead of speaking.

UNIT 4

RATIO AND PROPORTIONS



Key Unit Competence:

Solve problems involving ratios and proportions.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Describe ratio, proportion, and the difference between direct and inverse proportion.
- Express ratios in their simplest form and identify them in everyday situations.
- Identify and distinguish between direct and inverse proportions.
- Explain how ratios apply to sharing quantities, scale drawings, and mixtures.
- Recall and use the properties of a proportion.

Skills

- Apply ratio, unequal sharing, and division of quantities in a given ratio to solve problems.
- Differentiate and solve problems involving direct and inverse proportions.
- Interpret ratios and proportions in practical, real-life contexts.
- Solve real-life problems involving proportions, mixtures, and unit cost calculations.
- Calculate unit prices and costs in mixture-related problems.

Attitudes and values

- Develop accuracy and precision in calculations.
- Appreciate the importance of ratios and proportions in real-life contexts
- Show cooperation in group work and logical reasoning in solving problems on ratios.
- Develop interest and confidence in applying mathematics to daily life situations.
- Work cooperatively and respect others' ideas in group tasks.

INTRODUCTION

This unit introduces Students to **ratios, proportions, and mixtures**, aiming to develop their ability to compare quantities, share amounts fairly, and solve practical problems using mathematical relationships. The unit covers simplification of ratios, sharing quantities, applications in scale drawing, the concept and properties of proportion, direct and inverse proportions with their graphs, and solving related problems, including finding average prices and quantities in mixtures. These concepts are widely used in **daily life**, such as sharing food or money, mixing ingredients in cooking, calculating prices and discounts in markets, reading maps and building plans drawn to scale, understanding relationships between speed, time, and distance, and making informed decisions in business and household activities. Through this unit, Students gain essential skills for logical thinking and real-life problem solving.



Introductory activity

Three friends, Aline, Bosco, and Eric, buy **12 liters of juice** to share for a school event. They agree to share the juice in the ratio **1:2:3**.

Discuss:

- How many liters of juice should each friend receive?
- Why does the person with a larger number in the ratio get more juice?

4.1 Recall on ratios and simplification



Activity 4.1

A basket containing **8 oranges and 4 bananas**. Discuss in group:

- How many oranges are there compared to bananas?
- Can we write this comparison using numbers?
- Is there a simpler way to write this comparison?



Content Summary

A **ratio** is a way of comparing two or more quantities of the same kind.

A ratio is a mathematical statement which shows how two or more quantities or numbers are compared.

In general, if a and b are two quantities of the same units, then the ratio of a to b is written as **$a : b$** and can be expressed as a fraction as $\frac{a}{b}$

A ratio can be **simplified** by dividing both terms by their **Greatest common factor (GCF)**.

Simplifying a ratio makes it easier to understand and use in calculations.

Ratios do not have units because the quantities compared are of the same type.

Example.

- 1) A business lady mixes every fifteen kilograms of sorghum flour with nine kilograms of cassava flour in order to increase quality and maximize profit. Represent her mixture of flour with a ratio.

Solution:

Sorghum flour: Cassava flour 15 kg : 9 kg (are in the same units)

In ratio form we write as 15:9 (note that a ratio has no units).

- 2) 150 cm^3 of water are contained in 15 litres of a chemical. Find the simplest ratio of water to the chemical.

Solution:

Since quantities of a ratio must be of the same unit, we need to convert litres to cubic centimetres.

$$1 \text{ litre} = 1\,000 \text{ cm}^3$$

$$15 \text{ litres} = 1\,000 \times 15$$

$$= 15\,000 \text{ cm}^3$$

$$\text{Required ratio} = \frac{\text{Amount of water}}{\text{Amount of chemicals}} = \frac{150 \text{ cm}^3}{15\,000 \text{ cm}^3} = \frac{1}{100}$$

The ratio is 1:100.

- 3) Express the ratio 15 : 45 in its simplest form.

Solution:

15 : 45 Divide both values with their GCD, **The Greatest Common Factor (GCF)** of 15 and 45 is 15 DCD $15 : 45 = \frac{15}{45} = \frac{1}{3} = 1:3$ Therefore, the ratio 15:45 can be simplified to 1:3.



APPLICATION ACTIVITY 4.1

- 1) Express the following ratios in their simplest forms:
 - a) 28 : 42
 - b) 30 : 50
 - c) 24 kg : 30 kg
 - d) 150 cm to 3 m
 - e) 1 litre to 250 ml
 - f) 45 min : $1\frac{1}{2}$ hours
- 2) Mutanguha had 120 chicken in his farm. He sold 30 of them to buy a dairy goat. What is the simplest ratio of the remaining number of chicken to the original number of chicken?
- 3) A church has 325 males and 390 females. Find the simplest ratio of:
 - a) the number of females to the number of males.
 - b) the number of males to the total number of people in the church.
 - c) the number of females to the total number of people in the church.

4.2 Sharing quantities using ratios



Activity 4.2

Suppose two old men from your village have come to you to arbitrate after they disagreed over how to share 7 000 FRW such that for every 2 FRW that the first man gets, the second one gets 3 FRW. Discuss.

- i) In what ratio would you share the money between the two old men?
- ii) Tell your partner how you would share the money and how much each would get.



Content Summary

To share a quantity into two parts in the ratio $a : b$, the quantity is split into $a + b$ equal parts. The required parts become $\frac{a}{a + b}$ and $\frac{b}{a + b}$ of the quantity.

Examples:

- 1) Share 38 400 FRW between Linda and Jean in the ratio 5:7 respectively.

Solution:

38 400 FRW is to be shared in the ratio 5:7.

It is split into 12 equal parts i.e $5 + 7 = 12$ equal parts.

The amount Linda receives $\frac{5}{12} \times 38\,400 \text{ FRW} = 16\,000 \text{ FRW}$.

Jean receives $\frac{7}{12} \times 38\,400 \text{ FRW} = 22\,400 \text{ FRW}$.

- 2) Ingabire, Mugenzi and Shami have jointly invested in buying and selling of shares in the Rwanda stock exchange market. In one sale, they realised a gain of 1 080 000 FRW and intend to share it in the ratio 2:3:4 respectively. How much did Mugenzi get?

Solution:

$$\text{Mugenzi's share} = \frac{3}{2+3+4} \text{ of } 1\,080\,000 \text{ FRW} = \frac{3}{9} \times 1\,080\,000 \text{ FRW} = 360\,000 \text{ FRW}.$$



APPLICATION ACTIVITY 4.2

- 1) Divide the following quantities in the given ratio:
(a) 1600 FRW (3:5) (b) 24 ha (2:3:5) (c) 327 kg (1:5:6) (d) 775 chicken (14:11).
- 2) 8 720 000 000 FRW allocated to a certain department of a ministry from the national budget is to be shared between two sections A and B, in the ratio 3 : 5 respectively. How much will each section get?
- 3) 55% of students in a school are boys. (a) What is the ratio of boys to girls?
(b) How many girls are there if the school has 400 students?

4.3 Applications of ratios in Scale drawing



Activity 4.3

Measure the length and width of the classroom. Now try to draw the classroom on your exercise book. Discuss the following questions:

- i) Can you draw it using the same measurements?
- ii) What can we do to draw the classroom so that it fits on the page?
- iii) Can we reduce all the measurements in the same way?

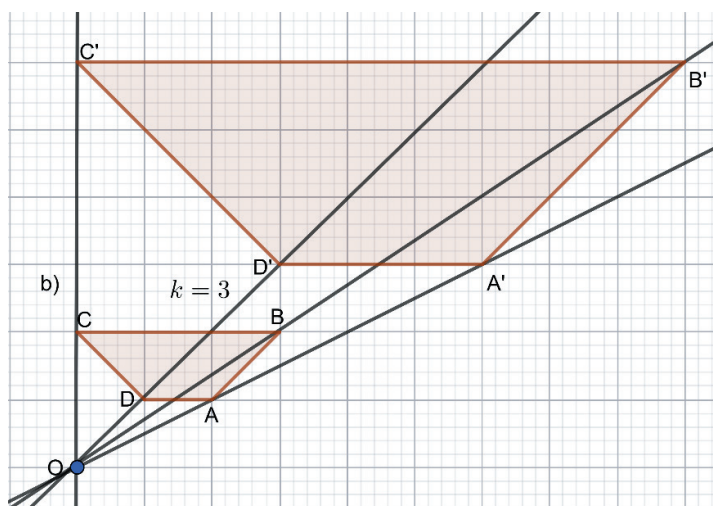


Content Summary

Applications of ratios in **scale drawing** refer to the use of ratios to represent real objects' distances, or sizes accurately on a drawing, map, or plan by **reducing or enlarging** them in the same proportion.

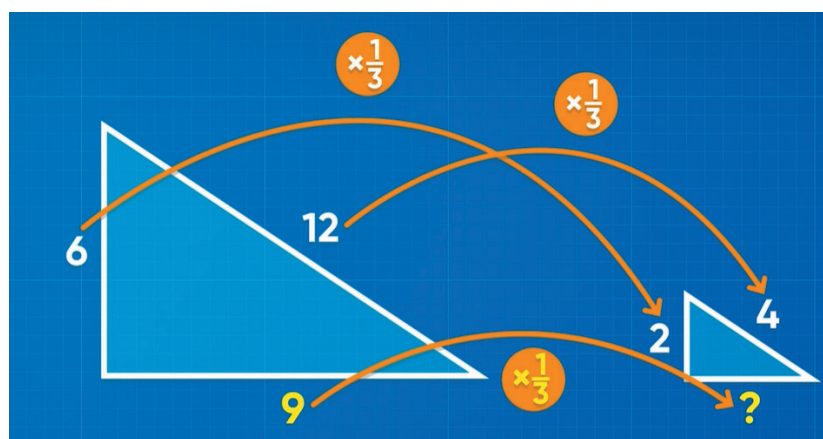


The big ball is 4 times bigger than the small ball.



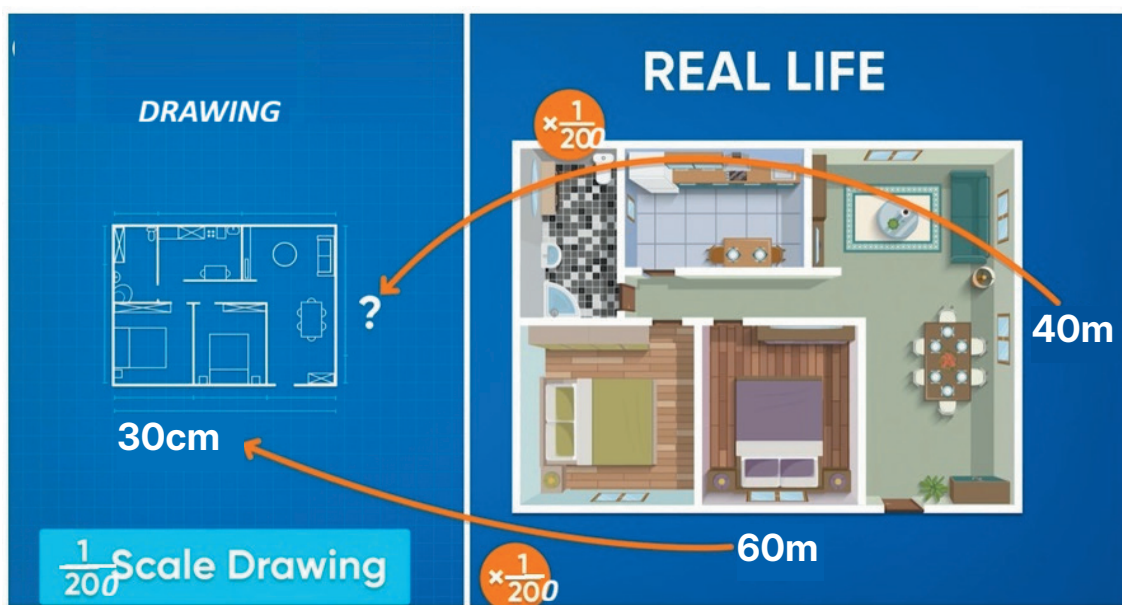
The big trapezium is 3 times bigger than the small trapezium.

The ratio (called the **scale**) shows the relationship between a length or size on the drawing and the corresponding actual length/size, ensuring that all measurements are proportional and realistic.



The small triangle is $\frac{1}{3}$ of the big one.

Example: A student drew the plan of their house to a scale of $\frac{1}{200}$ on a manila paper.



The real-life house is 200 times bigger than the drawing.

- If the house field is 40 m by 60 m, find its length and width on the drawing.
- If in the scale drawing of the kitchen is drawn as a rectangle 9 cm by 2.2 cm, find its actual length and width.

Solution:

1 : 200 means that 1 cm represents 200 cm or 1 cm represents 2 m.

a) 2 m is represented by 1 cm, 40 m is represented by $\frac{1}{2} \times 40 \text{ cm} = 20 \text{ cm}$

60 m is represented by $\frac{1}{2} \times 60 \text{ cm} = 30 \text{ cm}$.

The house is 30 cm long and 20 cm wide.

b) 1 cm represents 2 m

9 cm represents $2 \times 9 \text{ m} = 18 \text{ m}$ and 2.2 cm represents $2 \times 2.2 \text{ m} = 4.4 \text{ m}$

The kitchen is 18 m long and 4.4 m wide.



APPLICATION ACTIVITY 4.3

- Find the scale drawing length that represents: (a) 73 m when the scale is '1 cm represents 10 m'. (b) 73 m when the scale is '1 cm represents 5 m'. (c) 1 km when the scale is '10 cm represents 1 km'. (d) 1.53 km when the scale is '5 cm represents 2 km'.
- Find the actual length represented by a scale drawing length of, (a) 7.3 cm when the scale '1 cm represents 3 m'. (b) 8.5 cm when the scale is '2 cm represents 4 km'. (c) 5.3 cm when the scale is '5 cm represents 1 km'. (d) 15.6 cm when the scale is '2 cm represents 100 m'.
- A physical planner drew the plan of a dump site to a scale of 1 : 1 000. If the site measures 60 m by 45 m, find the dimensions of the site on the plan.

4.4 Concept of proportion



Activity 4.4

At the market, **2 kg of sugar costs 2,000 FRW**. We need to know the cost of **4 kg** of the same sugar. Think individually, then discuss the following:

- What is the cost of 1 kg of sugar?
- What is the cost of 4 kg of sugar?
- What happens to the cost when the quantity of sugar increases?
- Is the relationship between quantity and cost the same when one of them varies?



Content Summary

A proportion is a relationship between four numbers or quantities in which the ratio of the first pair equals the ratio of the second pair, and is written as $a:b = c:d$ and is read as "a is to b as c is to d"

Examples: 1) Determine whether the ratios 3 : 6 and 5 : 10 are in proportion.

Solution:

Simplify each ratio: $3 : 6 = 1 : 2$ and $5 : 10 = 1 : 2$

Since the two ratios are equal, they are in proportion.

Answer: Yes, the ratios are in proportion.

2) Check whether 4 : 8 and 6 : 12 form a proportion.

Solution:

$4 : 8 = 1 : 2$ and $6 : 12 = 1 : 2$

Both ratios are equal.

Answer: The ratios are in proportion.

3) If 2 exercise books cost 600 FRW, check whether 5 books cost 1,500 FRW.

Solution:

Compare the ratios: Books: 2 : 5 Cost: 600 : 1,500

Simplify: 2 : 5 stays 2 : 5 and $600 : 1,500 = 2 : 5$

Since the ratios are equal, the situation shows a proportion.

Answer: Yes, the quantities are in proportion.

3) The proportion $a:b = c:d$ can be written in the form of equivalent fractions as follows.

In this case, $a:b = c:d \Rightarrow \frac{a}{b} = \frac{c}{d}$ and $\frac{a}{b} = \frac{c}{d} \Rightarrow a \times d = b \times c$.

Example: The number 2 is to 3 as 4 is to x. Find the value of x.

Solution:

2 is to 3 as 4 is to x means $\frac{2}{3} = \frac{4}{x} \Rightarrow 3 \times 4 = 2 \times x$ or

$$2x = 12.$$

$$x = 6.$$

Other properties of proportions are going to be developed in the next lesson.



APPLICATION ACTIVITY 4.4

- 1) State whether the following ratios are in proportion:
 - a) 6 : 9 and 8 : 12 b) 10 : 20 and 3 : 6 c) 4 : 5 and 12 : 15
- 2) A shopkeeper sells:
 - 3 kg of rice for 2,400 FRW
 - 6 kg of rice for 4,800 FRW

Are the quantities of rice and their prices in proportion?
- 3) A car travels:
 - 40 km in 2 hours
 - 60 km in 3 hours

Check whether the distances and times are in proportion.

4.5 Properties of proportions



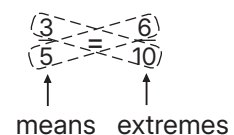
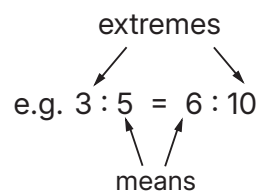
Activity 4.5

A chef is making fruit juice. For every 2 cups of orange juice, he uses 3 cups of water. If he wants to make more juice, he must keep the same ratio of orange juice to water. How can he do that? How many cups of water will he use for 8 cups of orange juice?



Content Summary

In a proportion, the second and third values are called means and the first and fourth values are called extremes.



The following are properties of proportions:

- a) **Cross product property:** The product of the means is equal to the product of the extremes $\frac{a}{b} = \frac{c}{d}$ is same as $a \times d = b \times c$.

Example: Tomatoes are sold at 3 tomatoes for 100 FRW. How much will 12 tomatoes cost?

Solution:

Proportion: $\frac{3}{12} = \frac{100}{x}$. Solve: $3x = 12 \times 100$ FRW

$$3x = 1\,200 \text{ FRW}$$

$$x = 400 \text{ FRW i.e. 12 tomatoes will cost 400 FRW.}$$

b) Means or extremes switching property: if $\frac{a}{b} = \frac{c}{d}$ then $\frac{a}{c} = \frac{b}{d}$

Example: If $\frac{x}{5} = \frac{y}{4}$, find the ratio of $\frac{x}{y}$

Solution:

Using switching property of proportion

$$\frac{x}{y} = \frac{5}{4}$$

c) Inverse property of proportions: if $\frac{a}{b} = \frac{c}{d}$ then $\frac{b}{a} = \frac{d}{c}$

Example: If $\frac{9}{a} = \frac{5}{b}$, find the ratio $\frac{a}{b}$

Solution:

Solution Applying inverse property $\frac{9}{a} = \frac{5}{b}$

Applying switching property $\frac{a}{b} = \frac{9}{5}$

d) Additional property of proportions: Adding 1 both sides we get $\frac{a}{b} + 1 = \frac{c}{d} + 1$

Simplifying we get $\frac{a+b}{b} = \frac{c+d}{d}$ Similarly, $\frac{a-b}{b} = \frac{c-d}{d}$

Example: If $\frac{5}{8} = \frac{x}{y}$, find the value of $\frac{13}{8} = ?$

Solution:

Applying additional property $\frac{5}{8} + 1 = \frac{x}{y} + 1$

$$\frac{5+8}{8} = \frac{x+y}{y} \text{ then } \frac{13}{8} = \frac{x+y}{y}$$



APPLICATION ACTIVITY 4.5

1) Which of these are true proportions?

(a) $\frac{2}{3} = \frac{4}{6}$

(b) $\frac{8}{10} = \frac{2}{3}$

(c) $\frac{8}{6} = \frac{6}{4}$

(d) $3 : 2 = 3 : 5$

(e) $6 : 4 = 2 : 1$

(f) $1 : 3 = 6 : 18$

2) Solve the proportions:

(a) $\frac{3}{7} = \frac{x}{14}$

(b) $\frac{8}{x} = \frac{16}{4}$

(c) $\frac{6}{4} = \frac{x}{12}$

(d) $3 : 4 = x : 8$

(e) $3 : 4 = 9 : x$

(f) $x : 12 = 6 : 8$

3) Given that $x : y = 3 : 5$, find the ratios

(a) $(2x - y) : (2x + 3y)$

(b) $(x + y) : (2x - y)$

4.6 Direct proportions



Activity 4.6

The table below shows the relationship between the number of pens and the cost of pens.

No. of pens	1	2	3	4	5
Cost (FRW)	120	240	360	480	600

- What happens to the cost when the number of pens is doubled?
- If you divide the number of pens by two, what do you notice about the cost?
- What can you say about the relation between the cost of pens and the quantity of pens?



Content Summary

Two quantities are said to be directly proportional when one quantity increases through a particular ratio and the other quantity increases in the same ratio, and vice versa.

Example: The cost of 5 kgs of sugar is 2 880 FRW. Find the cost of 3 kgs of the same sugar.

Solution:

We can use ratios or unitary method to solve this problem.

Ratio method:

Let the cost of 3 kg be x . Then, $x : 2\,880 = 3 : 5$ (the ratio of the cost is equal to the ratio of the sugar in kgs). Using the ratios in fraction form: $\frac{x}{2880} = \frac{3}{5}$

$$x = \frac{3}{5} \times 2\,880 \text{ FRW} = 1,728 \text{ FRW.}$$

Hence, the cost of 3 kg is 1,728 FRW.

Unitary method:

5 kg cost 2 880 FRW

1 kg cost $\frac{2880}{5}$ FRW

3 kg cost $\frac{2880 \times 3}{5}$ FRW = 1,728 FRW

Notice that a decrease in the number of kg of sugar bought leads to a decrease in the cost in the same ratio.



APPLICATION ACTIVITY 4.6

- 1) At a steady speed, a car uses 5 litres of petrol to travel 85 km. How much petrol is needed to travel 70 km at the same speed?
- 2) In a spring balance, the extension in the spring is proportional to the load. If the extension is 3.5 cm when the load is 10 kg, what is the extension when the load is 4.5 kg?
- 3) If the mass of 16 cm^3 of a metal is 20 g, what is the mass of 30 cm^3 ?

4.7 Inverse proportion



Activity 4.7

Consider the relationship between the speed and time taken by a car to cover a fixed distance of 320 km.

Speed (km/h)	20	40	80	160
Time (h)	16	8	4	2

Take 20 km/h to be the original speed.

- What do you notice when the speed is doubled?
- What happens for the time used when the speed increases?
- Plot the graph of speed against time.
- Describe the graph you drew to your friend.



Content Summary

Two quantities are **inversely proportional** (or in inverse variation) if an increase in one quantity causes a decrease in the other quantity, and vice versa, in such a way that their **product remains constant**.

Mathematical Definition: If x and y are inversely proportional, then: $x \times y = k$ where k is a constant.

This can also be written as: $y = \frac{k}{x}$

Example 1:

Quantity 1: Speed of a vehicle **Quantity 2:** Time taken to travel a fixed distance

If you need to travel **120 km**:

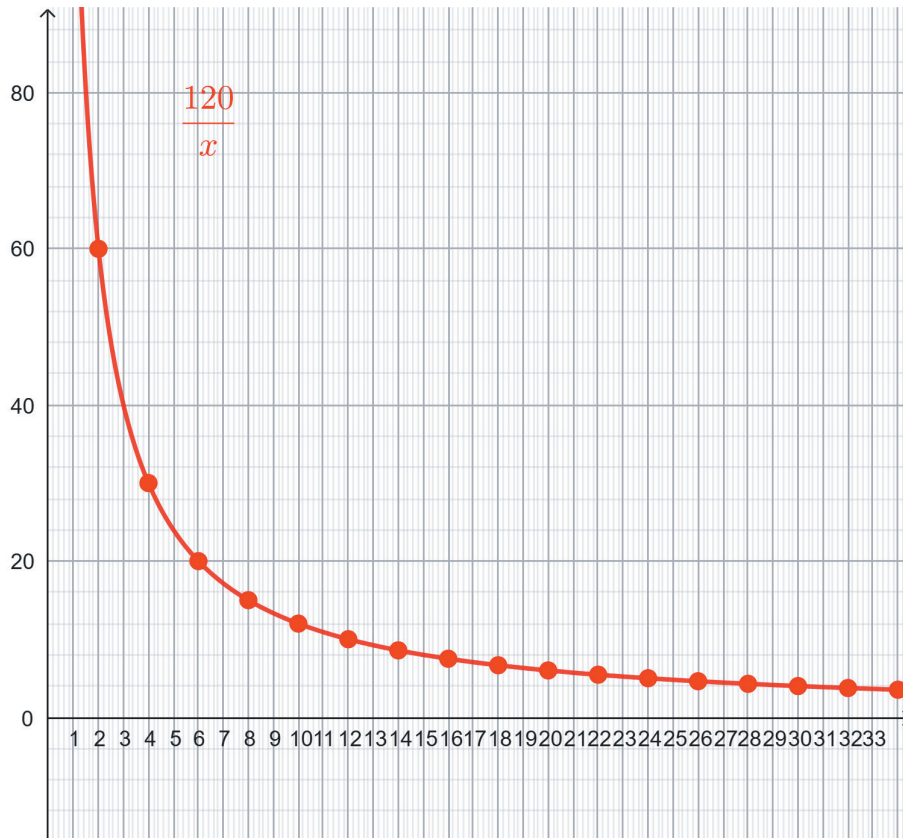
- At 40 km/h, time = $120/40 = 3$ hours.
- At 60 km/h, time = $120/60 = 2$ hours.
- At 80 km/h, time = $120/80 = 1.5$ hours

Here, **Speed $\times$ Time = Distance = constant k.**

We see that for the speed $v=40\text{km/h}$, the **speed used = distance /time.**

The function speed $=120/t$.

As time increases, the speed decreases proportionally, and their product (distance) stays the same (120 km). This is inverse proportion.



The graph of speed (S) against hour (h) is a curve, $x = \text{time}$.

Note: Problems on inverse proportion are solved using either unitary, ratio or graphical method.

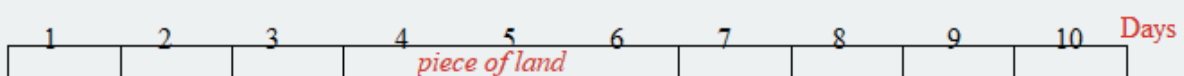
Example 2:

Four men working at the same rate can dig a piece of land in ten days. How long would it take five men to do the same job?

Solution:

Unitary method:

4 men can dig in 10 days



4 men dig the land in 10 days.

1 man working in the same ways takes: 10 days $\times 4$ to dig the entire land.

For 5 men working in the same ways take = $\frac{40}{5}$ days (taken by one man) = 8 days

Ratio method:

Five men take shorter time than four men, so we use the inverse ratio. The number of men increase in the ratio 5:4 The days decrease in the ratio x: 10.

Using inverse ratio, $\frac{x}{10} = \frac{4}{5}$

$$x = \frac{4}{5} \times 10 = 8 \text{ days.}$$



APPLICATION ACTIVITY 4.7

12 taps can fill a tank in three hours. How long would it take to fill the tank if only 4 taps are working?

- 1) If 8 people can cultivate a piece of land in 6 days. How long will it take 12 people to cultivate the same piece of land?
- 2) Three men can build a wall in 10 hours. How many men would be needed to build the wall in $7\frac{1}{2}$ hours?
- 3) If it takes 6 men 4 days to dig a hole 5 m deep, how long will it take 10 men to dig a hole 8 m deep?

4.8 Graphs on direct and inverse proportions



Activity 4.8

4 workers take 12 days to complete a job. Discuss these questions:

- If 6 workers do the same job, will it take more or less time?
- If 12 workers do the job, how many days might it take?
- What happens to the time when the number of workers increases?
- Plot number of workers on the x-axis and time on the y-axis.



Content Summary

Direct Proportion occurs when two quantities increase or decrease together at a constant rate. **The graph of a direct proportion:**

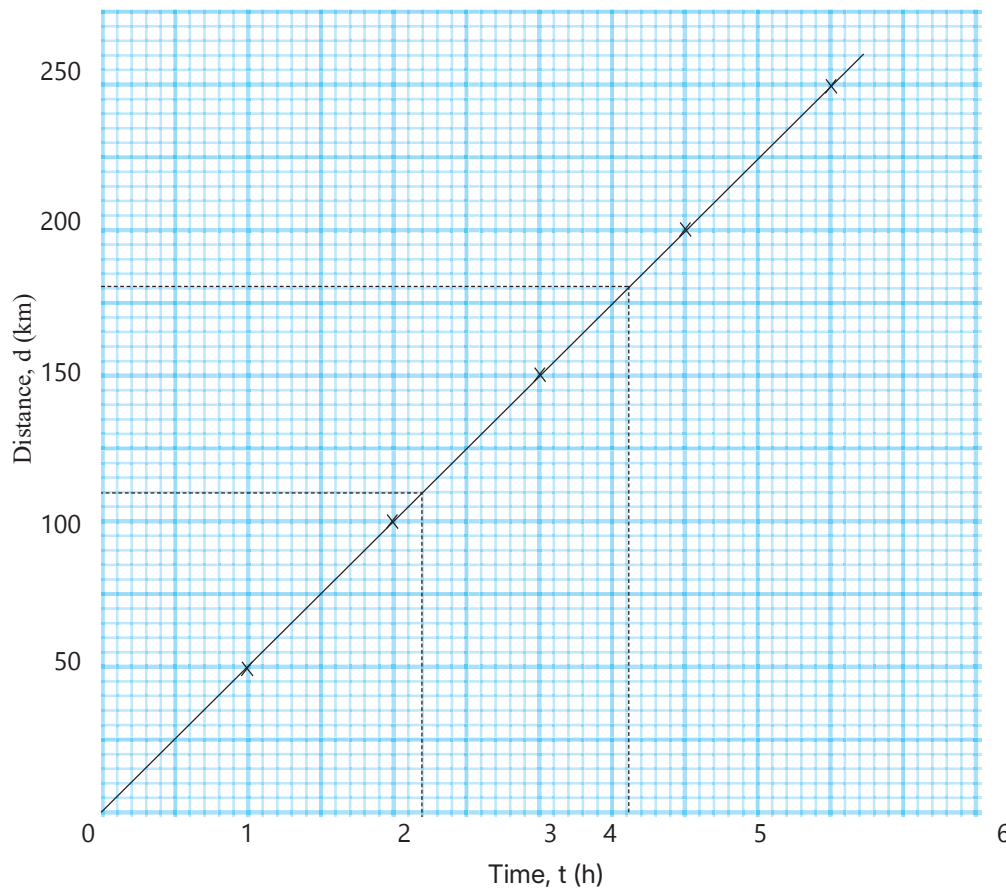
- Is a **straight line**
- **Passes through the origin (0,0)**
- Shows that when $x = 0$, $y = 0$
- Has a constant gradient (slope).

Example 1: Table below shows the distance (d) in kilometers covered and the time taken at constant speed.

Distance(d) covered in km	50	100	150	200	250
Time(t) taken in hours	1	2	3	4	5

Use a graph to find:

(a) The distance covered in 2hr 12 min. (b) The time taken to cover a distance of 180 km.



Solution:

- From the graph, we see that the distance covered in 2h 12 min is 110 km.
- From the graph we see that a distance of 180 km will take 3 hours 36 minutes.

Inverse Proportion occurs when one quantity increases while the other decreases so that their product remains constant.

The graph of an inverse proportion:

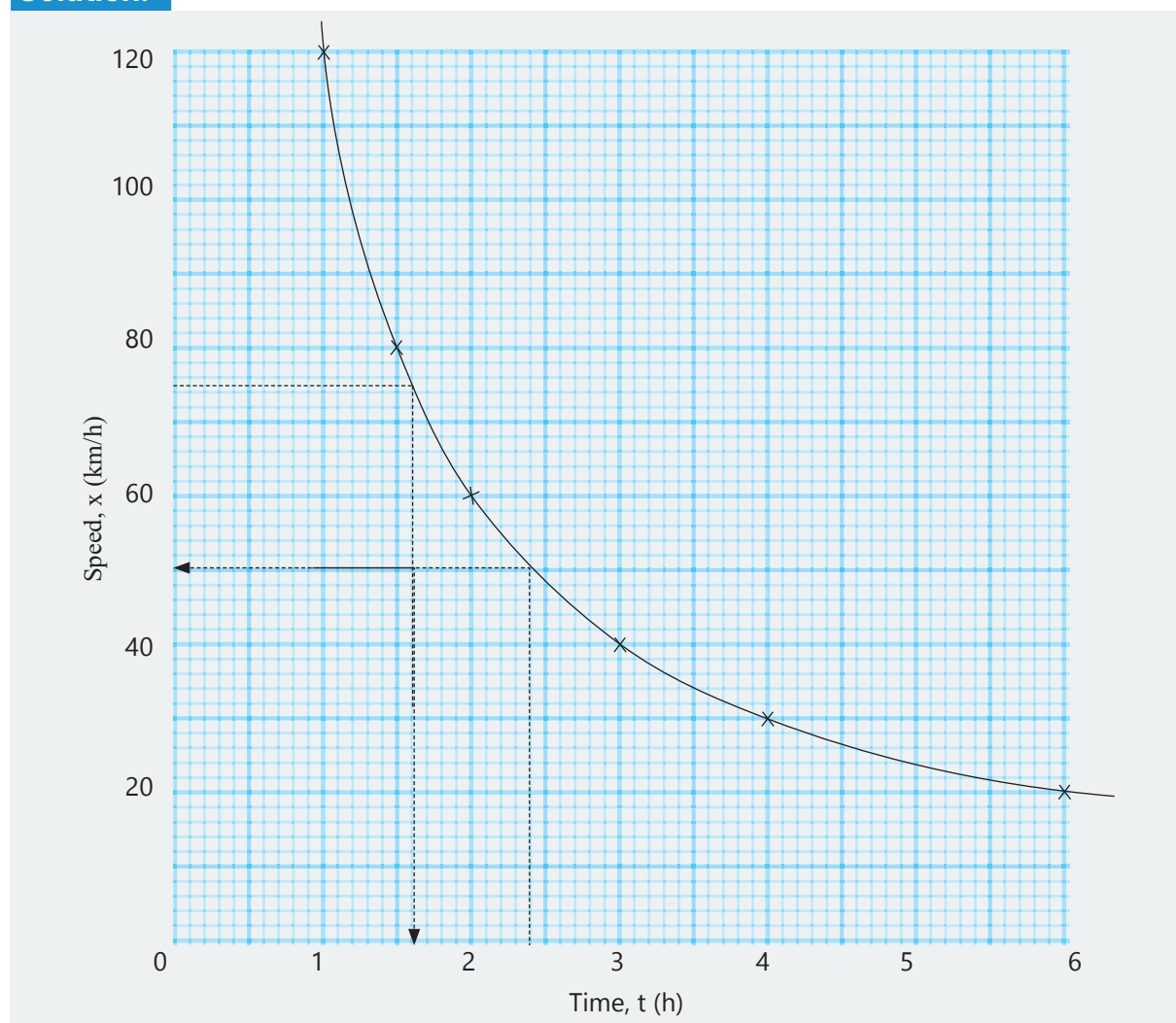
- Is a curve (hyperbola).
- Does not pass through the origin.
- Approaches both axes but never touches them.
- Shows that as x increases, y decreases.

Example 2: 2. The table below shows the time taken to cover a distance of 120 km at various speeds.

Speed, x (km/h)	20	30	40	60	80	120
Time, t (h)	6	4	3	2	1.5	1.0

- Draw a graph of speed against time taken.
- Use your graph to determine:
 - The time taken to cover the same distance at a speed of 75 km/h.
 - The speed required to cover the same distance in 2.4 hours.

Solution:



- From the graph:
 - The time taken to cover the same distance at a speed of 75 km/h is 1.6 h.
 - The speed required to cover the same distance in 2.4 h is 50 km/h.



APPLICATION ACTIVITY 4.8

- 1) The volume of a fixed mass of a gas at constant temperature is inversely proportional to its pressure. Table below shows the volume of a given mass of gas at different pressures when the temperature is constant.

Pressure, P (Pascals)	750	500	400	300	250	200	150	100
Volume, V (m^3)	2	3	3.75	5	6	7.5	10	15

- a) Draw a graph of V against P.
 b) Using the graph, find:
 i) P when $V = 8 \text{ m}^3$.
 ii) V when $P = 350$ pascals.
 c) Calculate V when $P = 600$ pascals.
- 2) Table below shows the acceleration of different masses when a certain constant force is applied to them. The masses m (kg) and the acceleration $a (\frac{m}{s^2})$ are shown.

m (kg)	20	15	12	10	8	6	5	2
A ($\frac{m}{s^2}$)	1	1.33	1.67	2	2.5	3.33	4	10

- a) Draw a graph of m against acceleration A.
 b) Use the graph to find:
 i) m when $a = 3 \frac{m}{s^2}$.
 ii) the acceleration when the mass is 9 kg.
 c) The mass when acceleration is $4.5 \frac{m}{s^2}$.

4.9 Problems involving direct and inverse proportions



Activity 4.9

"A school buys exercise books at 500 Frw each." Now discuss.

- How much will 1 book cost?
- How much will 2 books cost?
- How much will 5 books cost?



Content Summary

A. To solve a problem on direct proportion Step-by-Step Process

- Read and Understand the Problem
- Identify the two quantities involved and what is required.
- Decide the Type of Proportion Check whether both quantities increase or decrease together.
- If yes → Direct proportion
- Write the Proportional Statement
- Find the Constant of Proportionality
- Substitute the given values
- Calculate the Required Value
- Use the equation with the new value to find the unknown.
- Write the Final Answer Clearly Include correct units.

Example 1: If 5 books cost 2,500 FRW, find the cost of 8 books.

Solution:

Ratio method:

Let the cost of 5 books be 2500FRW Then 2500FRW: x = 5:8 (the ratio of the cost is equal to the ratio of the books). Using the ratios in fraction form: $\frac{x}{2500} = \frac{8}{5}$

$$x = \frac{8}{5} \times 2\,500 \text{ FRW} = 4000 \text{ FRW}$$

Hence x, the cost of 8 books = 4000 FRW

Unitary method:

5 books cost 2 500 FRW

1 book costs $\frac{2500}{5}$ FRW

8 books cost $\frac{2500 \times 8}{5}$ FRW = 4000 FRW

B. Solving a problem on inverse Proportion Step-by-Step Process:

- Read and Understand the Problem
- Identify the two related quantities.
- Decide the Type of Proportion
- Check whether one quantity increases while the other decreases. If yes → Inverse proportion
- Write the proportional statement $k = \frac{x}{y}$
- Calculate the Required Value Substitute into the formula to find the unknown.
- Write the Final Answer Clearly Include correct units.

Example: 6 workers take 12 days to complete a job. How long will 9 workers take?

Solution:

Unitary method:

6 workers take 12 days

1 worker takes many days: $6 \times 12 \text{ days} = 72 \text{ days}$.

For 9 men take = $\frac{72}{9}$ days (taken by one man) = 8 days.

Ratio method:

6 workers take shorter time than 9 so we use the inverse ratio. The number of men increase in the ratio $x : 12$ The days decrease in the ratio $6 : 9$ Using inverse ratio, $\frac{6}{9} = \frac{x}{12}$

$$x = \frac{12}{9} \times 6 = 8 \text{ days}$$



APPLICATION ACTIVITY 4.9

- 1) Given that x is inversely proportional to y , copy and complete the following table.

x	1	2	4	8	12	32	64
y		16		4		1	

- 2) A school bought 120 books costing 960 FRW each. If the price of each book was reduced by 280 FRW, how many more books could be bought?
- 3) A cyclist averages at a speed of 27 km/h for 4 hours. At what average speed would he need to cycle to cover the same distance in 3 hours?

4.10 Finding the average price of a mixture



Activity 4.10

A trader mixes 2 kg of rice at 1,200 FRW per kg with 3 kg of rice at 1,000 FRW per kg and sells the mixture at one price per kg. Discuss:

- Is the mixture price the same as 1,200 FRW per kg? Why?
- Is it the same as 1,000 FRW per kg? Why?
- How can we find one fair price for the whole mixture?



Content Summary

The average price of a mixture is the price per unit obtained after mixing items bought at different prices.

Formula:
$$\text{Average price} = \frac{\text{Total cost of all items}}{\text{Total quantity of items}}$$

Example: A shopkeeper mixes:

- 4 kg of sugar at 900 FRW per kg.
- 6 kg of sugar at 1,100 FRW per kg.

Solution:

- Cost of first sugar = $4 \times 900 = 3,600$ FRW
 - Cost of second sugar = $6 \times 1,100 = 6,600$ FRW
- Total cost = 3,600 RWF + 6,600 RWF = 10,200 FRW
- Total quantity = 4 kg + 6 kg = 10 kg
- $$\text{Average price} = \frac{10,200}{10} = 1,020 \text{ Frw per kg}$$



APPLICATION ACTIVITY 4.10

- 1) A farmer mixes 3 kg of maize at 500 FRW per kg with 7 kg at 600 FRW per kg. Find the average price per kg.
- 2) A shopkeeper mixes: 10 litres of milk at 450 FRW per litre and 5 litres at 550 FRW per litre. Find the average price of the mixture.
- 3) In groups, Students create a **market scenario** involving:
 - At least two items
 - Different prices
 - Different quantities
 Then calculate and present the average price.
- 4) Why is it **wrong** to find the average price by simply adding the prices and dividing by 2 when quantities are different?

4.11 Finding quantity of one type of the mixture



Activity 4.11

A trader mixes beans bought at 800 FRW per kg and beans bought at 1,200 FRW per kg. He sells the mixture at an average price of 1,000 FRW per kg. He used 10 kg of beans in total.

Discuss in groups:

- Which beans are cheaper? Which are more expensive?
- Do we need more of the cheap beans or the expensive beans to get 1,000 FRW?
- Given that the total number of kg is 10. If there are x kg of beans at 800Frw, How can you find the number of beans costing 1200kg?



Content Summary

We have: The prices of two items, the average price of the mixture, the total quantity. We are asked to find **how much of one item** was used.

Method (Using total cost).

Let: Price of item A = P_1 ; Price of item B = P_2 ; Average price = A

The total quantity = T .

Steps: Find total cost of the mixture: $\text{Total cost} = A \times T$

- 1) Let the quantity of one item be x
- 2) Find the quantity of the other item: $T - x$.
- 3) Form an equation using total cost.
- 4) Solve the equation to find x

Example

A trader mixes rice at **600 FRW per kg** and rice at **900 FRW per kg**. The average price of the mixture is **750 FRW per kg**. If the total quantity is **20 kg**, find the quantity of each type.

Solution:

Let quantity of rice at 600 FRW per kg = x kg

Then quantity of rice at 900 FRW per kg = $20 - x$

Total cost of mixture: $750 \text{ FRW} \times 20 \text{ FRW} = 15,000 \text{ FRW}$

Cost equation: $600x + 900(20 - x) = 15,000$

Simplify: $600x + 18,000 - 900x = 15,000$

$600x - 900x = 15,000 \text{ FRW} - 18,000 \text{ FRW}$

$x = 10$.

So: 10 kg at 600 FRW per kg and 10 kg at 900 FRW per kg.



APPLICATION ACTIVITY 4.11

- 1) A shopkeeper mixes maize at 500 FRW per kg and 700 FRW per kg. The average price of the mixture is 620 FRW per kg. If the total quantity is 15 kg, find the quantity of maize bought at 500 FRW per kg.
- 2) Oil at 2,000 FRW per litre is mixed with oil at 3,000 FRW per litre. The average price is 2,400 FRW per litre. If the total quantity is 20 litres, find how many litres of each type were mixed.
- 3) Create a market situation where: Two goods have different prices, The average price and total quantity are known

Then:

- Find the quantity of one type.
- Explain your steps to the class.

4.12 Finding the price of one type of ingredient in the mixture



Activity 4.12

A shopkeeper mixes **5 kg of beans at 800 FRW per kg** with **3 kg of beans at an unknown price**. The mixture is sold at an **average price of 900 FRW per kg**.

Discuss in pairs and suggest ways of finding the unknown price using total cost.

- Do we know the total quantity of the mixture?
- Do we know the average price?
- Which information is missing?
- How can we find the unknown price?



Content Summary

Sometimes in mixture problems:

- The **quantities** of items are known
- The **average price** of the mixture is known
- The **price of one ingredient is unknown**

We can find the unknown price by using **total cost**.

Let:

- Quantity of known item = Q_1
- Price of known item = P_1
- Quantity of unknown item = Q_2
- Price of unknown item = P
- Average price of mixture = A

Steps:

- 1) Find total quantity: $Q_1 + Q_2$
- 2) Find total cost of the mixture: $A \times (Q_1 + Q_2)$
- 3) Write cost equation: $Q_1P_1 + Q_2P_2 = \text{Total cost}$
- 4) Solve for the unknown price P .

Example: A trader mixes: **6 kg of sugar at 700 FRW per kg, 4 kg of sugar at an unknown price**, The average price of the mixture is **820 FRW per kg**. Find the price per kg of the second sugar.

Solution:

Total quantity: $6 \text{ kg} + 4 \text{ kg} = 10 \text{ kg}$

Total cost of mixture: $820 \text{ FRW} \times 10 = 8,200 \text{ FRW}$

Cost of known sugar: $6 \times 700 \text{ FRW} = 4,200 \text{ FRW}$

Let the unknown price be $x \text{ FRW per kg}$.

Cost equation: $4,200 \text{ FRW} + 4x = 8,200 \text{ FRW}$

Solve: $4x = 8,200 \text{ FRW} - 4,200 \text{ FRW} = 4,000 \text{ FRW}$

$x = 1,000 \text{ FRW}$

Answer: The price of the second sugar is **1,000 FRW per kg**.

**APPLICATION ACTIVITY 4.12**

- 1) A shopkeeper mixes: 8 kg of maize at 500 FRW per kg, 4 kg of maize at an unknown price, The average price of the mixture is 600 FRW per kg. Find the unknown price.
- 2) A mixture is made using: 10 litres of oil at 2,200 FRW per litre, 5 litres of oil at an unknown price The average price is 2,500 FRW per litre. Find the unknown price.
- 3) Why must the unknown price be higher than the average price if its quantity is small and the known price is low?

4.13 Finding both quantities of a mixture**Activity 4.13**

A trader mixes beans at 800 FRW per kg and beans at 1,200 FRW per kg. The mixture is sold at an average price of 1,000 FRW per kg, and the total quantity is 30 kg.

Discuss in groups and predict the quantities before calculating.

- Are the two prices equal?
- Is the average price exactly in the middle?
- Can we find how many kilograms of each type were mixed?
- Which type should be more: the cheaper or the expensive one?



Content Summary

In some mixture problems:

- The **prices of two ingredients** are known.
- The **average price of the mixture** is known.
- The **total quantity** is known.

We are required to find **how much of each ingredient** was mixed.

Method: Let: Price of ingredient A = P_1 ; Price of ingredient B = P_2 ; Average price = A ; Total quantity = T

Steps:

- 1) Let the quantity of ingredient A be x .
- 2) Then the quantity of ingredient B will be $T-x$
- 3) Find the total cost of the mixture: $A \times T$
- 4) Form the cost equation: $P_1 \times x + P_2 \times (T-x) = A \times T$
- 5) Solve for x , then find the other quantity.

Example: A trader mixes: Rice at **700 FRW per kg**, to Rice at **1,100 FRW per kg**, The average price of the mixture is **900 FRW per kg**. If the total quantity is **40 kg**, find the quantity of each type.

Solution:

Let quantity of rice at 700 FRW per kg = x kg

Quantity of rice at 1,100 FRW per kg = $(40-x)$ kg

Total cost of mixture: $900 \times 40 = 36,000$ FRW

Cost equation: $700x + 1,100(40-x) = 36,000$

Simplify: $700x + 44,000 - 1,100x = 36,000$

$-400x = -8,000$

$x = 20$

So:

- **20 kg** at 700 FRW per kg
- **20 kg** at 1,100 FRW per kg



APPLICATION ACTIVITY 4.13

- 1) A shopkeeper mixes maize at: 500 FRW per kg and 800 FRW per kg. The average price is 650 FRW per kg, and the total quantity is 30 kg. Find the quantity of each type.
- 2) A mixture contains fuel bought at: 1,400 FRW per litre and 2,000 FRW per litre. The average price is 1,700 FRW per litre, and the total quantity is 50 litres. Find the quantity of each type.
- 3) Why are the quantities equal when the average price is exactly halfway between the two prices?



END OF UNIT ASSESSMENT

Assessing the knowledge and understanding

- 1) Define a ratio.
- 2) Simplify the ratio: 12:18
- 3) What is meant by a scale in scale drawing?
- 4) Define proportion.
- 5) State **one** property of proportion.
- 6) What is direct proportion?
- 7) What is inverse proportion?
- 8) Write the formula for finding the average price of a mixture
- 9) (a) State the shape of the graph of direct proportion.
(b) State the shape of the graph of inverse proportion.

Assessing skills

- 10) Divide **60,000 FRW** in the ratio **2 : 3 : 5**.
- 11) A map is drawn to a scale of **1 : 500**. If a road measures **6 cm** on the map, find its actual length.
- 12) Five workers take **12 days** to complete a job. How long will **10 workers** take to complete the same job?
- 13) A trader mixes: 4 kg of rice at **900 FRW per kg** and 6 kg of rice at **1,100 FRW per kg**. Find the average price per kg of the mixture.
- 14) Maize at **500 FRW/kg** and **700 FRW/kg** is mixed. The average price is **620 FRW/kg** and the total quantity is **15 kg**. Find the quantity of maize bought at **500 FRW/kg**.
- 15) A mixture contains: 8 kg of beans at **600 FRW/kg** and 4 kg of beans at an unknown price. The average price is **800 FRW/kg**. Find the unknown price per kg.
- 16) Sugar at 700 FRW/kg and 1,100 FRW/kg is mixed to obtain an average price of 900 FRW/kg. If the total quantity is **40 kg**, find the quantity of each type.

Assessing attitudes and values

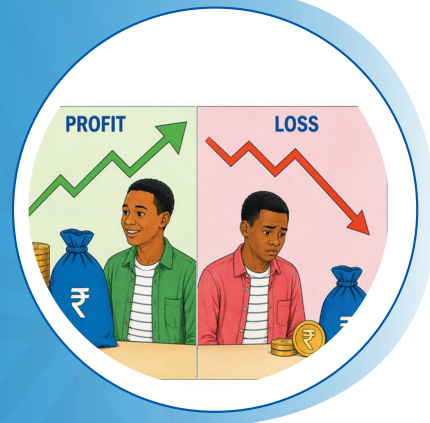
Choose the correct statement:

- 17) Explain why honesty is important when mixing goods for sale.
 - A) It speeds up the production process and reduces labor costs.
 - B) It builds customer trust, ensures safety, and maintains a reputable brand.
 - C) It is only necessary when the goods are being inspected by authorities.
 - D) It allows for more creative and varied product mixtures.

- 18) A trader shares profits using ratios correctly. What value does this show?
- A) Advanced mathematical skill.
 - B) A focus on maximizing personal profit above all else.
 - C) Fairness, transparency, and honoring agreements with partners.
 - D) Reluctance to take financial risks.
- 19) State one importance of scale drawing in real life.
- A) It guarantees that all artistic drawings are aesthetically pleasing.
 - B) It enables architects and engineers to plan buildings and machinery accurately before construction.
 - C) It is primarily used to teach geometry in schools.
 - D) It ensures that all drawings are completed quickly.
- 20) Explain how knowledge of direct and inverse proportion helps people in daily life.
- A) It primarily helps in solving abstract puzzles and math Olympiad problems.
 - B) It enables better decision-making in recipes, travel planning, shopping, and managing resources.
 - C) Its main use is to check the work of mathematicians and scientists.
 - D) It is mostly historical knowledge with little modern application.

UNIT 5

PERCENTAGE, DISCOUNT, PROFIT AND LOSS



Key Unit Competence:

Solve problems that involve percentage, discount, commission, profit, loss, simple interest, taxes.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain percentage as a fraction out of 100.
- Use multipliers to increase or decrease quantities.
- Explain discount, commission, profit, loss, loans, savings, simple interest, and tax.
- State the formulas for percentage change and simple interest.

Skills

- Calculate percentages correctly.
- Apply multipliers to increase or decrease values.
- Calculate discount, commission, profit, and loss in real-life situations.
- Use formulas to find simple interest, tax, and insurance.
- Interpret financial problems and present clear solutions.

Attitudes and values

- Develop confidence in solving real-life financial problems.
- Show honesty and responsibility in financial matters.
- Appreciate the importance of saving and wise money management.
- Work cooperatively and respect others' ideas in group tasks.

INTRODUCTION

In our day to day lives, we get involved in activities of either buying or selling goods or services using money. Such activities are called transactions. There are many terms that are used to describe various aspects of such transactions. In this unit, we will learn a number of these terms and how to calculate their values in financial transactions.



Introductory activity

A shop advertises a 20% discount on a shirt priced at 10,000 FRW, a salesperson earns 5% commission on sales, and a student saves money in a bank that gives simple interest. Now observe the picture and discuss the following questions:

- How much is the discount?
- How much commission is earned?
- Why do people save money in banks?
- Why do we pay taxes?



5.1 Percentages



Activity 5.1

- A shirt is sold at 20% off.
- A student scores 75% in a test.
- A bank offers 10% interest per year.
- VAT charged is 18%.

Discuss:

Where have you seen these numbers before?

What do you think about the meaning of percentage (%) in each case?



Content Summary

A percentage is a way of expressing a number as a fraction of 100. The symbol used for percentage is %. $\text{Percentage} = \frac{\text{Part}}{\text{Whole}} \times 100$

Percentages are widely used in daily life such as in marks, discounts, profit and loss, interest, taxes, and savings.

1) Converting Fractions and Decimals to Percentages

Fraction $\rightarrow$ multiply by 100%

Decimal $\rightarrow$ multiply by 100%

Example 1: Convert $\frac{3}{5}$ to a percentage: $\frac{3}{5} \times 100 = 60\%$

Example 2: Convert 0.45 to a percentage: $0.45 \times 100 = 45\%$

2) Finding a Percentage of a quantity

Percentage of a quantity

Percentage $\div 100 \times$ Quantity

Example: Find 20% of 250 = $20 \div 100 \times 250$ Answer: 20% of 250 is 50.

3) Finding What Percentage One Quantity Is of Another

Percentage

Part $\div$ Whole $\times 100$

Example: A student scored 36 marks out of 60. Find the percentage score.

Solution:

$$(36 \div 60) \times 100 = 60\%$$

4) Increasing a Quantity by a Given Percentage

- Find the increase.
- Add the increase to the original value.

Example: Increase 500 by 10%.

$$10 \div 100 \times 500 = 500 + 50 = 550$$

Answer: The new value is 550.

5) Decreasing a Quantity by a Given Percentage

Steps:

- Find the decrease.
- Subtract the decrease from the original value.

Example: Decrease 800 by 25%.

$$(25 \div 100) \times 800 = 200 \text{ then } 800 - 200 = 600.$$

Answer: The new value is 600.

6) Percentage Increase and Percentage Decrease

- Percentage increase = $(\text{Increase} \div \text{Original value}) \times 100$
- Percentage decrease = $(\text{Decrease} \div \text{Original value}) \times 100$

Example: The price of a book increased from 2,000 FRW to 2,500 FRW. Find the percentage increase.

$$\text{Increase} = 2,500 - 2,000 = 500$$

$$(500 \div 2,000) \times 100 = 25\%$$

Answer: The percentage increase is 25%.



APPLICATION ACTIVITY 5.1

- 1) Express the following fractions as percentages: (a) $\frac{5}{12}$ (b) $\frac{3}{4}$ (c) $\frac{27}{50}$
- 2) Express the following decimals as percentages: (a) 0.36 (b) 0.45 (c) 0.81 (d) 0.04
- 3) Express the first quantity as a percentage of the second quantity in each of the following: (a) 20 cm, 50 cm (b) 30 cm, 5 m (c) 54 mm, 25 cm
- 4) Mutanguha scored 27 marks out of 45 in a mathematics test. What percentage was this?
- 5) In a survey of 100 cars, 45 were white, 25 were blue and 30 were red. Express each of these numbers as a percentage of the total.

5.2 Increasing and decreasing a number or a quantity using a multiplier



Activity 5.2

You get a price tag for 6000 FRW

You Sale it on 25% off → Find new price.

When the price rise 10% more! → Find new price.

Think: Did you multiply by a number? What number?



Content Summary

A multiplier is one number you multiply with to:

- Increase a number (make it bigger).
- Decrease a number (make it smaller).

Formula to remember: New Amount = Old Amount $\times$ Multiplier

For increase by p %; Multiplier = $1 + \frac{p}{100}$

Example: Increase by 20% $\rightarrow 1 + 0.20 = 1.20$

For decrease by p % Multiplier = $1 - \frac{p}{100}$

Example: Decrease by 30% $\rightarrow 1 - 0.30 = 0.70$

Common Multipliers Table

Change	Multiplier
Increase 25%	1.25
Decrease 25%	0.75
Increase 50%	1.50
Decrease 50%	0.50
Increase 5%	1.05
Decrease 15%	0.85
Increase 100%	2.00
Decrease 100%	0.00

Remember:

- Multiplier $> 1 \rightarrow$ Increase
- Multiplier $< 1 \rightarrow$ Decrease
- Multiply once – quick and easy!
- Check if your answer makes sense: Increased number should be bigger. Decreased number should be smaller.

Examples

1. Increase 200 by 15%.

Step 1: Find multiplier: 15% increase $\rightarrow 1 + 0.15 = 1.15$

Step 2: Multiply: $200 \times 1.15 = 230$ **Answer:** 230

2. Decrease 80 kg by 40%.

Step 1: Find multiplier 40% decrease $\rightarrow 1 - 0.40 = 0.60$

Step 2: Multiply $80 \times 0.60 = 48$ **Answer:** 48 kg

3. Two Changes

A book costs 1000 FRW. First: 20% discount Then: 8% tax added

Method 1 – Step by step:

Discount: $1000 \text{ FRW} \times 0.80 = 800 \text{ FRW}$

Add tax: $800 \text{ FRW} \times 1.08 = 864 \text{ FRW}$

Method 2 – Combined multiplier: $0.80 \times 1.08 = 0.864$.

Then, $1000 \times 0.864 = 864$ **Answer:** 864 FRW.



APPLICATION ACTIVITY 5.2

- 1) Rise Kamana's salary: 2,000 of he gets 6% rise. Find new salary.
- 2) Phone Battery 100% full. Use phone: loses 35%. Find the percentage of battery left?
- 3) Sale Price T-shirt: 2500FRW. Shop: 30% off. Find the sale price?

5.3 Discount



Activity 5.3

You may have heard or even used the term discount. Do a research from business books and the internet on:

- The actual meaning of the term discount and the quantities used to calculate the discount.
- The possible reasons why a business person or service provider may decide to give some discount to the customer.
- Present your findings to your classmates in a class.



Content Summary

Applications of ratios in **scale drawing** refer to the use of ratios to represent real objects, distances, or sizes accurately on a drawing, map, or plan by **reducing or enlarging**. A **discount** is a reduction in the marked price (original price) of an item.

Key Terms:

Marked Price (MP): The original price of an item.

Discount (D): The amount by which the price is reduced.

Selling Price (SP): The price after discount.

Relationship: Selling Price = Marked Price – Discount

Discount Formula: $\text{Discount} = (\text{Discount Rate} \div 100) \times \text{Marked Price}$

Selling Price Formula: $\text{Selling Price} = \text{Marked Price} - \text{Discount}$

Examples

1. A shirt costs 20,000 FRW. It is sold at 10% discount. Find the discount and selling price.

Step 1: Find the discount

$$\text{Discount} = (10 \div 100) \times 20,000 = 2,000 \text{ FRW}$$

Step 2: Find the selling price

$$\text{Selling Price} = 20,000 - 2,000 = 18,000 \text{ FRW}$$

Answer: Discount = 2,000 FRW, Selling Price = 18,000 FRW

2. A book has a marked price of 5,000 FRW. It is sold for 4,500 FRW. Find the discount and discount rate.

Step 1: Find the discount

$$\text{Discount} = 5,000 - 4,500 = 500 \text{ FRW}$$

Step 2: Find the discount rate

$$\text{Discount Rate} = (\text{Discount} \div \text{Marked Price}) \times 100 = (500 \div 5,000) \times 100 = 10\%$$

Answer: Discount = 500 FRW, Discount Rate = 10%



APPLICATION ACTIVITY 5.3

- 1) The marked price for a sofa set is 24 000 FRW. If it is sold at a discount of 20%, calculate
 - a) The discount.
 - b) The sale price.
- 2) A radio costs 2 400 FRW. If it is sold at a discount for 1 920 FRW, find the:
 - a) Discount.
 - b) Rate of discount.
- 3) A shirt is sold for 4 000 FRW when a discount of 30% is allowed. Find the:
 - a) Marked price.
 - b) Discount allowed.
- 4) A television costs 54 000 FRW. If a customer paid 43 200 FRW after being allowed a discount, find the:
 - a) Discount allowed.
 - b) Rate of discount.

5.4 Commission



Activity 5.4

Suppose you are the manager in charge of sales in a company,

- 1) Discuss in your group, ways you would use to reward your salespeople who sell more than the target given to them.
- 2) How would calculate the extra reward? What name would you give to such a form of payment.



Content Summary

Commission is a **payment earned for performing a task or selling goods**, usually calculated as a **percentage of the total sales**. The main points include:

- **Definition:** Commission is a reward or income given to a salesperson or agent for selling goods or services.
- **Formula:** $\text{Commission} = \text{Rate (\%)} \times \text{Sales Amount}$
- **Applications:** Common in sales, banking, insurance, and real-life business transactions.
- **Example:** If a salesperson earns 5% commission on sales of 50,000 FRW, the commission is $0.05 \times 50,000 = 2,500$ FRW



APPLICATION ACTIVITY 5.4

- 1) Peter receives a monthly salary of 120 000 FRW plus a commission of 12% on all sales. Last month he made sales worth 1 200 000 FRW. How much did he earn that month?
- 2) Mrs. Uwamahoro sells charity tickets. She gets 160 FRW for every 8 tickets she sells. How much will she get for selling 480 tickets?
- 3) A mobile money agent received a commission of 40 000 FRW for a transaction worth 1 280 000 FRW. Find the rate of his commission.
- 4) Jane is an insurance agent and receives a monthly salary of 128 000 FRW and a commission of 15% on all the policies she sells in a month. In a certain month she sold policies worth 14 762 400 FRW. What was her total earnings that month?

5.5 Profit and loss, percentage profit and loss



Activity 5.5

- 1) Discuss what you understand by the words profit and loss.
- 2) In the following transactions, identify the buying price and the selling price. Determine whether a profit or loss was made in each of these transactions.
 - a) Habanabashaka bought a radio for 14 000 FRW and sold it for 17 500 FRW.
 - b) Charlote sold a dress for 24 500 FRW. She had bought it for 26 700 FRW.



Content Summary

- ❖ **Profit and Loss:** Profit occurs when the selling price of an item is higher than its cost price, while loss occurs when the selling price is lower than the cost price.
- ❖ **Formulas:**
 - Profit = Selling Price – Cost Price
 - Loss = Cost Price – Selling Price
- ❖ **Percentage Profit and Loss:** Shows profit or loss as a fraction of the cost price, expressed in percentage.
 - Percentage Profit = $\frac{\text{Profit}}{\text{Cost Price}} \times 100\%$
 - Percentage Loss = $\frac{\text{Loss}}{\text{Cost Price}} \times 100\%$

Examples

1. Calculate the percentage loss for an item bought at 2 000 FRW and sold at a price of 1 850 FRW.

Solution:

Buying price = 2 000 FRW, Selling price = 1 850 FRW

Loss = buying price (BP) – selling price (SP) = (2 000 – 1 850)FRW = 150 FRW

Percentage loss = $\frac{\text{Loss}}{\text{Cost Price}} \times 100\% = \frac{150}{2000} \times 100\% = 7.5\% \text{ loss}$

2. The total percentage profit of each cell phone bought at 48 000 FRW is 20%.
 - a) At what price does the business person sell each cell phone?
 - b) Find the total amount of money the business person will get after selling all the stock of 200 cell phones.

Solution:

(a) Percentage profit = 20%, Buying price = 48 000 FRW

Percentage profit = $\frac{S.P - B.P}{B.P} \times 100\%$

$20\% = \frac{S.P - 48000}{48000} \times 100\%$

$\frac{1}{5} = \frac{S.P - 48000}{48000}$

By cross multiplication $48\ 000 \times 1 = 5(\text{selling price} - 48\ 000)$

(Dividing by 5 both side) $9\ 600 = \text{selling price} - 48\ 000$

Selling price = $9\ 600 + 48\ 000 = 57\ 600$ FRW

(b) The total amount of money obtained is $57\ 600 \text{ FRW} \times 200 = 11\ 520\ 000 \text{ FRW}$

**APPLICATION ACTIVITY 5.5**

- 1) A student bought 10 Mathematics books from a book store selling each book at a price of 400 FRW. If each book was bought at a price of 260 FRW, find the percentage profit.
- 2) An item was bought at 2 000 FRW and sold at 2 600 FRW. Find the percentage profit.
- 3) A company buys and sell cars. During a certain month, the company sales were 80% of the buying price. Find the amount of money invested if the total sales were twenty-eight million Rwandan Francs.

5.6 Loans and savings



Activity 5.6

You are aware that successful people, businesses or institutions usually save money in banks and take loans to finance their operations.

- Discuss advantages of saving money: Why is it better to save money in a bank and financial institutions rather than keeping it elsewhere?
- Compare the advantages and the disadvantages of taking loans from a bank.



Content Summary

- **Loans:** Money borrowed from a bank, financial institution, or individual, usually with an **agreed interest** to be paid over a period of time.
- **Savings:** Money set aside in a bank or other financial institution to earn **interest** over time.
- The fee paid for borrowing or depositing money for a specific period of time is called **interest**, denoted by I.
- The total amount of money borrowed from a financial institution or deposited to earn interest is called the **principal**, denoted by P.
- The ratio of interest earned to the principal borrowed or deposited for a specific period of time is called the **rate** denoted by R.
- Rate is usually expressed as a percentage per annum (p.a) or year. The duration over which a given sum of money is borrowed or deposited to earn some interest is known as **time**, denoted by T.

Respond with true or false:

- 1) A loan is money borrowed that must be repaid with interest. – **True**
- 2) Savings in a bank do not earn any extra money. – **False**
- 3) Interest on a loan increases the total amount to be repaid. – **True**
- 4) You can borrow money without any plan for repayment. – **False.**



APPLICATION ACTIVITY 5.6

Respond with true or false:

- 1) Saving money regularly helps in achieving future goals.
- 2) Simple interest is calculated only on the original amount borrowed or saved.
- 3) Borrowing money always reduces financial responsibility.

- 4) If you borrow money from a bank, you can choose not to pay the interest.
- 5) Banks give interest on savings to encourage people to save.
- 6) A loan taken for an urgent need should always be repaid on time to avoid extra charges.
- 7) Savings in a bank account are unsafe and can be lost easily.
- 8) Borrowing more than you can repay is a good financial decision.
- 9) The longer you keep your savings in the bank, the more interest you earn.
- 10) Only adults can take loans or save money in banks.
- 11) Paying back a loan late can increase the total amount you owe.
- 12) You can use the interest earned from savings to buy something you need.
- 13) Loans and savings are unrelated; learning about one does not help in understanding the other.

5.7 Simple interest



Activity 5.7

A student saves money in a **Community Saving and Credit Group** and after some time the CSCG adds extra money to the original amount. Another person borrows money and later pays back more than they borrowed. Try to discuss:

- **Why is extra money added** in both situations?
- Name this extra money.



Content Summary

Simple interest is the extra money earned or paid on a loan or savings, calculated only on the **original amount** (called the principal).

It depends on three main factors: the **principal**, the **rate of interest**, and the **time**. Simple interest is commonly used in banks, loans, and savings for short periods.

The total amount after interest is the sum of the principal and the interest earned or paid.

$$I = P \times \frac{R}{100} \times T = \frac{PRT}{100}$$

Thus, amount (A) = principal (P) + interest (I)

Examples

1. Gatete borrowed 32 000 FRW from a **Community Saving and Credit Group** to start a business. If that credit group charged interest at a rate of 8% p.a, calculate the simple interest and the total amount she eventually paid back after 4 years.

Solution:

$$\text{Interest, } I = P \times \frac{R}{100} \times T = \frac{PRT}{100} = \frac{32000 \times 8 \times 4 \text{ FRW}}{100} = 10\,240 \text{ FRW}$$

$$\text{Amount, } A = 32\,000 + 10\,240 \text{ FRW} = 42\,240 \text{ FRW}$$

2. For how long, must 160 000 FRW be invested in a financial institution to earn a simple interest of 19 200 FRW at 4.5% p.a.?

Solution:

$$I = \frac{PRT}{100} = \frac{160\,000 \times 4.5 \times T}{100} = 10\,200 \text{ FRW}$$

$$19\,200 \text{ FRW} = 720\,000T$$

$$T = 2.666 = 2 \text{ years } 8 \text{ months.}$$

The whole number is **2 years**

Convert the decimal part to months: $0.666 \times 12 = 7.992$ months, This is approximately **8 months.**

**APPLICATION ACTIVITY 5.7**

- 1) Find the time, to the nearest year, in which:
 - a) 33 800 FRW will earn 2 000 FRW at 8% p.a.
 - b) 6 600 FRW will earn 24 000 FRW at 10% p.a.
 - c) 100 000 FRW will earn 20 800 FRW at 8.12 % p.a.
 - d) 300 000 FRW will earn 48 000 FRW at 7% p.a.
- 2) If 720 000 FRW is invested for 9 months at an annual simple interest rate of 15%.
 - a) How much interest will be earned?
 - b) What is the amount of investment after 9 months?
- 3) A businessman obtained a loan of 200 000 FRW from a bank that requires him to pay back 24 000 FRW after 6 months. What is the interest rate?

5.8 Solving Problems on Simple Interest



Activity 5.8

A man saves **200,000 FRW** at **10% p.a.** for **2 years**.
Find the interest earned.



Content Summary

Example 1: Finding Amount

A woman borrows **150,000 FRW** at **8% p.a.** for **3 years**.
Find the total amount she pays back.

$$I = 150,000 \times 0.08 \times 3 = 36,000$$

$$A = P + I = 150,000 + 36,000 = 186,000. \text{ Amount paid back is } \mathbf{186,000 \text{ FRW}}$$

Example 2: Finding Time

How long will **50,000 FRW** take to earn **10,000 FRW** at **10% p.a.**?

$$T = \frac{I}{PR} = \frac{10,000}{50,000 \times 0.10} \text{ 2 years}$$



APPLICATION ACTIVITY 5.8

- 1) A student saves 300,000 FRW at 12% p.a. for 1½ years. Find the interest earned.
- 2) A trader borrows 400,000 FRW at 15% p.a. for 8 months. Find:
 - a) The interest
 - b) The total amount paid back
- 3) A loan of 120,000 FRW earns 18,000 FRW interest in 2 years. Find the rate of interest.

5.9 Taxes



Activity 5.9

Discuss:

- Where does the government find money for building roads, schools, and hospitals?
- How does the government get that money?
- Did you understand someone talking about tax? What is it?



Content Summary

A tax is money that individuals or businesses are required to pay to the government. This money is used to provide public services such as **schools, hospitals, roads, security, and water supply**.

- **Tax Rate:** Percentage charged as tax.
- **Taxable amount:** Amount on which tax is calculated.
- **Tax amount:** Money paid as tax.
- **Total cost:** Price including tax.

Common Types of Taxes

1) Value Added Tax (VAT): Tax charged on goods and services at each stage of production or sale.

Example:

If a television costs **200,000 FRW** and VAT is **18%**: Find total cost.

Solution:

$$\text{VAT} = \frac{18}{100} \times 200,000 = 36,000 \text{ FRW},$$

$$\text{Total amount paid: } 200,000 + 36,000 = 236,000 \text{ FRW}$$

2) Sales tax: is a tax added to the price of goods at the point of sale. Find the total price.

Example: A pair of shoes costs **30,000 FRW**. If sales tax is **10%**:

Solution:

$$\text{Sales tax} = \frac{10}{100} \times 30,000 \text{ FRW} = 3,000 \text{ FRW}$$

$$\text{Total price: } 30,000 + 3,000 = 33,000 \text{ FRW}$$

3) Income Tax: is a tax paid on the income earned by individuals or businesses.

Find the net income

Example: A worker earns **100,000 FRW** per month. If income tax is **15%**, find the net income.

$$\text{Income tax} = \frac{15}{100} \times 100,000 \text{ FRW} = 15,000 \text{ FRW}$$

$$\text{Net income: } 100,000 - 15,000 = 85,000$$

4) Property Tax: Property tax is a tax paid on owned property such as land or buildings.

Example: A house is valued at **5,000,000 FRW**. If property tax is **1%** per year:

Solution:

$$\text{Property tax} = \frac{1}{100} \times 5,000,000 = 50,000 \text{ FRW}$$

5) Import Tax (Customs Duty): is a tax charged on goods brought into a country from another country.

Example: A trader imports goods worth **1,000,000 FRW**. If import tax is **20%**: find import tax

Solution:

$$\text{Import tax} = \frac{20}{100} \times 1,000,000 = 200,000 \text{ FRW}$$

6) Excise Tax: is a tax charged on specific goods such as fuel, alcohol, or cigarettes.

Example: If excise tax on fuel is **100 FRW per litre**, how many exercise tax is charged for **20 litres**:

Solution:

$$\text{Excise tax} = 100 \times 20 = 2,000 \text{ FRW}$$



APPLICATION ACTIVITY 5.9

- 1) A car worth 900 000 FRW was subjected to the import duty (Custom duty) at a rate of 40% followed by a VAT at 18%. Calculate the total amount of tax charged.
- 2) A manufacturer produces a product at a cost of 20 000 FRW and sold it at 28 000 FRW to a wholesaler. Calculate the value of VAT that he paid to the tax authority.
- 3) An author earned a royalty of 980 250 FRW before an advance tax was deducted. If the tax was charged at a rate of 20%, calculate: (a) How much tax was he charged? (b) His earning after the tax.

**END OF UNIT ASSESSMENT****Assessing the knowledge and understanding****Question 1: True or False**

Write True or False. If false, correct the statement. (2 marks each)

1. Percentage means a number expressed out of 100.
2. To increase a quantity by 15%, we multiply it by 0.15.
3. A discount reduces the marked price of an item.
4. Commission is paid to a worker based on the number of hours worked.
5. If the cost price is greater than the selling price, a profit is made.
6. Simple interest depends on the principal, rate, and time only.
7. VAT is a type of discount given to customers.
8. A loan always decreases a person's savings.

Question 2: Definitions (2 marks each)

Define the following:

- a) Discount
- b) Simple interest
- c) Tax

Assessing skills**Question 3: Calculations**

1. Find 25% of 80,000 FRW. (2 marks)
2. A price of a bag increased by 10%. If the original price was 40,000 FRW, find the new price using a multiplier. (3 marks)
3. A radio has a marked price of 60,000 FRW. A discount of 20% is allowed. Find the selling price. (3 marks)
4. A trader buys goods for 120,000 FRW and sells them for 150,000 FRW. Find:
(a) Profit (2 marks)
(b) Percentage profit (3 marks)
5. Calculate the simple interest on 200,000 FRW at 12% per year for 3 years. (4 marks)
6. A customer buys goods worth 100,000 FRW. VAT is 18%. Find the total amount paid. (3 marks)

7. A customer paid 59,000 FRW for a jacket including 18% VAT. Find the original price. (4 marks)
8. A trader sells a bicycle for 150,000 FRW excluding tax. If VAT is 18%, find:
 - a) Tax charged (2 marks)
 - b) Selling price including tax (2 marks)

Assessing attitudes and values

Question 4: Answer the following questions briefly.

1. Explain why paying taxes is important to a country. (5 marks)
2. State two benefits of saving money in a bank. (4 marks)
3. Why should traders be honest when calculating profit, commission, and tax? (3 marks)
4. How can understanding discounts and percentages help you in daily life? (3 marks)

UNIT 6

POINTS, LINES AND ANGLES



Key Unit Competence:

Explore properties of angles formed by lines.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Identify and explain points, lines, line segments, and rays.
- Describe and classify types of angles.
- State relationships between angles on a straight line and angles at a point.
- Identify angle types and relationships in polygons and in parallel lines cut by a transversal

Skills

- Construct and measure angles with a protractor.
- Identify and calculate unknown angles using angle relationships.
- Represent and interpret angles formed by intersecting lines.
- Identify and apply properties of transversal angles in problem solving.

Attitudes and values

- Develop accuracy and neatness in angle construction and measurement.
- Show cooperation and respect in group activities.
- Appreciate the importance of angles and lines in daily life.
- Show patience and accuracy in angle construction and measurement.
- Develop teamwork and curiosity through exploration activities.

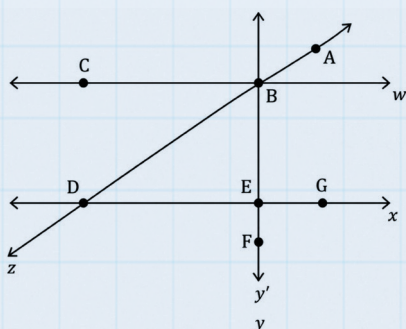
INTRODUCTION

This unit introduces Students to the basic ideas of geometry that help us understand shapes, directions, and positions in everyday life. The purpose of the unit is to help Students identify and use **points, lines, line segments, and rays**, and to understand **angles and their types**. Students will study **angles on a straight line and angles at a point, angles in polygons, and angles formed when a transversal cuts parallel lines**, including **corresponding, alternate, and co-interior angles**. These ideas are used in real life when drawing maps, designing buildings, making roads, carpentry, engineering, and art. By learning these concepts, Students will be able to recognize angle relationships, measure and calculate angles, and solve simple geometry problems. Key words in this unit include point, line, segment, ray, angle, polygon, parallel lines, and transversal.



Introductory activity

Here is the image:



Now discuss

1. Where can you see **points** (corners, intersections)?
2. Which parts look like **lines, line segments, or rays**?
3. Can you identify any **angles**? Are they the same or different?
4. Do any angles lie on a **straight line**?
5. Do you see angles around a point? How are they?
6. Can you see angles inside shapes (polygons)?
7. Where does a line cross two parallel lines? What do you notice about the angles formed?

6.1 Points, lines, segments and rays



Activity 6.1

Observe the classroom and point out examples of: Tiny spots or dots (corners of papers, nails, marks on the board), Straight edges (desk edges, book edges, blackboard border), Lines on the floor or walls. Discuss:

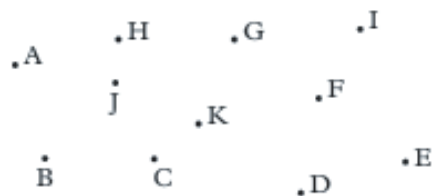
- What is the smallest thing you can see on the board?
- Which edges look like they could go on forever?
- Where do edges start and stop?



Content Summary

1. **A point** marks a particular position. A point has no size or dimension, but for the sake of illustration it is marked with a fine dot and labelled with a capital letter

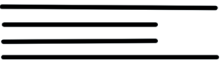
Example



2. **A line** is a collection of infinite points. A line is straight and goes on forever in both directions.

Lines are categorized by **orientation** (horizontal, vertical, diagonal), **relationship** (parallel, perpendicular, intersecting), and **form** (straight, curved, broken) The line is represented by two points on it with a line symbol above: $\overline{AB}$

Here are the types of line:

		
Thick line	Thin line	Dotted line
		
Straight line	Vertical lines	Parallel lines
		
Perpendicular lines	Horizontal lines	Zigzag lines
		
Wavy	Curved line	Intersecting lines

3. **Line Segment** is a part of a line with two endpoints. It does not go on forever, it is represented as $\overline{AB}$ when it joins point A and point B.

d) Ray

A line that starts at one point (endpoint) and goes on forever in one direction.

Represented as $\overrightarrow{AB}$ where A is the endpoint.

Line	Line Segment	Ray
A Line is a collection of points that go forever in two opposite directions indefinitely. It is an endless continuous and it has no finite endpoints.	Line Segment is a part of the line that can link two fixed or definite endpoints, in which all points exist between these end points.	When a line has a starting point but not a finite end point means end at infinity is called a Ray.
It is represented with arrows at both ends to show that it extends infinitely in both directions.	It has a definite length and it does not extend infinitely in both the direction.	It represents one start point and an arrow at the other end that means it moves forever in one direction.
Line ab or $\overleftrightarrow{ab}$	$[a,b]$ or $\overline{ab}$	$[a,\infty)$

Example 1: Identify each of the following: point, line, segment, ray.

A dot labelled P

$\overleftrightarrow{XY}$:

$\overline{MN}$:

Solution:

P $\rightarrow$ Point $\bullet P$

$\overleftrightarrow{XY}$: Line (extends in both directions) 

$\overline{MN}$: Line Segment (two endpoints M and N) 



APPLICATION ACTIVITY 6.1

- 1) Look around the classroom and identify:
 - a) A point
 - b) A line
 - c) A line segment
 - d) A ray
- 2) Draw and label
 - a) Draw a point P.
 - b) Draw a line through points A and B.
 - c) Draw a segment between points M and N.
 - d) Draw a ray starting from point X.

3) True or False

- a) A line has two endpoints.
- b) A ray goes on forever in one direction.
- c) A segment can go on forever.
- d) A point has length and width.

6.2 Angles and their types



Activity 6.2

Look around the classroom for corners, doors, windows, and clock hands.

Discuss:

- Where do you see two lines meeting?
- What shape is formed when two lines meet?
- Can you show examples of wide and narrow openings?



Content Summary

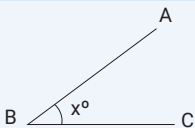
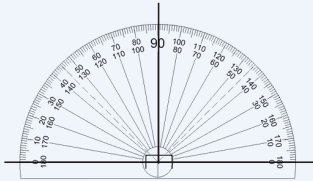
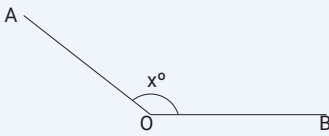
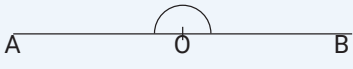
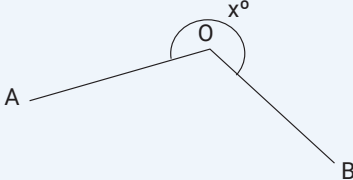
An angle is formed when two lines or rays meet at a point.

Parts of an angle:

Vertex: The point where the lines meet

Arms (sides): The two lines forming the angle

Types of Angles:

Type of Angle	Measure	Example	
Acute	$< 90^\circ$	Small opening in clock hands	
Right	$= 90^\circ$	Corner of a book	
Obtuse	$> 90^\circ$ and $< 180^\circ$	Wide-open door	
Straight	$= 180^\circ$	A flat line	
Reflex angle	$> 180^\circ$ but $< 360^\circ$ e.g. 270° , 200° , 344° and so on.	Open door	

Example 1: 1. Identify the type of angle:

- a) Angle measuring $45^\circ \rightarrow$ **Acute**
- b) Angle measuring $90^\circ \rightarrow$ **Right**
- c) Angle measuring $135^\circ \rightarrow$ **Obtuse**
- d) Angle measuring $180^\circ \rightarrow$ **Straight**

2. Real-life identification

- a) Corner of a notebook $\rightarrow$ **Right angle**.
- b) Partially open door $\rightarrow$ **Acute angle**.
- c) Open gate $\rightarrow$ **Obtuse angle**.
- d) Flat road $\rightarrow$ **Straight angle**.

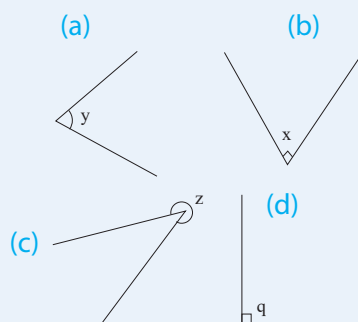


APPLICATION ACTIVITY 6.2

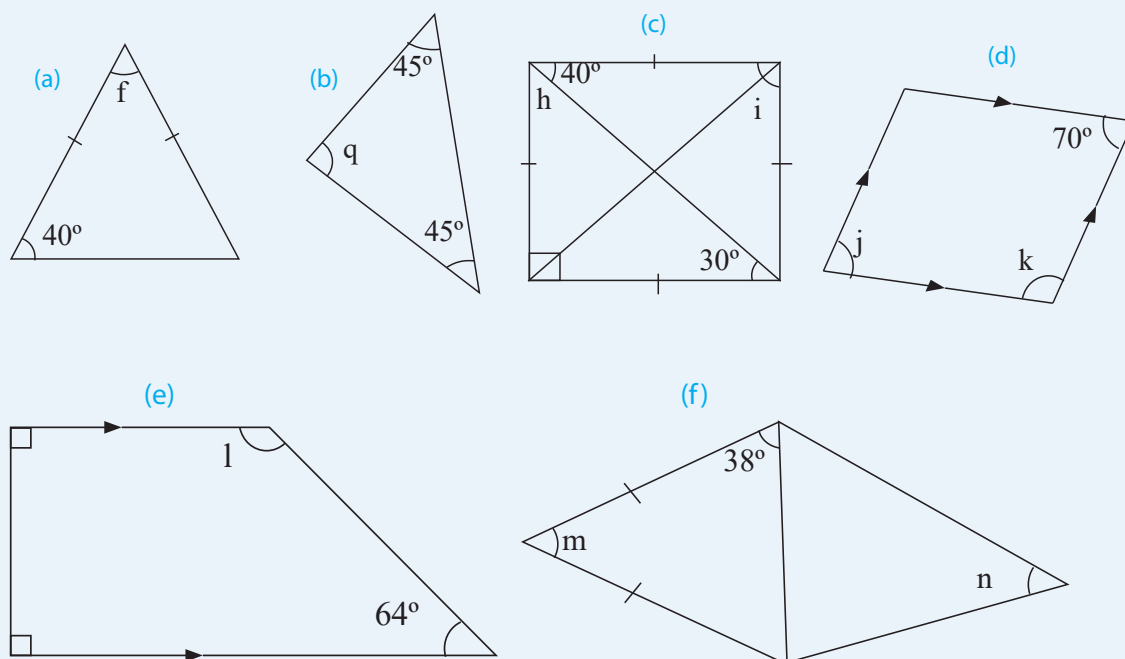
1. True or False

- a) A right angle is smaller than 90° .
- b) An obtuse angle is greater than 90° .
- c) A straight angle measures 90° .
- d) An acute angle is smaller than 90° .

2. Indicate the type of angle marked with letters in each of the following



3. State the type of angles marked with letters in each of the following

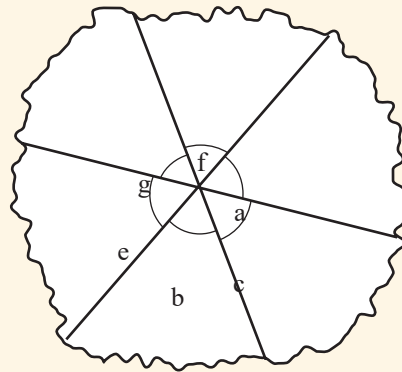


6.3 Angles on a straight line and angles at a point



Activity 6.3

Cut off all the angles of triangles ABC and EFG. Fit all the six angles on the point y as marked in adjacent to each other without overlapping as shown in the figure below.



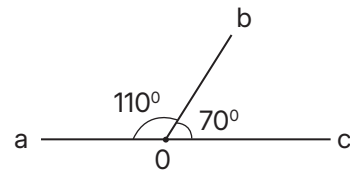
- What do you notice? Discuss.
- How do the six angles fit on one point?
- What is the sum of all the six angles?
- What is the sum of angles at a point?



Content Summary

a) Angles on a straight Line:

- When two or more angles lie on a straight line, they are called **angles on a straight line**.
- The sum of angles on a straight line is always **180°** .
- These angles are often called **adjacent angles** because they share a common side.



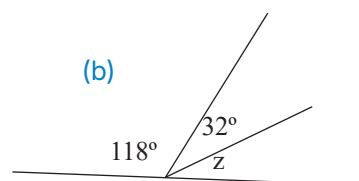
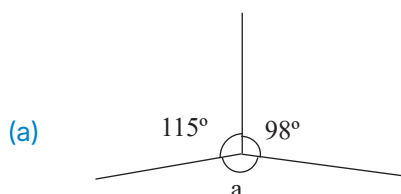
- Example:** If one angle is 70° , the other angle on the same straight line will be $180^\circ - 70^\circ = 110^\circ$.

Angle aob and angle boc form a straight angle of 180°

b) Angles at a Point:

- When two or more angles meet at a single point, they are called **angles at a point**.
- The sum of all angles around a point is always **360°** .
- These angles can be adjacent or non-adjacent, but together they complete a full turn around the point.
- Example:** If three angles at a point are 120° , 90° , and 50° , the fourth angle will be $360^\circ - (120^\circ + 90^\circ + 50^\circ) = 100^\circ$.

Example: Find the angles marked with letters in each of the following.



Solution:

a) $115^\circ + 98^\circ + a = 360^\circ$ (angles at a point add up to 360°) $213^\circ + a = 360^\circ$

$a = 360^\circ - 213^\circ = 147^\circ$

$a = 147^\circ$

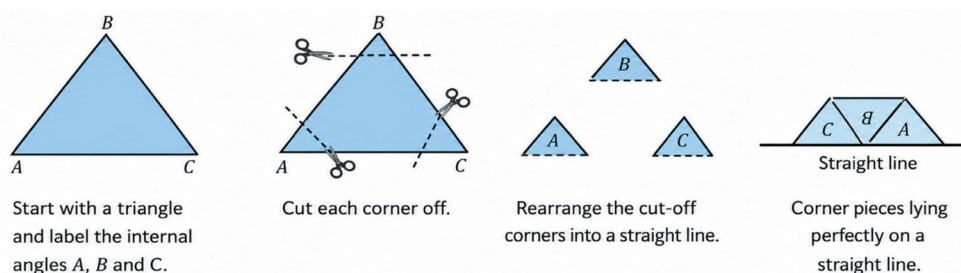
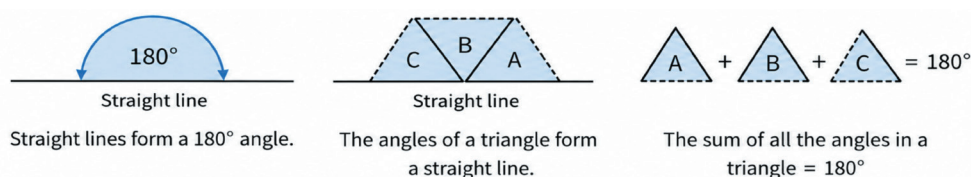
b) $118^\circ + 32^\circ + z = 180^\circ$ (angles on a straight line) $150^\circ + z = 180^\circ$

$z = 180^\circ - 150^\circ$

$z = 30^\circ$

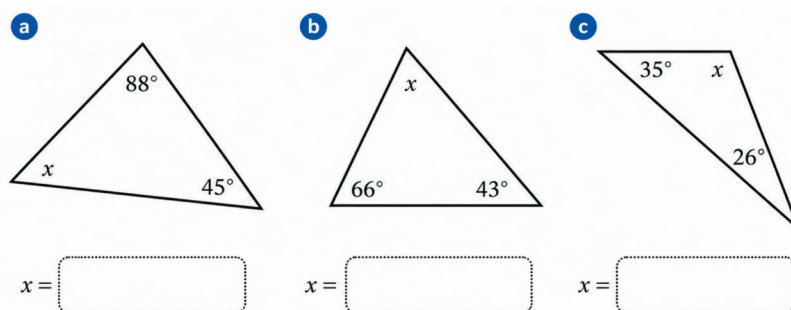
c) Angle sum of a triangle

Triangles have this great property where if all the angles are brought together, they form a straight line.

**We find the following:****Question:**

The unknown angle in each of these triangles is labelled with the letter x (also called a variable).

Calculate the size of each angle represented by the variable x.

**Solution:**

a) $x = 180 \text{ degrees} - (88 + 45) \text{ degrees} = (180 - 133) \text{ degrees} = 47 \text{ degrees}$

b) $x = 71^\circ$

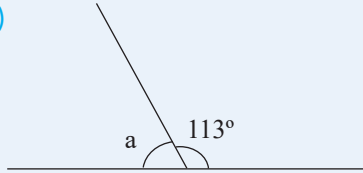
c) $x = 119^\circ$



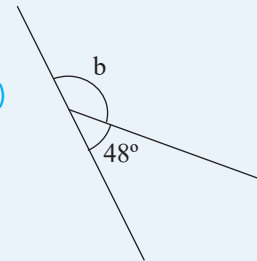
APPLICATION ACTIVITY 6.3

1. Calculate the size of each of the angles marked with letters

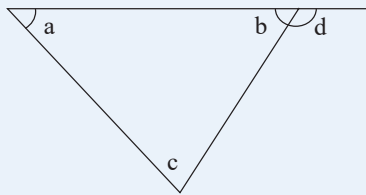
(a)



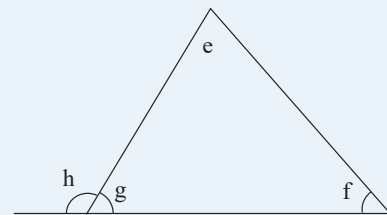
(b)



2. In each of the following triangles, use the relationship between the angles of a triangle and the straight angle properties to fill in the gaps.



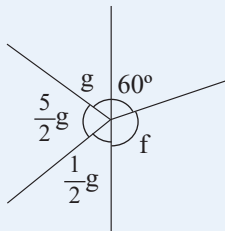
(a) $a + b + c =$
 $b + d =$



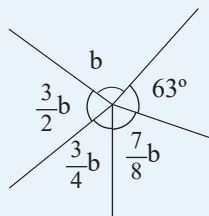
(b) $e + f + g =$
 $h + = 180^\circ$

3. Calculate the size of each of the following angles marked with letters

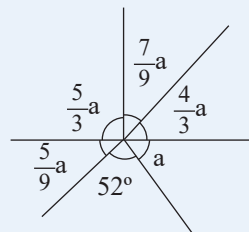
(i)



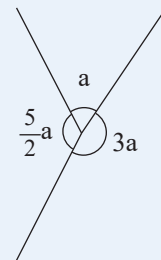
(ii)



(iii)

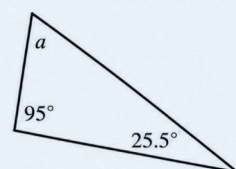


(iv)



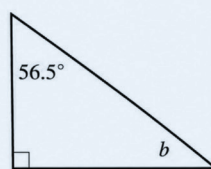
4. Calculate the size of the angle labelled with a letter for these triangles:

(a)



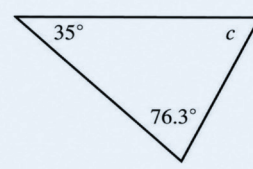
$a =$

(b)



$b =$

(c)



$c =$

6.4 Angles in the polygons



Activity 6.4

Here is the picture:



Discuss:

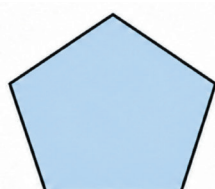
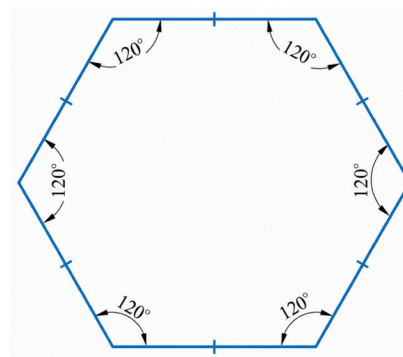
1. What shape do you see? How many sides does the shape have?
2. Can you identify the corners of the shape?
3. What do you think the sum of the angles inside the shape is?
4. Are all angles equal in this shape?



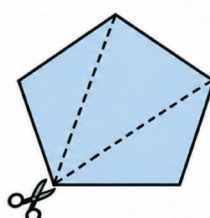
Content Summary

- A **polygon** is a closed figure with **three or more straight sides**.
- Examples: Triangle (3 sides), Quadrilateral (4 sides), Pentagon (5 sides), Hexagon (6 sides), etc.
- Regular polygons have sides of the same length. A regular polygon having six sides is called a regular hexagon ($n=6$):

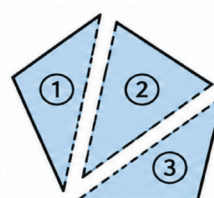
All polygons can be split into triangles by drawing all the diagonals possible from one corner (vertex) only.



Pentagon
(5 straight sides)



Can be cut into
three triangles



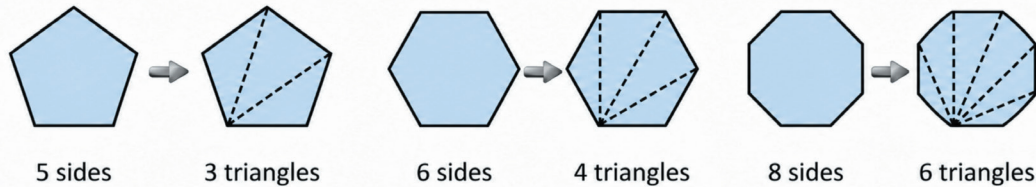
Each triangle has an internal
angle sum of 180°

$$\begin{aligned}
 \therefore \text{Internal angle sum of a pentagon} &= 3 \times \text{angle sum of a triangle} \\
 &= 3 \times 180^\circ \\
 &= 540^\circ
 \end{aligned}$$

Sum of interior angles:

It can be difficult to draw polygons with a lot of sides and then count the triangles formed.

To make things easier, there is a simple rule you can use from now on for any polygon



The number of triangles equals two less than the number of sides.

Internal angle sum of a polygon = The number of triangles $\times 180^\circ$

The sum of interior angles of a regular polygon with **n sides** is:

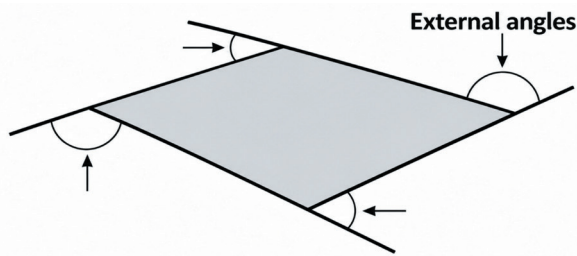
$$\text{Sum of interior angles} = (n-2) \times 180^\circ$$

Example: Triangle $\rightarrow (3 - 2) \times 180^\circ = 180^\circ$

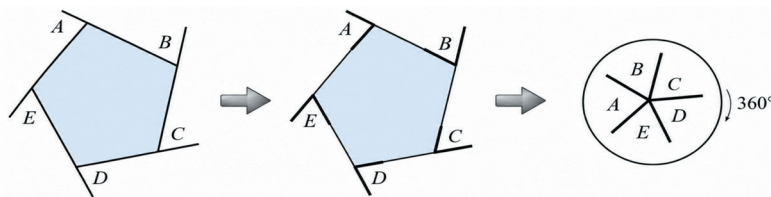
Quadrilateral $\rightarrow (4 - 2) \times 180^\circ = 360^\circ$

Exterior angles:

When you extend a side outside the border of a polygon, you create an external angle

**Sum of Exterior Angles:**

The sum of **exterior angles** of any polygon is always **360°**

**Examples**

1) Find the sum of interior angles of a pentagon.

Solution:

$$\text{Sum} = (5-2) \times 180^\circ = 3 \times 180^\circ = 540^\circ$$

2) A **triangle** has two angles of 50° and 60° . Find the third angle.

Solution:

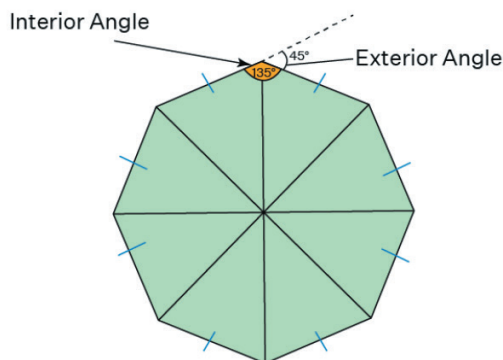
$$\text{Sum of angles in triangle} = 180^\circ \quad \text{Third angle} = 180^\circ - (50^\circ + 60^\circ) = 70^\circ$$

3) Find the sum of **exterior angles of a hexagon**.

Solution:

Sum of exterior angles = 360° (true for any polygon).

4) Interior and exterior angle of a **regular octagon** add up 180 degrees because $135^\circ + 45^\circ = 180^\circ$.



APPLICATION ACTIVITY 6.4

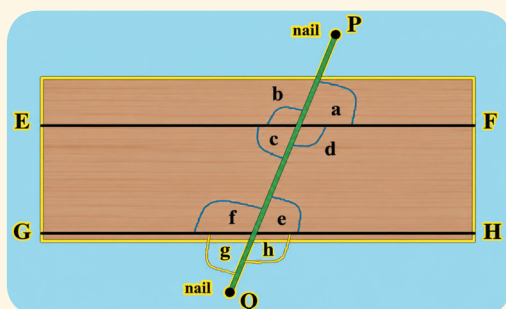
- 1) A rectangular classroom has four right angles. Find the sum of interior angles.
- 2) A pentagon-shaped garden has three known interior angles of 100° , 110° , and 120° . Find the sum of the remaining two angles.
- 3) A hexagonal floor tile has all interior angles equal. Find the measure of one interior angle.

6.5 Angles formed by a transversal on parallel lines: Corresponding and alternate angles



Activity 6.5

Work in pairs to observe the pictures and discuss the following questions:



1. Can you see two straight lines that never meet?
2. What happens when another line PQ crosses these two lines?
3. Can you see angles that look equal in size?
4. Where are these angles located?



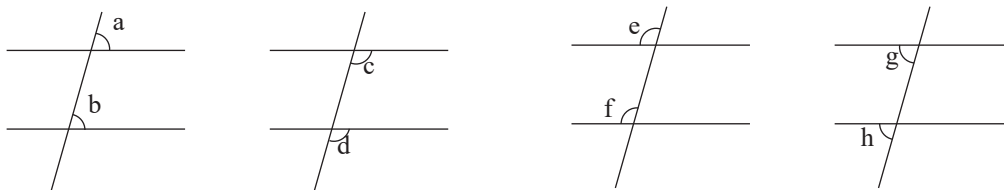
Content Summary

Parallel Lines and transversal:

- **Parallel lines** are lines that never meet and are always the same distance apart.
- A **transversal** is a line that cuts across two or more lines.

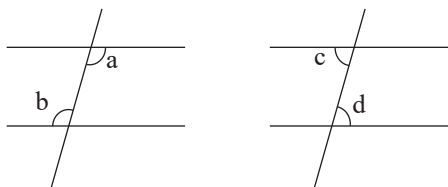
Corresponding Angles:

- Corresponding angles are angles in the **same position** on each parallel line when a transversal cuts them.
- **Corresponding angles are equal** when the lines are parallel: Angles a and b, c and d, e and f, g and h are corresponding angles.



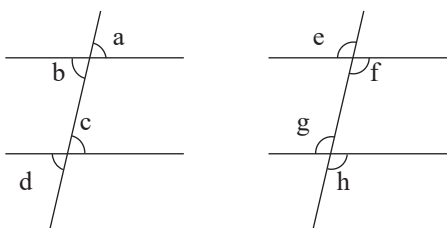
Alternate angles:

- Alternate angles are angles that lie on **opposite sides of the transversal** and **between or outside** the parallel lines.
- **Alternate angles are equal** when the lines are parallel.

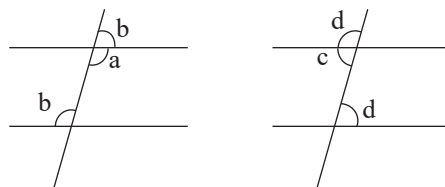
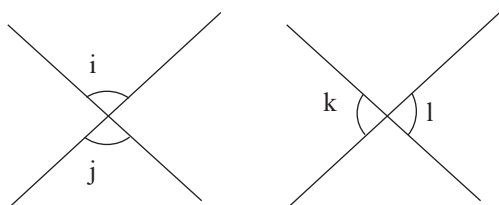


Angles a and b, c and d, e and f, g and h, i and j, k and l are vertically opposite angles. Vertically opposite angles are equal.

Angles which are opposite each other where two straight lines intersect or cuts each other are called vertically **opposite angles**.



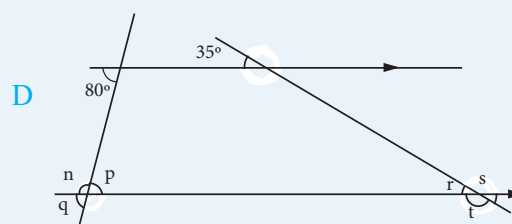
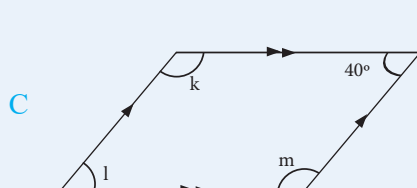
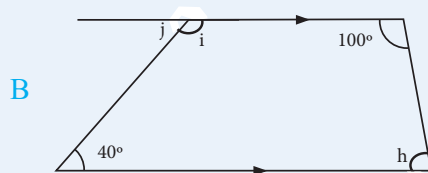
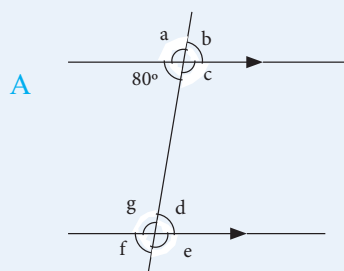
Angles that add up to 180° are called supplementary angles: Angles a and b, c and d, e and f and g and h are supplementary angles.





APPLICATION ACTIVITY 6.5

Observe the following figures:



a) On each figure, write down all the pairs of:

- corresponding angles.
- alternate angles.

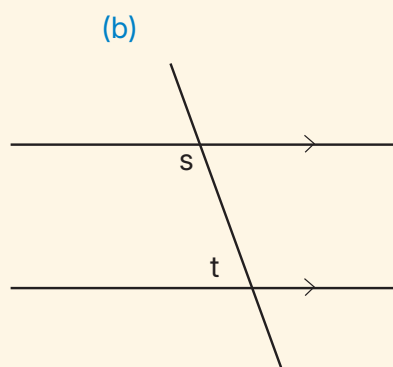
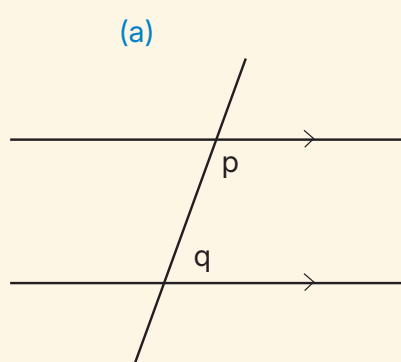
(b) On each figure write down the sizes of the angles marked with letters.

6.6 Angles formed by a transversal on parallel lines: Co-interior angles



Activity 6.6

Use a protractor to measure the angles marked with letters.



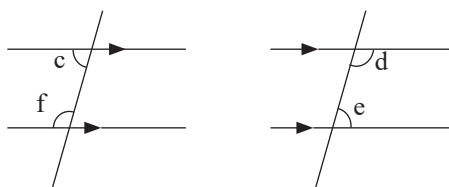
- Add angles $p + q$; Then add $s + t$.
- What do you notice with the sum of angle p and angle q ?
- What do you notice with the sum of angle s and t ? made by the transversal of two parallel lines? Explain.



Content Summary

- **Co-interior angles** are the angles that are on the same side of a transversal and inside the two lines it crosses.
- When the transversal crosses two parallel lines, the co-interior angles are supplementary angles. they add up to 180 degrees. $P + q = 180$ degrees.

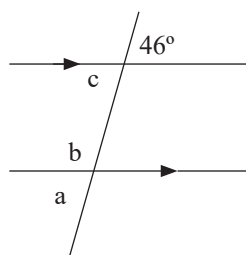
Property: **Sum of co-interior angles = 180°**



c and **f**, **d** and **e** are pairs of co-interior angles. $c + f = 180^\circ$; $d + e = 180^\circ$.

Example:

Calculate the angles marked with letters in each of the following



Solution:

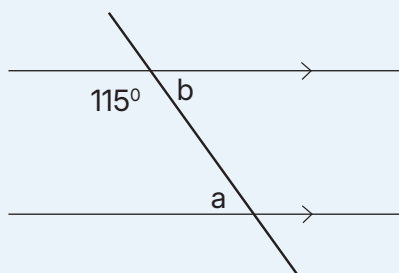
$c = 46^\circ$ (vertically opposite angles) $a = c = 46^\circ$ (corresponding angles) $b = 180^\circ - a$ (angles on a straight line/ supplementary angles) $b = 180^\circ - 46^\circ$ $b = 134^\circ$.



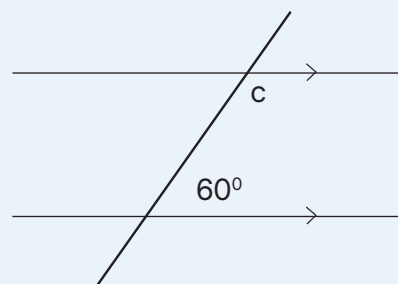
APPLICATION ACTIVITY 6.6

Find the value of the angles marked with letters

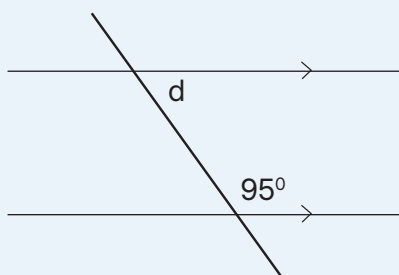
1)



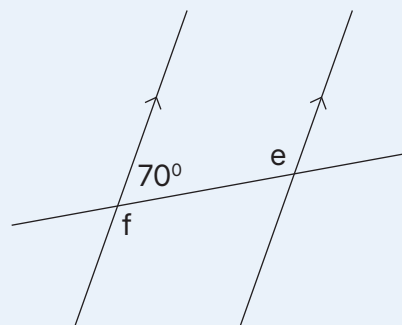
2)



3)



4)





END OF UNIT ASSESSMENT

Assessing the knowledge and understanding

- 1) Write **True** or **False**. If the statement is false, **correct it**.
 - a) A point shows a position but has no size.
 - b) A ray has two endpoints.
 - c) An obtuse angle is less than 90° .
 - d) The sum of angles on a straight line is 180° .
 - e) Corresponding angles are equal when a transversal cuts parallel lines.
- 2) Short Answer Questions
 - a) Define a line segment.
 - b) Name any **three types of angles**.
 - c) What is a transversal?

Assessing skills

- 3) Calculate:
 - a) One angle on a straight line is 48° . Find the other angle.
 - b) Three angles at a point are 90° , 110° , and 80° . Find the fourth angle.
 - c) Find the sum of interior angles of a **hexagon**.
- 4) A transversal cuts two parallel lines. One corresponding angle is 65° . Find:
 - a) The size of other corresponding angles
 - b) The size of alternate angles
- 5) One co-interior angle is 125° . Find the size of other co-interior angles.

Assessing values and attitudes

- 6) True or False
 - a) Accuracy is important when measuring angles.
 - b) Group work helps you to develop teamwork and communication.
 - c) Geometry is not useful in daily life.
- 7) Application Questions
 - a) Mention **two uses of angles in daily life**.
 - b) Why is patience important when constructing angles using a protractor?

UNIT 7

STATISTICS WITH UNGROUPED DATA



Key Unit Competence:

Organize real-world data using appropriate statistical methods and graphical representations.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain qualitative and quantitative data and distinguish between discrete and continuous data.
- Organize and present data using frequency distribution tables.
- Describe and interpret the mode, median, and quartiles of statistical data.
- Identify and use formulae for calculating the mean, median, quartiles, and frequencies.
- Read, interpret, and analyse statistical graphs and frequency diagrams.

Skills

- Collect, classify, and organize data accurately.
- Apply data collection methods to conduct simple research.
- Represent statistical data using frequency tables and various graphs.
- Calculate the mode, mean, median, and quartiles of statistical data.
- Interpret and analyze statistical graphs correctly.

Attitudes and values

- Show curiosity and teamwork during data collection activities.
- Demonstrate accuracy, honesty, and responsibility when recording and presenting data.
- Appreciate the importance of statistics in solving everyday life problems.
- Develop critical thinking skills when interpreting statistical information.
- Show patience and accuracy in angle construction and measurement.
- Develop teamwork and curiosity through exploration activities.

INTRODUCTION

This unit covers the meaning and importance of statistics, as well as the different types of data, including qualitative, quantitative, discrete, and continuous data. Student studies methods of collecting and organizing data, frequency distribution tables, and cumulative frequency tables. The unit also focuses on measures of central tendency and position such as the arithmetic mean, mode, median, quartiles, midrange, and inter-quartile range for ungrouped data. In addition, student learns how to represent data using pictograms, pie charts, line graphs, bar charts, frequency polygons, and cumulative frequency diagrams. Emphasis is placed on reading, interpreting, and converting statistical graphs and diagrams into frequency tables.



Introductory activity

- 1) Here are the questions, try to answer each question individually:
 - a) How many books did you read last month?
 - b) How many siblings do you have?
- 2) In group put your solutions in table then answer the following questions
 - a) Which number appears most often?
 - b) What is the smallest and the largest number?
 - c) Is it easy to understand the data in this raw form?

7.1 Meaning of statistics



Activity 7.1

Discuss in pairs whether the following statements are **True or False**:

- "Statistics is only about numbers."
- "Collecting data helps us make better decisions."
- "Organizing data is not necessary if we remember it."

After discussion, volunteers explain their answers to colleagues.



Content Summary

Meaning of Statistics:

Statistics is the science of collecting, organizing, analysing, and interpreting data to make it useful for decision-making.

Importance of statistics:

- It helps understand information in daily life.
- It assists in making decisions and solving problems.
- It is used in schools, businesses, governments, and research.
- It is used in making inferences, predictions and decisions based on the data available.

Key components of statistics:

- **Data Collection:** Gathering information carefully.
- **Organization:** Arranging data in tables, lists, or charts.
- **Analysis:** Summarizing data using averages, graphs, or frequency counts.
- **Interpretation:** Drawing conclusions or making decisions from the analysed data.



Examples:

1. Statistics is only useful for scientists. **Answer:** False
Explanation: Statistics is useful in everyday life, including schools, business, and government decisions.
2. Organizing data in tables or charts makes it easier to understand. **Answer:** True
Explanation: Presenting data in an organized form helps interpret it clearly.
3. Interpreting data means just reading numbers without understanding them.
Answer: False
Explanation: Interpreting data involves analysing it to make decisions or draw conclusions.



APPLICATION ACTIVITY 7.1

Discuss whether the following are **True or False**:

- a) Collecting data about classmates' favourite subjects is a form of statistics.
- b) Statistics can only be used for math problems.
- c) Organizing survey results in a chart helps everyone understand the data.
- d) Counting the number of students who bring lunch from home is a statistical activity.

- e) The mode of data can be found even if the data is not organized.
- f) Statistics helps a coach decide which player performs best.

7.2 Types of Data: Qualitative, quantitative, discrete and continuous data



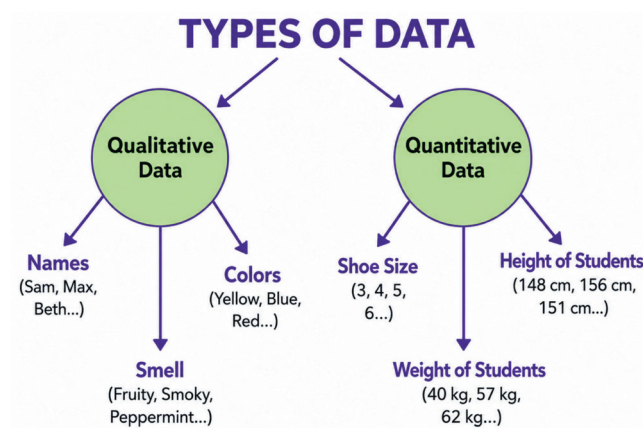
Activity 7.2

Use an English dictionary, a mathematical dictionary or search from the internet to find the meaning of the following terms: **Data, qualitative data, quantitative data, continuous data and discrete data.**



Content Summary

- **Qualitative Data:** Descriptive information that cannot be measured numerically (e.g., hair colour, favourite subject).
- **Quantitative Data:** Numerical information that can be measured or counted (e.g., age, number of siblings).
- **Discrete Data:** Countable numbers, usually whole numbers (e.g., number of books, number of goals).
- **Continuous Data:** Measurable quantities that can take any value within a range (e.g., height, weight, temperature).



Examples: Solved with True or False.

1. The number of students in a class is continuous data. **Answer:** False
Explanation: It is discrete data because students are counted in whole numbers.
2. The colour of a car is qualitative data. **Answer:** True
Explanation: Colour describes a characteristic and is not numerical.
3. The height of students is discrete data. **Answer:** False
Explanation: Height can take any value in a range, so it is continuous data.
4. The number of books read last week is quantitative and discrete. **Answer:** True
Explanation: The number can be counted and expressed numerically.



APPLICATION ACTIVITY 7.2

Mark each statement as True or False and explain:

1. Favourite subject of students is quantitative data.
2. The weight of a bag of rice is continuous data.
3. The number of goals scored in a football match is discrete data.
4. The colour of a shirt is quantitative data.
5. The amount of water in a bottle can be measured and is continuous.
6. The number of cars in a parking lot is discrete data.
7. The type of music a learner likes is quantitative.
8. Height and weight can be measured and are quantitative.

7.3 Collecting and organizing data



Activity 7.3

Discuss in pairs the following **statements** and decide if they are **True or False**:

- a) "Data can only be collected by asking questions to people."



- b) "Collecting data is unnecessary if we remember it in our heads."
 c) "Organizing data in a table makes it easier to analyse."



Content Summary

- **Collecting Data:** Gathering information from surveys, interviews, observations, or experiments.
- **Organizing Data:** Arranging collected data in **lists, tables, or charts** to make it easier to understand and analyse.

Importance of organising data: Makes patterns and trends clear and helps in calculating averages, identifying modes, and other statistics.

Ways of collecting data:

- **Survey:** A structured data collection method used to gather information from a predefined group of respondents (a sample) to learn about their characteristics, opinions, attitudes, or behaviours.
- **Questionnaire:** Often confused with «survey,» a questionnaire is specifically the **instrument or set of questions** used within a survey or other method. It is the physical or digital form that contains the questions. A survey is the **overall process** of collecting and analysing the data. A questionnaire is the **tool** used in that process.
- **Interview:** A direct, one-on-one (or sometimes one-to-few) conversation where an interviewer asks questions to collect in-depth information.
- **Focus Group:** A moderated discussion with a small group of participants (typically 6-10) selected for their shared characteristics or experiences relevant to the research topic.
- **Delphi Technique:** A structured, method used to achieve a consensus among a panel of experts on a complex issue.

Tools for data organization

1. **Tally Charts:** A tally chart is a simple way of recording and counting data using tally marks. Each mark represents one occurrence, and every fifth mark is drawn across the previous four to make counting easier. Tally charts are useful for organizing **small sets of data** quickly.

Example:

Tally	Frequency
	2
	8
	11

2. **Frequency Tables:** A frequency table is a table that shows **how often each value or category occurs** in a set of data. It summarizes raw data into a structured form, making it easier to analyze patterns and calculate statistics like **mode** and **mean**.

Example

Temperature	Tally marks	Frequency
21	I	1
22		0
23	II	2
24	III	3
25	I	1
Total		7

3. Simple Lists or Charts: Simple lists or charts are a basic way to organize data by writing values or categories in order, either in rows or columns. They help **present data clearly**, making it easier to read and interpret before creating more advanced tables or graphs.

Examples:

Discuss if the statement is True or False:

- We can collect data by observing people's behavior. **Answer:** True
Explanation: Observation is a valid method of data collection.
- It is not necessary to organize data before analysis. **Answer:** False
Explanation: Organizing data is important for proper analysis and interpretation.
- A frequency table helps to summarize collected data. **Answer:** True
Explanation: Frequency tables make it easier to see how often each value occurs.

**APPLICATION ACTIVITY 7.3**

The statement below talks about data collection and organization. Decide if it is True or False:

- Asking classmates about their favourite fruit is a way to collect data.
- Writing answers randomly without a table is an effective way to organize data.
- Organizing data helps us identify trends quickly
- A shopkeeper recording daily sales in a table is collecting and organizing data.
- Data from surveys should not be recorded; memory is enough.
- Tally marks can help organize data efficiently.
- Organized data is easier to represent in graphs.

7.4 Frequency distribution



Activity 7.4

Discuss in your group: “Which fruit do you like most: mango, banana, or apple?”
Mention one fruit only.

Then, answer the following questions: How many respondents of questions are there?

- For each fruit, how many students like it?
- Which fruit is the most popular?
- Try to organize the information recorded in you group in the form of table

Fruit	Tally	Number of students who likes it (frequency)
Mango		
Banana		
Apple		
Total		



Content Summary

A **frequency distribution** is a table that shows the number of times (frequency) each value or category occurs in a dataset.

The **tallying** method of recording makes counting easier and more accurate, especially when big numbers are involved. Bundling strokes into fives (||||) makes it easier to count them.

Example:

The table below shows the grades scored by a class of 30 students in a Mathematics examination.

C	B	C	A	C	B	B	D	A	A	B	B	C	C	B	D	B
C	B	D	A	B	A	C	B	A	C	C	B	C				

Make a frequency table for the information in this table.

Solution:

Here is the required frequency table:

Grade	Tally	Frequency
A	/	6
B	/	11
C	/	10
D		3
Total		30



APPLICATION ACTIVITY 7.4

- 1) The amount of milk in litres produced by 36 cows in Mugisha's farm in one day are represented in the following table

8	10	11	9	9	4	8	18	13	16	15	12	13	4	6	11	15	4	8
7	9	13	7	10	11	8	5	14	12	9	14	9	12	5	14	14		

Make a frequency distribution table for this data. Imagine this was your farm, is there any decision you might make based on the performance of your cows? Explain.

- 2) The following table shows the sizes of shoes worn by 40 students in a Senior 2 class in a local High school. (a) Make a frequency table for the data.

6	10	7	6	7	8	7	9	11	11	10	7	8	6	8
7	9	11	6	7	9	9	8	7	10	10	8	8	7	8
8	7	8	8	7	9	8	7	10	9					

- (b) Do you think that this information can be used by which person?

7.5 Cumulative frequency table



Activity 7.5

Suppose 10 students scored the following marks in a quiz: 2, 3, 5, 3, 4, 2, 5, 3, 4, 2.

Discuss: How can we **organize this data** to see how many students scored **up to a certain mark** (For example from 2 up to 4)?

Predict which score level has the **most students scoring below or equal** to it.



Content Summary

A cumulative frequency table is a table that shows the running total of frequencies up to each value or category.

Cumulative Frequency (CF): Running total of frequencies, calculated as: $CF_1 = f_1$, $CF_2 = CF_1 + f_2$, $CF_3 = CF_2 + f_3$, and so on.

Examples: 1 Marks of 8 Students: 2, 3, 5, 3, 4, 2, 5, 3. Show the cumulative table.

Marks	Tally	Frequency (f)	Cumulative Frequency (CF)
2		2	2
3		3	$2 + 3 = 5$
4		1	$5 + 1 = 6$
5		2	$6 + 2 = 8$

Interpretation:

How many students scored **3 or less**? **Answer:** 5 students

Why is cumulative frequency useful? **Answer:** It shows total counts up to each value, helping analyze distribution.

**APPLICATION ACTIVITY 7.5**

Survey Students on **number of books read last month**: 1, 2, 3, 2, 4, 1, 3, 2.

- 1) Create a **frequency table** first.
- 2) Calculate **cumulative frequency**.
- 3) Solve these questions:
 - a) How many students read **2 or fewer books**?
 - b) How many students read **3 or fewer books**?
 - c) Which book count has the highest frequency?
- 4) **True/False:** "Cumulative frequency is the total number of values up to a given point."
- 5) From the table, how many students read **less than 4 books**?
- 6) "How can cumulative frequency help?"

7.6 The arithmetic mean or average for raw data in ungrouped data**Activity 7.6**

- 1) Use your mathematics dictionary or internet to find the meaning of the following terms: Arithmetic mean and average of data.
- 2) A car travels the first 1 km of its journey at a speed of 40 km/h, the next 1 km at a speed of 60 km/h, and the final 1 km at a speed of 50 km/h. Calculate the average speed of the car without considering the time used.

**Content Summary**

Ungrouped data: Data in its original, raw form, not organized into classes or intervals

Example: Test scores: 85, 90, 78, 92, 88

Arithmetic Mean (Average) is the sum of all values divided by the number of values.

The mean of a numerical set of observations (data) is defined as "the sum of the observations divided by the number of observations".

$$\text{Mean} = \frac{\text{sum of all the values in the group}}{\text{the number of all data items in the group}}$$

Thus, $mean = \bar{x} = \frac{x_1 + x_2 + x_3 + x_4 + \dots + x_n}{n}$ where $\bar{x}$ = arithmetic mean (say «x-bar»).

This can be written as $\bar{x} = \frac{\sum x}{n}$ (where $\sum$ means "sum of").

$\sum x$ = sum of all data values (Sigma means «sum») and n = total number of observations

Example 1:

Calculate the mean/ average of the pocket money of some 5 students who get 2 500 FRW, 4 000 FRW, 5 500 FRW, 7 500 FRW and 3 000 FRW.

Solution:

$$mean = \frac{\sum x}{n} = \frac{(2500 + 4000 + 5500 + 7500 + 3000)FRW}{5} = \frac{22\ 500}{5}FRW = 4\ 500FRW$$

Example 2:

Find the mean of these ages: 18, 20, 19, 22, 21

Solution:

$$\sum x = 18+20+19+22+21 = 100 \quad n = 5 \quad \bar{x} = \frac{100}{5} = 20 \quad \text{then Average age} = 20 \text{ years.}$$



APPLICATION ACTIVITY 7.6

1) Calculate the mean of the following:

(a) 1, 3, 5, 7

(d) 6, 7, 8, 9, 10

(g) 3, 9, 4, 7, 2, 8, 7, 9

(b) 2, 4, 6, 8

(e) 2, 2, 3, 4, 4, 5

(h) 10, 4, 11, 13, 15, 19, 21, 5

(c) 1, 2, 3, 4, 5

(f) 3, 4, 4, 7, 8, 9

(i) 8, 0, 3, 3, 1, 7, 4, 1, 4

7.7 Determining the arithmetic mean using frequency table in ungrouped data



Activity 7.7

Observe the following data on the marks of 20 students on an exam that was conducted out of 20: 9; 9; 9; 10; 10; 10; 10; 12; 15; 15; 15; 15; 18; 18; 19; 19; 19; 19; 20; 20.

a) Find the arithmetic mean of the marks using the raw data.

b) Complete the following table:

#	Mark	Number of students (Frequency)
1	9	3
2	10	4
3	12	1
4	15	4
5	18	2
6	19	4
7	20	2
	Total	20

c) Can you find the mean using the frequencies given that the sum of frequencies $\sum f = 20$?



Content Summary

The arithmetic mean of raw data is a single central value that balances the sum of all deviations, effectively representing the distribution's center under the condition of equal weighting for each data point. It is calculated by summing all individual data values in a dataset and dividing this total by the number of observations.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Arithmetic mean from frequency table:

$$\text{Formula: } \bar{x} = \frac{\sum(x \times f)}{\sum f}$$

Where: x = each data value; f = frequency of that value; $\sum f$ = total number of observations (sum of all frequencies); $\sum(x \times f)$ = sum of (value $\times$ frequency).

Step-by-step method to find the mean:

1. Create a table with columns: x , f , and $x \times f$
2. Multiply each x by its f
3. Add up the f column = $\sum f$
4. Add up the $x \times f$ column = $\sum(x \times f)$

$$\text{Divide: Mean} = \bar{x} = \frac{\sum(x \times f)}{\sum f}$$

Example:

The table below shows the masses, in grams, of some 40 mangoes sold in a grocery shop.

Calculate the mean mass of the mangoes.

70 100 90 120 110 80 110 80 100 90
 120 110 120 80 110 100 110 90 110 100
 80 70 100 80 90 110 110 90 100 90
 100 100 90 100 90 90 80 100 100 80

Solution:

Make a frequency table:

Mass x (g)	Tally	Frequency f	fx
70		2	140
80		7	560
90		9	810
100		11	1100
110		8	880
120		3	360
		$\sum f = 40$	$\sum fx = 3850$

The column ' fx ' represents the total mass of mangoes. For example, if there are 2 mangoes each of mass 70 g, their total mass is $(70 \times 2) \text{ g} = 140 \text{ g}$. Similarly, there are 7 mangoes, each of mass 80 g, having a total mass of $(80 \times 7) \text{ g} = 560 \text{ g}$, etc.

The symbol Σ (Greek letter 'sigma'), stands for 'sum of'. Thus, Σf means 'sum of frequencies' and Σfx means 'sum of products of f by x '.

The **mean**, denoted by $\bar{x}$, is given by: $mean, \bar{x} = \frac{\sum fx}{\sum f}$.

In the example, $\Sigma f = 40$, $\Sigma fx = 3850$

$$mean, \bar{x} = \frac{\sum fx}{\sum f} = \frac{3850}{40} = 96.25 \text{ g}$$



APPLICATION ACTIVITY 7.7

1) Calculate the mean of the following data:

- 2.1; 1.4; 3.5; 2.7
- 4.8; 4.5; 3.2; 1.8; 2.2.
- 0.7; 0.9; 0.3; 0.8; 0.7; 0.9; 0.8; 0.6; 0.5; 0.2.
- 3.5; 4.25; 2.125; 3.75.

- 2) Eight ladies had masses as follows: 51 kg, 44 kg, 57 kg, 63 kg, 48 kg, 49 kg, 45 kg, 53 kg. Find the mean of their masses.
- 3) The mean mark scored by 5 students in a mathematics test was 19. Four students had the following scores 15, 18, 17, and 16. What was the score for 5th student?

7.8 Mode and median of ungrouped data



Activity 7.8

Observe the marks obtained by 7 students out of 10: 4, 6, 5, 6, 7, 5, 6. Discuss:

- 1) Which number appears **most often**?
- 2) If the data is arranged in order from the smallest to the biggest, which number is **in the middle**?



Content Summary

Mode: The mode is the value that occurs **most frequently** in a set of ungrouped data. A dataset may have one mode, more than one mode, or no mode.

Median: The median is the **middle value** when data is arranged in **ascending or descending order**.

The data item in the middle position when all the data values are arranged in increasing order is called the median.

Let n be the number of data values or number of observations,

(a) Median for odd number of data values: When the number of data values (n) is odd, the median is the data value in the $\left(\frac{n+1}{2}\right)^{th}$ position in the rank order.

$$\text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{ value or } \left(X_{\frac{n+1}{2}}\right)$$

(b) Median for even number of data values: When the number of data values (n) is even there are two middle data values i.e. one in the $\frac{1}{2}n^{th}$ position and the one in the $\left(\frac{1}{2}n+1\right)^{th}$ position.

The median is halfway between the two middle numbers. $\text{median} = \frac{1}{2} \left(X_{\frac{n}{2}} + X_{\frac{n}{2}+1} \right)$.

The average (mean) of these two data values gives the median.

Thus, we determine the two values, add them up then divide their sum by 2.

Examples:

- 1) Find the median of the following numbers:
 - a) 15; 12; 13; 13; 9; 10; 8; 11; 10; 8; 7; 9; 10; 10; 11.
 - b) 12; 8; 21; 11; 4; 12; 13; 18; 20; 19; 21; 11.

Solution:

Arrange the values in order from the smallest to the largest.

a) 7, 8, 8, 9, 9, 10, 10, 10, 10, 11, 11, 12, 13, 13, 15. There are 15 values, it is odd,

then median is the $\left(\frac{15+1}{2}\right)^{\text{th}} = 8^{\text{th}}$. This 10 which is the middle value.

b) 4, 8, 11, 11, 12, 12, 13, 18, 19, 20, 21, 21 In this case, there is no one value that is in the middle. In such a case the median is the mean of the two middle values.

$$\text{median} = \frac{1}{2}(X_{\frac{n}{2}} + X_{\frac{n}{2}+1}) \quad \text{median} = \frac{12+13}{2} = 12.5$$

2) The following table shows the masses of some tomatoes bought from a farmer.

Mass (in g)	58	59	60	61	62	63
Frequency	2	6	12	9	8	3

Find: (a) the mean, (b) the median, (c) the mode of the masses of the tomatoes.

Solution:

$$\text{a) Mean mass} = \frac{(58 \times 2) + (59 \times 6) + (60 \times 12) + (61 \times 9) + (62 \times 8) + (63 \times 3)}{(2 + 6 + 12 + 9 + 8 + 3)}$$

$$= \frac{2424}{40} = 60.6 \text{ g}$$

b) Since there are 40 tomatoes, the median mass is the mass between those of the 20th and 21st tomatoes. We see that there are 20 tomatoes with mass 60 g and less. The 21st tomato must, therefore, have a mass of 61 g.

Mass	58	59	60	61	62	63
Frequency	2	6	12	9	8	3
Cumulative frequency	2	8	20	29	37	40

The mass of the 20th tomato = 60g, and the mass of the 21st tomato = 61 g.

$$\text{median} = \frac{60+61}{2} = 60.5 \text{ g}$$

c) The mode is 60 g.



APPLICATION ACTIVITY 7.8

- 1) Find the mean, median and mode of each of the following groups of numbers:
 - a) 4, 5, 8, 6, 4, 12, 3
 - b) 9, 9, 7, 7, 4, 13, 3, 17, 2, 21, 7
 - c) 7, 9, 4, 9, 3, 9, 10, 1, 1, 12
 - d) 5, 4, 5, 5, 4, 2, 3, 2, 1, 3, 1, 4, 0, 2, 2, 0, 1
- 2) Write down 5 numbers such that the mean is 6, the median is 5 and the mode is 4.
- 3) Seven pieces of luggage have masses of 48 kg, 45 kg, 49 kg, 63 kg, 57 kg, 44 kg and 51 kg.
 - a) Find the mean mass of the seven pieces.
 - b) If the lightest and the heaviest pieces are taken away, what is the mean mass of the remaining ones?
 - c) What is the median mass if the lightest piece is removed?

7.9 Quartiles (1st, 2nd, 3rd quartiles, inter-quartile range), midrange



Activity 7.9

Begin this activity by using your dictionary to find out the meaning of the word quartile. Consider the following marks scored out of 100 for 11 students arranged in order of rank:

Order	1	2	3	4	5	6	7	8	9	10	11
Marks	45	46	48	52	54	56	63	64	77	77	78

- i) In this set, identify the median mark.
- ii) How many marks are there below the median mark?
- iii) How many marks are above the median mark?
- iv) Find the middle value of the lower half of the set.
- v) Find the middle value of the upper half of the set.
- vi) Together with the median mark, what do the middle values in (iv) and (v) do to the given data? Discuss.



Content Summary

Mode: The mode is the value that occurs **most frequently** in a set of ungrouped data. A dataset may have one mode, more than one mode, or no mode.

Median: The median is the **middle value** when data is arranged in **ascending or descending order**.

The data item in the middle position when all the data values are arranged in increasing order is called the median.

Let n be the number of data values or number of observations

(a) Median for odd number of data values: When the number of data values (n) is odd, the median is the data value in the $\left(\frac{n+1}{2}\right)^{th}$ position in the rank order.

$$\text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{ value or } \left(X_{\frac{n+1}{2}}\right)$$

(b) Median for even number of data values: When the number of data values (n) is even there are two middle data values i.e. one in the $\frac{1}{2}n^{th}$ position and the one in the $\left(\frac{1}{2}n+1\right)^{th}$ position.

The median is halfway between the two middle numbers. $\text{median} = \frac{1}{2} \left(X_{\frac{n}{2}} + X_{\frac{n}{2}+1} \right)$.

The average (mean) of these two data values gives the median.

Thus, we determine the two values, add them up then divide their sum by 2.

Examples:

1) Find the median of the following data:

a) 15; 12; 13; 13; 9; 10; 8; 11; 10; 8; 7; 9; 10; 10; 11.

b) 12; 8; 21; 11; 4; 12; 13; 18; 20; 19; 21; 11.

In general, the lower quartile, Q_1 takes the $\left[\frac{1}{4}(n+1)\right]^{th}$ position from the lower end on the rank order. • The upper quartile, Q_3 , takes the $\left[\frac{3}{4}(n+1)\right]^{th}$ position on the rank order. •

For large population, it is enough to use $\left(\frac{1}{4}n\right)^{th}$ and $\left(\frac{3}{4}n\right)^{th}$ positions for the lower and upper quartiles respectively.

The difference between the upper Quartile and the lower Quartile is called the interquartile range. i.e. $Q_3 - Q_1$.

Example

The heights in cm of 13 boys are: 163, 162, 170, 161, 165, 163, 162, 163, 164, 160, 158, 153, 165. Determine: (a) The median. (b) The interquartile range.

Solution:

We begin by arranging the heights in ascending rank or order.

Rank order	1	2	3	4	5	6	7	8	9	10	11	12	13
Height in cm	153	158	160	161	162	162	163	163	163	164	165	165	170

a) The middle height lies in the 7th position. median = 163 cm

b) The lower quartile lies between the third and the fourth height.

3rd height = 160 cm and 4th height = 161 cm

The lower quartile, $Q_1 = \frac{1}{2} (160 + 161) = \frac{1}{2} \times 321 = 160.5$ cm

The upper quartile, Q_3 lies between the 10th and the 11th height.

The 10th height = 164 cm and The 11th height = 165 cm

The upper quartile, $Q_3 = \frac{1}{2} (164 + 165) = \frac{1}{2} (329) = 164.5$ cm

The interquartile range = $(Q_3 - Q_1) = 164.5 - 160.5 = 4$ cm.

**APPLICATION ACTIVITY 7.9**

- In a certain weather station, the annual rainfall in cm for 15 consecutive years were recorded as follows: 24 38 26 37 41 33 26 25 23 22 32 31 34 34 43 29. Use the rank order to determine:

(a) The median. (b) The quartiles. (c) Interquartile range.

- The marks scored by a group of students in a test.

Marks	38	39	40	41	42	43	44
F	1	2	3	1	3	3	2

Use the information to determine: (a) The number of students in the group. (b) The median. (c) The upper and the lower quartiles. (d) The interquartile range.

7.10 Pictogram



Activity 7.10

Carry out the following activities:

- Look at the following list which shows the number of computers allocated to each class:
 1. Senior class 1 – 6 computers
 2. Senior class 2 – 4 computers
 3. Senior class 3 – 4 computers
 4. Senior class 4 – 6 computers
 5. Senior class 5 – 5 computers
 6. Senior class 6 – 3 computers •
- On a piece of paper, draw computers for each class until all classes are represented.
- Each class and the computer drawn should be on one row: You should then have 6 rows, one for each class.
- What is the name given to such a presentation?



Content Summary

A pictogram (pictograph) is a diagram that represents statistical data in a pictorial form. Each picture or drawing represents a certain number or value from the data.

The picture to be used is chosen to represent the data subject as closely as possible, e.g. a computer to represent the number of computers, a car to represent the number of cars etc.

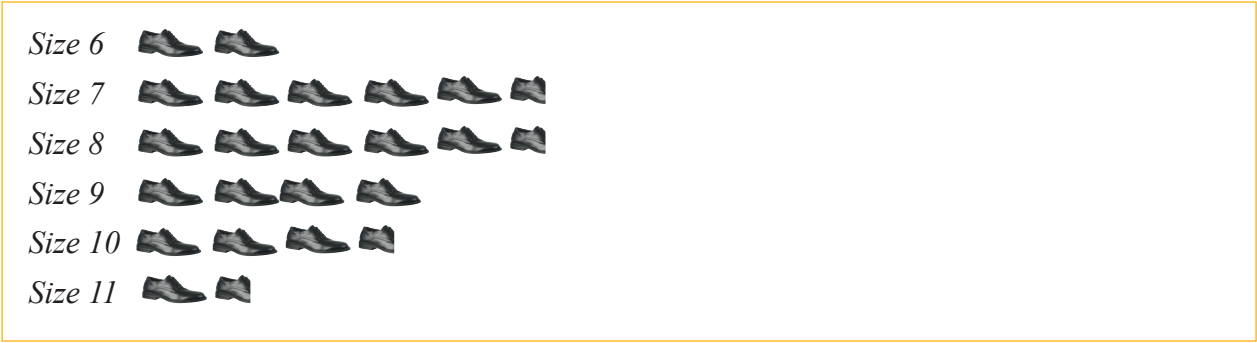
A fraction of 2 is represented by a fraction of the drawing and so on.

A pictogram is used to display information on posters, magazines and newspapers to attract attention. They are means of advertisements.

Example: Represent the data in a pictogram.

Size	6	7	8	9	10	11
Number of shoes	4	11	11	8	7	3

Let  represent 2 pairs of shoes. Since 4 Students wore size 6 shoes, then 4 will be represented by   , etc



APPLICATION ACTIVITY 7.10

In a national safari rally, there were several contestants. They all had different types of vehicles namely, Nissan, Toyota, Isuzu and Daihatsu. The following pictograph shows how many vehicles of each type were used during the safari rally.

Type of vehicle	Icon
Nissan	   
Toyota	 
Isuzu	  
Daihatsu	 

Key

 represents 6 vehicles.

 represents 3 vehicles.

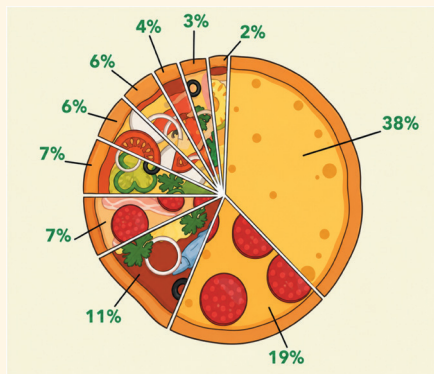
- a) How many Toyota participated in the rally?
- b) Which type of vehicle participated most in the rally?
- c) How many vehicles participated in the rally altogether?
- d) Draw a frequency table to display the information above.

7.11 Pie chart



Activity 7.11

A **pizza** is divided into unequal parts:



- 1) If this pizza is shared among different groups, how many groups are there? Do groups get the same or equal portion?
- 2) Which group gets the largest share? Which gets the smallest?
- 3) Is it possible to find the angle corresponding to each part out of the total of 360 degrees?



Content Summary

A pie-chart is a graph or a diagram in which different proportions of a given data distribution are represented by sectors of a circle. The diagram is looked at as a circular 'pie', hence the name pie chart.

Usually there are no numbers on a pie chart beside the angles sizes. The sizes of the sectors give a comparison between the quantities represented. The order in which the sectors are presented does not matter. Sectors may be shaded with different patterns (or colours) to give a better visual impression.

Example:

- 1) The following table shows grades scored by 15 candidates who sat for a certain test.

Grade	A	B	C	D	E
Number of candidates	2	5	4	1	3

Draw a pie chart for this data.

Solution:

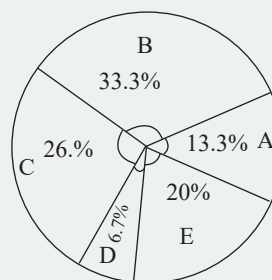
Work out the fractions of numbers of candidates who scored each grade. For example, for grade A we have $\frac{2}{15}$. Since the angle at the centre of a circle is 360° ,

we calculate the angle to represent grade A as $\frac{2}{15}$ of 360°

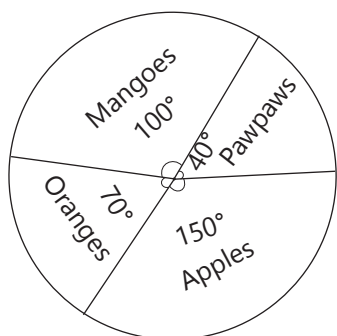
i.e. $\frac{2}{15} \times 360^\circ = 48^\circ$. A is represented by an angle of 48° .

Grade	No. of candidates	Fraction of total	Angle at the centre of the circle	%
A	2	$\frac{2}{15}$	$\frac{2}{15} \times 360^\circ = 48^\circ$	13.3%
B	5	$\frac{5}{15}$	$\frac{5}{15} \times 360^\circ = 120^\circ$	33.30%
C	4	$\frac{4}{15}$	$\frac{4}{15} \times 360^\circ = 96^\circ$	26.7%
D	1	$\frac{1}{15}$	$\frac{1}{15} \times 360^\circ = 24^\circ$	6.7%
E	3	$\frac{3}{15}$	$\frac{3}{15} \times 360^\circ = 72^\circ$	20%

In pie chat.



- 2) After selling fruits in a market, Asha had a total of 144 fruits remaining. The pie chart below shows each type of fruit that remains.



- Find the total cost of mangoes and pawpaws remaining if a mango sells at 30 FRW and a pawpaw at 160 FRW each.
- Which type of fruit remained the most?
- What was the median number of fruit that remained?
- Draw a frequency table to display the information on the pie chart.

Solution:

(a) Numbers of mangoes = $\frac{100}{360} \times 144 = 40$ mangoes

Number of pawpaws = $\frac{40}{360} \times 144 = 16$ pawpaws

Cost of mangoes = $40 \times 30 = 1\,200$ FRW

Cost of paw paws = $16 \times 160 = 2\,560$ FRW

Total cost = $1\,200 + 2\,560 = 3\,760$ FRW

(b) Apples remained the most unsold i.e. $\frac{150}{360} \times 144 = 60$ apples

(c) Median number of remaining fruits. $n=4$; it is even.

Mangoes = 40

Pawpaws = 16

Apples = 60

Oranges = $\frac{70}{360} \times 144 = 28$.

Median = $(16, 28, 40, 60) = \frac{28 + 40}{2} = \frac{68}{2} = 34$ fruits

Type of fruit	Frequency (number remaining)
Mangoes	40
Oranges	28
Pawpaws	16
Apples	60
Total	144



APPLICATION ACTIVITY 7.11

In a survey on soft-drinks, 180 people were asked to state the brand they preferred. 35 chose brand A, 30 chose brand B, 100 chose brand C and 15 chose brand D. Draw a pie-chart to display this information.

7.12 Line graph



Activity 7.12

Observe in the following table how the temperature changes from day to day:

Day	Temperature (°C)
Mon	20
Tue	22
Wed	25
Thu	24
Fri	26

Then, answer the following questions:

- 1) Which day is the hottest?
- 2) Does the temperature increase or decrease during the week?
- 3) From the table, form ordered pairs of the form (day, temperature).
- 4) Draw a cartesian plan with days on x-axis and temperature on y-axis. How could you represent ordered pairs found in 3) to represent the change of temperature clearly using a drawing?



Content Summary

A **line graph** is a graph that uses **points connected by straight lines** to show how a quantity changes over time.

When to use a line graph:

- To show increase or decrease.
- To show trends over time.
- To compare changes in data.

Parts of a line graph:

- **Horizontal axis (x-axis):** shows time or independent data
- **Vertical axis (y-axis):** shows values or measurements
- **Scale:** shows the value of each unit.
- **Points:** represent data values.
- **Line:** connects points to show changes.
- **Title:** Describes the graph.

Steps to draw a line graph:

1. Draw the x-axis and y-axis.
2. Choose a suitable scale.
3. Plot the data points.
4. Join the points with straight lines.
5. Give the graph a title.

Examples 1: The following table shows the temperature in degrees Celsius, observed at 2-hourly intervals of a patient who was admitted at a hospital.

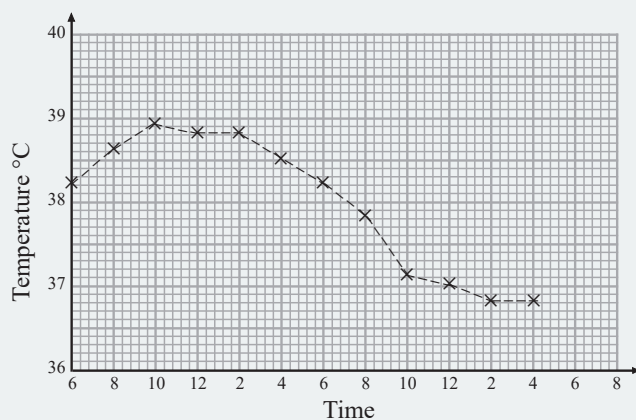
Time	6:00 A.M	8:00 A.M	10:00 A.M	12:00 noon	2:00 P.M	4:00 P.M
Temp °C	38.2	38.6	38.9	38.8	38.8	38.5

Time	6:00 P.M	8:00 P.M	10:00 P.M	0:00-night	2:00 A.M	4:00 A.M
Temp °C	38.2	37.8	37.1	37.0	36.8	36.8

- a) Plot a graph of time against temperature.
- b) Estimate the patient's temperature at:
 - i) 9 a.m (ii) 3 p.m
 - ii) 11 p.m (iv) 3 a.m
- c) What can you say about the patient's temperature;
 - i) during the day?
 - ii) during the night?

Solution:

- a) When a graph of time against temperature is plotted using the data in Table and the points joined by dotted line segments, is obtained.



- b) (i) 38.4°C (ii) 38.6°C
 (iii) 37.1°C (iv) 36.8°C
- c) (i) The temperature rises during the day in the morning hours and in the afternoon it starts to fall down.
 (ii) During the night, the temperature falls back to normal temperature.

**APPLICATION ACTIVITY 7.12**

The following table shows the population of a certain country, in thousands, between the year 1901 to 1966.

Year	1901	1911	1921	1931	1941	1951	1961	1966
Population	5 400	7 200	8 800	10 400	11 500	14 000	18 200	20 000

- a) Draw a line graph for the data.
 b) Estimate the population in: (i) 1926 (ii) 1971.

7.13 Bar chart/ graph



Activity 7.13

The teacher asks students of the class to say the Favourite fruits. The table below summarizes the number of students who mentioned a given fruit:

Fruit	Number of Students
Banana	8
Orange	5
Apple	6
Mango	11

- 1) Which fruit is liked by most Students?
- 2) Which fruit is liked by the fewest students?
- 3) For each fruit, draw a vertical bar whose length equals the number of students. If you compare the use of table and the use of bars, which one is easy to interpret?



Content Summary

A **bar chart** is a diagram that uses **rectangular bars of equal width** to represent data.

Uses of a bar chart:

- To compare quantities
- To show differences between categories
- To represent discrete data

Types of bar charts:

- Vertical bar chart
- Horizontal bar chart
- Double bar chart (for comparison)

Parts of a Bar Chart:

- Horizontal axis (x-axis): categories
- Vertical axis (y-axis): values or frequency
- Bars: represent quantities
- Scale: shows the value of each unit
- Title: describes the chart

Rules for Drawing a Bar Chart:

- Bars must have equal width
- Bars must have equal spacing
- Bars must not touch each other
- Use a clear scale.

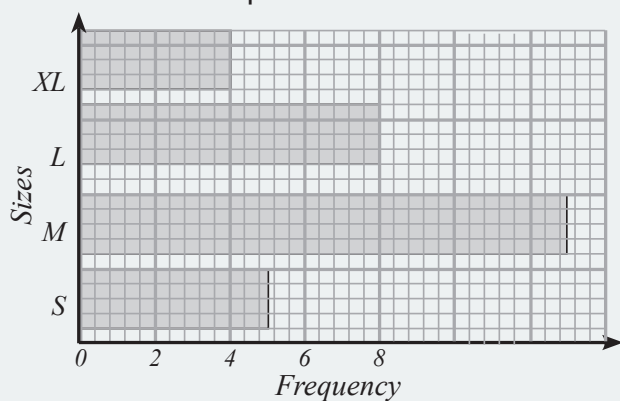
Example: The following table shows the sizes of sweaters worn by 30 Senior 1 students in a certain school.

Size	Small (S)	Medium(M)	Large(L)	Extra-large (XL)
Number of Students	5	13	8	4

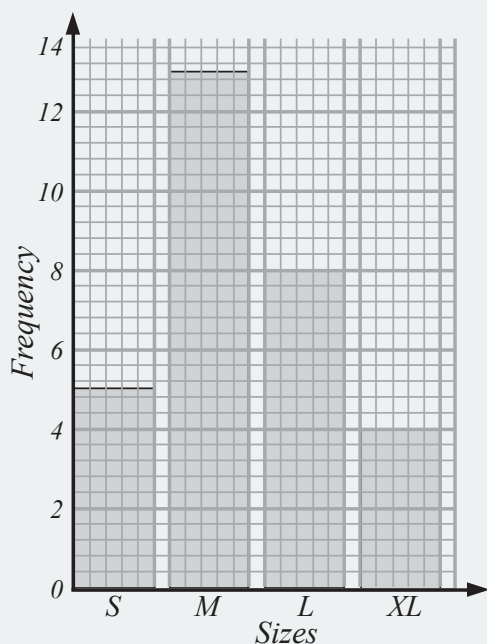
Represent the data on: (a) a horizontal bar chart (b) a vertical bar chart.

Solution:

a) Horizontal bar chart: frequency is represented on the horizontal axis. Bars are drawn with spaces between them



b) In a vertical bar graph, frequency is represented on the vertical axis.





APPLICATION ACTIVITY 7.13

Kanyange scored the following marks in her end of term examination. English 72, Biology 78, Mathematics 94, Physics 84 and History 86.

- Draw the table to illustrate this information.
- Draw a bar graph to represent the given information.
- What is the difference between the highest and the lowest marks?
 - What is the total mark for Kanyange?

7.14 Cumulative frequency diagram or ogive



Activity 7.14

The distribution Table below shows a group of 60 children whose ages are recorded to the nearest year.

Age (yrs)	7	8	9	10	11	12	13	Total
Frequency (number of students)	0	10	24	13	6	4	3	60
Running totals								

Consider the last row in the table which shows running totals of the frequencies; Complete it as follows: for the age 7, total = 0 For the age 8, total = 0 + 10, For the age 9, total = 0 + 10 + 24 and so on.

What is the other name you would call the running totals?

Draw a graph by plotting age against the corresponding running totals i.e. (7,0) (8,10), (9,34), (10, 47), (11, 53), (12, 57) and (13, 60).

What is the name of such a graph?



Content Summary

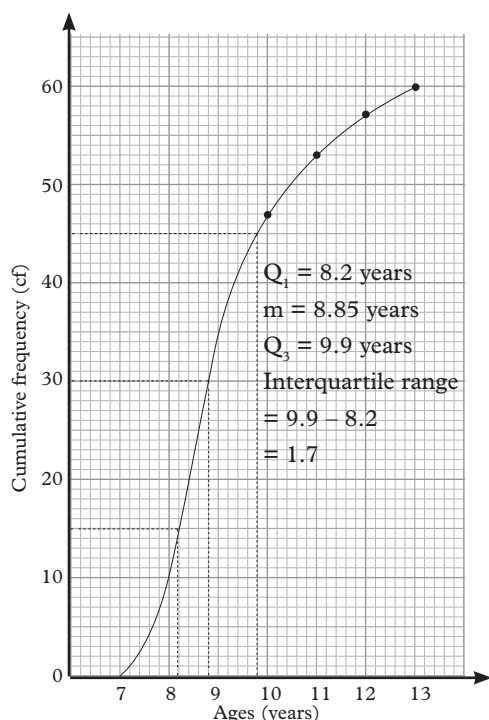
A **cumulative frequency diagram** is a graph that shows the **total frequency up to a given value**.

Example:

- 1) From the activity 7.14 you should have obtained a cumulative frequency table as the following

Age (yrs)	frequency (f)	Cumulative frequency	Age range represented by cf
7	0	0	up to 8 years
8	10	10	up to 9 years
9	24	34	up to 10 years
10	13	47	up to 11 years
11	6	53	up to 12 years
12	4	57	up to 13 years
13	3	60	up to 13 years

The graph you plotted is called a **cumulative frequency graph** and should be as follows:



Using the cumulative frequency graph, we can read the Cumulative frequency (Cf) and this can help us to identify quartiles.

Example:

From the graph given above,

$Q_1 = \frac{Cf}{4} = \frac{60}{4} = 15$ then find the correspond of 15 on axe of age = 8.2 years.

$Median = Q_2 = \frac{Cf}{2} = \frac{60}{2} = 30$ then find the correspond of 30 on axe of age = 8.85 years.

$Q_3 = \frac{Cf \times 3}{4} = \frac{60 \times 3}{4} = 45$ then find the correspond of 45 on axe of age = 9.9 years.

- 2) The following table shows the length of a metal bar according to the estimates by a group of 32 students:

Length(cm)	35	36	37	38	39	40	41	42
F	1	3	4	8	6	5	3	2

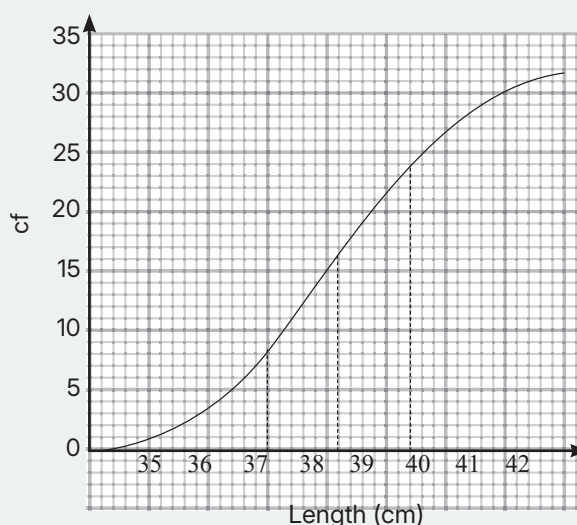
Find:

- The range of these estimates.
- Draw a cumulative frequency table and plot a cumulative frequency curve from this information.
- From the curve, estimate the median and the interquartile range.
- Use the information to estimate the mode.

Solution:

- The longest metal bar is 42 cm. The shortest metal bar is 35 cm. The range = $42 - 35 = 7\text{cm}$
- The cumulative frequency table and cumulative frequency curve.

Length (cm)	f	cf
35	1	1
36	3	4
37	4	8
38	8	16
39	6	22
40	5	27
41	3	30
42	2	32



- Median Length = 38 cm LQ = $Q_1 = 37$, UQ = $Q_3 = 39.4$.
Interquartile range = $39.4 - 37 = 2.4$
- the mode is 38 cm because it has a highest frequency.

**APPLICATION ACTIVITY 7.14**

A tour company recorded the number of people travelling to a holiday resort each day for 30 consecutive days as in table below

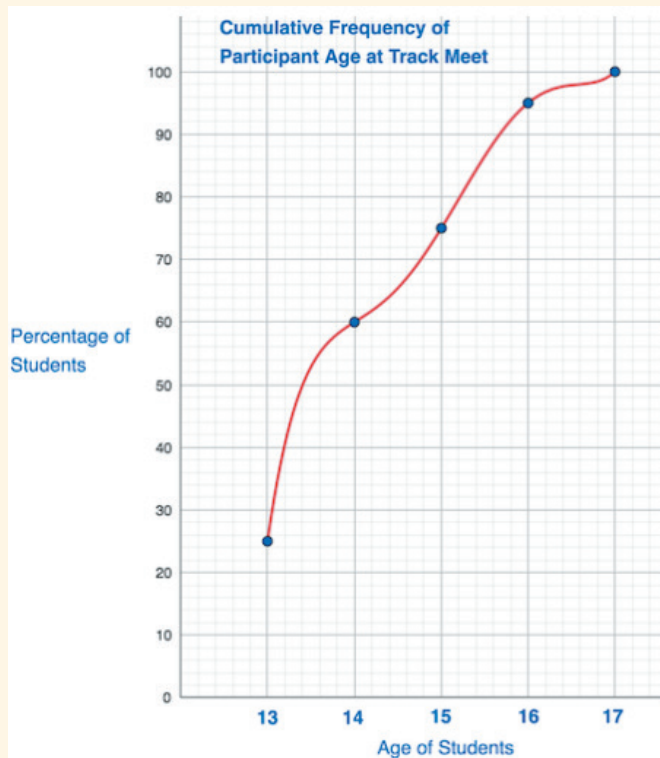
Number of people	125	175	225	275	325	375	425
f	0	6	7	8	5	3	1

- Make a cumulative frequency table for the data. (b) Draw a cumulative frequency curve and use it to estimate: (i) The median. (ii) The lower quartile. (iii) The upper quartile. (iv) The interquartile range.

7.15 Reading and converting statistical graphs: Line graph into frequency table



Activity 7.15



Discuss in pairs

What do the points on the graph represent?

Why are the points joined by straight lines?

Can this information be written using numbers in a frequency table?



Content Summary

a) Reading statistical graphs

- **Reading a graph** means understanding the information shown.
- To read a graph:
 - Read the **title**
 - Identify the **horizontal axis (x-axis)** and **vertical axis (y-axis)**
 - Note the **scale** used
 - Interpret the values shown

b) Line Graph

- A **line graph** shows how data changes over time.
- Points are plotted and joined using straight lines.
- Commonly used for time data (days, weeks, months, years).

c) Converting graphs into frequency tables:

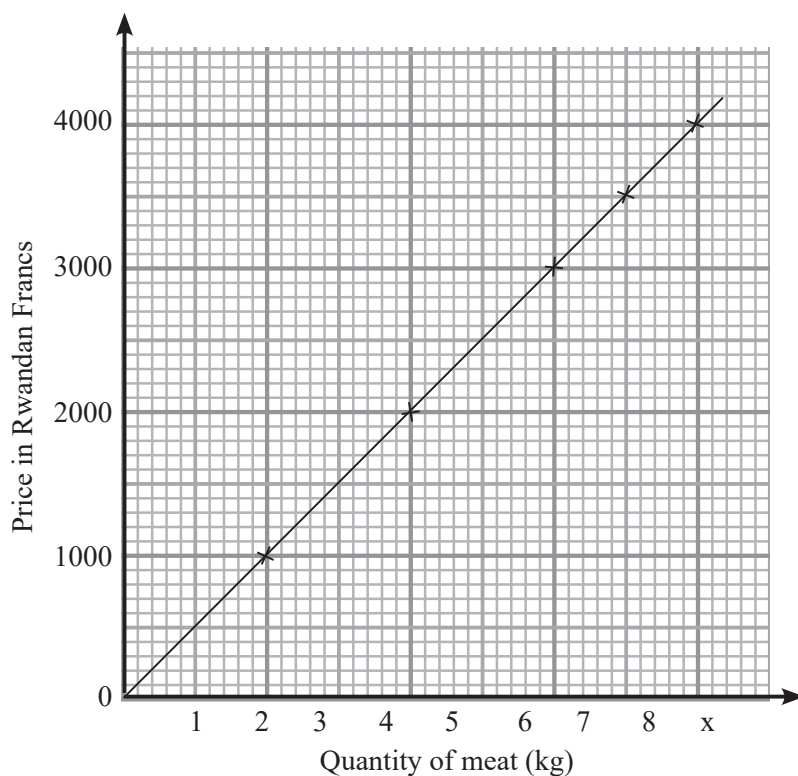
A **frequency table** shows: Categories or values and corresponding frequencies

Steps:

- Identify categories from the graph
- Read the value for each category
- Write them in table form

Example:

The following line graph shows the relationship between the price of a kilogram of meat and the quantity of meat. Use it to answer the questions that follow



From the graph:

- What is the price of:
 - 1 kg of meat?
 - 5 kg of meat?
- Draw a frequency table to show the quantity of meat and price per quantity.

Solution:

From the graph:

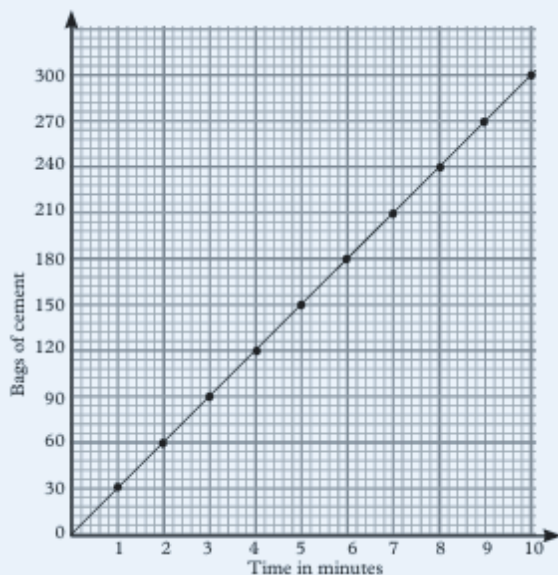
- (a) 1kg of meat costs 500 FRW.
(b) 5kg of meat costs 2 500 FRW.

Quantity (kg)	1	2	3	4	5	6	7	8
Price (FRW)	500	1000	1500	2000	2500	3000	3500	4000



APPLICATION ACTIVITY 7.15

The line graph below shows bags of cement produced by cement factory in a minute.



- Find how many bags of cement will be produced in: (i) 8 minutes, (ii) 3 minutes 12 seconds, (iii) 5 minutes, (iv) 7 minutes 24 seconds, (v) 10 minutes
- Find how long will it take to produce: (i) 78 bags of cement, (ii) 120 bags of cement, (iii) 300 bags of cement, (iv) 54 bags of cement, (v) 246 bags of cement
- Draw a frequency table to show the number of bags produced and the time taken

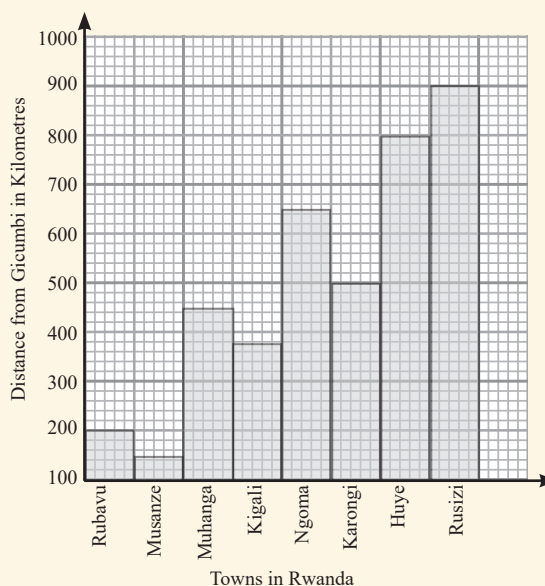
7.16 Reading and converting statistical graphs: Bar graph into frequency table



Activity 7.16

The bar graph below shows how far it is by road from Gicumbi to nine other towns in Rwanda. Study the bar graph and answer the questions that follow.

- On which axis of the bar graph is distance shown?
- On which axis of the graph are towns shown?
- How far is it from Gicumbi to Rusizi?
- Write down the names of the towns on the bar graph which are over 500 km from Gicumbi?
- If someone makes a return journey from Gicumbi to Karongi, what is the total distance the person will have to cover?
- Which is the farthest town on the bar graph from Gicumbi? (g) Draw a frequency table to display the information on the chart?





Content Summary

A bar graph uses bars of different heights/lengths to show data,

On X-axis: Categories (not necessarily in order)

On Y-axis: Frequency or count for each category

Note that bars don't touch each other (unlike histograms)

Why do we need to convert bar graphs to frequency tables?

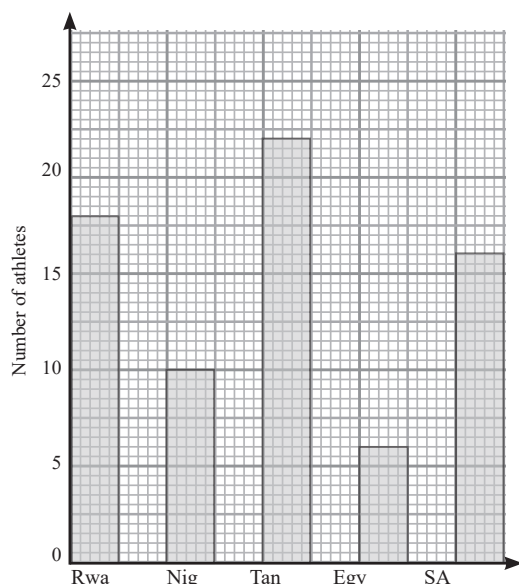
- To reorganize data by frequency count
- To find patterns in how often values occur
- To prepare data for further calculations (mean, median, mode)
- To compare different bar graphs more easily

Conversion Steps:

- 1) Read the height of each bar
- 2) Record the category and its frequency
- 3) Reorder the table by frequency if needed
- 4) Verify: Total frequency should match total observations

Example:

The bar graph shows the number of athletes who represented five African countries in an international championship.



Key words of the graph: Rwa : Rwanda, Nig: Nigeria; Tan: Tanzania; Egy: Egypt; SA: South Africa.

Discuss in group the following questions

- a) What was the total number of athletes representing the five countries?
- b) What was the smallest number of athletes representing one country?
- c) What was the most number of athletes representing a country?
- d) Represent the information on the graph on a frequency table.

Solution:

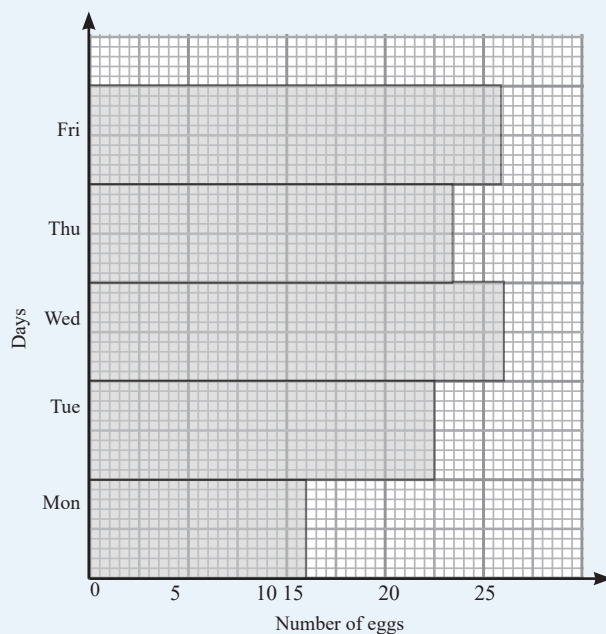
Total number of athletes are:

- a) $18 + 10 + 22 + 6 + 16 = 72$ athletes
- b) 6 athletes
- c) 22 athletes

Country	Number of Athletes
Rwanda	18
Nigeria	10
Tanzania	22
Egypt	6
South Africa	16
Total	72

**APPLICATION ACTIVITY 7.16**

The bar graph below represents the number of eggs collected in five days. Use it to answer the questions that follow.



- a) What scale has been used on the horizontal axis?
- b) Which day was:
 - i) The highest number of eggs collected?
 - ii) The least number of eggs collected?
- c) Find the mean daily collection of eggs for that week.
- d) If one egg was sold for 60 FRW, what was the total amount of money collected that week? (e) Represent the data in the bar chart on a frequency table.

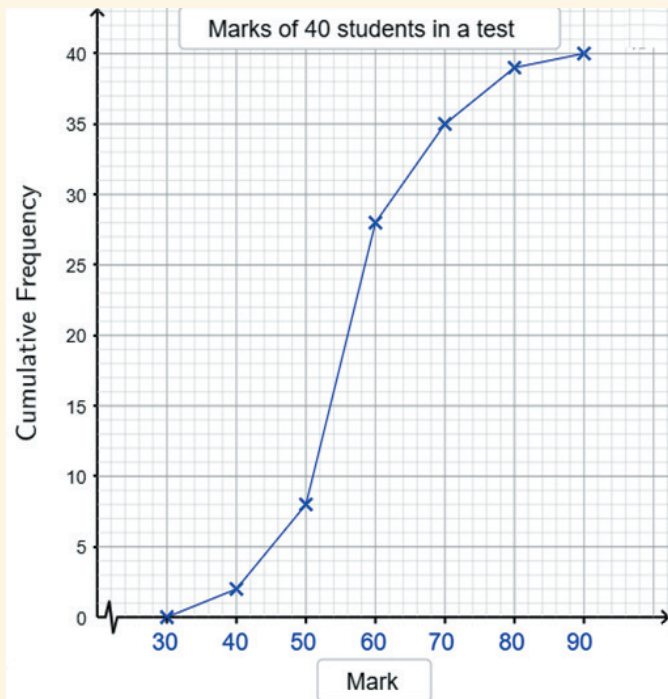
7.17 Reading and Converting Cumulative frequency diagram into frequency tables



Activity 7.17

Discuss in group the following questions;

- What information does this graph show?
- What do the points on the graph represent?
- Which mark has the highest number of Students?
- Can this information be written using numbers in a table?



Content Summary

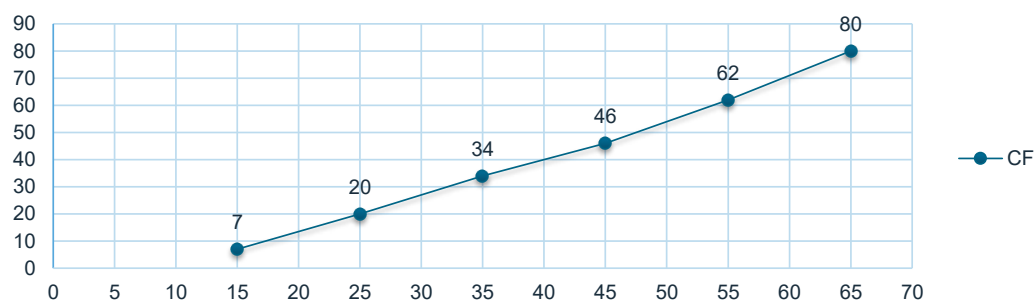
a) Reading a graph involves:

- Understanding the title
- Identifying the axes
- Observing the scale
- Interpreting the data values

b) Steps for converting a cumulative Frequency Polygon into a Frequency Table

- Identify the values or class intervals on the horizontal axis.
- Read the frequency corresponding to each point.
- Record the values and frequencies in a table.
- Ensure all points on the graph are included.

Example: Converting the following Cumulative frequency diagram into frequency tables.



Solution:

The frequency table:

Height (cm)	15	25	35	45	55	65
Cumulative frequency	7	20	34	46	62	80

Find each frequency by subtracting consecutive cumulative frequencies:

$$f_1 = 7; f_2 = 20 - 7 = 13; f_3 = 34 - 20 = 14; f_4 = 46 - 34 = 12$$

$$f_5 = 62 - 46 = 16; f_6 = 80 - 62 = 18$$

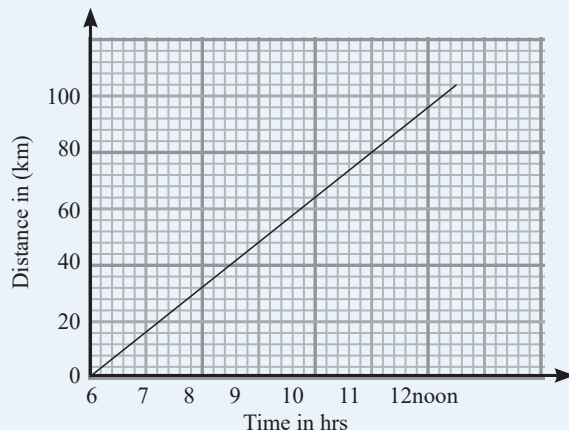
Therefore, the reconstructed frequency table is:

Height (cm)	15	25	35	45	55	65
Frequency	7	13	14	12	16	18



APPLICATION ACTIVITY 7.17

The travel graph below shows the journey made by a traveler between 6:00 am and 12:00.



Use the graph to answer the following questions.

- Find: (i) The vertical scale. (ii) The horizontal scale.
- How far had the traveler moved by 8:00 am?
- At what time did she cover 92 km?

PROJECT: STATISTICAL EXPLORATION OF OUR SCHOOL

1. Project Overview

Big Idea: Students will become data detectives for their own school community. They will collect, organize, analyze, and present real-world data about their classmates, making statistics personal and relevant.

Learning Objective: To analyze real-world data using appropriate statistical methods (measures of central tendency, range) and graphical representations (bar charts, pictograms, pie charts).

Final Product: A «Data Poster» and a short, structured report.

2. Suggested Data Collection Topics

Students can pick one area to investigate:

- **Free Time & Hobbies:** «Favorite hobby/sport after school»
- **Digital Life:** «Favorite type of app/game» or «Average screen time on a weekday»
- **School Life:** «Favorite subject» or «Mode of transport to school»
- **Tastes and Preferences:** «Favorite type of music genre» or «Favorite snack»
- **Personal Metrics:** «Number of siblings» or «Shoe size»

3. Step-by-Step project Guide

Phase 1: Ask a Question and Collect Data (Saturday)

- **Task:** Choose your topic and decide on specific categories (e.g., for hobbies: Football, Gaming, Reading, Music, Other).
- **Action:** Survey **at least 20 classmates** (a simple, pre-approved tally sheet).
- **Skill:** Data collection, creating a frequency table.

Phase 2: Organize the Data (Sunday)

- **Task:** Create a **Frequency Table** from your tallies.

Phase 3: Analyze the Data (Calculations) (one Saturday)

- **Task:** Calculate the following from your frequency table:
 1. **Mode:** The hobby with the *highest frequency*.
 2. **Median:** The *middle value* when frequencies are considered. For nominal categories use mode only.
 3. **Mean (Average):** (For categories like “number of siblings,” you can calculate a true mean. For favorites like hobbies, discuss why the mean isn’t meaningful here—this is a key insight!).

4. Range: The spread of your data (e.g., for «shoe sizes»: Largest size - Smallest size).

Phase 4: Represent the Data (Saturday)

- **Task:** Create **two different graphical representations** of your data.
- Choices:
 - **Bar Chart:** Perfect for comparing frequencies of different categories. Must have labeled axes, a title, and equal bar spacing.
 - **Pictogram:** Use simple, relevant icons (e.g., a ball for football).
 - **Pie Chart:** Show how the data breaks into parts of a whole.
- **Phase 5: Interpret and Present (Day 5)**
- **Task:** Prepare your **"Data Poster"** and **3-4 conclusion sentences**.
- **Poster Must Include:**
 1. A title (e.g., "How S1B Spends Its Free Time").
 2. Your frequency table.
 3. Your two graphs.
 4. Your calculated statistics (Mode, Median, etc.).
- **5. Conclusions:** What did you discover? What was the most popular choice? Was the data spread out or similar? Was there anything surprising?



END OF UNIT ASSESSMENT

Assessing the knowledge and understanding

State whether each statement is **TRUE or FALSE**. If false, correct the statement.

- 1) Statistics is the study of collecting, organizing, presenting, analyzing, and interpreting data.
- 2) Qualitative data can be measured numerically.
- 3) The number of students in a class is an example of continuous data.
- 4) A frequency distribution table shows how often each value or group of values occurs.
- 5) Cumulative frequency is obtained by subtracting frequencies successively.
- 6) The arithmetic mean is found by dividing the sum of observations by the number of observations.
- 7) The median is always the most frequent value in a data set.
- 8) The mode is the value that occurs most frequently.
- 9) The inter-quartile range is found by subtracting Q_1 from Q_3 .
- 10) The midrange is the average of the highest and lowest values.

Assessing skills

- 11) Mr Keza has a monthly income of 40 000 FRW. He spends 10 000 FRW on house rent, 8000 FRW on food, 4 000 FRW on transport, 9 000 FRW on entertainment and saves the remaining amount. Draw a pie chart to represent this information.
- 12) A die was thrown 25 times and the face that appeared at the top was recorded

Face	Face 1	Face 2	Face 3	Face 4	Face 5	Face 6
Number of times	3	6	3	2	4	7

Draw a bar chart to represent the information above.

Assessing values and attitudes

- 13) Explain one importance of statistics in daily life.
- 14) Why should data be collected accurately?

UNIT 8

ELEMENTARY PROBABILITY



Key Unit Competence:

Calculate theoretical and experimental probabilities in order to evaluate the likelihood of events in everyday situations.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain the terms: experiment, outcome, event, and sample space.
- Identify and classify events as certain, impossible, or uncertain.
- Demonstrate that the more data collected, the better the estimate of the probability.
- Differentiate between theoretical and experimental probability.
- State and apply the basic rules of probability

Skills

- Calculate the theoretical probability of simple events.
- Design and conduct a simple probability experiment.
- Collect, record, and analyse data from experiments.
- Compare theoretical and experimental results, recognizing the Law of Large Numbers.
- Use the basic rules of probability to solve simple problems.

Attitudes and values

- Appreciate the role of probability in making informed decisions in daily life.
- Develop a systematic and organized approach to problem-solving.
- Value accuracy and honesty in collecting and reporting experimental data.
- Develop patience and persistence when experimental results do not immediately match theoretical expectations.

INTRODUCTION

In everyday life, we often make statements such as *"It will rain today,"* *"Our team may win the match,"* or *"I am sure the sun will rise tomorrow."* Some events are certain, some are impossible, and others are uncertain. The study of probability helps us to measure how likely an event is to happen.

In this unit, you will explore the basic language of probability, including important terms such as experiment, outcome, event, and sample space. You will learn to identify different types of events, such as certain, impossible, and uncertain events, using real-life examples. In addition, you will calculate theoretical and experimental probability in order to evaluate the likelihood of different events.



Introductory activity

Discuss in group and answer the following questions:

- 1) When you toss a coin, what can happen?
- 2) Do you think getting heads is guaranteed every time? Why?
- 3) Is it possible to get the number 7 when rolling a normal die?
- 4) If you roll a die once, are you sure you will get 6? Getting it is a chance, or you can predetermine and ensure you get it?

8.1 Concepts used to describe probability: experiment, event, outcome, sample space



Activity 8.1

Roll the die several times and record all results on the paper. Answer the following questions:

- a) Is rolling a die an action or a result?
- b) How can we name each number obtained?
- c) What are all the possible numbers that can appear?



Content Summary

An **experiment** is an action or process that leads to one or more results, and the result cannot be predicted with certainty.

Example: Tossing a coin, rolling a die.

An **outcome** is a single possible result of an experiment.

Example: Getting a 4 when rolling a die.

The **sample space** is the set of all possible outcomes of an experiment.

Example: For rolling a die: $S = \{1, 2, 3, 4, 5, 6\}$

An **event** is a subset of the sample space, it is a collection of outcomes that satisfy some condition.

Example: Getting an even number when rolling a die (2, 4, 6).

1) Tossing a Coin

Experiment: Tossing a coin

Sample Space: $S = \{H, T\}$

Possible Outcomes: H or T

Event: Getting a head $\rightarrow \{H\}$

2) Rolling a Die

Experiment: Rolling a die

Sample Space: $S = \{1, 2, 3, 4, 5, 6\}$

Outcome: 3

Event: Getting a number greater than 4 $\rightarrow \{5, 6\}$

3) Drawing a Ball: A bag contains red, blue, and green balls. One ball is drawn.

Experiment: Drawing one ball

Sample Space: $S = \{\text{Red, Blue, Green}\}$

Outcome: Blue

Event: Drawing a coloured ball $\rightarrow \{\text{Red, Blue, Green}\}$



APPLICATION ACTIVITY 8.1

1) A coin is tossed once.

- Identify the experiment.
- State one outcome.

- Write the sample space.
- Give one event.

2) A die is rolled once.

- What is the experiment?
- State an outcome.

- List the sample space.
- Write the event of getting an odd number.

3) (True or False)

- An experiment is the result obtained.
- The sample space contains all possible outcomes.
- An event can have more than one outcome.
- Getting a 6 when rolling a die is an experiment.

8.2 Types of events and examples



Activity 8.2

Discuss whether the following events are possible or not. Estimate the likelihood in each case

1. When a stone is thrown upwards into the air, it will finally drop to the ground.
2. When a Rwanda coin is thrown at a random, it will show a head (H). It shows two sides at the same time.
3. When a dice is thrown it will give a 10.



Content Summary

A certain event is an event that will definitely happen when an experiment is performed. Its probability is 1.

Example: Getting a number less than 7 when rolling a die.

Getting either head or tail when tossing a coin.

An impossible event is an event that cannot happen at all. Its probability is 0.

Example: Getting a 7 when rolling a die.

Getting both head and tail at the same time when tossing a coin.

An uncertain event is an event that may or may not happen. Its probability lies between 0 and 1.

Example: Getting a head when tossing a coin.

Getting an even number when rolling a die.

Examples:

Say the type of event:

- 1) Tossing a coin, Event: Getting a head. **Answer:** Uncertain event
- 2) Experiment: Rolling a die, Event: Getting a number less than 7. **Answer:** Certain event.
- 3) Experiment: Rolling a die, Event: Getting the number 8. **Answer:** Impossible event.
- 4) Experiment: Drawing one ball from a bag containing red and blue balls, Event: Getting a green ball, **Answer:** Impossible event.



APPLICATION ACTIVITY 8.2

1) Identify the Type of Event

For each event, state whether it is **certain, impossible, or uncertain**.

- Getting a number between 1 and 6 when rolling a die.
- Getting a tail when tossing a coin.
- Getting a number 10 when rolling a die.
- The sun will rise tomorrow.

2) Complete the Table

Experiment	Event	Type of Event
Tossing a coin	Getting head	
Rolling a die	Getting 7	
Rolling a die	Getting a number ≤ 6	
Drawing a card from 1–5	Getting 3	

3) True or False

- A certain event has probability 0.
- An impossible event can occur sometimes
- An uncertain event may or may not happen.
- Getting a number less than 7 when rolling a die is an impossible event.

4) Write the type of event:

A bag contains 3 red balls and 2 blue balls.

- Drawing a red or blue ball.
- Drawing a green ball.
- Drawing a red ball.

8.3 Experimental probability



Activity 8.3

Toss a coin **30 times** in front of the students. count how many times **Heads** appears. write the results.

- Did heads appear exactly 15 times?
- Will the result be the same if we repeat the experiment?



Content Summary

Experimental probability is the probability calculated based on **actual experiments or trials**.

It depends on the **number of times an event occurs** during an experiment.

Formula: $P(E) = \frac{\text{Number of times the event occurs}}{\text{Total number of trials}}$

Important notes:

- Experimental probability may change when the experiment is repeated.
- As the number of trials increases, experimental probability becomes closer to theoretical probability.
- Experimental probability is expressed as a fraction, decimal, or percentage.

Example 1: A coin is tossed **20 times** and heads appears **12 times**. Find the experimental probability of getting heads.

Solution:

Number of times heads occurred = 12

Total number of trials = 20

$$P(\text{Heads}) = \frac{12}{20} = \frac{3}{5}$$

2. A die is rolled **30 times** and the number 4 appears **6 times**. Find the experimental probability of getting 4.

Solution:

$$P(4) = \frac{6}{30} = \frac{1}{5}$$



APPLICATION ACTIVITY 8.3

- 1) A coin was tossed **50 times** in a classroom experiment. The results are:
Heads: 28 times and Tails: 22 times
 - a) Find the experimental probability of heads.
 - b) Find the experimental probability of tails.
- 2) A bag contains colored balls. A ball is picked at random and replaced each time. The experiment is done 40 times. Red ball appeared 18 times
- 3) True or False
 - a) Experimental probability is based on actual experiments.
 - b) Experimental probability is always equal to theoretical probability.
 - c) Increasing the number of trials makes experimental probability more reliable.

8.4 Numerical expression of probability: Theoretical probability



Activity 8.4

- 1) Toss a coin once. List all possible outcomes. Can we be sure it will land on head?
- 2) Throw a die: List all possible outcomes. How many are they? The event is fixed: "getting number 6". How many favourable ways of getting 6 are there? What is the chance of getting number 6?



Content Summary

Probability of an event is a measure of how likely this event is to occur.

Probability is expressed as a fraction, decimal, or percentage.

The theoretical probability of an event E is calculated using the formula:

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Important notes:

- Probability ranges from 0 to 1.
- $P(E) = 0$ represents an impossible event.
- $P(E) = 1$ represents a certain event.
- If the probability is between 0 and 1, there are uncertain events.

In any probability question, when the following terms are used, great care must be taken. To understand the terms, let us take an example: a family of 4 children, where B represents a boy and G represents a girl.

- i) **at least:** 'At least two are boys', $B \geq 2$. Therefore, B can take values 2, 3 or 4.
- ii) **'at most':** 'At most two are boys', i.e. $B \leq 2$. Therefore, B can take values 2, 1 or 0.
- iii) **Not more than:** 'Not more than two boys' and 'at most two boys' mean the same.
- iv) **Not less than:** 'Not less than two boys' and 'at least two boys' mean the same.
- v) **More than:** 'More than two boys', i.e. $B > 2$. Therefore, B can take values 3 or 4.
- vi) **Less than:** 'Less than two boys', i.e. $B < 2$. Therefore, B can take values 0 or 1
- vii) **Equally likely:** Have equal probabilities.

Examples:

- 1) A fair coin is tossed once. Find the probability of getting Heads.

Solution:

Possible outcomes: Heads (H), Tails (T) → 2 outcomes

Favourable outcome: Heads → 1 outcome

$$P(H) = \frac{1}{2}$$

- 2) A fair die is rolled. Find the probability of getting an even number.

Solution:

Possible outcomes: {1, 2, 3, 4, 5, 6}

Even numbers: 2, 4, 6 → 3 outcomes.

$$P(\text{even}) = \frac{3}{6} = \frac{1}{2}$$

Example 3: Experiment: Toss a fair coin 50 times. **Result:** Heads = 28, Tails = 22.
Find: Experimental probability of Heads and Tails.

Solution:

$$P(\text{Heads}) = \frac{28}{50} = 0.56 \quad P(\text{Tails}) = \frac{22}{50} = 0.44$$

Observation: With more trials, the experimental probability (0.56) is closer to the theoretical probability (0.5).

Example 4: Experiment: Roll a die 60 times. **Result:** Number 4 appeared 8 times.
Find: Experimental probability of getting 4.

Solution:

$$P(4) = \frac{8}{60} = 0.133 \approx 0.13.$$

The theoretical probability of getting 4 = $\frac{1}{6} \approx 0.167$. As trials increase, the experimental probability tends to approach 0.167.



APPLICATION ACTIVITY 8.4

- 1) A bag contains 3 red balls, 5 blue balls, and 2 green balls. One ball is picked at random.
 - a) Find the probability of picking a red ball.
 - b) Find the probability of picking a blue ball.
- 2) In a class of 20 Students, 12 are girls. One learner is chosen at random. Find the probability that the learner is a girl.
- 3) True or False
 - a) The probability of an impossible event is 1.
 - b) A probability can be greater than 1.
 - c) If all outcomes are equally likely, probability can be expressed using a fraction.

8.5 Estimation of probability after increasing trials



Activity 8.5

Imagine you have a spinner that is divided into *four equal sections*: red, blue, yellow, and green. You suspect it might be a fair spinner, meaning each color has an equal chance of being landed on. You spin it 10 times and get the following results:

Color	Number of times
Red	6
Blue	2
Yellow	1
Green	1

- a) Based on these 10 spins, would you estimate the probability of landing on red to be exactly 60%?
- b) Is the spinner unfair? What do you think would happen to your estimate if you continued spinning and performed 100 spins instead of just 10? Would your confidence in your estimate change?



Content Summary

When we estimate probability through experiments, we are dealing with **experimental probability** (also called relative frequency). This is different from **theoretical probability**, which is what we expect to happen based on known possible outcomes.

Theoretical probability: What *should* happen. (e.g., The chance of landing on red with a fair spinner is $\frac{1}{4}$ or 25%.) because $P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$

Experimental probability: What *did* happen. Calculated as:

$$\text{Experimental Probability} = P(E) = \frac{\text{Number of times the event occurs}}{\text{Total number of trials}}.$$

The **Law of large numbers** states that as the number of trials in an experiment increases, the experimental probability of an event gets closer and closer to its theoretical probability.

In our initial 10 spins, the experimental probability for red is $\frac{6}{10} = 60\%$. This is far from the theoretical 25% because 10 is a small number of trials; the results are easily swayed by random chance. However, if we spin the spinner 100, 1,000, or 10,000 times, the overall proportion of reds will settle down and get very close to 25%. The early “lucky streak” of reds becomes less significant in the overall average.

Comparison of theoretical probability and experimental Probability

In general, if N is number of all possible outcomes of A and n is number of times a particular outcome appears, then $P(A) = \frac{n}{N}$

The more times an experiment is repeated, the closer the experimental probability is to the theoretical probability.

Increasing trials reduces the effect of chance fluctuations and gives a more reliable estimate.

Why is the law of large numbers important?

- **Predictability:** It shows that even though individual events are random, large-scale patterns are predictable.
- **Confidence:** It tells us that our estimates become more reliable and accurate when they are based on a larger amount of data.
- **Real-World Application:** This is why pollsters need to survey thousands of people, not just a few, to accurately predict an election result.

Examples:

- 1) A student flips a coin 20 times and gets 15 heads and 5 tails.
 - a) What is the experimental probability of getting heads based on these 20 flips?
 - b) If the coin is fair, what is the theoretical probability of getting heads?

- c) If the student flips the coin 500 more times (520 total), and the total number of heads becomes 270, what is the new experimental probability for heads? How does this compare to the theoretical probability?

Solution:

- a) Experimental Probability (Heads) = $15/20=0.75$ or 75%.
 b) Theoretical Probability (Heads) = $1/2=0.5$ or 50%.
 c) With 520 total flips and 270 heads:

New Experimental Probability = $270/520 \approx 0.519$ or 51.9%.

This new estimate (51.9%) is much closer to the theoretical probability of 50% than the initial estimate of 75% was. By increasing the number of trials, our experimental probability became a much more accurate reflection of the true probability.

- 2) A factory produces light bulbs. An inspector tests a batch of 50 bulbs and finds that 2 are defective.
- a) Based on this test, estimate the probability that a bulb from this factory is defective.
- b) The inspector then tests a much larger sample of 2,000 bulbs from the same production run and finds 110 defective bulbs. What is the new, more reliable estimate for the probability of a defective bulb?
- c) Why is the second estimate considered more reliable?

Solution:

- a) Estimated Probability (Defective) = $2/50=0.04$ or 4%.
 b) New Estimated Probability (Defective) = $110/2000= 0.055$ or 5.5%.
 c) The second estimate is more reliable because it is based on a much larger number of trials (2,000 vs. 50). According to the Law of Large Numbers, as the sample size increases, the experimental probability (the defect rate found in the sample) is a better estimator of the true population parameter (the actual defect rate of the entire factory's production).

- 3) A basketball player makes 7 out of her first 10 free throws in a game.
- a) A commentator says, "She's a 70% free-throw shooter tonight!" Is this a fair statement based on the data? Why or why not?
- b) By the end of the game, she has taken 40 free throws and made 28 of them. What is her final free-throw percentage for the game?

Solution:

- a) It is not a fair statement. Ten attempts is a very small sample size. A player's performance can fluctuate significantly over just a few number of shots due to random chance, fatigue, or pressure. You cannot reliably state her true ability for the entire game from just 10 shots.
- b) Final free-throw percentage = $28/40=0.70$ or 70%. (Interestingly, it remained at 70%).

**APPLICATION ACTIVITY 8.5**

- 1) A new video is posted online. After the first hour, it has been viewed 150 times and 120 "likes".
 - a) Based on the first hour, what is the experimental probability that a viewer will "like" the video?
 - b) After one week, the video has 15,000 views and 11,250 "likes". What is the experimental probability of a "like" after one week?
 - c) Which estimate—the one from the first hour or the one from the first week—do you think is a better predictor of the video's future "like" percentage? Explain your reasoning using the key concept from this lesson.
- 2) A pharmaceutical company is testing a new drug to see if it helps people recover from an illness faster.

Trial 1: They test the drug on 5 patients. All 5 recover quickly. The company claims the drug has a 100% success rate.

Trial 2: They expand the trial to 500 patients. Of these, 425 recovered quickly.

 - a) What is the experimental probability of recovery based on Trial 2?
 - b) Why is the result from Trial 2 a more realistic and trustworthy estimate of the drug's true effectiveness than the result from Trial 1?
 - c) If the recovery rate without any treatment is known to be 80%, does the drug appear to be effective? Justify your answer using the probability you calculated in (a).

8.6 Basic rules of probability



Activity 8.6

Discuss in pairs:

- What is the probability for getting heads when tossing a coin?
- What is the probability for getting a number greater than 6 when you roll a die?
- Discuss the small and the greater probability that exists. Can the probability of an event be greater than 1?



Content Summary

Rule 1: Probability range: The probability of any event lies between **0 and 1**.

This means for every event E , $0 \leq P(E) \leq 1$

Rule 2: Probability of an impossible event: An impossible event has probability 0.

Example: Getting number 7 on a standard die. $P(7)=0$

Rule 3: Probability of a certain event: A certain event has probability 1.

Example: Getting a number less than 7 on a die. $P(\text{number} < 7)=1$

Rule 4: Sum of probabilities of all outcomes: The sum of probabilities of all possible outcomes in a sample space is **1**.

Rule 5: Complementary events: The probability that an event does **not** occur is:

$$P(\text{not } E) = 1 - P(E)$$

Examples

- Can the probability of an event be **1.3**?

Solution:

No. Probability must lie between 0 and 1.

- The probability that it rains today is 0.40. Find the probability that it does **not** rain.

Solution:

$$P(\text{not rain}) = 1 - 0.4 = 0.6$$

- The probabilities of three outcomes are 0.20, 0.50, and x . Find x .

Solution:

$$0.2 + 0.5 + x = 1 \text{ then } x = 1 - 0.7 = 0.3.$$

Rule 6: For any two events A and B (even if they can occur together): The probability that A or B occurs is equal to the probability of A plus the probability of B, minus the probability that both A and B occur." It expresses Probability of union of two events A or B noted as

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Explanation

P(A) means «the probability of event A happening.»

P(B) means «the probability of event B happening.»

A $\cup$ B means «A **union** B» — that is, the event that A happens or B happens (or both).

A $\cap$ B means «A **intersection** B» — that is, the event that both A and B happen.

P(A $\cup$ B) means «the probability that A or B (or both) occurs.»

P(A $\cap$ B) means «the probability that both A and B occur.»

Example:

If A is "it rains today" and B is "it is cloudy today":

- P(A) = probability of rain
- P(B) = probability of cloudiness
- P(A $\cap$ B) = probability that it rains **and** is cloudy
- Then P(A $\cup$ B) = probability it rains **or** is cloudy (or both)

We don't want to count "rainy and cloudy" twice, so we subtract it once.

This is the important rule. It will be well developed in S2 together with other rules of probability.



APPLICATION ACTIVITY 8.6

- 1) A fair die is rolled:
 - a) Find the probability of getting a number greater than 6.
 - b) Find the probability of getting a number from 1 to 6.
- 2) The probability that a learner passes an exam is 0.75. Find the probability that the learner fails.
- 3) Answer by True or False:
 - a) The probability of a certain event is 0.
 - b) The sum of probabilities of all outcomes is always 1.
 - c) If P(E)=0.6, then P(not E)=0.4

**END OF UNIT ASSESSMENT****Assessing the knowledge and understanding**

- 1) Explain the following terms:
 - a) Experiment
 - b) Outcome
 - c) Event
 - d) Sample Space
- 2) State whether each of the following events is certain, impossible, or uncertain.
 - a) Getting a number less than 7 when rolling a die.
 - b) Getting number 8 when rolling a die.
 - c) Getting heads when tossing a coin.
- 3) Answer by True or False
 - a) The probability of an impossible event is 1.
 - b) The sum of probabilities of all outcomes in a sample space is 1.
 - c) Probability values can be greater than 1.

Assessing skills

- 4) A bag contains 4 red balls and 6 blue balls. One ball is picked at random. Find the probability of picking a red ball.
- 5) A coin is tossed 40 times and heads appears 18 times. Find the experimental probability of getting heads.
- 6) The probability that a learner passes a test is 0.65.
 - a) Find the probability that the learner fails.
 - b) State the rule of probability used.
- 7) A die is rolled once.
 - (a) Write the sample space.
 - (b) Find the probability of getting an odd number.

Assessing values and attitudes

- 8) Why is it important to record results honestly when carrying out a probability experiment?
- 9) During a class experiment, Students work in groups to toss a coin many times.
 - a) State one value Students develop by working in groups.
 - b) Explain how this value helps in learning probability.
 - c) Explain why increasing the number of trials gives a better estimate of probability.

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